# Kube-OVN Document

**v1.17.0**

*Kube-OVN Team*

*2025 Kube-OVN Team*

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# 1. Kube-OVN




## 1.1 What is Kube-OVN?

Kube-OVN ■■■ CNCF ■■■■■■■■■■■■■■■■■■■■ SDN ■■■■■■■■■■■■■■■■■■■

### 1.3.1 □□□ eBPF □□□

□□ Cilium □ Calico eBPF□

Kube-OVN □□ Open vSwitch □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

### 1.3.2 □□□ CNI, Ingress, Service Mesh □ Observability All in One □□□

□□ Cilium□

Kube-OVN □□□□□ CNI □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

### 1.3.3 □□□□ OpenShift □□□

□□ ovn-kubernetes□

□ OpenShift □□□□□□ CNI □□□□ Cluster Network Operator □□□Kube-OVN □□□□□□□□□□□□□□□□□□□□□□□□ RedHat □□□□□□□□□□□□ Kubernetes □□□□□□□□□□□□□□□□□□□□

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# 1.5 ⬚⬚

# 2. □□□□

## 2.1 □□□□

Kube-OVN □□□□□ CNI □□□□□□□□□□□□□□□□ Kubernetes □□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

### 2.1.1 □□□□

- Kubernetes >= 1.29□
- Docker >= 1.12.6, Containerd >= 1.3.4□
- □□□□: CentOS 7/8, Ubuntu 16.04/18.04/20.04□
- □□ Linux □□□□□□□□□□□□□□□□□□□□□ `geneve`, `openvswitch`, `ip_tables` □ `iptable_nat` □Kube-OVN □□□□□□□□□□□□□

□□□□□

1. □□□□□□□□ 3.10.0-862□□□ `netfilter` □□□□□ bug □□□ Kube-OVN □□□□□□□□□□□□□□□□□□□□□□□□□□□□ CentOS □□□□□□□□□□□□□□□□□□□□□□□□□ bug □□ Floating IPs broken after kernel upgrade to Centos/RHEL 7.5 - DNAT not working□
2. Rocky Linux 8.6 □□□ 4.18.0-372.9.1.el8.x86_64 □□ TCP □□□□□ TCP connection failed in Rocky Linux 8.6□□□□□□□□ 4.18.0-372.13.1.el8_6.x86_64 □□□□□□□
3. □□□□□□□□ 4.4□□□□□□□□ `openvswitch` □□□□□□□□□□□□□□□□□□ `openvswitch` □□□□□□□□□□□□□
4. Geneve □□□□□□□□□ IPv6□□□□□ `cat /proc/cmdline` □□□□□□□□□□ □□□□□ bug □□□ Geneve tunnels don't work when ipv6 is disabled□

### 2.1.2 □□□□

- □□□□□□□□□ IPv6□□□□□□□□□□□□□ `ipv6.disable=1` □□□□□□□□ 0□
- `kube-proxy` □□□□□□Kube-OVN □□□□□ Service ClusterIP □□□ `kube-apiserver` □
- □□ kubelet □□□□□□□□ CNI□□□□□□□□□□□□□, kubelet □□□□□□□□□□□□ `--network-plugin=cni --cni-bin-dir=/opt/cni/bin --cni-conf-dir=/etc/cni/net.d` □
- □□□□□□□□□□□□□□□□□□□□

```
#    IPv4 Masquerade
firewall-cmd --add-masquerade --permanent
#   Kube-OVN IPv6/       Masquerade
firewall-cmd --permanent --add-rich-rule 'rule family="ipv6" source address="fd00:10:16::/112" masquerade'

firewall-cmd --reload
```

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## 2.1.4 ⬚⬚

## 2.2 ████

Kube-OVN ██████████ Charts ████████████████████████ Kube-OVN ██████████ Overlay ████

██████████ Underlay/Vlan ██████ Underlay ████████████ ████████

██████████████████████████ Kube-OVN ████

### 2.2.1 ████

███████

████████████ release ████████████████████

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/refs/tags/v1.17.0/dist/images/install.sh
```

███ master ████████████████████████████

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/images/install.sh
```

███████

████████████████

```
REGISTRY="docker.io/kubeovn"           #
VERSION="v1.17.0"                      #     /Tag
POD_CIDR="10.16.0.0/16"                #      CIDR     SVC/NODE/JOIN CIDR
SVC_CIDR="10.96.0.0/12"                #     apiserver   service-cluster-ip-range
JOIN_CIDR="100.64.0.0/16"              # Pod         CIDR     SVC/NODE/POD CIDR
LABEL="node-role.kubernetes.io/control-plane" #    OVN DB
IFACE=""                               #                       Kubernetes    Node IP
TUNNEL_TYPE="geneve"                   #          geneve, vxlan   stt stt        ovs
```

████████████████ `IFACE=enp6s0f0,eth.*` ▯

███████

> ███████ root ▯▯

`bash install.sh`

████████

▯▯

████████ Kube-OVN ██████████

1. ████ `[Step 4/6]` ██████████ Pod████████████████████████████████████
2. ██████ ███ Kube-OVN ████████████████████████████████████████████████

### 2.2.2 Helm Chart ▯▯

▯▯ Kube-OVN ██████████████ Helm ▯▯ Kube-OVN ████████████

████ **IP** ▯▯

```
# kubectl get node -o wide
NAME                   STATUS     ROLES           AGE   VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE             KERNEL-VERSION      CONTAINER-RUNTIME
kube-ovn-control-plane   NotReady   control-plane   20h   v1.26.0   172.18.0.3    <none>        Ubuntu 22.04.1 LTS   5.10.104-linuxkit   containerd://1.6.9
kube-ovn-worker          NotReady   <none>          20h   v1.26.0   172.18.0.2    <none>        Ubuntu 22.04.1 LTS   5.10.104-linuxkit   containerd://1.6.9
```



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⬚⬚⬚⬚ **label**

```
# kubectl label node -lbeta.kubernetes.io/os=linux kubernetes.io/os=linux --overwrite
node/kube-ovn-control-plane not labeled
node/kube-ovn-worker not labeled

# kubectl label node -lnode-role.kubernetes.io/control-plane kube-ovn/role=master --overwrite
node/kube-ovn-control-plane labeled

#    label    dpdk        dpdk
# kubectl label node -lovn.kubernetes.io/ovs_dp_type!=userspace ovn.kubernetes.io/ovs_dp_type=kernel --overwrite
node/kube-ovn-control-plane labeled
node/kube-ovn-worker labeled
```

⬚⬚ **Helm Repo** ⬚⬚

```
# helm repo add kubeovn https://kubeovn.github.io/kube-ovn/
"kubeovn" has been added to your repositories

# helm repo list
NAME            URL
kubeovn         https://kubeovn.github.io/kube-ovn/

# helm repo update kubeovn
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "kubeovn" chart repository
Update Complete. ⎈Happy Helming!⎈

# helm search repo kubeovn
NAME                    CHART VERSION   APP VERSION     DESCRIPTION
kubeovn/kube-ovn        v1.17.0         v1.17.0         Helm chart for Kube-OVN
```

⬚⬚ **helm install ⬚⬚ Kube-OVN**

Chart ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `values.yaml` ⬚⬚⬚⬚⬚⬚⬚⬚

```
# helm install kube-ovn kubeovn/kube-ovn --wait -n kube-system --version v1.17.0
NAME: kube-ovn
LAST DEPLOYED: Thu Apr 24 08:30:13 2025
NAMESPACE: kube-system
STATUS: deployed
REVISION: 1
TEST SUITE: None
```

⬚⬚

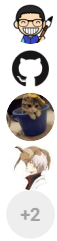
⬚⬚⬚⬚⬚ ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ **Helm** ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `values.yaml` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
helm upgrade -f values.yaml kube-ovn kubeovn/kube-ovn --wait -n kube-system --version v1.17.0
```

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## 2.3 Underlay ▯▯▯▯

▯▯▯▯▯ Kube-OVN ▯▯▯▯▯▯▯ Geneve ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Overlay ▯▯▯

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Kube-OVN ▯▯▯▯▯▯▯▯▯ Underlay ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

6. ■■■■■■■■ Linux Bridge ■■■■■■

■■■■■■■■■■ Kube-OVN Underlay ■■■■■■■■■■

▒▒▒**ProviderNetwork** ▒▒▒▒▒▒▒▒▒▒ **12** ▒▒▒▒

- `defaultInterface` : ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ Bond ▒▒▒▒ Vlan ▒▒▒▒ProviderNetwork ▒▒▒▒▒▒▒▒▒▒▒▒ excludeNodes ▒▒▒▒▒▒▒▒▒ br-net1▒▒▒▒ `br-NAME` ▒▒ OVS ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒
- `customInterfaces` : ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒
- `nodeSelector` : ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ OVS ▒▒▒▒▒▒▒▒▒ `matchLabels` ▒ `matchExpressions` ▒▒▒▒▒▒▒▒
- `excludeNodes` : ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ `net1.provider-network.ovn.kubernetes.io/exclude=true` ▒▒▒▒▒▒▒▒▒▒▒▒▒ **`nodeSelector`** ▒ **`excludeNodes`** ▒▒▒▒▒▒▒▒▒▒▒▒▒ **`nodeSelector`** ▒
- `autoCreateVlanSubinterfaces` : ▒▒▒▒▒▒▒▒ `false` ▒▒▒▒ `true` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ VLAN ▒▒▒▒▒▒▒ `<interface>.<vlanid>` ▒▒▒▒▒▒▒▒▒

▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

| Key | Value | ▒▒ |
| --- | --- | --- |
| net1.provider-network.ovn.kubernetes.io/ready | true | ▒▒▒▒▒▒▒▒▒▒▒▒▒▒ProviderNetwork ▒▒▒▒▒▒▒ |
| net1.provider-network.ovn.kubernetes.io/interface | eth1 | ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ |
| net1.provider-network.ovn.kubernetes.io/mtu | 1500 | ▒▒▒▒▒▒▒▒▒▒▒▒ MTU |

> ▒▒▒▒▒▒▒▒▒▒▒▒▒▒ IP▒▒ IP ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ OVS ▒▒▒▒

### ▒▒▒▒ VLAN ▒▒▒▒

▒ ProviderNetwork ▒▒▒▒ `autoCreateVlanSubinterfaces: true` ▒▒▒▒▒▒▒▒▒▒ VLAN ▒▒▒▒▒▒▒▒ VLAN ▒▒▒▒▒▒▒▒▒▒

▒▒▒▒

```yaml
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: vlan-provider
spec:
  defaultInterface: eth0.341
  autoCreateVlanSubinterfaces: true
  customInterfaces:
    - interface: eth2.341
      nodes:
        - worker-node-1
        - worker-node-2
```

▒▒▒▒

▒▒▒▒▒▒▒▒ `<interface>.<vlanid>` ▒▒▒▒ VLAN ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ VLAN ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ ProviderNetwork ▒▒▒▒▒▒▒▒▒▒▒▒▒

### ▒▒ VLAN

Vlan ▒▒▒▒ Vlan Tag ▒ ProviderNetwork ▒▒▒▒▒▒▒▒▒

▒▒▒▒ VLAN ▒▒▒▒

```yaml
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan1
spec:
  id: 0
  provider: net1
```

- `id` : ▒ VLAN ID/Tag▒Kube-OVN ▒▒▒▒ Vlan ▒▒▒▒▒▒▒ Vlan ▒▒▒▒ 0 ▒▒▒▒▒▒▒▒▒▒vlan ▒▒▒▒▒▒▒▒ localnet ▒▒▒
- `provider` : ▒▒▒▒▒▒▒ ProviderNetwork ▒▒▒▒▒▒▒▒▒ VLAN ▒▒▒▒▒▒▒▒ ProviderNetwork▒

### ▒▒ Subnet

▒ Vlan ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  vlan: vlan1
  disableGatewayCheck: false
```

- `vlan` □□□□□□ VLAN □□□□□ Subnet □□□□□□□ VLAN□
- `disableGatewayCheck` □□□□ Underlay □□□□□□□□□□□□□□□□□ `true` □□□□□□□□

### 2.3.6 □□□□

□□□□□□□□□□□□□□□□□□□□ IP □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

□□□□ IP □□□□□□ Pod □□ IP □ Mac

### 2.3.7 □□□□□□□

□□□□□□□□□□□□□□□Kube-OVN □□□ Underlay □□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□ `spec.logicalGateway` □ `true` □□□

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  vlan: vlan1
  logicalGateway: true
```

□□□□□□□□Pod □□□□□□□□□□□□□ Kube-OVN □□□□□□□□□□Logical Router□□□□□□□□□□□□□□□□

### 2.3.8 Underlay □ Overlay □□□□

□□□□□□□□□□□ Underlay □ Overlay □□□□□□□□□ Overlay □□□□□ Pod □□□□□□□□ NAT □□□□□□ Underlay □□□□□ Pod IP□□ □ Underlay □□□ Pod □□ Overlay □□□□□□□□□□□□□□□□□□□□

```
      dhcp4: no
      addresses:
        - 172.16.143.129/24
  version: 2
```

▯▯▯▯▯▯▯▯▯ IP ▯▯▯▯▯▯▯▯▯▯▯▯ netplan ▯▯▯▯▯▯▯▯▯▯▯

```
netplan generate

nmcli connection reload netplan-eth0
nmcli device set eth0 managed yes
```

▯▯▯▯▯▯▯▯▯Kube-OVN ▯▯▯▯▯▯▯ IP ▯▯▯▯▯▯▯▯▯ OVS ▯▯▯

▯▯▯▯▯▯▯▯▯▯▯▯▯ NetworkManager ▯▯▯▯▯▯▯ CentOS▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
nmcli connection reload eth0
nmcli device set eth0 managed yes
nmcli -t -f GENERAL.STATE device show eth0 | grep -qw unmanaged || nmcli device reapply eth0
```

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ IP ▯▯▯▯▯▯▯▯ MAC ▯▯▯▯▯▯▯

## 2.3.10 ▯▯▯▯

▯▯▯▯▯▯▯ **hairpin** ▯ **Pod** ▯▯▯▯▯

▯▯▯▯▯▯▯▯ hairpin ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Pod ▯▯▯▯▯▯▯▯▯Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ OVS ▯▯▯▯▯▯ MAC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

▯▯▯▯▯▯▯▯▯▯▯▯▯ hairpin▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Kube-OVN ▯▯▯▯

**Pod ▯▯▯▯▯▯▯▯ Pod ▯▯▯▯▯▯▯**

▯▯▯▯▯▯▯▯▯▯▯▯ Pod ▯▯▯▯▯▯▯▯▯ 300▯▯▯▯▯▯▯▯▯ ARP ▯▯▯▯▯ OVS ▯▯ resubmit ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
2022-11-13T08:43:46.782Z|00222|ofproto_dpif_upcall(handler5)|WARN|Flow: arp,in_port=331,vlan_tci=0x0000,dl_src=00:00:00:25:eb:
39,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.213.131.240,arp_tpa=10.213.159.254,arp_op=1,arp_sha=00:00:00:25:eb:39,arp_tha=ff:ff:ff:ff:ff:ff

bridge("br-int")
----------------
 0. No match.
     >>>> received packet on unknown port 331 <<<<
    drop

Final flow: unchanged
Megaflow: recirc_id=0,eth,arp,in_port=331,dl_src=00:00:00:25:eb:39
Datapath actions: drop
2022-11-13T08:44:34.077Z|00224|ofproto_dpif_xlate(handler5)|WARN|over 4096 resubmit actions on bridge br-int while processing
arp,in_port=13483,vlan_tci=0x0000,dl_src=00:00:00:59:ef:
13,dl_dst=ff:ff:ff:ff:ff:ff,arp_spa=10.213.152.3,arp_tpa=10.213.159.254,arp_op=1,arp_sha=00:00:00:59:ef:13,arp_tha=ff:ff:ff:ff:ff:ff
```

▯▯▯▯▯▯▯▯▯▯▯ OVN NB ▯▯ `bcast_arp_req_flood` ▯ `false` ▯

```
kubectl ko nbctl set NB_Global . options:bcast_arp_req_flood=false
```

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## 2.3.11 ▯▯

# 2.4 ▯▯ CNI ▯▯

▯▯▯▯▯▯▯▯ Kube-OVN ▯▯▯▯ CNI ▯▯▯▯ CNI ▯▯▯▯ Cilium▯Calico ▯ Flannel▯▯▯▯▯▯

## 2.4.1 ▯▯

▯▯▯ CNI ▯▯▯▯Kube-OVN ▯▯▯▯ CNI ▯▯▯▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ CNI ▯▯▯▯▯▯▯▯eth0▯▯▯▯▯▯▯

- ▯▯▯ CNI ▯▯▯▯▯ Kube-OVN ▯▯▯▯▯▯▯▯VPC▯▯▯▯▯▯▯▯
- ▯▯▯▯▯▯▯▯▯▯▯
- ▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯
- ▯▯ Kube-OVN ▯ VPC NAT ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

## 2.4.2 ▯▯▯▯

▯▯▯▯

1. ▯ **CNI**▯▯▯▯ Kubernetes ▯▯▯ CNI▯Cilium▯Calico▯Flannel ▯▯
2. **Multus CNI**▯▯▯▯▯▯▯▯▯▯
3. **Kube-OVN**▯▯▯▯▯▯▯▯▯

▯▯▯▯

1. ▯▯▯▯▯▯▯ CNI
2. ▯▯ Multus CNI
3. ▯▯▯▯▯▯▯▯ Kube-OVN

## 2.4.3 ▯▯

**▯▯ Helm Chart v2**

```
# values.yaml
cni:
  nonPrimaryCNI: true
```

▯▯ Helm ▯▯▯

```
helm install kube-ovn ./charts/kube-ovn-v2 \
  --namespace kube-system \
  --set cni.nonPrimaryCNI=true
```

**▯▯ Helm Chart v1**

```
# values.yaml
cni_conf:
  NON_PRIMARY_CNI: true
```

▯▯ Helm ▯▯▯

```
helm install kube-ovn ./charts/kube-ovn \
  --namespace kube-system \
  --set cni_conf.NON_PRIMARY_CNI=true
```

▯▯▯▯

▯ `kube-ovn-controller` Deployment ▯ `kube-ovn-cni` DaemonSet ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯



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```
# kube-ovn-controller Deployment
containers:
- name: kube-ovn-controller
  args:
  - --non-primary-cni-mode=true
```

```
# kube-ovn-cni DaemonSet
containers:
- name: kube-ovn-cni
  args:
  - --non-primary-cni-mode=true
```

## 2.4.4 ▯▯

▯▯▯▯▯▯▯**NADs**▯

▯▯ NAD ▯▯▯▯▯▯▯▯▯▯▯▯

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: kube-ovn-vpc-network
  namespace: default
spec:
  config: |
    {
      "cniVersion": "0.3.1",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "kube-ovn-vpc-network.default.ovn"
    }
```

**VPC** ▯▯▯▯▯

▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc-secondary
spec:
  namespaces:
  - default
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet-secondary
spec:
  vpc: vpc-secondary
  cidr: "10.100.0.0/16"
  gateway: "10.100.0.1"
  provider: kube-ovn-vpc-network.default.ovn
```

**Pod** ▯▯

▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
apiVersion: v1
kind: Pod
metadata:
  name: multi-network-pod
  annotations:
    k8s.v1.cni.cncf.io/networks: default/kube-ovn-vpc-network
spec:
  containers:
  - name: app
    image: nginx
```

## 2.4.5 ▯▯▯▯▯▯

▯▯▯▯**eth0**▯

- ▯▯ CNI▯Cilium▯Calico ▯▯▯▯
- ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯



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- ███ Kubernetes ██████████

██████**net1**、**net2...**、

- ▯ Kube-OVN ▯▯
- ▯▯▯▯▯▯▯▯▯
- ▯▯▯▯▯▯▯▯ VPC ▯▯▯
- ▯▯ Kube-OVN ▯▯▯▯▯▯▯▯QoS ▯ NAT

## 2.4.6 ▯▯▯▯

**1.** ▯▯▯▯

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```
#    Pod
apiVersion: v1
kind: Pod
metadata:
  name: frontend
  annotations:
    k8s.v1.cni.cncf.io/networks: |
      [
        {"name": "default/public-network"},
        {"name": "default/internal-network"}
      ]
```

**2.** ▯▯▯

▯▯▯▯▯▯▯▯▯▯

```
#    A Pod
apiVersion: v1
kind: Pod
metadata:
  name: tenant-a-app
  annotations:
    k8s.v1.cni.cncf.io/networks: default/tenant-a-network

---
#    B Pod
apiVersion: v1
kind: Pod
metadata:
  name: tenant-b-app
  annotations:
    k8s.v1.cni.cncf.io/networks: default/tenant-b-network
```

**3. VPC NAT** ▯▯

▯▯▯▯▯▯▯▯▯▯▯▯▯

```
apiVersion: kubeovn.io/v1
kind: VpcNatGateway
metadata:
  name: vpc-nat-gw
spec:
  vpc: vpc-secondary
  subnet: subnet-secondary
  lanIp: "10.100.0.254"
```

## 2.4.7 ▯▯▯▯

▯▯▯▯▯▯▯

Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
metadata:
  annotations:
    k8s.v1.cni.cncf.io/networks: |
      [
```

```
        {"name": "default/kube-ovn-network-1", "interface": "net1"},
        {"name": "default/kube-ovn-network-2", "interface": "net2"}
      ]
```

**IP** ■■■■

■■■■■■■ IP ■■■

```
#    IP
metadata:
  annotations:
    k8s.v1.cni.cncf.io/networks: |
      [{
        "name": "default/kube-ovn-network",
        "ips": ["10.100.0.100/16"]
      }]
```

■■■■■**QoS**■

■■■■■■■ QoS ■■■

```
metadata:
  annotations:
    ovn.kubernetes.io/ingress_rate: "1000"
    ovn.kubernetes.io/egress_rate: "1000"
```

## 2.4.8 ■■■■

■■■■

1. **Pod** ■■■■■■■■ - ■■ NAD ■■■■■■■■■■■■■■■ - ■■ kube-ovn-cni ■■■■■■■ - ■■ Multus ■■■■■■
2. ■■■■■■■■ - ■■■■■■ VPC ■■ - ■■ Pod ■■■■■■ - ■■■■■■■■■■■■■
3. **IP** ■■■■■ - ■■■■■ CIDR ■■■■■■ - ■■■■■ IP ■■ - ■■ IPAM ■■

■■■■

```
#    Pod
kubectl get pod <pod-name> -o yaml | grep -A 10 "networks-status"

#    Pod
kubectl exec <pod-name> -- ip addr show

#
kubectl exec <pod-name> -- ip route show

#    kube-ovn-cni
kubectl logs -n kube-system daemonset/kube-ovn-cni
```

## 2.4.9 ■■

1. ■■■■■■■■■■■■■■ Kubernetes ■■■■■
2. ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
3. **DNS**■Pod DNS ■■■■■■■■■■

## 2.4.10 ■■■■■

1. ■■■■■■■■■■■■■■■■■■■■ IP ■■
2. ■■■■■■■■■■■■■■■■■■■

## 2.4.11 ⬚⬚

- Multus CNI ⬚⬚
- ⬚⬚⬚⬚⬚⬚⬚⬚
- Kube-OVN ⬚⬚

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## 2.4.12 ⬚⬚

# 2.5 Talos ⬚⬚

Talos Linux ⬚⬚ Kubernetes ⬚⬚⬚⬚ Linux ⬚⬚⬚

## 2.5.1 ⬚⬚ Helm Chart ⬚⬚ Kube-OVN

⬚⬚⬚⬚⬚⬚⬚⬚⬚ Talos Linux ⬚⬚⬚⬚ Kube-OVN⬚

```
helm install kube-ovn kubeovn/kube-ovn --wait \
    -n kube-system \
    --version v1.17.0 \
    --set OVN_DIR=/var/lib/ovn \
    --set OPENVSWITCH_DIR=/var/lib/openvswitch \
    --set DISABLE_MODULES_MANAGEMENT=true \
    --set cni_conf.MOUNT_LOCAL_BIN_DIR=false
```

⬚⬚⬚⬚⬚⬚ Underlay ⬚⬚⬚⬚⬚⬚⬚⬚⬚ Helm ⬚⬚⬚⬚⬚⬚ Chart ⬚⬚⬚⬚⬚⬚⬚

```
helm install kubeovn kubeovn/kube-ovn --wait \
    -n kube-system \
    --version v1.17.0 \
    --set OVN_DIR=/var/lib/ovn \
    --set OPENVSWITCH_DIR=/var/lib/openvswitch \
    --set DISABLE_MODULES_MANAGEMENT=true \
    --set cni_conf.MOUNT_LOCAL_BIN_DIR=false \
    --set networking.NETWORK_TYPE=vlan \
    --set networking.vlan.VLAN_INTERFACE_NAME=enp0s5f1 \
    --set networking.vlan.VLAN_ID=0 \    # VLAN_ID=0      VLAN tag    untagged      chart      100
    --set networking.NET_STACK=ipv4 \
    --set-json networking.EXCLUDE_IPS='"172.99.99.11..172.99.99.99"' \
    --set-json ipv4.POD_CIDR='"172.99.99.0/24"' \
    --set-json ipv4.POD_GATEWAY='"172.99.99.1"'
```

**Note**

VLAN⬚Bond⬚Bridge ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ Underlay ⬚⬚⬚⬚⬚⬚⬚⬚Underlay ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ Talos ⬚⬚⬚⬚⬚⬚ `ignore=true` ⬚⬚⬚⬚⬚⬚

```
machine:
  network:
    interfaces:
      - interface: enp0s5f1
        ignore: true
```

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## 2.5.2 ⬚⬚



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# 2.6 □□□□□

Kube-OVN □□□□□

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## 2.6.7 ⬜⬜

# 2.7 ▯▯

▯▯▯▯▯▯ Kube-OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Kube-OVN ▯▯▯▯ OVS ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯ issue ▯▯▯▯▯▯▯▯▯▯ Kube-OVN ▯▯▯▯▯▯▯▯▯▯▯▯

## 2.7.1 ▯▯▯ Kubernetes ▯▯▯▯▯▯

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

**Script Uninstall** **Helm Uninstall**

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/cleanup.sh
bash cleanup.sh

helm uninstall kube-ovn -n kube-system
```

## 2.7.2 ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ovsdb ▯▯ openvswitch ▯▯▯▯▯▯▯

```
rm -rf /var/run/openvswitch
rm -rf /var/run/ovn
rm -rf /etc/origin/openvswitch/
rm -rf /etc/origin/ovn/
rm -rf /etc/cni/net.d/01-kube-ovn.conflist
#       v1.10         00-kube-ovn.conflist
rm -rf /etc/cni/net.d/00-kube-ovn.conflist
rm -rf /var/log/openvswitch
rm -rf /var/log/ovn
rm -fr /var/log/kube-ovn
```

## 2.7.3 ▯▯▯▯

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯iptable/ipset ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

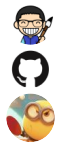
```
reboot
```

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## 2.7.4 ▯▯



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# 3. □□□□

## 3.1 □□□□

□□□ Kube-OVN □□□□□□□□□□□□□□□□Kube-OVN □□□□□□□ IP □□□□□□□□ Namespace □□□□□□□□□□□□ Namespace □□ Pod □□□□□□□□□□□□□ IP □□□□□□□□□□□□CIDR□□□□□□□□□□□□□NAT □□□□□

□□□ CNI □□□□□□□□□□□□□□□□□□□□ Kube-OVN □□□□□□□□□□□□□□□□□□□□

```
  selfLink: /apis/kubeovn.io/v1/subnets/ovn-default
  uid: 7e2451f8-fb44-4f7f-b3e0-cfd27f6fd5d6
spec:
  cidrBlock: 10.16.0.0/16
  default: true
  excludeIps:
  - 10.16.0.1
  gateway: 10.16.0.1
  gatewayType: distributed
  natOutgoing: true
  private: false
  protocol: IPv4
```

## 3.1.2 Join ⬚⬚

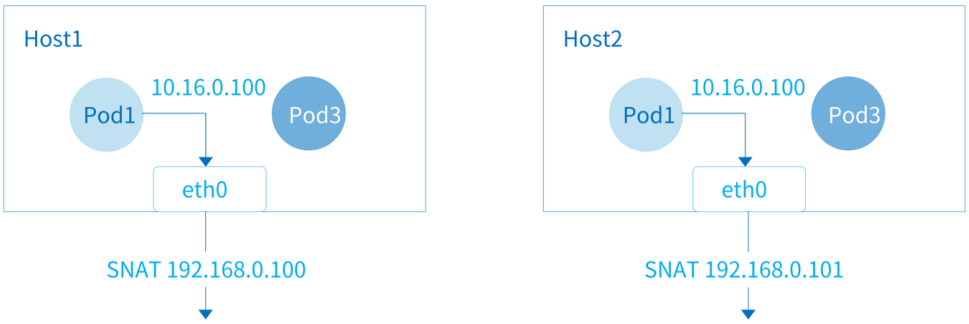
⬚ Kubernetes ⬚⬚⬚⬚⬚⬚⬚⬚ Node ⬚⬚⬚⬚⬚⬚ Pod ⬚⬚⬚⬚⬚ ⬚⬚⬚ Overlay ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ Kube-OVN ⬚⬚⬚⬚⬚ `join` ⬚⬚⬚ ⬚⬚⬚⬚ Node ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ ovn0 ⬚⬚ `join` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ ⬚ Pod ⬚⬚⬚⬚⬚⬚⬚⬚

⬚⬚ Pods ⬚ Nodes ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ ovn0 ⬚⬚⬚⬚⬚Node ⬚⬚ Pod ⬚⬚⬚ ovn0 ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ ovn0 ⬚⬚⬚⬚⬚⬚⬚⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

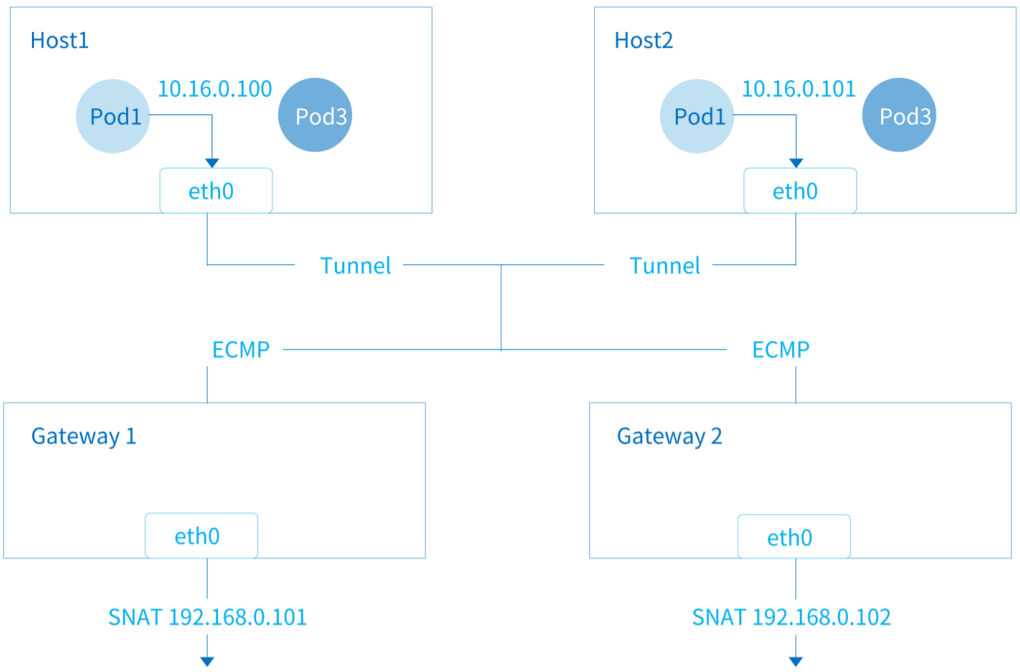
```
  - 10.66.0.1..10.66.0.10
  - 10.66.0.101..10.66.0.151
  gateway: 10.66.0.1
  gatewayType: distributed
  natOutgoing: true
  routeTable: ""
  namespaces:
  - ns1
  - ns2
EOF
```

- `cidrBlock` : ▒▒ CIDR ▒▒▒▒▒▒ VPC ▒▒▒▒ Subnet CIDR ▒▒▒▒▒
- `excludeIps` : ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ IP ▒▒▒▒▒▒▒▒▒ Underlay ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒
- `gateway` ▒▒▒▒▒▒▒▒▒▒Overlay ▒▒▒ Kube-OVN ▒▒▒▒▒▒▒▒▒▒▒▒▒▒Underlay ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒
- `namespaces` : ▒▒▒▒▒▒ Namespace ▒▒▒▒▒▒ Namespace ▒▒ Pod ▒▒▒▒▒▒▒▒▒▒▒▒▒
- `routeTable` : ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒




⬚⬚⬚⬚⬚⬚⬚ gatewayType ⬚⬚⬚ distributed ⬚

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: distributed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: distributed
  natOutgoing: true
```




IP Pod ovn0 natOutgoing true Pod IP

gatewayType centralized gatewayNode Kubernetes NodeName gatewayNode

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: centralized
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNode: "node1,node2"
  natOutgoing: true
```

- gatewayNode kube-ovn-worker:172.18.0.2, kube-ovn-control-plane:172.18.0.3
- ECMP ECMP
- Kube-OVN v1.12.0 subnet crd spec enableEcmp ECMP ECMP kube-ovn-controller Deployment enable-ecmp v1.12.0

> 

ECMP kube-ovn-controller ping 5s 5s-10s

Node Ready

gatewayNodeSelectors



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**gatewayNodeSelectors □□□□**

- □□ `gatewayNode` □□□□□□□□□ gatewayNode □ `gatewayNodeSelectors` □□□□□
- □□□□□□□□ OR □□□□□□□□□□□□□□□□□□□□□□□□□□□□□
- □□□□□□□□□□□□□□□□□□□□□□□□□□□□□

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: centralized-selector
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNodeSelectors:
    - matchLabels:
        role: gateway
    - matchExpressions:
        - key: node-type
          operator: In
          values: ["gateway", "egress"]
  natOutgoing: true
```

## 3.1.5 □□ ACL □□

**Warning**

Kube-OVN □□□□ NetworkPolicy□Network Policy API□Subnet ACL □ Security Group □□□□□□□□□□□□□□□□□□ OVN ACL □□□□□□ NetworkPolicy □ NetworkPolicy API □□□□ □□□□□□□□□□□□□□□□□□□□

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: private
spec:
  protocol: IPv4
  default: false
  namespaces:
  - ns1
  - ns2
  cidrBlock: 10.69.0.0/16
  private: true
  allowSubnets:
  - 10.16.0.0/16
  - 10.18.0.0/16
```

### 3.1.7 Underlay ▯▯▯▯

> ▯▯▯▯▯▯▯ Underlay ▯▯▯▯▯▯▯

- `vlan`：▯▯▯▯ Underlay ▯▯▯▯▯▯▯▯▯▯▯▯ Subnet ▯▯▯ Vlan CR ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Underlay ▯▯▯
- `logicalGateway`：▯▯ Underlay ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Underlay ▯ Overlay ▯▯▯▯▯▯▯▯▯▯▯ `false` ▯

### 3.1.8 ▯▯▯▯▯▯▯

▯▯▯▯▯ `kube-ovn-cni` ▯▯▯ Pod ▯▯▯▯▯ ICMP ▯ ARP ▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Underlay ▯▯▯▯▯▯▯▯▯ ICMP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: disable-gw-check
spec:
  disableGatewayCheck: true
```

### 3.1.9 Multicast-Snoop ▯▯

▯▯▯▯▯ subnet ▯▯ Pod ▯▯▯▯▯▯▯▯▯▯OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Pod▯▯▯▯▯ subnet ▯ multicast snoop ▯▯▯OVN ▯▯▯ `South Database` ▯▯▯▯▯ `Multicast_Group` ▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sample1
spec:
  enableMulticastSnoop: true
```

### 3.1.10 Subnet MTU ▯▯

▯▯ Subnet ▯ Pod ▯ MTU▯▯▯▯▯▯▯▯▯ Subnet ▯▯ Pod ▯▯▯

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sample1
spec:
  mtu: 1300
```

### 3.1.11 ▯▯▯▯▯▯▯

- IP ▯▯▯
- ▯▯ VPC NAT ▯▯▯▯
- QoS ▯▯
- ▯▯▯▯▯▯
- DHCP ▯▯
- ▯▯▯▯▯▯▯



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- ⬚⬚⬚⬚⬚⬚⬚
- ⬚⬚ IP ⬚⬚

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## 3.1.12 ⬚⬚

## 3.2 □□□□

Kube-OVN □□□□□ Pod □□ Namespace □□□□□□□□□□ IP □ Mac□ □□□□□□□□□□□□□□□□□Kube-OVN □□□□□□□□□□□□□□□□□□□□□□□□

- □□ Pod □□ IP/Mac□
- □□□□□□□□□□□□□□ IP/MAC□□□□□□□□□□□□□□□□□□□□□□
- Workload □□ IP Pool □□□□□□□□□□□□
- StatefulSet □□□□□□
- KubeVirt VM □□□□□□
- □□ Multus □□□□□□□□□□□□

### 3.2.1 □□ Pod □□ IP □ Mac

□□□□□ Pod □□□ annotation □□□ Pod □□□□□□□ IP/Mac, `kube-ovn-controller` □□□□□□□□□□□□□□□□□□□□

## 3.2.3 Workload □□ IP Pool □□□□

Kube-OVN □□□□ annotation `ovn.kubernetes.io/ip_pool` □ Workload□Deployment/StatefulSet/DaemonSet/Job/CronJob□□□□ IP□ `kube-ovn-controller` □□□□□ `ovn.kubernetes.io/ip_pool` □□□□ IP □□□□□□□□□

IP Pool □ Annotation □□□□ `template` □□ `annotation` □□□□□ Kubernetes □□□ Workload □□□ □□□□□□□□ Workload □□□□□□□□□□□□□□□□□□□□□□□

**Deployment □□ IP □□**

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ippool
  labels:
    app: ippool
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ippool
  template:
    metadata:
      labels:
        app: ippool
      annotations:
        ovn.kubernetes.io/ip_pool: 10.16.0.15,10.16.0.16,10.16.0.17 #             10.16.0.15,fd00:10:16::000E;10.16.0.16,fd00:10:16::000F;
10.16.0.17,fd00:10:16::0010
    spec:
      containers:
      - name: ippool
        image: docker.io/library/nginx:alpine
```

□ Workload □□□□ IP □□□□□□□□□

1. `ovn.kubernetes.io/ip_pool` □□ IP □□□□□□□□□ CIDR □□
2. `ovn.kubernetes.io/ip_pool` □□ IP □□□□□□□ IP □□□
3. □ `ovn.kubernetes.io/ip_pool` □□ IP □□□□ replicas □□□□□□□ Pod □□□□□□□□□□□ Workload □□□□□□□□□□□□□□□ `ovn.kubernetes.io/ip_pool` □ IP □□□□

## 3.2.4 StatefulSet □□□□

StatefulSet □□□□□□ IP□□□□□□ Workload □□□□□□□ `ovn.kubernetes.io/ip_pool` □□□ Pod □□□ IP □□□

□□ StatefulSet □□□□□□□□□□□□□□□□□□□□□□□□□□□Kube-OVN □□□□□□□□□

1. Pod □□□□□□ `ovn.kubernetes.io/ip_pool` □□ IP□□□ StatefulSet □□□□ web□□ web-0 □□□ `ovn.kubernetes.io/ip_pool` □□□□□ IP□ web-1 □□□□□□ IP□□□□□□
2. StatefulSet Pod □□□□□□□□□□ OVN □□ logical_switch_port □□□□□□□□□ Pod □□□□□□ interface □□□□□ Pod □□□□ IP/Mac □□□□□□□□□□□ StatefulSet Volume □□□□□ □□□□□
3. □□ 2 □□□□□□□□ `ovn.kubernetes.io/ip_pool` □□□ StatefulSet□Pod □□□□□□□□□□□ IP/Mac□□□□□□ StatefulSet □□□□□□□□□□□□□□□□□□□□

**StatefulSet □□**

```yaml
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: web
spec:
  serviceName: "nginx"
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: docker.io/library/nginx:alpine
        ports:
        - containerPort: 80
          name: web
```



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▯▯▯▯▯▯ StatefulSet ▯ Pod ▯▯ Pod IP ▯▯▯▯▯

▯▯ **StatefulSet Pod IP**

▯▯ StatefulSet ▯▯ IP ▯ Pod Name ▯▯▯▯▯▯▯▯▯▯▯▯ Statefulset ▯ `ovn.kubernetes.io/ip_pool` Annotation ▯▯▯▯ Pod ▯ IP▯

▯▯▯▯▯▯▯▯ StatefulSet Pod IP ▯▯▯▯▯▯ StatefulSet ▯▯▯▯ scale ▯ 0▯▯▯▯▯▯▯ Annotation ▯▯▯ StatefulSet ▯▯▯▯

## 3.2.5 KubeVirt VM ▯▯▯▯

▯▯ KubeVirt ▯▯▯ VM ▯▯▯ `kube-ovn-controller` ▯▯▯▯▯▯ StatefulSet Pod ▯▯▯▯▯ IP ▯▯▯▯▯▯▯▯ ▯▯▯ VM ▯▯▯▯▯▯▯▯▯▯▯▯

## 3.3 IP ■■■

IP ■■IPPool■■■■■■Subnet■■■■■■ IPAM ■■■■■■■■■■ IP ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■Namespace■■■ Workload ■■■

### 3.3.1 ■■■■

■■■■■

```
apiVersion: kubeovn.io/v1
kind: IPPool
metadata:
  name: pool-1
spec:
  subnet: ovn-default
  ips:
  - "10.16.0.201"
  - "10.16.0.210/30"
  - "10.16.0.220..10.16.0.230"
  namespaces:
  - ns-1
  enableAddressSet: true
```

■■■■■ Workload■

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ippool
  labels:
    app: ippool
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ippool
  template:
    metadata:
      labels:
        app: ippool
      annotations:
        ovn.kubernetes.io/ip_pool: pool-1
    spec:
      containers:
      - name: ippool
        image: docker.io/library/nginx:alpine
```

■■■■■

| ■■ | ■■ | ■■ |
| --- | --- | --- |
| subnet | ■■■■■■■ | ■■ |
| ips | ■■■■■■ IP ■■ | ■■ ■ ■■ .. ■■■■■■■■ IPv6■ |
| namespaces | ■■■■■■■ | ■■■■■■■■■■■■■■ Pod ■■■■■■■■ IP ■■■■■ IP■■■■■■■■■■■■■■■■■■ |
| enableAddressSet | ■■■■■■■■■■ AddressSet | ■■ false■■■■ true ■ ACL ■■■■■■■■■■■■ AddressSet ■■■■■■■ |

### 3.3.2 ■■■■

1. ■■■■ Workload ■■ IP Pool ■■■■ ■■■■■■IP ■■■■■■■■■■ IP ■■■
2. IP ■■ `.spec.ips` ■■■■■■■■■■■ IP ■■■■■■■■■■ IP ■■■ `.spec.ips` ■■■ CIDR ■■■■
3. ■■■■■■■■■ IP ■■■■■■■■■■■■■■■IP ■■■
4. IP ■■ `.spec.ips` ■■■■■■■
5. IP ■■■■■■■■■■ IP■■ IP ■■■■■■ IP ■■■■■■■■■■■■ IP ■■■■■■ IP■
6. ■■■■■■■■ IP ■■■■■■■■■■■■■ IP ■■■■■■■■■■■
7. ■■ IP ■■■■■■■■■ Namespace■



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8. IP ▯▯ `.spec.enableAddressSet` ▯▯▯ `false` ▯▯▯▯ `true` ▯▯▯▯▯▯▯▯ IP ▯▯▯▯ OVN NB ▯▯▯ AddressSet ▯▯▯▯▯ IP ▯▯▯▯▯ IP ▯▯▯▯▯▯ AddressSet ▯▯▯▯▯▯ NetworkPolicy ▯ VPC ▯▯▯▯▯▯▯▯▯▯ AddressSet ▯▯▯

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## 3.3.3 ▯▯

## 3.4 ⬚⬚⬚⬚⬚

⬚⬚⬚⬚ Pod ⬚⬚⬚ Annotations ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: custom-routes
  annotations:
    ovn.kubernetes.io/routes: |
      [{
        "dst": "192.168.0.101/24",
        "gw": "10.16.0.254"
      }, {
        "gw": "10.16.0.254"
      }]
spec:
  containers:
  - name: nginx
    image: docker.io/library/nginx:alpine
```

> `dst` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

⬚⬚⬚⬚⬚⬚⬚ Deployment⬚DaemonSet ⬚ StatefulSet⬚⬚⬚⬚ Annotation ⬚⬚⬚⬚⬚⬚⬚⬚ `.spec.template.metadata.annotations` ⬚⬚⬚⬚⬚⬚⬚⬚

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: custom-routes
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
      annotations:
        ovn.kubernetes.io/routes: |
          [{
            "dst": "192.168.0.101/24",
            "gw": "10.16.0.254"
          }, {
            "gw": "10.16.0.254"
          }]
    spec:
      containers:
      - name: nginx
        image: docker.io/library/nginx:alpine
```

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### 3.4.1 ⬚⬚

# 3.5 EIP 和 SNAT 配置

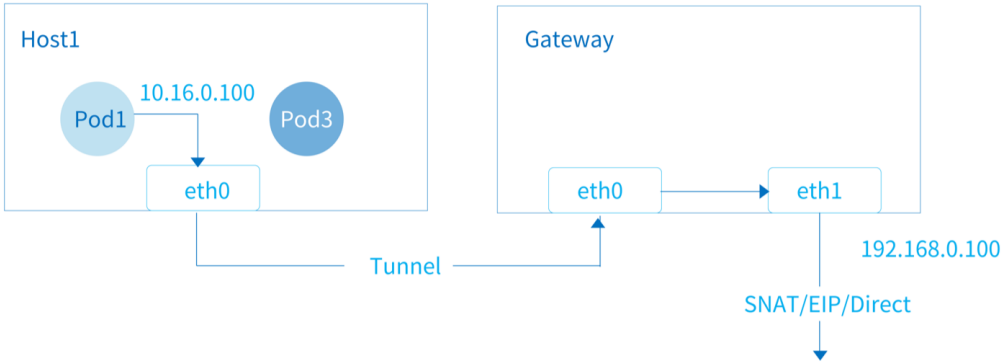
> 介绍如何在 VPC 网络中使用 VPC 内部的 NAT网关。

- VPC Iptables NAT 网关
- VPC OVN NAT 网关

> 在 VPC 网络中使用浮动公网IP, 实现公网访问。

- 介绍如何在 VPC 内部使用 OVN NAT 功能为 pod annotation 配置浮动公网IP
- 介绍如何在 VPC 网络中使用 OVN NAT 功能为 provider-network、vlan、subnet CRD 资源对象配置 ovn-eip、ovn-dnat、ovn-fip、ovn-snat CRD 资源对象 EIP 和 NAT规则。VPC OVN NAT 网关
- 介绍如何在 VPC 网络中使用 Iptables NAT 功能为集群VPC Iptables NAT 网关

Kube-OVN 支持 OVN 的 L3 Gateway 功能为 Pod 提供 SNAT 和 EIP 等多种 SNAT规则 Pod 访问外部 IP 时会将源地址 EIP 为指定 Pod、Pod 访问外部 EIP 为指定




## 3.5.1 使用场景

- 支持 OVN 的 L3 Gateway 功能，通过 OVS 隧道将 Overlay 和 Underlay 网络打通
- 支持 NAT 网关访问 Underlay 网络中的外部资源
- 支持 EIP 和 SNAT 规则的动态配置

## 3.5.2 配置说明

在 `kube-system` 中创建 ConfigMap `ovn-external-gw-config`：

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-external-gw-config
  namespace: kube-system
data:
  type: "centralized"
  enable-external-gw: "true"
  external-gw-nodes: "kube-ovn-worker"
  external-gw-nic: "eth1"
  external-gw-addr: "172.56.0.1/16"
  nic-ip: "172.56.0.254/16"
  nic-mac: "16:52:f3:13:6a:25"
```

- `enable-external-gw`：是否启用 SNAT 和 EIP 功能
- `type`：`centralized` 或 `distributed`，默认为 `centralized`，表示 `distributed` 模式
- `external-gw-nodes`：`centralized` 模式下的网关节点



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- `external-gw-nic` : ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚
- `external-gw-addr` : ⬚⬚⬚⬚⬚⬚⬚ IP ⬚⬚⬚⬚
- `nic-ip`,`nic-mac` : ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ IP ⬚ Mac⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ IP ⬚ Mac⬚

### 3.5.3 ⬚⬚ OVN ⬚ OVS ⬚⬚⬚⬚⬚⬚⬚⬚

⬚⬚ OVN-NB ⬚⬚⬚⬚⬚ `ovn-external` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `ovn-cluster-ovn-external` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ chassis⬚

```
# kubectl ko nbctl show
switch 3de4cea7-1a71-43f3-8b62-435a57ef16a6 (external)
    port localnet.external
        type: localnet
        addresses: ["unknown"]
    port external-ovn-cluster
        type: router
        router-port: ovn-cluster-external
router e1eb83ad-34be-4ed5-9a02-fcc8b1d357c4 (ovn-cluster)
    port ovn-cluster-external
        mac: "ac:1f:6b:2d:33:f1"
        networks: ["172.56.0.100/16"]
        gateway chassis: [a5682814-2e2c-46dd-9c1c-6803ef0dab66]
```

⬚⬚ OVS ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `br-external` ⬚⬚⬚

```
# kubectl ko vsctl ${gateway node name} show
e7d81150-7743-4d6e-9e6f-5c688232e130
    Bridge br-external
        Port br-external
            Interface br-external
                type: internal
        Port eth1
            Interface eth1
        Port patch-localnet.external-to-br-int
            Interface patch-localnet.external-to-br-int
                type: patch
                options: {peer=patch-br-int-to-localnet.external}
```

### 3.5.4 Pod ⬚⬚ EIP ⬚ SNAT

⬚⬚⬚⬚ Pod ⬚⬚⬚ `ovn.kubernetes.io/snat` ⬚ `ovn.kubernetes.io/eip` annotation ⬚⬚⬚⬚⬚ SNAT ⬚ EIP⬚

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-snat
  annotations:
    ovn.kubernetes.io/snat: 172.56.0.200
spec:
  containers:
  - name: pod-snat
    image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  name: pod-eip
  annotations:
    ovn.kubernetes.io/eip: 172.56.0.233
spec:
  containers:
  - name: pod-eip
    image: docker.io/library/nginx:alpine
```

⬚⬚⬚ kubectl ⬚⬚⬚⬚⬚⬚⬚⬚ Pod ⬚⬚⬚⬚ EIP ⬚ SNAT ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `ovn.kubernetes.io/routed` annotation ⬚⬚⬚⬚⬚⬚⬚⬚

```
kubectl annotate pod pod-eip ovn.kubernetes.io/eip=172.56.0.221 --overwrite
kubectl annotate pod pod-eip ovn.kubernetes.io/routed-
```

⬚ EIP ⬚ SNAT ⬚⬚⬚⬚⬚⬚ `ovn.kubernetes.io/routed` annotation ⬚⬚⬚⬚⬚⬚⬚⬚

### 3.5.5 ⬚⬚⬚⬚

`kube-ovn-controller` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ SNAT ⬚ EIP ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

- `--external-gateway-config-ns` : Configmap `ovn-external-gw-config` ⬚⬚ Namespace⬚⬚⬚⬚ `kube-system` ⬚



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- `--external-gateway-net` : □□□□□□□□□□□□□□□□ `external` □
- `--external-gateway-vlanid` : □□□□ Vlan Tag □□□□□ 0□□□□□ Vlan□

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# 3.6 □□□□ QoS □□

Kube-OVN □□□□□□ Pod □□□□□□□□ QoS□

- □□□□□□ QoS□
- `linux-netem` □□□□□□□□□□□□□□□ QoS□□□□□□□□□□

> □□□□□ Pod □□ QoS □□□ Namespace □ Subnet □□□ QoS □□□

## 3.6.1 □□□□□□□□□□ QoS

□□□□ QoS □□□□ Pod annotation □□□□□□□□□□□ Pod □□□□□□□□□□□□ □□□□□□□□ `Mbit/s` □

```
apiVersion: v1
kind: Pod
metadata:
  name: qos
  namespace: ls1
  annotations:
    ovn.kubernetes.io/ingress_rate: "3"
    ovn.kubernetes.io/egress_rate: "1"
spec:
  containers:
  - name: qos
    image: docker.io/library/nginx:alpine
```

□□ annotation □□□□ QoS□

```
kubectl annotate --overwrite  pod nginx-74d5899f46-d7qkn ovn.kubernetes.io/ingress_rate=3
```

□□ **QoS** □□

□□□□□□□□□□□□□

```
kind: DaemonSet
apiVersion: apps/v1
metadata:
  name: perf
  namespace: ls1
  labels:
    app: perf
spec:
  selector:
    matchLabels:
      app: perf
  template:
    metadata:
      labels:
        app: perf
    spec:
      containers:
      - name: nginx
        image: docker.io/kubeovn/perf
```

□□□□□□ Pod □□□ iperf3 server□

```
# kubectl exec -it perf-4n4gt -n ls1 sh
# iperf3 -s
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
```

□□□□□ Pod □□□□□ Pod□

```
# kubectl exec -it perf-d4mqc -n ls1 sh
# iperf3 -c 10.66.0.12
Connecting to host 10.66.0.12, port 5201
[  4] local 10.66.0.14 port 51544 connected to 10.66.0.12 port 5201
[ ID] Interval           Transfer     Bandwidth       Retr  Cwnd
[  4]   0.00-1.00   sec  86.4 MBytes   725 Mbits/sec    3    350 KBytes
[  4]   1.00-2.00   sec  89.9 MBytes   754 Mbits/sec  118    473 KBytes
[  4]   2.00-3.00   sec   101 MBytes   848 Mbits/sec  184    586 KBytes
[  4]   3.00-4.00   sec   104 MBytes   875 Mbits/sec  217    671 KBytes
[  4]   4.00-5.00   sec   111 MBytes   935 Mbits/sec  175    772 KBytes
```



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```
[  4]   5.00-6.00   sec   100 MBytes   840 Mbits/sec  658    598 KBytes
[  4]   6.00-7.00   sec   106 MBytes   890 Mbits/sec  742    668 KBytes
[  4]   7.00-8.00   sec   102 MBytes   857 Mbits/sec  764    724 KBytes
[  4]   8.00-9.00   sec  97.4 MBytes   817 Mbits/sec  1175   764 KBytes
[  4]   9.00-10.00  sec   111 MBytes   934 Mbits/sec  1083   838 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-10.00  sec  1010 MBytes   848 Mbits/sec  5119             sender
[  4]   0.00-10.00  sec  1008 MBytes   846 Mbits/sec                  receiver

iperf Done.
```

□□□□□ Pod □□□□□ QoS□

```
kubectl annotate --overwrite  pod perf-4n4gt -n ls1 ovn.kubernetes.io/ingress_rate=30
```

□□□□□□ Pod □□□□□ Pod □□□

```
# iperf3 -c 10.66.0.12
Connecting to host 10.66.0.12, port 5201
[  4] local 10.66.0.14 port 52372 connected to 10.66.0.12 port 5201
[ ID] Interval           Transfer     Bandwidth       Retr  Cwnd
[  4]   0.00-1.00   sec  3.66 MBytes  30.7 Mbits/sec    2   76.1 KBytes
[  4]   1.00-2.00   sec  3.43 MBytes  28.8 Mbits/sec    0    104 KBytes
[  4]   2.00-3.00   sec  3.50 MBytes  29.4 Mbits/sec    0    126 KBytes
[  4]   3.00-4.00   sec  3.50 MBytes  29.3 Mbits/sec    0    144 KBytes
[  4]   4.00-5.00   sec  3.43 MBytes  28.8 Mbits/sec    0    160 KBytes
[  4]   5.00-6.00   sec  3.43 MBytes  28.8 Mbits/sec    0    175 KBytes
[  4]   6.00-7.00   sec  3.50 MBytes  29.3 Mbits/sec    0    212 KBytes
[  4]   7.00-8.00   sec  3.68 MBytes  30.9 Mbits/sec    0    294 KBytes
[  4]   8.00-9.00   sec  3.74 MBytes  31.4 Mbits/sec    0    398 KBytes
[  4]   9.00-10.00  sec  3.80 MBytes  31.9 Mbits/sec    0    526 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-10.00  sec  35.7 MBytes  29.9 Mbits/sec    2             sender
[  4]   0.00-10.00  sec  34.5 MBytes  29.0 Mbits/sec                  receiver

iperf Done.
```

## 3.6.2 linux-netem QoS

> RHEL □□□□□□□□□□ netem □□□□□ yum install -y kernel-modules-extra && modprobe sch_netem

Pod □□□□□□ annotation □□ `linux-netem` □□ QoS□□□ netem □□□□□□□□□□ QoS □□□ Pod □ Ingress □□□□□□□

- `ovn.kubernetes.io/latency` □□□ Pod □□□□□□□□□□□□□□□ ms□
- `ovn.kubernetes.io/jitter` □□□ Pod □□□□□□□□□□□□□□□□□□ ms□
- `ovn.kubernetes.io/limit` □ □ `qdisc` □□□□□□□□□□□□□□□□□□□□□□□□□□□ 1000□
- `ovn.kubernetes.io/loss` □ □□□□□□□□□□□□□□□ float □□□□□□□□ 20□□□□□ 20% □□□□□□

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## 3.6.3 □□

# 3.7 ▒▒▒▒

Kube-OVN ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ IPv4▒IPv6 ▒▒▒▒▒▒▒▒▒▒ ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

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## 3.7.1 ▒▒▒▒▒▒▒

▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ CIDR ▒▒▒▒ `cidr=<IPv4 CIDR>,<IPv6 CIDR>` ▒▒▒▒ CIDR ▒▒▒▒▒ IPv4 ▒▒▒IPv6 ▒▒▒▒▒▒▒▒▒

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-test
spec:
  cidrBlock: 10.16.0.0/16,fd00:10:16::/64
  excludeIps:
  - 10.16.0.1
  - fd00:10:16::1
  gateway: 10.16.0.1,fd00:10:16::1
```

▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ `install.sh` ▒▒▒▒▒▒▒▒▒▒ `/112` ▒▒▒▒▒▒▒▒ ULA ▒▒▒▒▒▒▒▒ `/64` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒/Service CIDR ▒▒▒▒▒

```
POD_CIDR="10.16.0.0/16,fd00:10:16::/112"
JOIN_CIDR="100.64.0.0/16,fd00:100:64::/112"
```

## 3.7.2 ▒▒ Pod ▒▒

▒▒▒▒▒▒▒▒ Pod ▒▒▒▒▒▒▒▒▒▒▒▒ IPv4 ▒ IPv6 ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ Pod ▒ annotation ▒:

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/allocated: "true"
    ovn.kubernetes.io/cidr: 10.16.0.0/16,fd00:10:16::/64
    ovn.kubernetes.io/gateway: 10.16.0.1,fd00:10:16::1
    ovn.kubernetes.io/ip_address: 10.16.0.9,fd00:10:16::9
    ovn.kubernetes.io/logical_switch: ovn-default
    ovn.kubernetes.io/mac_address: 00:00:00:14:88:09
    ovn.kubernetes.io/routed: "true"
...
podIP: 10.16.0.9
  podIPs:
  - ip: 10.16.0.9
  - ip: fd00:10:16::9
```

## 3.7.3 ▒▒▒ Pod ▒▒▒▒▒ IP ▒▒▒

▒▒▒▒▒▒▒▒▒▒▒ Pod ▒▒▒▒▒▒▒▒▒▒▒ IPv4 ▒ IPv6 ▒▒▒▒▒▒▒▒▒ Pod ▒▒▒▒ `ip_family` annotation▒ ▒▒▒▒▒ annotation ▒▒Kube-OVN ▒▒▒▒▒▒▒▒▒▒▒▒▒ ▒▒▒▒▒▒▒▒▒ Pod ▒▒▒▒▒▒▒ Subnet ▒▒▒▒▒▒▒

▒▒▒▒▒▒▒ `ovn.kubernetes.io/ip_family` ▒

```
apiVersion: v1
kind: Pod
metadata:
  name: ipv4-only
  annotations:
    ovn.kubernetes.io/ip_family: ipv4
spec:
  containers:
  - name: nginx
    image: docker.io/library/nginx:alpine
```

▒▒▒▒▒▒▒ `ipv4` ▒ `ipv6` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

```
  name: attach-ipv6-only
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
    attachnet.default.ovn.kubernetes.io/ip_family: ipv6
spec:
  containers:
  - name: nginx
    image: docker.io/library/nginx:alpine
```

⬚⬚⬚⬚ NetworkAttachmentDefinition ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `interface` ⬚⬚⬚⬚Kube-OVN ⬚⬚⬚⬚⬚⬚⬚⬚ provider ⬚⬚ annotation⬚ ⬚⬚ `net1` ⬚⬚⬚ IPv4⬚ `net2` ⬚⬚⬚ IPv6⬚

```
apiVersion: v1
kind: Pod
metadata:
  name: attach-mixed-family
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{"name": "attachnet", "namespace": "default", "interface": "net1"}, {"name": "attachnet", "namespace": "default",
"interface": "net2"}]'
    attachnet.default.ovn.net1.kubernetes.io/ip_family: ipv4
    attachnet.default.ovn.net2.kubernetes.io/ip_family: ipv6
spec:
  containers:
  - name: nginx
    image: docker.io/library/nginx:alpine
```

⬚⬚⬚⬚⬚⬚⬚⬚⬚ IP⬚⬚⬚ `ovn.kubernetes.io/ip_address` ⬚ `<provider>.kubernetes.io/ip_address` ⬚⬚⬚⬚ NetworkAttachmentDefinition ⬚⬚⬚⬚⬚⬚⬚ `<nadName>.<nadNamespace>.kubernetes.io/ip_address.<interfaceName>` ⬚⬚⬚ IP ⬚⬚⬚⬚⬚⬚⬚ `ip_family` ⬚⬚⬚

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## 3.7.4 ⬚⬚



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# 3.8 Webhook ⬚⬚

⬚⬚ Webhook ⬚⬚⬚ Kube-OVN ⬚⬚ CRD ⬚⬚⬚⬚⬚⬚⬚⬚ Webhook ⬚⬚⬚⬚ ⬚⬚ IP ⬚⬚⬚⬚⬚⬚⬚ Subnet CIDR ⬚⬚⬚⬚⬚⬚⬚⬚

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## 3.8.4 ⬜⬜

# 3.9 □□□□

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## 3.9.3 □□□□

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### 1. Pod to Pod in the same Nodes

□□□□□□□

| Size | TCP Latency | TCP Bandwidth | UDP Latency | UDP Lost Rate | UDP Bandwidth |
|---|---|---|---|---|---|
| 64 | 12.7 us | 289 Mbits/sec | 12.6 us | (1.8%) | 77.9 Mbits/sec |
| 128 | 15.5 us | 517 Mbits/sec | 12.7 us | (0%) | 155 Mbits/sec |
| 512 | 12.2 us | 1.64 Gbits/sec | 12.4 us | (0%) | 624 Mbits/sec |
| 1k | 13 us | 2.96 Gbits/sec | 11.4 us | (0.53%) | 1.22 Gbits/sec |
| 4k | 18 us | 7.67 Gbits/sec | 25.7 us | (0.41%) | 1.50 Gbits/sec |

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| Size | TCP Latency | TCP Bandwidth | UDP Latency | UDP Lost Rate | UDP Bandwidth |
|---|---|---|---|---|---|
| 64 | 11.9 us | 324 Mbits/sec | 12.2 us | (0.22%) | 102 Mbits/sec |
| 128 | 10.5 us | 582 Mbits/sec | 9.5 us | (0.21%) | 198 Mbits/sec |
| 512 | 11.6 us | 1.84 Gbits/sec | 9.32 us | (0.091%) | 827 Mbits/sec |
| 1k | 10.5 us | 3.44 Gbits/sec | 10 us | (1.2%) | 1.52 Gbits/sec |
| 4k | 16.7 us | 8.52 Gbits/sec | 18.2 us | (1.3%) | 2.42 Gbits/sec |

### 2. Pod to Pod in the different Nodes

□□□□□□□

| Size | TCP Latency | TCP Bandwidth | UDP Latency | UDP Lost Rate | UDP Bandwidth |
|---|---|---|---|---|---|
| 64 | 258 us | 143 Mbits/sec | 237 us | (61%) | 28.5 Mbits/sec |
| 128 | 240 us | 252 Mbits/sec | 231 us | (64%) | 54.9 Mbits/sec |
| 512 | 236 us | 763 Mbits/sec | 256 us | (68%) | 194 Mbits/sec |
| 1k | 242 us | 969 Mbits/sec | 225 us | (62%) | 449 Mbits/sec |
| 4k | 352 us | 1.12 Gbits/sec | 382 us | (0.71%) | 21.4 Mbits/sec |

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| Size | TCP Latency | TCP Bandwidth | UDP Latency | UDP Lost Rate | UDP Bandwidth |
|---|---|---|---|---|---|
| 64 | 278 us | 140 Mbits/sec | 227 us | (24%) | 59.6 Mbits/sec |
| 128 | 249 us | 265 Mbits/sec | 265 us | (23%) | 114 Mbits/sec |
| 512 | 233 us | 914 Mbits/sec | 235 us | (21%) | 468 Mbits/sec |
| 1k | 238 us | 1.14 Gbits/sec | 240 us | (15%) | 891 Mbits/sec |
| 4k | 370 us | 1.25 Gbits/sec | 361 us | (0.43%) | 7.54 Mbits/sec |



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### 3. Node to Node

□□□□□□□

| Size | TCP Latency | TCP Bandwidth | UDP Latency | UDP Lost Rate | UDP Bandwidth |
|---|---|---|---|---|---|
| 64 | 205 us | 162 Mbits/sec | 183 us | (11%) | 74.2 Mbits/sec |
| 128 | 222 us | 280 Mbits/sec | 206 us | (6.3%) | 155 Mbits/sec |
| 512 | 220 us | 1.04 Gbits/sec | 177 us | (20%) | 503 Mbits/sec |
| 1k | 213 us | 2.06 Gbits/sec | 201 us | (8.6%) | 1.14 Gbits/sec |
| 4k | 280 us | 5.01 Gbits/sec | 315 us | (37%) | 1.20 Gbits/sec |

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| Size | TCP Latency | TCP Bandwidth | UDP Latency | UDP Lost Rate | UDP Bandwidth |
|---|---|---|---|---|---|
| 64 | 204 us | 157 Mbits/sec | 204 us | (8.8%) | 81.9 Mbits/sec |
| 128 | 213 us | 262 Mbits/sec | 225 us | (19%) | 136 Mbits/sec |
| 512 | 220 us | 1.02 Gbits/sec | 227 us | (21%) | 486 Mbits/sec |
| 1k | 217 us | 1.79 Gbits/sec | 218 us | (29%) | 845 Mbits/sec |
| 4k | 275 us | 5.27 Gbits/sec | 336 us | (34%) | 1.21 Gbits/sec |

### 4. Pod to the Node where the Pod is located

□□□□□□□

| Size | TCP Latency | TCP Bandwidth | UDP Latency | UDP Lost Rate | UDP Bandwidth |
|---|---|---|---|---|---|
| 64 | 12.2 us | 295 Mbits/sec | 12.7 us | (0.27%) | 74.1 Mbits/sec |
| 128 | 14.1 us | 549 Mbits/sec | 10.6 us | (0.41%) | 153 Mbits/sec |
| 512 | 13.5 us | 1.83 Gbits/sec | 12.7 us | (0.23%) | 586 Mbits/sec |
| 1k | 12 us | 2.69 Gbits/sec | 13 us | (1%) | 1.16 Gbits/sec |
| 4k | 18.9 us | 4.51 Gbits/sec | 21.8 us | (0.42%) | 1.81 Gbits/sec |

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| Size | TCP Latency | TCP Bandwidth | UDP Latency | UDP Lost Rate | UDP Bandwidth |
|---|---|---|---|---|---|
| 64 | 10.4 us | 335 Mbits/sec | 12.2 us | (0.75%) | 95.4 Mbits/sec |
| 128 | 12.1 us | 561 Mbits/sec | 11.3 us | (0.25%) | 194 Mbits/sec |
| 512 | 11.6 us | 1.87 Gbits/sec | 10.7 us | (0.66%) | 745 Mbits/sec |
| 1k | 12.7 us | 3.12 Gbits/sec | 10.9 us | (1.2%) | 1.46 Gbits/sec |
| 4k | 16.5 us | 8.23 Gbits/sec | 17.9 us | (1.5%) | 2.51 Gbits/sec |



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### 5. Pod to the Node where the Pod is not located

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| Size | TCP Latency | TCP Bandwidth | UDP Latency | UDP Lost Rate | UDP Bandwidth |
|---|---|---|---|---|---|
| 64 | 234 us | 153 Mbits/sec | 232 us | (63%) | 29.4 Mbits/sec |
| 128 | 237 us | 261 Mbits/sec | 238 us | (49%) | 76.1 Mbits/sec |
| 512 | 231 us | 701 Mbits/sec | 238 us | (57%) | 279 Mbits/sec |
| 1k | 256 us | 1.05 Gbits/sec | 228 us | (56%) | 524 Mbits/sec |
| 4k | 330 us | 1.08 Gbits/sec | 359 us | (1.5%) | 35.7 Mbits/sec |

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| Size | TCP Latency | TCP Bandwidth | UDP Latency | UDP Lost Rate | UDP Bandwidth |
|---|---|---|---|---|---|
| 64 | 283 us | 141 Mbits/sec | 230 us | (26%) | 55.8 Mbits/sec |
| 128 | 234 us | 255 Mbits/sec | 234 us | (25%) | 113 Mbits/sec |
| 512 | 246 us | 760 Mbits/sec | 234 us | (22%) | 458 Mbits/sec |
| 1k | 268 us | 1.23 Gbits/sec | 242 us | (20%) | 879 Mbits/sec |
| 4k | 326 us | 1.20 Gbits/sec | 369 us | (0.5%) | 7.87 Mbits/sec |

### 6. Pod to the cluster ip service

□□□□□□□

| Size | TCP Latency | TCP Bandwidth | UDP Latency | UDP Lost Rate | UDP Bandwidth |
|---|---|---|---|---|---|
| 64 | 237 us | 133 Mbits/sec | 213 us | (65%) | 25.5 Mbits/sec |
| 128 | 232 us | 271 Mbits/sec | 222 us | (62%) | 54.8 Mbits/sec |
| 512 | 266 us | 800 Mbits/sec | 234 us | (60%) | 232 Mbits/sec |
| 1k | 248 us | 986 Mbits/sec | 239 us | (50%) | 511 Mbits/sec |
| 4k | 314 us | 1.03 Gbits/sec | 367 us | (0.6%) | 13.2 Mbits/sec |

| TCP-Conn-Number | QPS | Avg-Resp-Time | Stdev-Resp-Time | Max-Resp-Time |
|---|---|---|---|---|
| 10 | 14305.17 | 0.87ms | 1.48ms | 24.46ms |
| 100 | 29082.07 | 3.87ms | 4.35ms | 102.85ms |

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| Size | TCP Latency | TCP Bandwidth | UDP Latency | UDP Lost Rate | UDP Bandwidth |
|---|---|---|---|---|---|
| 64 | 241 us | 145 Mbits/sec | 225 us | (19%) | 60.2 Mbits/sec |
| 128 | 245 us | 261 Mbits/sec | 212 us | (15%) | 123 Mbits/sec |
| 512 | 252 us | 821 Mbits/sec | 219 us | (14%) | 499 Mbits/sec |
| 1k | 253 us | 1.08 Gbits/sec | 242 us | (16%) | 852 Mbits/sec |
| 4k | 320 us | 1.32 Gbits/sec | 360 us | (0.47%) | 6.70 Mbits/sec |

| TCP-Conn-Number | QPS | Avg-Resp-Time | Stdev-Resp-Time | Max-Resp-Time |
|---|---|---|---|---|
| 10 | 13634.07 | 0.96ms | 1.72ms | 30.07ms |
| 100 | 30215.23 | 3.59ms | 3.20ms | 77.56ms |

**7. Host to the Node port service where the Pod is not located on the target Node**

▯▯▯▯▯▯▯

| TCP-Conn-Number | QPS | Avg-Resp-Time | Stdev-Resp-Time | Max-Resp-Time |
|---|---|---|---|---|
| 10 | 14802.73 | 0.88ms | 1.66ms | 31.49ms |
| 100 | 29809.58 | 3.78ms | 4.12ms | 105.34ms |

▯▯▯▯▯▯▯

| TCP-Conn-Number | QPS | Avg-Resp-Time | Stdev-Resp-Time | Max-Resp-Time |
|---|---|---|---|---|
| 10 | 14273.33 | 0.90ms | 1.60ms | 37.16ms |
| 100 | 30757.81 | 3.62ms | 3.41ms | 59.78ms |

**8. Host to the Node port service where the Pod is located on the target Node**

▯▯▯▯▯▯▯

| TCP-Conn-Number | QPS | Avg-Resp-Time | Stdev-Resp-Time | Max-Resp-Time |
|---|---|---|---|---|
| 10 | 15402.39 | 802.50us | 1.42ms | 30.91ms |
| 100 | 29424.66 | 4.05ms | 4.31ms | 90.60ms |

▯▯▯▯▯▯▯

| TCP-Conn-Number | QPS | Avg-Resp-Time | Stdev-Resp-Time | Max-Resp-Time |
|---|---|---|---|---|
| 10 | 14649.21 | 0.91ms | 1.72ms | 43.92ms |
| 100 | 32143.61 | 3.66ms | 3.76ms | 67.02ms |

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## 3.9.4 ▯▯

# 3.10 LoadBalancer Service

Kube-OVN VPC VPC VPC

VPC VPC VPC VPC `LoadBalancer` Service LoadBalancerIP VPC Service

1. `multus-cni` `macvlan cni`
2. LoadBalancer Service VPC `vpc-nat-gw` macvlan
3. `VPC` VPC LoadBalancer VPC VPC

## 3.10.1 VPC LoadBalancer Service

kube-system namespace deployment `kube-ovn-controller` `args` `--enable-lb-svc=true` false

```
containers:
- args:
  - /kube-ovn/start-controller.sh
  - --default-cidr=10.16.0.0/16
  - --default-gateway=10.16.0.1
  - --default-gateway-check=true
  - --enable-lb-svc=true                 #       true
```

**NetworkAttachmentDefinition CRD**

yaml `net-attach-def`

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: lb-svc-attachment
  namespace: kube-system
spec:
  # master                    eth0 / ens192 / bond0
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth0",
      "mode": "bridge"
    }'
```

`eth0` `master`

**Subnet**

Subnet LoadBalancer Service LoadBalancerIP Underlay Subnet

yaml

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attach-subnet
spec:
  protocol: IPv4
  provider: lb-svc-attachment.kube-system    # provider             net-attach-def     Name.Namespace
  cidrBlock: 172.18.0.0/16
  gateway: 172.18.0.1
  excludeIps:
  - 172.18.0.0..172.18.0.10
```

Subnet `provider` `ovn` `.ovn` Kube-OVN `logical switch`

`provider` `ovn` `.ovn` Kube-OVN IPAM IP



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### □□ LoadBalancer Service

□□□□ yaml□□□ LoadBalancer Service□

```
apiVersion: v1
kind: Service
metadata:
  annotations:
    lb-svc-attachment.kube-system.kubernetes.io/logical_switch: attach-subnet   #
    ovn.kubernetes.io/attachmentprovider: lb-svc-attachment.kube-system          #
  labels:
    app: dynamic
  name: test-service
  namespace: default
spec:
  loadBalancerIP: 172.18.0.18                                                    #
  ports:
    - name: test
      protocol: TCP
      port: 80
      targetPort: 80
  selector:
    app: dynamic
  sessionAffinity: None
  type: LoadBalancer
```

□ yaml □□annotation `ovn.kubernetes.io/attachmentprovider` □□□□□□□□□□□□□□□ `net-attach-def` □□□ Name.Namespace □□□□ annotation □□□□□ Pod □□ □□ `net-attach-def` □□□

□□□□ annotation □□□□□□□□□□□□□□□□annotation key □□□ net-attach-def □□□ `Name.Namespace.kubernetes.io/logical_switch` □□□□□ □□ □□□□□□□□ LoadBalancerIP □□□□□□□□□□□□□□□□□□□□□□□□□ LoadBalancerIP □□□

□□□□□□□□ LoadBalancerIP □□□□□□□ `spec.loadBalancerIP` □□□□□□□□□□□□□□□□□□□□□□□

□□□ yaml □□ Service □□□ Service □ Namespace □□□□□□ Pod □□□□□

```
# kubectl get pod
NAME                                    READY   STATUS    RESTARTS   AGE
lb-svc-test-service-6869d98dd8-cjvll    1/1     Running   0          107m
# kubectl get svc
NAME             TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
test-service     LoadBalancer   10.109.201.193   172.18.0.18   80:30056/TCP   107m
```

□□ `service.spec.loadBalancerIP` □□□□□□□□□□□□□□□ service external-ip □□□□□□□□□□□□□□□□□□□□□□□

□□□□ Pod □ yaml □□□□□□□□□□□□□□□□□□

```
# kubectl get pod -o yaml lb-svc-test-service-6869d98dd8-cjvll
apiVersion: v1
kind: Pod
metadata:
  annotations:
    k8s.v1.cni.cncf.io/network-status: |-
      [{
          "name": "kube-ovn",
          "ips": [
              "10.16.0.2"
          ],
          "default": true,
          "dns": {}
      },{
          "name": "default/test-service",
          "interface": "net1",
          "mac": "ba:85:f7:02:9f:42",
          "dns": {}
      }]
    k8s.v1.cni.cncf.io/networks: default/test-service
    k8s.v1.cni.cncf.io/networks-status: |-
      [{
          "name": "kube-ovn",
          "ips": [
              "10.16.0.2"
          ],
          "default": true,
          "dns": {}
      },{
          "name": "default/test-service",
          "interface": "net1",
          "mac": "ba:85:f7:02:9f:42",
          "dns": {}
      }]
    ovn.kubernetes.io/allocated: "true"
    ovn.kubernetes.io/cidr: 10.16.0.0/16
    ovn.kubernetes.io/gateway: 10.16.0.1
```

```
    ovn.kubernetes.io/ip_address: 10.16.0.2
    ovn.kubernetes.io/logical_router: ovn-cluster
    ovn.kubernetes.io/logical_switch: ovn-default
    ovn.kubernetes.io/mac_address: 00:00:00:45:F4:29
    ovn.kubernetes.io/pod_nic_type: veth-pair
    ovn.kubernetes.io/routed: "true"
    test-service.default.kubernetes.io/allocated: "true"
    test-service.default.kubernetes.io/cidr: 172.18.0.0/16
    test-service.default.kubernetes.io/gateway: 172.18.0.1
    test-service.default.kubernetes.io/ip_address: 172.18.0.18
    test-service.default.kubernetes.io/logical_switch: attach-subnet
    test-service.default.kubernetes.io/mac_address: 00:00:00:AF:AA:BF
    test-service.default.kubernetes.io/pod_nic_type: veth-pair
```

▯▯ Service ▯▯▯▯

```
# kubectl get svc -o yaml test-service
apiVersion: v1
kind: Service
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","kind":"Service","metadata":{"annotations":{"test-service.default.kubernetes.io/logical_switch":"attach-subnet"},"labels":
{"app":"dynamic"},"name":"test-service","namespace":"default"},"spec":{"ports":[{"name":"test","port":80,"protocol":"TCP","targetPort":80}],"selector":
{"app":"dynamic"},"sessionAffinity":"None","type":"LoadBalancer"}}
    ovn.kubernetes.io/vpc: ovn-cluster
    test-service.default.kubernetes.io/logical_switch: attach-subnet
  creationTimestamp: "2022-06-15T09:01:58Z"
  labels:
    app: dynamic
  name: test-service
  namespace: default
  resourceVersion: "38485"
  uid: 161edee1-7f6e-40f5-9e09-5a52c44267d0
spec:
  allocateLoadBalancerNodePorts: true
  clusterIP: 10.109.201.193
  clusterIPs:
  - 10.109.201.193
  externalTrafficPolicy: Cluster
  internalTrafficPolicy: Cluster
  ipFamilies:
  - IPv4
  ipFamilyPolicy: SingleStack
  ports:
  - name: test
    nodePort: 30056
    port: 80
    protocol: TCP
    targetPort: 80
  selector:
    app: dynamic
  sessionAffinity: None
  type: LoadBalancer
status:
  loadBalancer:
    ingress:
    - ip: 172.18.0.18
```

## 3.10.2 ▯▯ LoadBalancerIP ▯▯

▯▯▯▯ yaml, ▯▯▯▯ Pod▯▯▯ Service ▯ Endpoints ▯▯▯▯▯

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: dynamic
  name: dynamic
  namespace: default
spec:
  replicas: 2
  selector:
    matchLabels:
      app: dynamic
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: dynamic
    spec:
      containers:
      - image: docker.io/library/nginx:alpine
```

```
        imagePullPolicy: IfNotPresent
        name: nginx
      dnsPolicy: ClusterFirst
      restartPolicy: Always
```

▒▒▒

```
    kubernetes.io/os: linux
kind: ConfigMap
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
```

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## 3.10.3 ⋯

# 3.11 ████████

Kube-OVN ████████████████████████████████ Prometheus ████████████

████ kube-prometheus ████ CRD ██████ Prometheus ████ ████████ kube-prometheus ██████ CRD，Kube-OVN ████████████ Kube-OVN ████。

## 3.11.1 ██ Prometheus Monitor

Kube-OVN ██ Prometheus Monitor CRD ████████

```
#
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/pinger-monitor.yaml
# kube-ovn-controller
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/controller-monitor.yaml
# kube-ovn-cni
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/cni-monitor.yaml
# ovn
kubectl apply -f https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovn-monitor.yaml
```

Prometheus ████████████ 15s████████████ yaml ██ `interval` ███

## 3.11.2 ██ Grafana ██

Kube-OVN ██████████ Grafana Dashboard ██████████████████

████ Dashboard ███

```
#
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/pinger-grafana.json
# kube-ovn-controller
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/controller-grafana.json
# kube-ovn-cni
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/cni-grafana.json
# ovn
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovn-grafana.json
# ovs
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/master/dist/monitoring/ovs-grafana.json
```

█ Grafana ██████████████████ Prometheus ██████ Dashboard，

`kube-ovn-controller` ██████████



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`kube-ovn-pinger`




`kube-ovn-cni`




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## 3.11.3 ▯▯

# 3.12 ▯▯

## 3.12.1 NetworkPolicy ▯▯

NetworkPolicy ▯ Kubernetes ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Pod ▯▯▯▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Kube-OVN ▯▯ OVN ▯ ACL ▯▯▯▯▯▯ Kubernetes NetworkPolicy ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

Kube-OVN ▯ Named Port ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ **Named Port** ▯▯▯▯▯▯▯▯▯▯▯

▯▯▯▯▯▯▯▯▯▯ Pod ▯▯▯▯▯▯ Named Port ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯NetworkPolicy ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯ Pod A ▯ `http` ▯▯▯▯▯▯ 8080▯▯ Pod B ▯ `http` ▯▯▯▯▯▯ 8081▯▯▯▯ `port: http` ▯ NetworkPolicy ▯▯▯▯▯▯▯▯▯

▯▯▯▯▯▯ Named Port ▯▯▯▯▯▯▯▯▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-from-client
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
    ovn.kubernetes.io/log_acl_actions: "allow"
spec:
  podSelector:
    matchLabels:
      app: web
  policyTypes:
  - Ingress
  ingress:
  - from:
    - podSelector:
        matchLabels:
          app: client
```

`ovn.kubernetes.io/log_acl_actions` ⬚⬚⬚⬚⬚⬚

- `drop` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚
- `allow` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚
- `allow,drop` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
# tail -f /var/log/ovn/ovn-controller.log
2024-08-14T09:27:49.590Z|00004|acl_log(ovn_pinctrl0)|INFO|name="np/allow-from-client.default/ingress/IPv4/0", verdict=allow, severity=info, direction=to-
lport: icmp,vlan_tci=0x0000,dl_src=96:7b:b0:2f:a0:1a,dl_dst=a6:e5:1b:c2:1b:f8,nw_src=10.16.0.7,nw_dst=10.
16.0.12,nw_tos=0,nw_ecn=0,nw_ttl=64,nw_frag=no,icmp_type=8,icmp_code=0
```

⬚⬚⬚⬚⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `ovn.kubernetes.io/acl_log_meter_rate` annotation ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-from-client
  namespace: default
  annotations:
    ovn.kubernetes.io/enable_log: "true"
    ovn.kubernetes.io/log_acl_actions: "allow"
    ovn.kubernetes.io/acl_log_meter_rate: "100"
spec:
  podSelector:
    matchLabels:
      app: web
  policyTypes:
  - Ingress
  ingress:
  - from:
    - podSelector:
        matchLabels:
          app: client
```

`ovn.kubernetes.io/acl_log_meter_rate` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚/⬚⬚⬚⬚⬚⬚ `100` ⬚⬚⬚⬚⬚⬚⬚⬚ 100 ⬚⬚⬚⬚

⬚⬚⬚⬚⬚⬚

⬚ annotation `ovn.kubernetes.io/enable_log` ⬚⬚⬚ `false` ⬚⬚⬚⬚⬚⬚

```
kubectl annotate networkpolicy -n default allow-from-client ovn.kubernetes.io/enable_log=false --overwrite
```

⬚⬚⬚⬚⬚⬚

Kube-OVN ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

- **standard**⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ NetworkPolicy ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ IP ⬚⬚⬚⬚⬚⬚⬚⬚
- **lax**⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: example-policy
  namespace: default
  annotations:
    ovn.kubernetes.io/network_policy_enforcement: "lax"
spec:
  podSelector: {}
  policyTypes:
  - Ingress
```

⬚⬚⬚⬚ Kube-OVN ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `--np-enforcement` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

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- ⬚⬚⬚ NetworkPolicy

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## 3.12.2 ▯▯▯ NetworkPolicy

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▯▯▯▯▯▯Kube-OVN ▯▯ NetworkPolicy ▯▯▯ `podSelector` ▯▯▯ Pod ▯▯▯▯ OVN ▯▯▯

▯▯▯▯▯ Pod▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Provider▯

- `ovn.kubernetes.io/network_policy_for`

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Pod ▯▯▯▯▯▯▯

▯▯▯▯

▯▯▯▯▯▯▯▯▯▯

```
metadata:
  annotations:
    ovn.kubernetes.io/network_policy_for: "ovn,default/net-a,default/net-b"
```

▯▯▯▯▯▯▯▯▯

- `ovn` ▯▯▯ OVN Provider▯
- `<namespace>/<net-attach-def>`

▯▯▯

- `ovn`
- `default/net-a`
- `ovn,default/net-a`

**Provider** ▯▯▯▯

- ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ OVN Provider▯
- ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯
- ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Provider▯▯▯▯▯▯▯▯▯▯▯▯
- ▯▯▯▯▯▯▯▯▯▯▯▯

`<namespace>/<net-attach-def>` ▯ Kube-OVN ▯▯▯▯▯▯▯ Provider ▯▯▯▯▯

- `<nad-name>.<nad-namespace>.ovn`

▯▯

▯▯ Pod ▯▯▯▯▯▯▯

- ▯▯ OVN Provider▯ `ovn` ▯
- `default/net-a`
- `default/net-b`

▯▯▯▯

- ▯▯▯ `network_policy_for` ▯
- ▯▯▯▯▯▯ `ovn` ▯ `net-a` ▯ `net-b`
- `network_policy_for: default/net-a` ▯
- ▯▯▯▯▯▯▯ `net-a`
- `network_policy_for: ovn,default/net-b` ▯

- □□□□□ `ovn` □ `net-b`

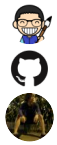
□□

- □□□□□□□□□□□□□ Provider/□□□□□□□□ Kubernetes NetworkPolicy □□□□□
- □□□□□□□□□□□□□□□□□□□□□□□□□□□

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□□

## 3.12.3 ■■■■■■■■■■

Kubernetes ■■■ NetworkPolicy ■■■■■ L3 ■ L4 ■■■■■■■■■■■■■■■ AdminNetworkPolicy (ANP) ■■■■■■■■■■■■■■■

[illegible]

[illegible]

```
apiVersion: policy.networking.k8s.io/v1alpha1
kind: AdminNetworkPolicy
metadata:
  name: deny-external-domains
spec:
  priority: 55
  subject:
    namespaces:
      matchLabels:
        kubernetes.io/metadata.name: kube-system
  egress:
  - action: Deny
    name: deny-baidu-google
    to:
    - domainNames:
      - '*.baidu.com.'
      - '*.google.com.'
```

[illegible]

| [illegible] | [illegible] |
| --- | --- |
| `priority` | [illegible] |
| `subject` | [illegible]Pod [illegible] |
| `egress` | [illegible] |
| `action` | [illegible] `Allow` / `Deny` / `Pass` |
| `domainNames` | [illegible] `.` [illegible] |

[illegible]

[illegible] kube-ovn-pinger [illegible]

```
#
kubectl exec -it -n kube-system kube-ovn-pinger-xxxxx -- ping baidu.com
```

> [illegible] DNS [illegible] ACL [illegible]

[illegible] DNSNameResolver [illegible]

```
# kubectl get dnsnameresolver
NAME                                DNS NAME        RESOLVED IPS
anp-deny-external-domains-88dc32ab  *.google.com.
anp-deny-external-domains-fb3029ce  *.baidu.com.    220.181.7.203
```

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[illegible]

## 3.12.4 ■■■■

■■■■■■■■■■ Pod ■■■■■■■■■■■■■■■■■■■■■■Pod ■■■■■■ Kube-OVN IPAM ■■■■ MAC ■■■■ IP ■■■■■■■■■■■■■■■■■■■

# 4. KubeVirt

## 4.1 VM □□ IP

□□□□□□□□□□ IP □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ VM □□□□□□□□□□ VM □ IP □□□□□□□□□□□□□□□□□□□□□□□

□□□□□□□□ CNI □□□□□□□□□□

- □□□ IP □□□ VM □□□□□□□□VM □□□□□□□ IP □□□□□□□
- IP □□□ Node □□□VM □□□□□□□□□□□□□□□□□ IP□
- □□□□□□□□ IP □□□□□□□□□□□□ VM □ IP □□□

□□□□□□□□ KubeVirt □ `masquerade` □□□□□□□□ iptables □ VM □□□□□□□□□□□□□□□□□□□ VM □ IP □□□□□ `masquerade` □□ `bridge` □□□□□□□□

- Pod IP □ VM IP □□□□□□□□□□□□□□□□□□□□□□□□□□ Pod IP □□□□□□□□□□□□□□□□□□□□□□□
- `masquerade` □□ iptables □□□□□□□□□□□□□ `bridge` □□□□□□
- `masquerade` □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
- `masquerade` □□□□□ conntrack □□□□□□□□□□□□□□□□□□□□□□□□□□

Kube-OVN □□□ KubeVirt □ `bridge` □ `managedTap` □□□□□□□ IP □ VM □□□□□□□□□ IP □□□ VM □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ annotation □□□□□ VM □□□□□ IP □□□□□□□□□□□□ Kube-OVN □□□□□ Multus □□□□□□□□□□□□□□□□□□□□

```
          cloudInitNoCloud:
            userDataBase64: SGkuXG4=
```

1. ▒▒ VM ▒▒

```
kubectl get vmi testvm
```

1. ▒▒ VM

```
virtctl restart testvm
```

1. ▒▒▒ VM

```
virtctl migrate testvm
```

▒▒▒▒▒ bridge ▒▒▒ VM ▒▒▒▒▒▒▒▒IP ▒▒▒▒▒▒▒▒

## 4.1.2 ▒▒ IP/Mac ▒▒

▒▒▒▒▒▒ VM ▒ IP/Mac ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ VM ▒▒▒ annotation ▒▒▒▒▒ VM ▒▒ IP ▒▒▒▒▒▒▒▒▒▒▒▒▒ KubeVirt ▒▒▒▒▒ VM ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

### 4.1.3 VM □□ IP

Kube-OVN □□□□□□□□□ VM IP□□□□ IP □□□□□ VM □□□□□□

□□□□□□□□□ VM IP□

1. □□ VM Annotation□□□□□□□□ IP □□□

2. □□ `virtctl restart <vm name>` □□ VM□□□□ IP □□□□□

### 4.1.4 □□□□

□□□□□ VM □□□□□□□ IP□□□□□□□□ VM □□□□□□□□□□□□ `ovn.kubernetes.io/logical_switch` □□□□

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        ovn.kubernetes.io/logical_switch: subnet1 #(1)
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
      networks:
      - name: default
        pod: {}
      volumes:
        - name: containerdisk
          containerDisk:
            image: quay.io/kubevirt/cirros-container-disk-demo
        - name: cloudinitdisk
          cloudInitNoCloud:
            userDataBase64: SGkuXG4=
```

1. 🙋 □□□□□ VM □□□□□

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### 4.1.5 □□

## 4.2 ⬚⬚⬚⬚⬚

KubeVirt ⬚ Clone ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ Label ⬚ Annotation⬚⬚⬚⬚⬚⬚⬚⬚ Kube-OVN ⬚ Annotation ⬚⬚⬚ IP⬚MAC ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

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## 4.2.3 ⬜⬜

# 4.3 ▒▒▒▒

▒ KubeVirt ▒ Bridge ▒▒▒▒▒▒DHCP ▒▒▒ virt-launcher ▒▒▒▒▒▒▒▒ KubeVirt ▒▒▒▒ IPv4 ▒▒▒ DHCP▒▒▒ Bridge ▒▒▒ KubeVirt VM ▒▒▒▒ RA ▒▒▒▒▒▒ IPv6 ▒▒▒▒▒ Kube-OVN ▒▒▒ DHCP ▒ RA ▒▒▒▒▒▒ KubeVirt ▒▒▒▒▒ DHCP/RA ▒▒▒▒▒▒▒▒▒▒▒▒▒

▒ KubeVirt 1.4.0 ▒▒▒▒▒▒▒▒ Network Binding Plugin ▒▒▒▒▒ Bridge ▒▒▒▒▒ `managedTap` ▒▒▒▒▒ KubeVirt ▒▒▒ DHCP ▒▒▒▒▒▒▒▒▒▒ `managedTap` ▒▒▒ Kube-OVN ▒ DHCP/RA ▒▒▒▒▒▒▒▒ VM ▒▒▒▒▒▒▒▒▒▒▒▒▒

## 4.3.1 ▒▒▒▒ DHCP

▒ Kube-OVN ▒ Subnet ▒▒▒ DHCP ▒ IPv6 RA ▒▒▒▒▒▒▒▒ YAML ▒▒▒▒▒

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: dual-stack-subnet
spec:
  cidrBlock: "10.244.0.0/16,fd00:10:244::/64"
  enableDHCP: true
  enableIPv6RA: true
```

## 4.3.2 ▒▒ `managedTap` ▒▒▒▒

▒ KubeVirt ▒▒▒ `managedTap` ▒ Network Binding Plugin:

```
# kubectl patch kubevirts -n kubevirt kubevirt --type=json -p=\
'[{"op": "add", "path": "/spec/configuration/network",    "value": {
    "binding": {
        "managedtap": {
            "domainAttachmentType": "managedTap"
        }
    }
}}]'
```

## 4.3.3 ▒▒▒▒▒▒▒▒▒▒ `managedTap` ▒▒▒▒

```yaml
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: dual-stack-vm
  namespace: default
spec:
  running: false
  template:
    spec:
      domain:
        devices:
          interfaces:
            - name: default
              binding:
                name: managedtap
      networks:
      - name: default
        pod: {}
```

▒▒▒▒▒▒▒▒▒▒▒▒ VM ▒▒ DHCP ▒ IPv6 RA ▒▒▒▒▒▒▒▒▒ IPv4/IPv6 ▒▒▒

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## 4.3.4 ▯▯

## 4.4 ...

...

KubeVirt ...

- KubeVirt ... Bridge ...
- KubeVirt ...
- ... IP ...
- ...

Kube-OVN ... 0.5 ... TCP ...

### 4.4.1 ...

... VM Spec ... `kubevirt.io/allow-pod-bridge-network-live-migration: "true"` ... annotation...Kube-OVN ...

1. ... VM

```
kubectl apply -f - <<EOF
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: testvm
spec:
  runStrategy: Always
  template:
    metadata:
      labels:
        kubevirt.io/size: small
        kubevirt.io/domain: testvm
      annotations:
        kubevirt.io/allow-pod-bridge-network-live-migration: "true"
    spec:
      domain:
        devices:
          disks:
            - name: containerdisk
              disk:
                bus: virtio
            - name: cloudinitdisk
              disk:
                bus: virtio
          interfaces:
          - name: default
            bridge: {}
        resources:
          requests:
            memory: 64M
      networks:
      - name: default
        pod: {}
      volumes:
        - name: containerdisk
          containerDisk:
            image: quay.io/kubevirt/cirros-container-disk-demo
        - name: cloudinitdisk
          cloudInitNoCloud:
            userDataBase64: SGkuXG4=
EOF
```

1. SSH ...

```
# password: gocubsgo
virtctl ssh cirros@testvm
ping 8.8.8.8
```

1. ...

```
virtctl migrate testvm
```

... VM ... SSH ... ping ...

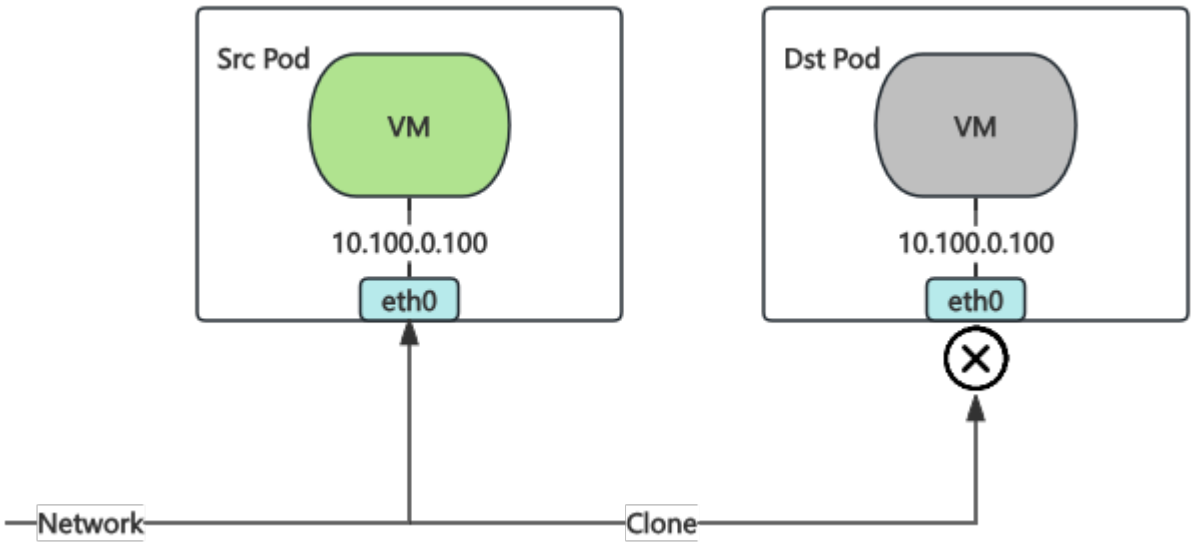


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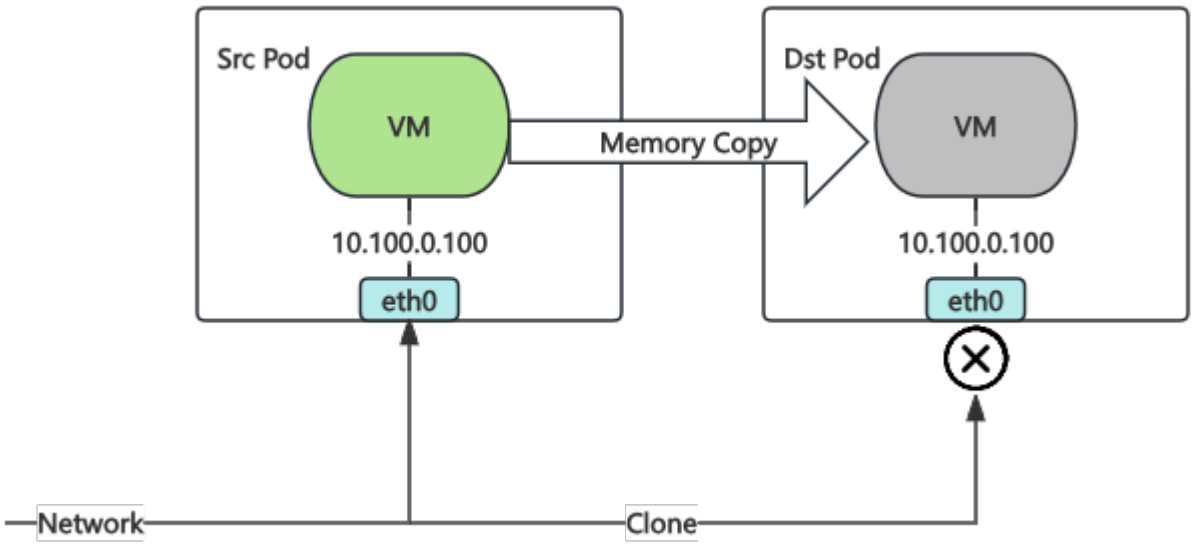
## 4.4.2 □□□□□

Kube-OVN □□□□□□□□□□□□□□□□□□□Live migration - Reducing downtime with multchassis port bindings□

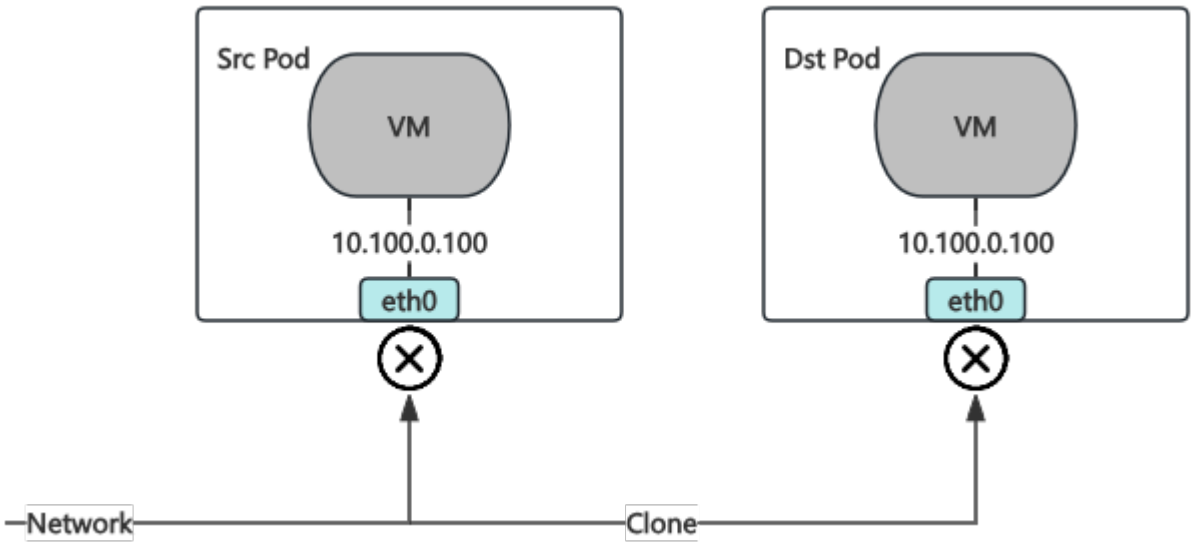
□□□□□□□□□□□□□□□□□□



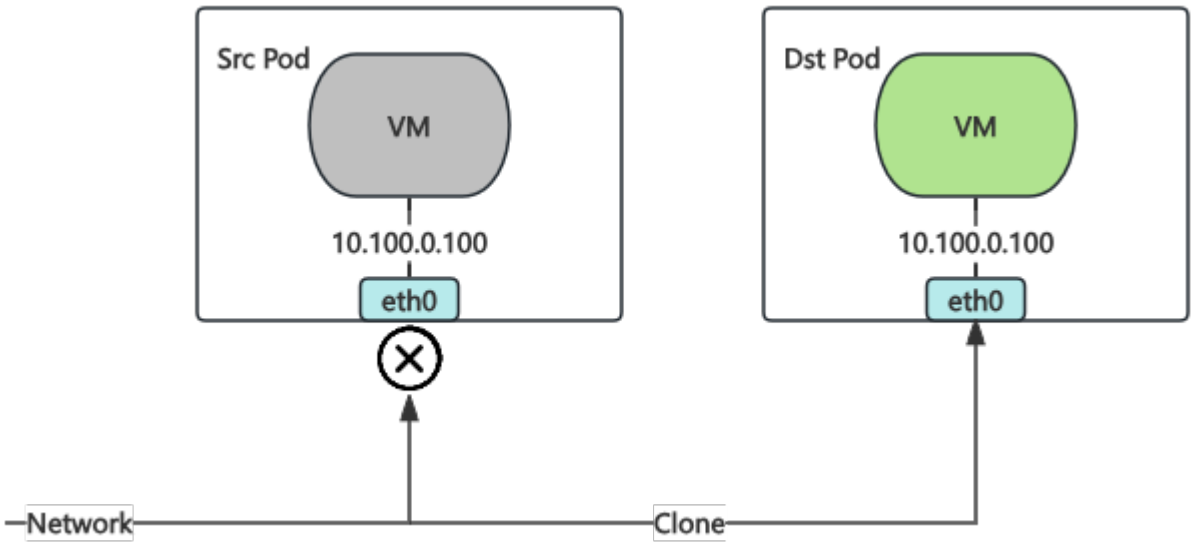

1. KubeVirt □□ VM □□□□□



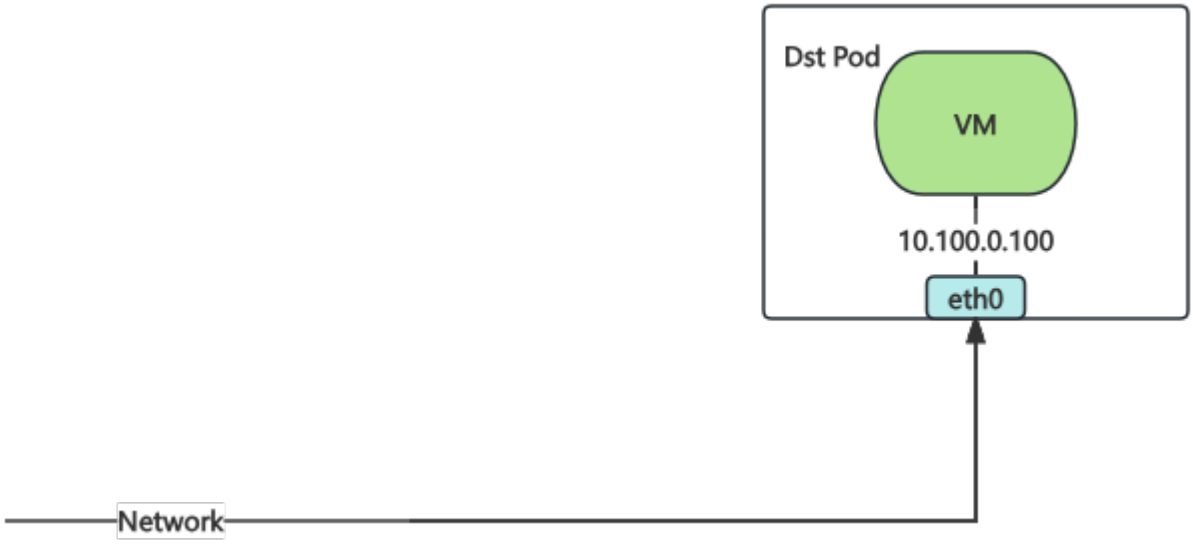

1. KubeVirt □□□□□□□□□□□ Pod□□□□ Pod □□□□□□□□□□□




1. KubeVirt □□□□ Pod□□□ libvirt □□□ RARP □□□□□□ Pod □□□□□□□□ Pod □□□□□□□




1. KubeVirt □□□ Pod□□□□□□□Kube-OVN □□ Watch Migration CR □□□□□□□□□□□□□□□□□□□□□□□□□




⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ 5 ⬚⬚⬚ 6 ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ libvirt ⬚ RARP ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ 0.5 ⬚⬚⬚⬚⬚ TCP ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

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4.4.3 ⬚⬚

# 4.5 ⬚⬚⬚⬚⬚

Kube-OVN ⬚ Multus Dynamic Networks Controller ⬚⬚⬚⬚⬚⬚⬚ KubeVirt ⬚ v1.4.0 ⬚⬚⬚⬚⬚⬚⬚

```
        devices:
          disks:
          - disk:
              bus: virtio
            name: containerdisk
          interfaces:
          - masquerade: {}
            name: defaultnetwork
          rng: {}
        resources:
          requests:
            memory: 1024M
      networks:
      - name: defaultnetwork
        pod: {}
      terminationGracePeriodSeconds: 0
      volumes:
      - containerDisk:
          image: quay.io/kubevirt/fedora-with-test-tooling-container-disk:devel
        name: containerdisk
```

▯▯▯▯

▯▯ VM ▯ Spec ▯▯▯▯▯▯▯▯▯

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
template:
  spec:
    domain:
      devices:
        interfaces:
        - name: defaultnetwork
          masquerade: {}
          # new interface
        - name: dyniface1
          bridge: {}
    networks:
    - name: defaultnetwork
      pod: {}
      # new network
    - name: dyniface1
      multus:
        networkName: attachnet
```

▯▯▯▯

▯▯▯ `interface` ▯ `state` ▯▯▯ `absent` ▯▯▯▯▯▯▯▯▯

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: vm-fedora
template:
  spec:
    domain:
      devices:
        interfaces:
          - name: defaultnetwork
            masquerade: {}
          # set the interface state to absent
          - name: dyniface1
            state: absent
            bridge: {}
    networks:
      - name: defaultnetwork
        pod: {}
      - name: dyniface1
        multus:
          networkName: attachnet
```

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4.5.4 ⬚⬚

# 4.6 DHCP

□□□ `managedTap` □ `SR-IOV` □ `DPDK` □□□□□□KubeVirt □□□ DHCP □□□□□□□□□□□□□Kube-OVN □□□□ OVN □ DHCP □□□□□□□□ Pod □□□□ DHCP □□□□□□□□□□□□□□ KubeVirt □□□□□□ DHCP □□□□□ IP □□□□□□ Kube-OVN □ DHCP □□□□ DHCPv6, IPv6RA, DNS□TFTP □ DHCP □□□□□□□□□□□□□□□□□□□□ DHCP □□□□□□□□□

> **Warning**
>
> □□ `bridge` □□□□□□KubeVirt □ DHCP □□□ Kube-OVN □□□□□ DHCP □□□□□ Kube-OVN □□□ DHCP □□□□□□□□□□□□□ Kube-OVN □□□□□ DHCP □□□ □□□□`managedTap` □□□□□□ `bridge` □□□□□`managedTap` □□□□□□□□□□□□ managedTap □□□□□

## 4.6.1 □□□□ DHCP □□

□□ DHCP □□□□□□

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: sn-dual
spec:
  cidrBlock: "10.0.0.0/24,240e::a00/120"
  default: false
  disableGatewayCheck: true
  disableInterConnection: false
  excludeIps:
    - 10.0.0.1
    - 240e::a01
  gateway: 10.0.0.1,240e::a01
  gatewayNode: ''
  gatewayType: distributed
  natOutgoing: false
  private: false
  protocol: Dual
  provider: ovn
  vpc: vpc-test
  enableDHCP: true
  dhcpV4Options: "lease_time=3600,router=10.0.0.1,server_id=169.254.0.254,server_mac=00:00:00:2E:2F:B8"
  dhcpV6Options: "server_id=00:00:00:2E:2F:C5"
  enableIPv6RA: true
  ipv6RAConfigs: "address_mode=dhcpv6_stateful,max_interval=30,min_interval=5,send_periodic=true"
```

- `enableDHCP` : □□□□□□□ DHCP □□□
- `dhcpV4Options` , `dhcpV6Options` : □□□□□□□ ovn-nb □ DHCP □□□□□□□□ DHCP Options□ □□□□□□□ `"lease_time=3600, router=$ipv4_gateway, server_id=169.254.0.254, server_mac=$random_mac"` □ `server_id=$random_mac` □
- `enableIPv6RA` : □□□□ DHCPv6 □□□□□□□□
- `ipv6RAConfigs` □□□□□□□□ ovn-nb □ Logical_Router_Port □□□□□□□□ Logical Router Port□□□□□□ `address_mode=dhcpv6_stateful, max_interval=30, min_interval=5, send_periodic=true` □

## 4.6.2 Pod □□ DHCP □□

□□□□□□□ DHCP □□□□Kube-OVN □□□□□□ Pod □□□□□□ Pod □□□□□□□□□□ DHCP □□□Pod □□□ DHCP □□□□□□□□□□□□□□□□□□□□ DHCP □□□□□□□□□□ `enableDHCP` □□□□□□□

□□□□□

```yaml
#     provider   "ovn"
ovn.kubernetes.io/dhcp-v4-options: "lease_time=3600,router=10.0.0.1,dns_server=8.8.8.8"
ovn.kubernetes.io/dhcp-v6-options: "server_id=00:00:00:00:00:01"

#      provider   "net1.ns1.ovn"
net1.ns1.ovn.kubernetes.io/dhcp-v4-options: "lease_time=7200"
```

□□□ key □□□ `<provider>.kubernetes.io/dhcp-v4-options` □ `<provider>.kubernetes.io/dhcp-v6-options` □□□ `<provider>` □□□□□□□□□□□□□□□□□□ provider □ `ovn` □□□□□□ Multus □□□□□□□□□provider □□□ `<net-attach-def-name>.<namespace>.ovn` □



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▯▯▯▯

▯▯▯▯ Pod ▯▯▯▯▯▯ DHCPv4 ▯▯▯

```
apiVersion: v1
kind: Pod
metadata:
  name: dhcp-pod
  annotations:
    ovn.kubernetes.io/dhcp-v4-options: "lease_time=3600,router=10.0.0.1,dns_server=8.8.8.8"
    ovn.kubernetes.io/dhcp-v6-options: "server_id=00:00:00:00:00:01"
spec:
  containers:
    - name: test
      image: docker.io/library/nginx:alpine
```

▯▯ Multus ▯▯▯▯▯▯▯▯▯ provider ▯▯▯▯▯▯▯▯▯▯▯▯ DHCP ▯▯▯

```
apiVersion: v1
kind: Pod
metadata:
  name: multi-nic-dhcp-pod
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{"name": "net1", "namespace": "ns1"}]'
    ovn.kubernetes.io/dhcp-v4-options: "lease_time=3600,router=10.0.0.1"
    net1.ns1.ovn.kubernetes.io/dhcp-v4-options: "lease_time=7200,router=10.0.1.1"
spec:
  containers:
    - name: test
      image: docker.io/library/nginx:alpine
```

▯▯▯▯▯

DHCP ▯▯▯▯▯▯▯▯▯▯▯▯▯

1. **Pod** ▯▯▯▯▯ `<provider>.kubernetes.io/dhcp-v4-options` ▯ `<provider>.kubernetes.io/dhcp-v6-options` ▯▯▯ Pod ▯▯ DHCP ▯▯▯
2. ▯▯▯▯▯▯▯ Subnet CRD ▯ `dhcpV4Options` ▯ `dhcpV6Options` ▯▯▯▯▯▯▯▯▯ DHCP ▯▯▯

▯ Pod ▯▯▯ DHCP ▯▯▯▯▯▯▯▯▯ Pod ▯▯▯▯ DHCP ▯▯▯▯▯▯▯▯▯▯▯▯▯

> **Note**
>
> ▯▯▯▯▯▯▯ Pod ▯▯ DHCP ▯▯▯▯▯▯▯▯ DHCP ▯▯▯▯▯▯▯ Pod ▯▯▯▯▯

PDF Slack Support

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GitHub

# 5. VPC ⊠⊠

## 5.1 VPC ⊠⊠

Kube-OVN ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ VPC ⊠⊠⊠⊠⊠ VPC ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ Subnet ⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠EIP ⊠⊠⊠⊠

> VPC ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ Kubernetes ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠ Pod ⊠⊠⊠NodePort ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ DNS ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠ Kubernetes ⊠⊠⊠⊠⊠⊠⊠ Kube-OVN ⊠⊠ VPC ⊠⊠⊠⊠⊠⊠⊠⊠⊠ VPC ⊠⊠ Subnet ⊠⊠⊠⊠⊠ Kubernetes ⊠⊠⊠⊠⊠⊠⊠⊠⊠ VPC ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠EIP ⊠ NAT ⊠⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ VPC ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ ACL ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ VPC ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ VPC ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ VPC ⊠⊠⊠⊠⊠⊠ ⊠ Underlay ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯ CIDR:

```
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net1
spec:
  vpc: test-vpc-1
  cidrBlock: 10.0.1.0/24
  protocol: IPv4
  namespaces:
    - ns1
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: test-vpc-2
  cidrBlock: 10.0.1.0/24
  protocol: IPv4
  namespaces:
    - ns2
```

▯▯▯▯▯ Namespace ▯▯▯ Pod:

```
apiVersion: v1
kind: Pod
metadata:
  namespace: ns1
  name: vpc1-pod
spec:
  containers:
    - name: vpc1-pod
      image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  namespace: ns2
  name: vpc2-pod
spec:
  containers:
    - name: vpc2-pod
      image: docker.io/library/nginx:alpine
```

▯▯▯▯▯▯▯▯▯▯ Pod ▯▯▯▯▯▯▯ CIDR▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC▯▯▯ Pod ▯▯▯▯▯▯▯

## 5.1.3 ▯▯ VPC ▯▯

VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯ IP▯SNAT ▯ DNAT ▯▯▯

VPC ▯▯▯▯▯▯▯ Multus-CNI ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ multus-cni▯

> **Note**
>
> ▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯
>
> ▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯VPC NAT ▯▯▯OVN ▯▯▯ Egress Gateway▯▯▯ VPC NAT ▯▯▯ Kube-OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC NAT ▯▯▯▯▯▯▯▯▯▯ Pod▯▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯ Macvlan ▯▯▯▯▯▯▯▯▯▯▯▯▯ Pod ▯▯ iptables ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
  - 192.168.0.1..192.168.0.10
---
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-vpc-external-network
  namespace: kube-system
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth1",
      "mode": "bridge",
      "ipam": {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "ovn-vpc-external-network.kube-system"
      }
    }'

         `ipam`        VPC     net1     Kube-OVN    IP
---
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-vpc-external-network
  namespace: kube-system
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth1",
      "mode": "bridge"
    }'
```

- ▯ Subnet ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Macvlan ▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ IP▯
- VPC ▯▯▯▯ Macvlan ▯▯▯▯▯▯▯▯ `NetworkAttachmentDefinition` ▯ `master` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯
- `name` ▯▯▯▯▯▯▯▯

▯ Macvlan ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯L2/L3 ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯Vlan ▯▯▯▯▯▯▯▯▯▯

1. ▯▯ OpenStack ▯ VM ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `PortSecurity` ▯▯▯
2. ▯▯ VMware ▯ vSwitch ▯▯▯▯▯▯▯ `MAC Address Changes`, `Forged Transmits` ▯ `Promiscuous Mode Operation` ▯▯▯ `allow` ▯
3. ▯▯ Hyper-V ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `MAC Address Spoofing` ▯
4. ▯▯▯▯▯▯▯ AWS▯GCE▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Mac ▯▯▯▯ Macvlan ▯▯▯▯▯
5. ▯▯ Macvlan ▯▯▯▯▯▯▯Macvlan ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ VpcNATGateway Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Pod▯
6. ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Trunk ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Macvlan ▯▯▯

### ▯▯ **VPC** ▯▯▯▯

VPC ▯▯▯▯▯▯▯▯▯ `kube-system` ▯▯▯▯▯▯▯ `ovn-vpc-nat-gw-config` ▯▯▯▯▯▯▯▯▯ `ovn-vpc-nat-config` ConfigMap ▯▯▯▯▯▯ `image` ▯▯▯ Pod ▯▯▯▯▯▯▯▯▯▯ `nodeSelector` ▯▯▯▯▯▯ `nodeSelector` ▯▯▯▯ `LoadBalancer` ▯▯ Service ▯ lb-svc ▯▯▯VpcNATGateway Pod ▯▯▯▯▯ `VpcNatGateway.spec.selector` ▯▯▯▯▯▯▯

```
---
kind: ConfigMap
apiVersion: v1
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
data:
  image: 'docker.io/kubeovn/vpc-nat-gateway:v1.17.0'
  nodeSelector: |
    kubernetes.io/hostname: kube-ovn-control-plane
---
kind: ConfigMap
apiVersion: v1
metadata:
  name: ovn-vpc-nat-gw-config
  namespace: kube-system
data:
  enable-vpc-nat-gw: 'true'
```

- `image` : ▯▯ Pod ▯▯▯▯▯▯▯▯
- `enable-vpc-nat-gw` ▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯



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▯▯ **VPC ▯▯**

```
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
  externalSubnets:
    - ovn-vpc-external-network
  noDefaultEIP: false
```

- `vpc` ▯▯ VpcNatGateway ▯▯▯ VPC▯
- `subnet` ▯▯ VPC ▯▯▯ Subnet ▯▯VPC ▯▯ Pod ▯▯▯▯▯▯▯ `lanIp` ▯▯▯▯▯▯▯▯
- `lanIp` ▯ `subnet` ▯▯▯▯▯▯▯▯ IP▯VPC ▯▯ Pod ▯▯▯▯▯▯ IP▯▯ VPC ▯▯▯▯▯▯▯▯▯▯ VpcNatGateway ▯ `nextHopIP` ▯▯▯▯▯▯▯ `lanIp` ▯
- `selector` ▯VpcNatGateway Pod ▯▯▯▯▯▯▯▯▯▯ Kubernetes ▯▯ NodeSelector ▯▯▯▯▯
- `externalSubnets` ▯VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `ovn-vpc-external-network` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯
- `noDefaultEIP` ▯VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ EIP▯▯▯▯ `false` ▯▯ v1.15 ▯▯▯▯▯▯▯▯ BGP ▯▯▯▯▯▯▯▯▯▯▯ `true` ▯▯▯ Underlay ▯▯▯

▯▯▯▯▯▯▯▯

- `tolerations` ▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯
- `affinity` ▯▯ VPC ▯▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯

▯▯▯ VPC-NAT-GW ▯▯▯▯▯▯▯

1. nat gw pod ▯▯▯▯▯▯net1 arp ▯▯▯▯▯▯ping ▯▯▯▯▯▯▯▯▯▯ eip ▯▯▯▯▯▯▯ arp▯▯▯ ping ▯▯

▯▯ **EIP**

EIP ▯▯▯▯▯▯▯▯▯▯ IP ▯▯▯ VPC ▯▯▯▯▯▯▯ DNAT▯SNAT ▯▯▯ IP ▯▯▯

▯▯▯▯▯▯▯▯▯▯ EIP▯

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-random
spec:
  natGwDp: gw1
```

▯▯ EIP ▯▯▯▯▯▯

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  natGwDp: gw1
  v4ip: 192.168.0.100
```

▯▯ EIP ▯▯▯▯▯▯▯▯▯

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-random
spec:
  natGwDp: gw1
  externalSubnet: ovn-vpc-external-network
```

- `externalSubnet` ▯EIP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `ovn-vpc-external-network` ▯IptablesEIP ▯▯▯▯▯▯▯ VpcNatGateway ▯ `externalSubnets` ▯▯▯▯▯▯▯▯EIP ▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

□□ **DNAT** □□

□□ DNAT □□□□□□□□□□□ EIP □□□□□□□□□ VPC □□□□ IP □□□□

```
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eipd01
spec:
  natGwDp: gw1

---
kind: IptablesDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: dnat01
spec:
  eip: eipd01
  externalPort: '8888'
  internalIp: 10.0.1.10
  internalPort: '80'
  protocol: tcp
```

□□ **SNAT** □□

□□ SNAT □□□VPC □□ Pod □□□□□□□□□□□□□□□ EIP □□ SNAT□

```
---
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eips01
spec:
  natGwDp: gw1
---
kind: IptablesSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat01
spec:
  eip: eips01
  internalCIDR: 10.0.1.0/24
```

□□□□ **IP**

□□□□ IP □□□VPC □□□□ IP □□ EIP □□□□□□□□□□□□□□□□ EIP □□ VPC □□ IP□VPC □□□□ IP □□□□□□□□□ SNAT □□□ EIP□

```
---
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eipf01
spec:
  natGwDp: gw1

---
kind: IptablesFIPRule
apiVersion: kubeovn.io/v1
metadata:
  name: fip01
spec:
  eip: eipf01
  internalIp: 10.0.1.5
```

## 5.1.4 □□□□□

□□□□ VPC □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ Kube-OVN □□□□□□□□□□□□□□□□□□□

□□□□

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  staticRoutes:
    - cidr: 0.0.0.0/0
      nextHopIP: 10.0.1.254
      policy: policyDst
```



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```
    - cidr: 172.31.0.0/24
      nextHopIP: 10.0.1.253
      policy: policySrc
      routeTable: "rtb1"
```

- `policy`：[illegible] `policyDst` [illegible] `policySrc`。
- [illegible]CIDR [illegible]
- `routeTable`：[illegible]

[illegible]

[illegible] OVN [illegible] Logical Router Policy。

[illegible]

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  policyRoutes:
    - action: drop
      match: ip4.src==10.0.1.0/24 && ip4.dst==10.0.1.250
      priority: 11
    - action: reroute
      match: ip4.src==10.0.1.0/24
      nextHopIP: 10.0.1.252
      priority: 10
```

## 5.1.5 [illegible]

Kubernetes [illegible] Service [illegible] Kubernetes [illegible] Service [illegible] IP [illegible] VPC [illegible] [illegible] VPC [illegible] Kubernetes [illegible] Service [illegible]

[illegible]Kube-OVN [illegible] `SwitchLBRule` [illegible]

[illegible] `SwitchLBRule` [illegible]

```
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name:  cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
  - name: dns
    port: 8888
    targetPort: 80
    protocol: TCP
```

- `vip` [illegible]
- `namespace` [illegible] Pod [illegible] Namespace。
- `sessionAffinity` [illegible] Service [illegible] `sessionAffinity` [illegible]
- `selector` [illegible] Service [illegible] `selector` [illegible]
- `ports` [illegible] Service [illegible] `port` [illegible]

[illegible]

```
# kubectl get slr
NAME               VIP         PORT(S)                 SERVICE                            AGE
vpc-dns-test-cjh2  10.96.0.3   53/UDP,53/TCP,9153/TCP  kube-system/slr-vpc-dns-test-cjh2  88m
```

## 5.1.6 [illegible] vpc-dns

[illegible] VPC [illegible] VPC [illegible]VPC [illegible] Pod [illegible] coredns [illegible] [illegible] VPC [illegible] CoreDNS [illegible] Service [illegible] Kube-OVN [illegible] vpc-dns [illegible]



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□□□□□□□

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-nad
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "ovn-nad.default.ovn"
    }'
```

□□ **vpc-dns □ ConfigMap**

□ kube-system □□□□□□□ configmap□□□ vpc-dns □□□□□□□□□□□ vpc-dns □□□

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-config
  namespace: kube-system
data:
  coredns-vip: 10.96.0.3
  enable-vpc-dns: "true"
  nad-name: ovn-nad
  nad-provider: ovn-nad.default.ovn
```

- `enable-vpc-dns` □□□□□□ `true` □□□□□ `false` □□□□□□□ `true` □
- `coredns-image` □□□□□□□dns □□□□□□□□□□ coredns □□□□□
- `coredns-vip` □□ coredns □□ lb □□□ vip□
- `nad-name` □□□□ `network-attachment-definitions` □□□□□
- `nad-provider` □□□□ provider □□□
- `k8s-service-host` □□□□□□□□ coredns □□ k8s apiserver □□□ ip□
- `k8s-service-port` □□□□□□□□ coredns □□ k8s apiserver □□□ port□

□□ **vpc-dns □□□□**

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: system:vpc-dns
rules:
  - apiGroups:
    - ""
    resources:
    - endpoints
    - services
    - pods
    - namespaces
    verbs:
    - list
    - watch
  - apiGroups:
    - discovery.k8s.io
    resources:
    - endpointslices
    verbs:
    - list
    - watch
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  annotations:
    rbac.authorization.kubernetes.io/autoupdate: "true"
  labels:
    kubernetes.io/bootstrapping: rbac-defaults
  name: vpc-dns
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: system:vpc-dns
subjects:
```

```
- kind: ServiceAccount
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: vpc-dns
  namespace: kube-system
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-corefile
  namespace: kube-system
data:
  Corefile: |
    .::53 {
        errors
        health {
          lameduck 5s
        }
        ready
        kubernetes cluster.local in-addr.arpa ip6.arpa {
          pods insecure
          fallthrough in-addr.arpa ip6.arpa
        }
        prometheus :9153
        forward . /etc/resolv.conf {
          prefer_udp
        }
        cache 30
        loop
        reload
        loadbalance
    }
```

**██ vpc-dns**

```
kind: VpcDns
apiVersion: kubeovn.io/v1
metadata:
  name: test-cjh1
spec:
  vpc: cjh-vpc-1
  subnet: cjh-subnet-1
```

- `vpc` ██████ dns ███ vpc ███
- `subnet` ██████ dns ██████████

████████

```
[root@hci-dev-mst-1 kubeovn]# kubectl get vpc-dns
NAME        ACTIVE   VPC         SUBNET
test-cjh1   false    cjh-vpc-1   cjh-subnet-1
test-cjh2   true     cjh-vpc-1   cjh-subnet-2
```

- `ACTIVE`：`true` ██████████ dns ███ `false` ███

██

- ██ VPC ████████████ DNS ███
- ███ VPC ██████ vpc-dns ████████ VPC ███ subnet██████ vpc-dns ██████ `true` ████ `false` ▯
- ▯ `true` ▯ vpc-dns ████████████ `false` ▯ vpc-dns ██████

## 5.1.7 ████

██████ VPC ████████████████████████ VPC ████████

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: test-vpc-1
spec:
  namespaces:
```

```
  - ns1
  defaultSubnet: test
```

- `defaultSubnet` ⬚VPC ⬚⬚⬚⬚⬚⬚⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ Namespace ⬚⬚ `ovn.kubernetes.io/logical_switch` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `ovn.kubernetes.io/logical_switch` ⬚⬚⬚ Pod ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

### ⬚⬚⬚ VPC Pod ⬚⬚ livenessProbe ⬚ readinessProbe

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚ Pod ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ kubelet ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚ Pod⬚Kube-OVN ⬚⬚ TProxy ⬚ kubelet ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚ Pod⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚ DaemonSet `kube-ovn-cni` ⬚⬚⬚⬚⬚⬚ `--enable-tproxy=true` ⬚

```
spec:
  template:
    spec:
      containers:
      - args:
        - --enable-tproxy=true
```

⬚⬚⬚⬚⬚⬚⬚⬚⬚

1. ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

## 5.2 VPC Egress Gateway

> **Note**
>
> ### VPC VPC
>
> VPC VPCNAT OVN Egress Gateway VPC NAT Kube-OVN VPC NAT Pod VPC Macvlan Pod iptables
>
> OVN OVN NAT OVN BFD OVN OVN OVN
>
> Egress Gateway VPC NAT

VPC Egress Gateway VPCVPC Pod

- ECMP Active-Active
- BFD <1s
- IPv6
- Namespace Pod
- Node Egress Gateway

VPC Egress Gateway BGP EVPN/VXLAN Egress Gateway BGP/EVPN

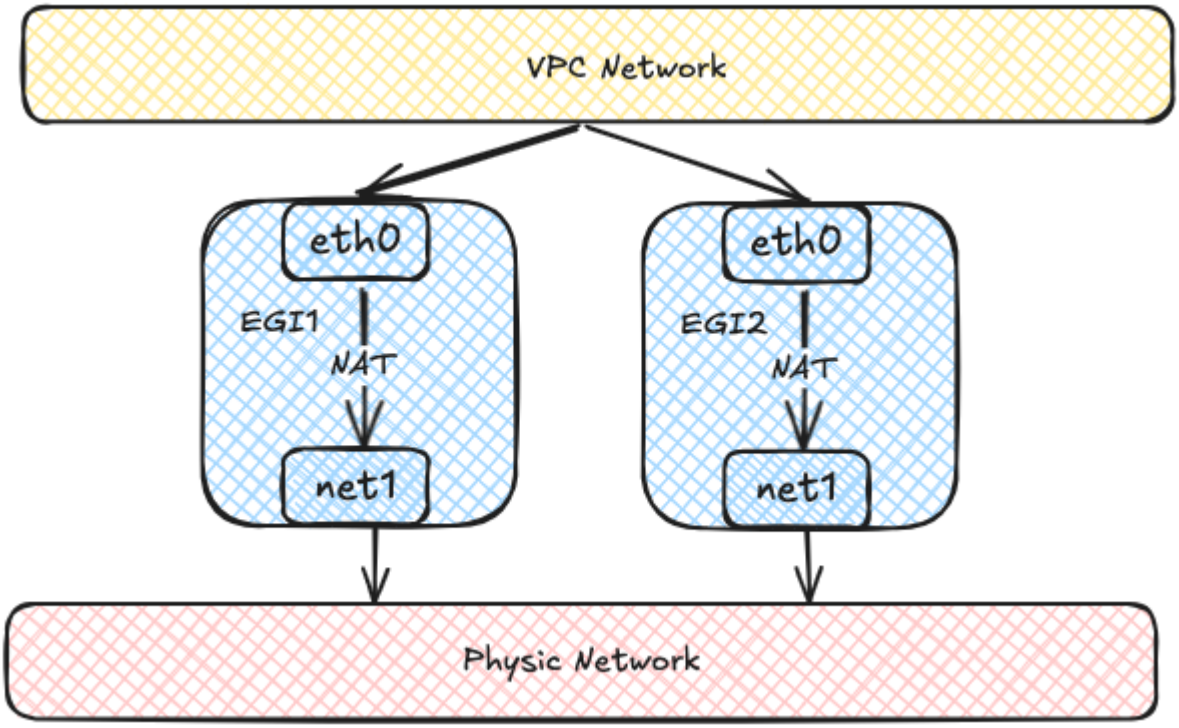
conntrack NAT JSON VPC Egress Gateway

VPC Egress Gateway

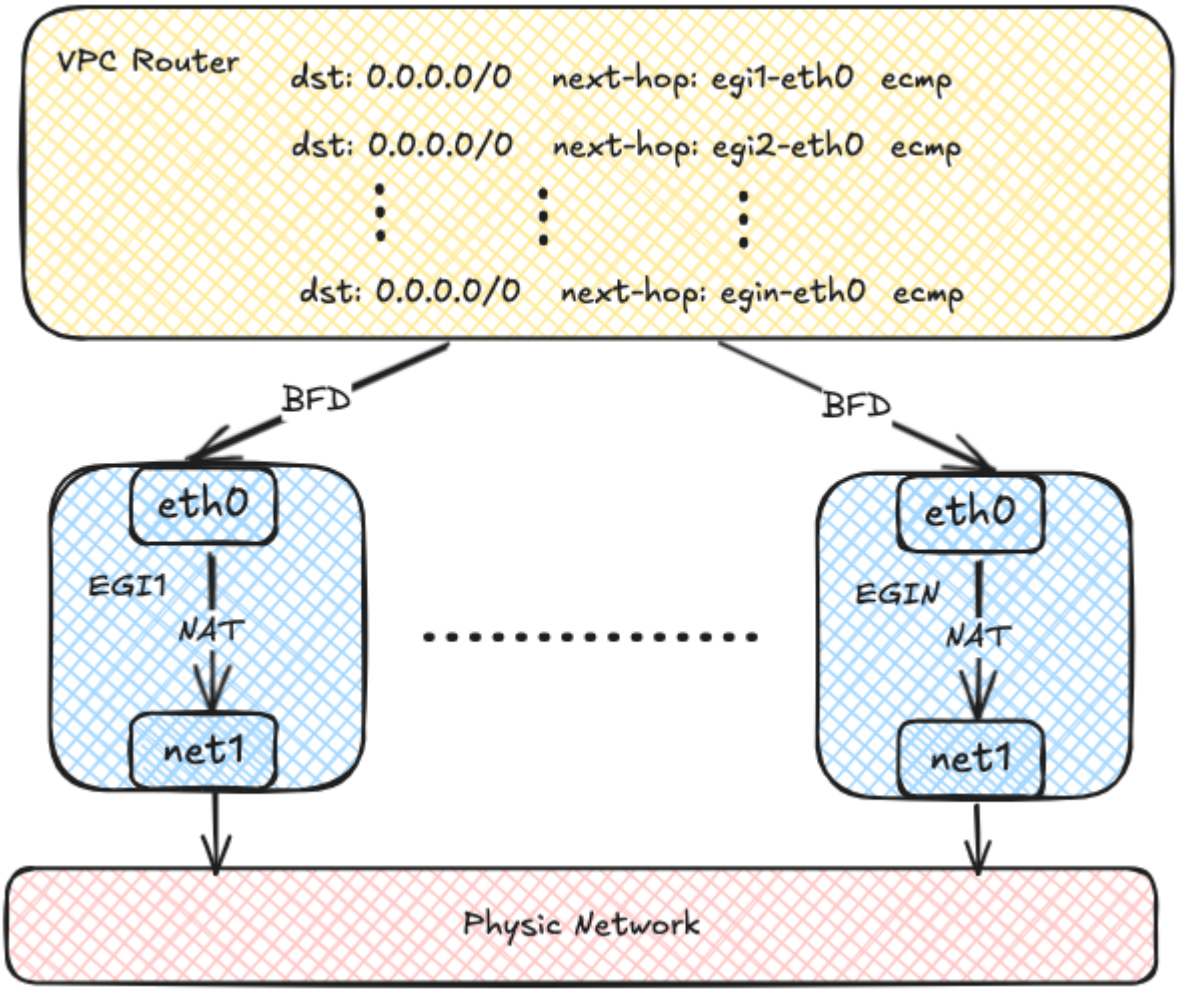
- Macvlan Underlay
- Gateway Egress IP
- SNAT EIP DNAT
-

### 5.2.1

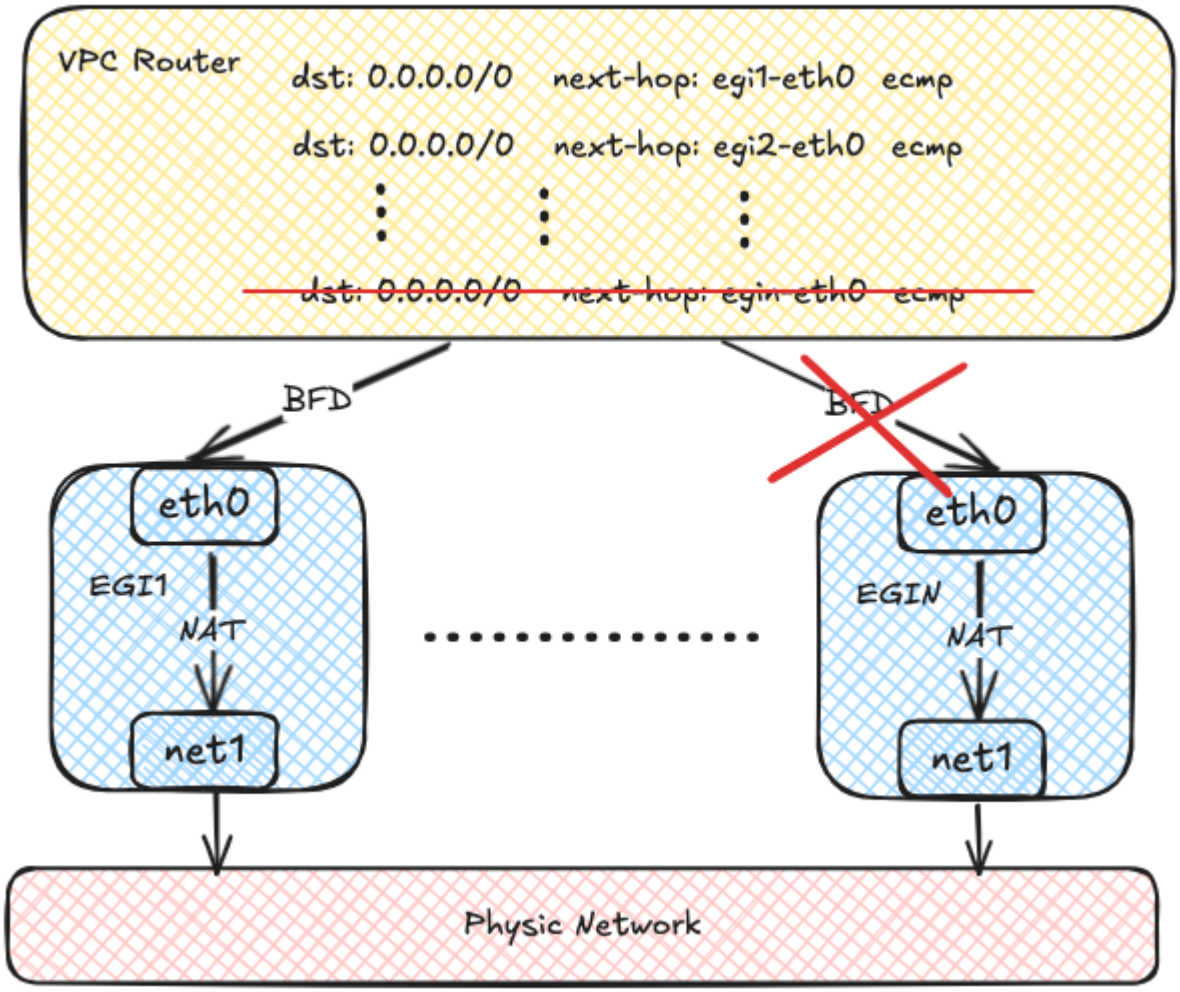
Egress Gateway Pod Pod VPC Macvlan Egress Gateway NAT




□□ Egress Gateway □□□□□□□□□□□□□ OVN □□□□□□ VPC □ Pod □□□□□□□□□□□OVN □□□□□□□□□□□□□□□□□□□ Egress Gateway □□□□□□□□□□□□□□□□□□□ Egress Gateway □□□□□□□□□□ □□□□□□□□□□□□□□




OVN ■■ BFD ■■■■■ Egress Gateway ■■■■■■■■■■ Egress Gateway ■■■■■■OVN ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■




### 5.2.2 ...

VPC Egress Gateway ... VPC NAT Gateway ... ... Multus-CNI...

> ... VPC Egress Gateway ... ConfigMap...

### 5.2.3 ...

**... NetworkAttachmentDefinition**

VPC Egress Gateway ... VPC ... NetworkAttachmentDefinition ... `macvlan` ... Kube-OVN ... IPAM ...

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: eth1
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth1",
      "mode": "bridge",
      "ipam": {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "eth1.default"
      }
    }'
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan1
```

```
spec:
  protocol: IPv4
  provider: eth1.default
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
    - 172.17.0.0..172.17.0.10
```

> ██████ CNI ████ NetworkAttachmentDefinition ▯ VPC Egress Gateway ██████████

██████████████████ ██████▯

### ▯▯ VPC Egress Gateway

▯▯ VPC Egress Gateway ████████

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalSubnet: macvlan1
  policies:
    - snat: true
      subnets:
        - ovn-default
```

██████ default ██████ VPC `ovn-cluster` ██████ gateway1 ████ VPC Egress Gateway▯ `ovn-cluster` ▯ `ovn-default` ███10.16.0.0/16▯▯▯▯ Pod ███ `macvlan1` ███ SNAT ██████████

████████ VPC Egress Gateway▯

```
$ kubectl get veg gateway1
NAME       VPC           REPLICAS   BFD ENABLED   EXTERNAL SUBNET   PHASE       READY   AGE
gateway1   ovn-cluster   1          false         macvlan1          Completed   true    13s
```

███████

```
kubectl get veg gateway1 -o wide
NAME       VPC           REPLICAS   BFD ENABLED   EXTERNAL SUBNET   PHASE       READY   INTERNAL IPS     EXTERNAL IPS      WORKING NODES         AGE
gateway1   ovn-cluster   1          false         macvlan1          Completed   true    ["10.16.0.12"]   ["172.17.0.11"]   ["kube-ovn-worker"]   82s
```

███████

```
$ kubectl get deployment -l ovn.kubernetes.io/vpc-egress-gateway=gateway1
NAME       READY   UP-TO-DATE   AVAILABLE   AGE
gateway1   1/1     1            1           4m40s

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway1 -o wide
NAME                        READY   STATUS    RESTARTS   AGE     IP           NODE              NOMINATED NODE   READINESS GATES
gateway1-b9f8b4448-76lhm   1/1     Running   0          4m48s   10.16.0.12   kube-ovn-worker   <none>           <none>
```

▯▯ Pod ▯▯ IP▯▯▯▯ iptables ▯▯▯

```
$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: net1@if13: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 62:d8:71:90:7b:86 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.17.0.11/16 brd 172.17.255.255 scope global net1
       valid_lft forever preferred_lft forever
    inet6 fe80::60d8:71ff:fe90:7b86/64 scope link
       valid_lft forever preferred_lft forever
17: eth0@if18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc noqueue state UP group default
    link/ether 36:7c:6b:c7:82:6b brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.16.0.12/16 brd 10.16.255.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::347c:6bff:fec7:826b/64 scope link
       valid_lft forever preferred_lft forever

$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- ip route show
default via 172.17.0.1 dev net1
10.16.0.0/16 dev eth0 proto kernel scope link src 10.16.0.12
```

```
172.17.0.0/16 dev net1 proto kernel scope link src 172.17.0.11

$ kubectl exec gateway1-b9f8b4448-76lhm -c gateway -- iptables -t nat -S
-P PREROUTING ACCEPT
-P INPUT ACCEPT
-P OUTPUT ACCEPT
-P POSTROUTING ACCEPT
-A POSTROUTING -s 10.16.0.0/16 -j MASQUERADE --random-fully
```

... Gateway Pod ...

```
$ kubectl exec -ti gateway1-b9f8b4448-76lhm -c gateway -- bash
nobody@gateway1-b9f8b4448-76lhm:/kube-ovn$ tcpdump -i any -nnve icmp and host 172.17.0.1
tcpdump: data link type LINUX_SLL2
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
06:50:58.936528 eth0  In  ifindex 17 92:26:b8:9e:f2:1c ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 63, id 30481, offset 0, flags [DF], proto ICMP (1),
length 84)
    10.16.0.9 > 172.17.0.1: ICMP echo request, id 37989, seq 0, length 64
06:50:58.936574 net1  Out ifindex 2 62:d8:71:90:7b:86 ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 62, id 30481, offset 0, flags [DF], proto ICMP (1),
length 84)
    172.17.0.11 > 172.17.0.1: ICMP echo request, id 39449, seq 0, length 64
06:50:58.936613 net1  In  ifindex 2 02:42:39:79:7f:08 ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 64, id 26701, offset 0, flags [none], proto ICMP
(1), length 84)
    172.17.0.1 > 172.17.0.11: ICMP echo reply, id 39449, seq 0, length 64
06:50:58.936621 eth0  Out ifindex 17 36:7c:6b:c7:82:6b ethertype IPv4 (0x0800), length 104: (tos 0x0, ttl 63, id 26701, offset 0, flags [none], proto ICMP
(1), length 84)
    172.17.0.1 > 10.16.0.9: ICMP echo reply, id 37989, seq 0, length 64
```

OVN Logical Router ...

```
$ kubectl ko nbctl lr-policy-list ovn-cluster
Routing Policies
     31000                           ip4.dst == 10.16.0.0/16           allow
     31000                          ip4.dst == 100.64.0.0/16           allow
     30000                             ip4.dst == 172.18.0.2         reroute               100.64.0.3
     30000                             ip4.dst == 172.18.0.3         reroute               100.64.0.2
     30000                             ip4.dst == 172.18.0.4         reroute               100.64.0.4
     29100                           ip4.src == 10.16.0.0/16         reroute               10.16.0.12
     29000 ip4.src == $ovn.default.kube.ovn.control.plane_ip4        reroute               100.64.0.2
     29000       ip4.src == $ovn.default.kube.ovn.worker2_ip4        reroute               100.64.0.4
     29000        ip4.src == $ovn.default.kube.ovn.worker_ip4        reroute               100.64.0.3
```

... `.spec.replicas` ...

```
$ kubectl scale veg gateway1 --replicas=2
vpcegressgateway.kubeovn.io/gateway1 scaled

$ kubectl get veg gateway1
NAME       VPC           REPLICAS   BFD ENABLED   EXTERNAL SUBNET   PHASE       READY   AGE
gateway1   ovn-cluster   2          false         macvlan           Completed   true    39m

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway1 -o wide
NAME                        READY   STATUS    RESTARTS   AGE   IP           NODE               NOMINATED NODE   READINESS GATES
gateway1-b9f8b4448-76lhm    1/1     Running   0          40m   10.16.0.12   kube-ovn-worker    <none>           <none>
gateway1-b9f8b4448-zd4dl    1/1     Running   0          64s   10.16.0.13   kube-ovn-worker2   <none>           <none>

$ kubectl ko nbctl lr-policy-list ovn-cluster
Routing Policies
     31000                           ip4.dst == 10.16.0.0/16           allow
     31000                          ip4.dst == 100.64.0.0/16           allow
     30000                             ip4.dst == 172.18.0.2         reroute               100.64.0.3
     30000                             ip4.dst == 172.18.0.3         reroute               100.64.0.2
     30000                             ip4.dst == 172.18.0.4         reroute               100.64.0.4
     29100                           ip4.src == 10.16.0.0/16         reroute               10.16.0.12, 10.16.0.13
     29000 ip4.src == $ovn.default.kube.ovn.control.plane_ip4        reroute               100.64.0.2
     29000       ip4.src == $ovn.default.kube.ovn.worker2_ip4        reroute               100.64.0.4
     29000        ip4.src == $ovn.default.kube.ovn.worker_ip4        reroute               100.64.0.3
```

### ... Egress Gateway IP ...

... `externalIPs` ... `nodeSelector` ... Egress Gateway Pods ... Egress IP ...

```yaml
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 2
  externalSubnet: macvlan1
  policies:
    - snat: true
      subnets:
        - ovn-default
```



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```
  externalIPs:
    - 172.17.0.10
    - 172.17.0.11
  nodeSelector:
    - matchLabels:
        kubernetes.io/hostname: kube-ovn-worker
```

▒▒▒▒▒▒ IP ▒▒▒▒▒▒▒▒▒▒ `internalIPPool` ▒ `externalIPPool` ▒▒▒▒▒▒ `IPPool` ▒▒▒IPPool ▒▒▒▒▒▒▒▒▒ `internalSubnet` ▒ `externalSubnet` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ `internalIPs` ▒ `externalIPs` ▒▒▒▒▒▒

```
spec:
  internalIPPool: veg-internal
  externalIPPool: veg-external
```

▒▒ **BFD** ▒▒▒▒

BFD ▒▒▒▒▒▒ VPC ▒ BFD LRP ▒▒▒▒▒▒▒▒▒▒▒ VPC ▒▒▒▒▒▒ BFD Port▒▒▒▒▒▒▒

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc1
spec:
  bfdPort:
    enabled: true
    ip: 10.255.255.255
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  vpc: vpc1
  protocol: IPv4
  cidrBlock: 192.168.0.0/24
```

▒▒ BFD Port ▒▒▒▒▒▒ OVN LR ▒▒▒▒▒▒▒▒▒▒▒▒ BFD ▒ LRP▒

```
$ kubectl ko nbctl show vpc1
router 0c1d1e8f-4c86-4d96-88b2-c4171c7ff824 (vpc1)
    port bfd@vpc1
        mac: "8e:51:4b:16:3c:90"
        networks: ["10.255.255.255"]
    port vpc1-subnet1
        mac: "de:c9:5c:38:7a:61"
        networks: ["192.168.0.1/24"]
```

▒▒▒▒▒▒ VPC Egress Gateway ▒ `.spec.bfd.enabled` ▒▒▒▒ `true` ▒▒▒▒▒▒▒▒▒

```
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway2
  namespace: default
spec:
  vpc: vpc1
  replicas: 2
  internalSubnet: subnet1
  externalSubnet: macvlan
  bfd:
    enabled: true
  policies:
    - snat: true
      ipBlocks:
        - 192.168.0.0/24
```

▒▒ VPC Egress Gateway ▒▒▒▒

```
$ kubectl get veg gateway2 -o wide
NAME       VPC    REPLICAS   BFD ENABLED   EXTERNAL SUBNET   PHASE       READY   INTERNAL IPS                    EXTERNAL IPS                    WORKING
NODES                            AGE
gateway2   vpc1   2          true          macvlan           Completed   true    ["192.168.0.2","192.168.0.3"]   ["172.17.0.13","172.17.0.14"]   ["kube-ovn-
worker","kube-ovn-worker2"]   58s

$ kubectl get pod -l ovn.kubernetes.io/vpc-egress-gateway=gateway2 -o wide
NAME                        READY   STATUS    RESTARTS   AGE     IP            NODE               NOMINATED NODE   READINESS GATES
gateway2-fcc6b8b87-8lgvx    1/1     Running   0          2m18s   192.168.0.3   kube-ovn-worker2   <none>           <none>
gateway2-fcc6b8b87-wmww6    1/1     Running   0          2m18s   192.168.0.2   kube-ovn-worker    <none>           <none>

$ kubectl ko nbctl lr-policy-list vpc1
Routing Policies
     29100                       ip4.src == 192.168.0.0/24         reroute               192.168.0.2, 192.168.0.3
```

```
     29090                    ip4.src == 192.168.0.0/24           drop

$ kubectl ko nbctl list bfd
_uuid               : 223ede10-9169-4c7d-9524-a546e24bfab5
detect_mult         : 3
dst_ip              : "192.168.0.2"
external_ids        : {af="4", vendor=kube-ovn, vpc-egress-gateway="default/gateway2"}
logical_port        : "bfd@vpc1"
min_rx              : 1000
min_tx              : 1000
options             : {}
status              : up

_uuid               : b050c75e-2462-470b-b89c-7bd38889b758
detect_mult         : 3
dst_ip              : "192.168.0.3"
external_ids        : {af="4", vendor=kube-ovn, vpc-egress-gateway="default/gateway2"}
logical_port        : "bfd@vpc1"
min_rx              : 1000
min_tx              : 1000
options             : {}
status              : up
```

... Pod ... BFD ...

```
$ kubectl exec gateway2-fcc6b8b87-8lgvx -c bfdd -- bfdd-control status
There are 1 sessions:
Session 1
 id=1 local=192.168.0.3 (p) remote=10.255.255.255 state=Up

$ kubectl exec gateway2-fcc6b8b87-wmww6 -c bfdd -- bfdd-control status
There are 1 sessions:
Session 1
 id=1 local=192.168.0.2 (p) remote=10.255.255.255 state=Up
```

...

**VPC BFD PORT**

| ... | ... | ... | ... | ... | ... |
|---|---|---|---|---|---|
| `enabled` | `boolean` | ... | `false` | ... BFD Port... | `true` |
| `ip` | `string` | ... | - | BFD Port ... IP ... ... IPv6 ... ... | `169.255.255.2` `fdff::1` / `169.255.255.2` `1` |
| `nodeSelector` | `object` | ... | - | ... BFD Port ... ...BFD Port ... ... OVN HA Chassis Group... Active/ Backup ... Active ... nodeSelector... Kube-OVN ... ... `kubectl ko nbctl list` `ha_chassis_group` ... OVN HA Chassis Group ... | - |
| `nodeSelector.matchLabels` | `dict/map` | ... | - | ... | - |
| `nodeSelector.matchExpressions` | `object array` | ... | - | ... | - |



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VPC EGRESS GATEWAY

Spec▯

| ... | ... | ... | ... | ... | ... |
| --- | --- | --- | --- | --- | --- |
| `vpc` | `string` | ... | ... VPC ...ovn-cluster... | VPC ... | `vpc1` |
| `replicas` | `integer/int32` | ... | `1` | ... | `2` |
| `prefix` | `string` | ... | - | ... Deployment ... | `veg-` |
| `image` | `string` | ... | - | ... Deployment ... | `docker.io/kubeovn/kube-ovn:v1.14.0-debug` |
| `internalSubnet` | `string` | ... | VPC ... | ... VPC ... | `subnet1` |
| `externalSubnet` | `string` | ... | - | ... | `ext1` |
| `internalIPs` | `string array` | ... | - | ... VPC ... IP ... IPv6 ... IP ... `<replicas> + 1` ... Pod ... | `10.16.0.101` / `fd00::11` / `10.16.0.101,fd00::11` |
| `externalIPs` | `string array` | ... | - | ... IP ... IPv6 ... IP ... `<replicas> + 1` ... Pod ... | `10.16.0.101` / `fd00::11` / `10.16.0.101,fd00::11` |
| `internalIPPool` | `string` | ... | - | ... `IPPool` ...IPPool ... `internalSubnet` ... `internalIPs` ... | `veg-internal` |
| `externalIPPool` | `string` | ... | - | ... `IPPool` ...IPPool ... `externalSubnet` ... `externalIPs` ... | `veg-external` |
| `bfd` | `object` | ... | - | BFD ... | - |
| `policies` | `object array` | ... | - | Egress ... `selectors` ... | - |
| `selectors` | `object array` | ... | - | ... Namespace Selector ... Pod Selector ... Egress ... Pod ... SNAT/MASQUERADE... ... `policies` ... | - |
| `nodeSelector` | `object array` | ... | - | ...Deployment/Pod... | - |



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| ... | ... | ... | ... | ... | ... |
| --- | --- | --- | --- | --- | --- |
| `trafficPolicy` | `string` | ... | `Cluster` | ... `Cluster` / `Local` ... **BFD** ... ... ... `Local` ... Egress ... ... VPC Egress Gateway ... ... VPC Egress Gateway ... Egress ... | `Local` |
| `bandwidth` | `object` | ... | - | ... | - |
| `bandwidth.ingress` | `integer` `string` | ... | - | ... VPC ... ... | `1024` / `1Gi` |
| `bandwidth.egress` | `integer` `string` | ... | - | ... VPC ... ... | `1024` / `1Gi` |

...

... Mbps ... `M` ... `Mi` ... `G` ... `Gi` ... quantity ... bit/s... Mbps... `100M` ... `100Mi` ... `1G` ... `1Gi` ... 100...105...1000 ... 1074 Mbps...

... 9223372036854 Mbps... `ingress` ... `egress` ...

```yaml
spec:
  bandwidth:
    ingress: 1024
    egress: 1Gi
```

> ...
>
> ...0 ... 9223372036854 ... Mbps ... Kube-OVN ... `vpc-egress-gateways.kubeovn.io` CRD ... `x-kubernetes-int-or-string: true` ...
>
> Helm ... Chart ... `crds/` ... CRD... `kube-ovn-v2` Chart ... `KUBE_OVN_VERSION` ... Kube-OVN ... CRD bundle...
>
> ```
> kubectl apply -f "https://raw.githubusercontent.com/kubeovn/kube-ovn/${KUBE_OVN_VERSION}/charts/kube-ovn-v2/crds/kube-ovn-crd.yaml"
> ```
>
> ... quantity ... Mbps ...

BFD ...

| ... | ... | ... | ... | ... | ... |
| --- | --- | --- | --- | --- | --- |
| `enabled` | `boolean` | ... | `false` | ... BFD... | `true` |
| `minRX` | `integer/int32` | ... | `1000` | BFD minRX ... ...ms... | `500` |
| `minTX` | `integer/int32` | ... | `1000` | BFD minTX ... ...ms... | `500` |
| `multiplier` | `integer/int32` | ... | `3` | BFD multiplier ... | `1` |



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Egress ■■■

| ■■ | ■■ | ■■ | ■■■ | ■■ | ■■ |
|---|---|---|---|---|---|
| `snat` | `boolean` | ■ | `false` | ■■■■ SNAT/ MASQUERADE■ | `true` |
| `ipBlocks` | `string array` | ■ | - | ■■■■ Gateway ■ IP ■■■■■■ IPv6■ | `192.168.0.1` / `192.168.0.0/24` |
| `subnets` | `string array` | ■ | - | ■■■■ Gateway ■ VPC ■■■■■■■ IPv6 ■■■■■■■■ | `subnet1` |

Selectors■

| ■■ | ■■ | ■■ | ■■■ | ■■ | ■■ |
|---|---|---|---|---|---|
| `namespaceSelector` | `object` | ■ | - | Namespace ■■■■■ ■■■■■■■ Namespace■ | - |
| `namespaceSelector.matchLabels` | `dict/map` | ■ | - | ■■■■■■■■■■■■■ | - |
| `namespaceSelector.matchExpressions` | `object array` | ■ | - | ■■■■■■■■■■■■■ | - |
| `podSelector` | `object` | ■ | - | Pod ■■■■■■■■■■■■■ Pod■ | - |
| `podSelector.matchLabels` | `dict/map` | ■ | - | ■■■■■■■■■■■■■ | - |
| `podSelector.matchExpressions` | `object array` | ■ | - | ■■■■■■■■■■■■■ | - |

■■■■■■■

| ■■ | ■■ | ■■ | ■■■ | ■■ | ■■ |
|---|---|---|---|---|---|
| `matchLabels` | `dict/map` | ■ | - | ■■■■■■■■■■■■■ | - |
| `matchExpressions` | `object array` | ■ | - | ■■■■■■■■■■■■■ | - |
| `matchFields` | `object array` | ■ | - | ■■■■■■■■■■■■■ | - |



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Status□

| □□ | □□ | □□ | □□ |
| --- | --- | --- | --- |
| `ready` | `boolean` | Gateway □□□□□ | `true` |
| `phase` | `string` | Gateway □□□□□ | `Pending` / `Processing` / `Completed` |
| `internalIPs` | `string array` | □□ VPC □□□□□ IP □□□ | - |
| `externalIPs` | `string array` | □□□□□□□□□ IP □□□ | - |
| `workload` | `object` | □□□□□□□ | - |
| `workload.apiVersion` | `string` | □□□□ API □□□ | `apps/v1` |
| `workload.kind` | `string` | □□□□□□□ | `Deployment` |
| `workload.name` | `string` | □□□□□□□ | `gateway1` |
| `workload.nodes` | `string array` | □□□□□□□□□□□□ | - |
| `conditions` | `object array` | - | - |

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# 5.3 VPC Egress Gateway ▯▯▯▯

VPC Egress Gateway ▯▯▯▯▯▯▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯ sidecar▯▯ sidecar ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ conntrack NAT ▯▯▯▯▯▯▯▯ NAT ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

.....ConfigMap . Service ..... condition..... observer ...volume ..... Pod.

..... `/kube-ovn/vpc-egress-gateway-observer` .launcher ..... `sleep infinity` ..... binary ..... observer ..... `CAP_NET_ADMIN` file capability..... conntrack .....

### 5.3.3 ...

observer . TCP .. `10666` ... `/metrics` . `/healthz` .observer .... Prometheus registry.... `go_*` . `process_*` . `promhttp_*` ..... `namespace` . `name` . `pod` . `node` .....

.....liveness probe ..... observer binary.... loopback .. `/healthz` ... observer ..... Pod ..... observer binary ... `sleep infinity` .....

....

.. scrape ... `/proc/net/dev` .....observer ..... scrape ... Pod `network-status` annotation ..... Multus .....

| .. | .... | .. |
| --- | --- | --- |
| `kube_ovn_vpc_egress_gateway_interface_rx_bytes_total` | `interface` . `type` | ..... |
| `kube_ovn_vpc_egress_gateway_interface_tx_bytes_total` | `interface` . `type` | ..... |
| `kube_ovn_vpc_egress_gateway_interface_rx_packets_total` | `interface` . `type` | ..... |
| `kube_ovn_vpc_egress_gateway_interface_tx_packets_total` | `interface` . `type` | ..... |
| `kube_ovn_vpc_egress_gateway_interface_drops_total` | `interface` . `type` . `direction` | ..... |
| `kube_ovn_vpc_egress_gateway_interface_errors_total` | `interface` . `type` . `direction` | ..... |

`type` ... `internal` . `external` . `direction` . `rx` . `tx` .

**Conntrack ..**

Conntrack ..... IP .....

| .. | .... | .. |
| --- | --- | --- |
| `kube_ovn_vpc_egress_gateway_conntrack_nat_flows_active` | `address_family` . `protocol` . `nat_type` | .. NAT .... conntrack table dump ..... |
| `kube_ovn_vpc_egress_gateway_conntrack_nat_flows_started_total` | `address_family` . `protocol` . `nat_type` | observer ..... NAT ..... |
| `kube_ovn_vpc_egress_gateway_conntrack_nat_flows_ended_total` | `address_family` . `protocol` . `nat_type` | .... NAT ..... |
| `kube_ovn_vpc_egress_gateway_conntrack_nat_packets_total` | `address_family` . `protocol` . `nat_type` . `direction` | conntrack accounting ..... ..... |
| `kube_ovn_vpc_egress_gateway_conntrack_nat_bytes_total` | `address_family` . `protocol` . `nat_type` . `direction` | conntrack accounting ..... ..... |

`address_family` . `ipv4` . `ipv6` ... `protocol` .. `tcp` . `udp` . `sctp` . `icmp` . `icmpv6` ..... `other` . `nat_type` . `snat` . `dnat` . `snat_dnat` . `direction` . `original` . `reply` .

..... `kube_ovn_vpc_egress_gateway_observability_*` .....conntrack .....cache ..... conntrack accounting ..... `/metrics` .....



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### 5.3.4 ServiceMonitor ⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `publishNotReadyAddresses: true` ⬚ headless Service⬚⬚⬚⬚ `targetPort` ⬚ `10666` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ ServiceMonitor⬚ `observability.serviceMonitor` ⬚⬚⬚⬚ label ⬚ annotation ⬚⬚⬚⬚ ServiceMonitor ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ selector label⬚

⬚⬚ ServiceMonitor CRD ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `ServiceMonitorReady=False` ⬚⬚⬚⬚⬚⬚CRD ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ ServiceMonitor⬚⬚ condition ⬚⬚⬚⬚⬚⬚ `Ready` condition ⬚⬚⬚⬚⬚

### 5.3.5 JSON ⬚⬚⬚

⬚⬚⬚⬚ JSON Lines ⬚⬚⬚⬚⬚⬚ `observability` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

# 5.4 Egress Gateway BGP 和 EVPN 支持

**实验目标**

BGP 和 EVPN 支持为租户网络提供了与外部网络进行动态路由交换的能力。

**学习内容：**

- 理解 L3VPN和以太网 L2VPN
- 掌握 FRR 在容器环境中的部署
- 掌握 BGP BFD

## 5.4.1 概述

VPC Egress Gateway 支持通过 BGP 和 EVPN/VXLAN 两种方式将流量从 Egress Gateway Pod 中的 FRR（Free Range Routing）转发出去。BGP 和 EVPN 两种模式均支持 Gateway 的 init 容器自动生成 BgpConf 和 EvpnConf 并启动 FRR 进程。

在 BGP 模式下，Egress Gateway 通过 BGP 协议将到达外部网络的路由发布给外部路由器。EVPN 模式下，init 容器为 Pod 创建 Linux VRF、bridge 和 VXLAN 设备，以 L2VPN EVPN 协议将流量通过 VXLAN 隧道转发至 Egress Gateway Pod 所在Kubernetes 节点。当前支持 Geneve。

> 本实验将使用 BGP 模式演示 kube-ovn-speaker 将 Pod/子网/Service 路由发布

## 5.4.2 实验环境

本实验将使用 BGP/EVPN 模式配置 VPC Egress Gateway 并通过 Multus-CNI 进行

实验环境中 BGP模式和 EVPN模式共用同一套 BGP 环境

## 5.4.3 实验步骤

**步骤 BgpConf 配置**

BgpConf 用于配置与外部 BGP 对等体的

```
apiVersion: kubeovn.io/v1
kind: BgpConf
metadata:
  name: bgp-conf-6502
spec:
  localASN: 65002
  peerASN: 65001
  neighbours:
    - 10.0.1.1
  holdTime: 90s
  keepaliveTime: 30s
  connectTime: 10s
  ebgpMultiHop: true
```

本实验环境中 AS 号为 65002，对端 AS 号为 65001 ，BGP 对等体地址为 10.0.1.1，开启了 EBGP Multi-Hop。

**步骤 EvpnConf 配置**

EvpnConf 用于配置与外部 EVPN 对等体EvpnConf 需与 BgpConf 配合使用

```
apiVersion: kubeovn.io/v1
kind: EvpnConf
metadata:
  name: evpn-conf-1016
spec:
  vni: 1016
  routeTargets:
    - "65000:1016"
```



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... VNI ... 1016 ... EVPN ...Route Target ... `65000:1016` ...

### ... BGP/EVPN ... VPC Egress Gateway

... VPC Egress Gateway ... NetworkAttachmentDefinition ...

... VpcEgressGateway ... `.spec` ... `bgpConf` ... `evpnConf` ...

```yaml
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway1
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalIPs:
    - 10.0.1.13
  internalIPs:
    - 10.16.0.13
  externalSubnet: macvlan1
  bgpConf: bgp-conf-6502
  evpnConf: evpn-conf-1016
  policies:
    - snat: false
      subnets:
        - ovn-default
    - snat: false
      ipBlocks:
        - 10.17.0.0/16
```

... BGP ... EVPN ... VPC Egress Gateway...Gateway Pod ... FRR ... BGP EVPN ... VXLAN ... BGP/EVPN ... `snat` ... `false` ...

### ... BGP... EVPN...

... BGP ... EVPN/VXLAN... `bgpConf` ... `evpnConf` ... FRR ... BGP IPv4 Unicast ...

```yaml
apiVersion: kubeovn.io/v1
kind: VpcEgressGateway
metadata:
  name: gateway-bgp-only
  namespace: default
spec:
  vpc: ovn-cluster
  replicas: 1
  externalSubnet: macvlan1
  bgpConf: bgp-conf-6502
  policies:
    - snat: true
      subnets:
        - ovn-default
```



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## 5.4.4 □□□□

### BgpConf

| □□ | □□ | □□ | □□□ | □□ | □□ |
|---|---|---|---|---|---|
| `localASN` | `uint32` | □ | - | □□ AS □□ | `65002` |
| `peerASN` | `uint32` | □ | - | □□ AS □□ | `65001` |
| `routerId` | `string` | □ | Pod IP | BGP Router ID□ | `10.0.1.13` |
| `neighbours` | `string array` | □ | - | BGP □□ IP □□□□□ | `10.0.1.1` |
| `password` | `string` | □ | - | BGP □□□□□ | `secret` |
| `holdTime` | `string (duration)` | □ | - | BGP Hold □□□ | `90s` |
| `keepaliveTime` | `string (duration)` | □ | - | BGP Keepalive □ □□ | `30s` |
| `connectTime` | `string (duration)` | □ | - | BGP □□□□□□ | `10s` |
| `ebgpMultiHop` | `boolean` | □ | `false` | □□□□ EBGP Multi-Hop□ | `true` |

### EvpnConf

| □□ | □□ | □□ | □□□ | □□ | □□ |
|---|---|---|---|---|---|
| `vni` | `uint32` | □ | - | VXLAN Network Identifier□ | `1016` |
| `routeTargets` | `string array` | □ | - | □□□□□□□□ Route Target □□□ | `65000:1016` |

### VpcEgressGateway □□□□

□□□□□ VpcEgressGateway `.spec` □□□□ BGP/EVPN □□□□□□□□ VpcEgressGateway □□□□□□□ Egress Gateway □□□□□□□□□□□□

| □□ | □□ | □□ | □□□ | □□ | □□ |
|---|---|---|---|---|---|
| `bgpConf` | `string` | □ | - | □□□ BgpConf □□□ □□□□□□□ Egress Gateway □□□ BGP□ | `bgp-conf-6502` |
| `evpnConf` | `string` | □ | - | □□□ EvpnConf □□□ □□□□□□□□ `bgpConf` □□□□□□ Egress Gateway □ □□ EVPN/VXLAN□ | `evpn-conf-1016` |

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## 5.4.5 □□

# 5.5 VPC QoS

Kube-OVN [illegible] QoSPolicy CRD [illegible] VPC [illegible]

## 5.5.1 [illegible]

QoSPolicy [illegible] `priority` [illegible] QoS [illegible]

[illegible]

| [illegible] | [illegible] | [illegible] |
|---|---|---|
| EIP QoS | 1 | [illegible] EIP [illegible] |
| NATGW [illegible] QoS | 2 | [illegible] NATGW [illegible] IP [illegible] |
| NATGW net1 [illegible] QoS | 3 | [illegible] |

[illegible]

- `shared=false` [illegible] QoS [illegible] `shared=true` [illegible]
- EIP [illegible] QoS [illegible] NATGW [illegible] QoS

## 5.5.2 EIP QoS

[illegible] EIP [illegible] 1Mbps[illegible] 1[illegible] `shared=false` [illegible] QoSPolicy [illegible] EIP [illegible] QoSPolicy [illegible] QoS [illegible]

QoSPolicy [illegible]

```yaml
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-eip-example
spec:
  shared: false
  bindingType: EIP
  bandwidthLimitRules:
  - name: eip-ingress
    rateMax: "1" # Mbps
    burstMax: "1" # Mbps
    priority: 1
    direction: ingress
  - name: eip-egress
    rateMax: "1" # Mbps
    burstMax: "1" # Mbps
    priority: 1
    direction: egress
```

IptablesEIP [illegible]

```yaml
kind: IptablesEIP
apiVersion: kubeovn.io/v1
metadata:
  name: eip-1
spec:
  natGwDp: gw1
  qosPolicy: qos-eip-example
```

`.spec.qosPolicy` [illegible]

## 5.5.3 [illegible] QoS [illegible] EIP

[illegible] `label` [illegible] qos [illegible] eip[illegible]

```
# kubectl get eip  -l ovn.kubernetes.io/qos=qos-eip-example
NAME    IP            MAC               NAT   NATGWDP   READY
eip-1   172.18.11.24   00:00:00:34:41:0B   fip   gw1       true
```



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## 5.5.4 VPC NATGW net1 □□ QoS

□ VPC NATGW □ net1 □□□□□□□□□□□□□□ 10Mbps□□□□□ 3□□□ `shared=true` □□□□□ QoSPolicy □□□□□□□□□□□□□□□□□□□□□□□□ QoSPolicy □□□□

QoSPolicy □□□□□

```yaml
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-natgw-example
spec:
  shared: true
  bindingType: NATGW
  bandwidthLimitRules:
  - name: net1-ingress
    interface: net1
    rateMax: "10" # Mbps
    burstMax: "10" # Mbps
    priority: 3
    direction: ingress
  - name: net1-egress
    interface: net1
    rateMax: "10" # Mbps
    burstMax: "10" # Mbps
    priority: 3
    direction: egress
```

VpcNatGateway □□□□□

```yaml
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  qosPolicy: qos-natgw-example
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
```

`.spec.qosPolicy` □□□□□□□□□□□□□□□□□□□

## 5.5.5 net1 □□□□□□ QoS

□ net1 □□□□□□□□□□□□□□□□□ 5Mbps□□□□□ 2□□□ `shared=true` □□□□□ QoSPolicy □□□□□□□□□□□□□□□□□□□□□ QoSPolicy □□□□

QoSPolicy □□□□□

```yaml
apiVersion: kubeovn.io/v1
kind: QoSPolicy
metadata:
  name: qos-natgw-example
spec:
  shared: true
  bindingType: NATGW
  bandwidthLimitRules:
  - name: net1-extip-ingress
    interface: net1
    rateMax: "5" # Mbps
    burstMax: "5" # Mbps
    priority: 2
    direction: ingress
    matchType: ip
    matchValue: src 172.18.11.22/32
  - name: net1-extip-egress
    interface: net1
    rateMax: "5" # Mbps
    burstMax: "5" # Mbps
    priority: 2
    direction: egress
    matchType: ip
    matchValue: dst 172.18.11.23/32
```

VpcNatGateway □□□□□

```yaml
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: gw1
spec:
```

```
  vpc: test-vpc-1
  subnet: net1
  lanIp: 10.0.1.254
  qosPolicy: qos-natgw-example
  selector:
    - "kubernetes.io/hostname: kube-ovn-worker"
    - "kubernetes.io/os: linux"
```

## 5.5.6 □□□□□ QoS □ NATGW

□□ `label` □□□□□□□□ qos □ eip□

```
# kubectl get vpc-nat-gw  -l ovn.kubernetes.io/qos=qos-natgw-example
NAME   VPC          SUBNET   LANIP
gw1    test-vpc-1   net1     10.0.1.254
```

## 5.5.7 □□ qos □□

```
# kubectl get qos -A
NAME               SHARED   BINDINGTYPE
qos-eip-example    false    EIP
qos-natgw-example  true     NATGW
```

## 5.5.8 □□

- □□□□□□□□□□□ QoS □□□□□□□□□ QoS □□□□□□□□□□□□□ QoS □ EIP □ NATGW□□□□□□□ `spec.qosPolicy` □□□

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## 5.5.9 □□

# 5.6 ▯▯▯ VPC ▯▯▯▯▯▯

Kubernetes ▯▯▯ Service ▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯ VPC ▯▯▯▯ ▯▯ Service ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

1. Service IP ▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯
2. ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ IP ▯▯▯

▯▯▯▯▯▯▯▯▯Kube-OVN ▯ 1.11 ▯▯ `SwitchLBRule` CRD▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯

`SwitchLBRule` ▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯

## 5.6.1 `Selector` ▯▯▯▯▯▯▯▯▯▯▯

▯▯ `selector` ▯▯▯▯▯ `label` ▯▯▯▯▯ `pod` ▯▯▯▯▯▯▯▯▯▯▯▯

`SwitchLBRule` ▯▯▯▯▯▯

```
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name:  cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
  - name: dns
    port: 8888
    targetPort: 80
    protocol: TCP
```

- `selector`, `sessionAffinity` ▯ `port` ▯▯▯▯▯▯ Kubernetes Service▯
- `vip` ▯▯▯▯▯▯▯▯▯▯ IP ▯▯▯
- `namespace` ▯ `selector` ▯▯▯ Pod ▯▯▯▯▯▯▯▯

Kube-OVN ▯▯▯ `SwitchLBRule` ▯▯▯▯▯▯ Pod ▯▯ Pod ▯▯ VPC ▯▯▯▯▯▯▯ L2 LB▯

## 5.6.2 `Endpoints` ▯▯▯▯▯▯▯▯▯▯▯

▯▯ `endpoints` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `selector` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `kubevirt` ▯▯▯▯ `vm` ▯

`SwitchLBRule` ▯▯▯▯▯▯

```
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name:  cjh-slr-nginx
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  endpoints:
    - 192.168.0.101
    - 192.168.0.102
    - 192.168.0.103
  ports:
  - name: dns
    port: 8888
    targetPort: 80
    protocol: TCP
```

- `sessionAffinity` ▯ `port` ▯▯▯▯▯▯ Kubernetes Service▯
- `vip` ▯▯▯▯▯▯▯▯▯▯ IP ▯▯▯
- `namespace` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯
- `endpoints` ▯▯▯▯▯▯▯▯ IP ▯▯▯



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████████ selector 和 endpoints,██████ selector ████

## 5.6.3 ████

OVN ██ IPv4 ████████████████████ ████████████████████████████████████████████████████████████

[Health Checks](https://www.ovn.org/support/dist-docs/ovn-nb.5.html)

██ ovn ████████████████ SwitchLBRule ████████████ SwitchLBRule ████████ VPC 和 subnet ██████████ vip ████████████████ ip_port_mappings 和 load_balancer_health_check ██████████████

- ████ vip ████████ subnet ██████████████ subnet ████████████████████████████ SwitchLBRule ████████████
- ████████ Selector ██████████████

█████████

```
root@server:~# kubectl get po -o wide -n vulpecula
NAME                     READY   STATUS    RESTARTS   AGE     IP          NODE     NOMINATED NODE   READINESS GATES
nginx-78d9578975-f4qn4   1/1     Running   3          4d16h   10.16.0.4   worker   <none>           <none>
nginx-78d9578975-t8tm5   1/1     Running   3          4d16h   10.16.0.6   worker   <none>           <none>

#    slr
root@server:~# cat << END > slr.yaml
apiVersion: kubeovn.io/v1
kind: SwitchLBRule
metadata:
  name:  nginx
  namespace:  vulpecula
spec:
  vip: 1.1.1.1
  sessionAffinity: ClientIP
  namespace: default
  selector:
    - app: nginx
  ports:
  - name: dns
    port: 8888
    targetPort: 80
    protocol: TCP
END
root@server:~# kubectl apply -f slr.yaml
root@server:~# kubectl get slr
NAME              VIP       PORT(S)    SERVICE                       AGE
vulpecula-nginx   1.1.1.1   8888/TCP   default/slr-vulpecula-nginx   3d21h
```

██████ subnet ████ vip ███████

```
#       vip

root@server:~# kubectl get vip
NAME          NS    V4IP        MAC                V6IP    PMAC   SUBNET        READY   TYPE
vulpecula-subnet    10.16.0.2   00:00:00:39:95:C1   <nil>          vulpecula-subnet   true
```

████████████ Load_Balancer_Health_Check 和 Service_Monitor 的

```
root@server:~# kubectl ko nbctl list Load_Balancer
_uuid               : 3cbb6d43-44aa-4028-962f-30d2dba9f0b8
external_ids        : {}
health_check        : [5bee3f12-6b54-411c-9cc8-c9def8f67356]
ip_port_mappings    : {"10.16.0.4"="nginx-78d9578975-f4qn4.default:10.16.0.2", "10.16.0.6"="nginx-78d9578975-t8tm5.default:10.16.0.2"}
name                : cluster-tcp-session-loadbalancer
options             : {affinity_timeout="10800"}
protocol            : tcp
selection_fields    : [ip_src]
vips                : {"1.1.1.1:8888"="10.16.0.4:80,10.16.0.6:80"}

root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
_uuid               : 5bee3f12-6b54-411c-9cc8-c9def8f67356
external_ids        : {switch_lb_subnet=vulpecula-subnet}
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
vip                 : "1.1.1.1:8888"

root@server:~# kubectl ko sbctl list Service_Monitor
_uuid               : 1bddc541-cc49-44ea-9935-a4208f627a91
external_ids        : {}
ip                  : "10.16.0.4"
logical_port        : nginx-78d9578975-f4qn4.default
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
port                : 80
protocol            : tcp
```

```
src_ip              : "10.16.0.2"
src_mac             : "c6:d4:b8:08:54:e7"
status              : online

_uuid               : 84dd24c5-e1b4-4e97-9daa-13687ed59785
external_ids        : {}
ip                  : "10.16.0.6"
logical_port        : nginx-78d9578975-t8tm5.default
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
port                : 80
protocol            : tcp
src_ip              : "10.16.0.2"
src_mac             : "c6:d4:b8:08:54:e7"
status              : online
```

□□□□□□□□□ `vip` □□□□□□□□□□□□□

```
root@server:~# kubectl exec -it -n vulpecula nginx-78d9578975-t8tm5 -- curl 1.1.1.1:8888
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

□□□□□□□□□□□

□□□□□ `pod` □□□□□□□□□□□□□□□□

```
kubectl delete po nginx-78d9578975-f4qn4
kubectl get po -o wide -n vulpecula
NAME                    READY   STATUS    RESTARTS   AGE     IP          NODE     NOMINATED NODE   READINESS GATES
nginx-78d9578975-lxmvh   1/1     Running   0          31s     10.16.0.8   worker   <none>           <none>
nginx-78d9578975-t8tm5   1/1     Running   3          4d16h   10.16.0.6   worker   <none>           <none>
```

□□□□□□□□□□□□□ `Load_Balancer_Health_Check` □ `Service_Monitor` □□□□□□□□□□□□

```
root@server:~# kubectl ko nbctl list Load_Balancer
_uuid               : 3cbb6d43-44aa-4028-962f-30d2dba9f0b8
external_ids        : {}
health_check        : [5bee3f12-6b54-411c-9cc8-c9def8f67356]
ip_port_mappings    : {"10.16.0.4"="nginx-78d9578975-f4qn4.default:10.16.0.2", "10.16.0.6"="nginx-78d9578975-t8tm5.default:10.16.0.2",
"10.16.0.8"="nginx-78d9578975-lxmvh.default:10.16.0.2"}
name                : cluster-tcp-session-loadbalancer
options             : {affinity_timeout="10800"}
protocol            : tcp
selection_fields    : [ip_src]
vips                : {"1.1.1.1:8888"="10.16.0.6:80,10.16.0.8:80"}

root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
_uuid               : 5bee3f12-6b54-411c-9cc8-c9def8f67356
external_ids        : {switch_lb_subnet=vulpecula-subnet}
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
vip                 : "1.1.1.1:8888"

root@server:~# kubectl ko sbctl list Service_Monitor
_uuid               : 84dd24c5-e1b4-4e97-9daa-13687ed59785
external_ids        : {}
ip                  : "10.16.0.6"
logical_port        : nginx-78d9578975-t8tm5.default
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
port                : 80
protocol            : tcp
src_ip              : "10.16.0.2"
src_mac             : "c6:d4:b8:08:54:e7"
status              : online

_uuid               : 5917b7b7-a999-49f2-a42d-da81f1eeb28f
external_ids        : {}
ip                  : "10.16.0.8"
logical_port        : nginx-78d9578975-lxmvh.default
options             : {failure_count="3", interval="5", success_count="3", timeout="20"}
port                : 80
protocol            : tcp
src_ip              : "10.16.0.2"
src_mac             : "c6:d4:b8:08:54:e7"
status              : online
```

□□ `SwitchLBRule` □□□□□□□□□□□□□□□ `Load_Balancer_Health_Check` □ `Service_Monitor` □□□□□□□□□□□□□□ `vip` □□□□□□

```
root@server:~# kubectl delete -f slr.yaml
switchlbrule.kubeovn.io "vulpecula-nginx" deleted
root@server:~# kubectl get vip
No resources found
```

```
root@server:~# kubectl ko sbctl list Service_Monitor
root@server:~#
root@server:~# kubectl ko nbctl list Load_Balancer_Health_Check
root@server:~#
```

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## 5.6.4 ⅛⅛

# 5.7 ▯▯▯ VPC ▯▯ DNS

▯▯▯▯▯▯▯ VPC ▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯ coredns▯ ▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯ Kubernetes ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `vpc-dns` CRD ▯▯▯▯

▯ CRD ▯▯▯▯▯▯▯ coredns▯▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
        loop
        reload
        loadbalance
    }
```

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ nat-gw-pod ⬚⬚⬚⬚⬚⬚⬚⬚⬚

## 5.7.2 ⬚⬚⬚⬚⬚⬚

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-nad
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "ovn-nad.default.ovn"
    }'
```

## 5.7.3 ⬚⬚ vpc-dns ⬚ Configmap

⬚ kube-system ⬚⬚⬚⬚⬚⬚ configmap⬚⬚⬚ vpc-dns ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ vpc-dns ⬚⬚⬚

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: vpc-dns-config
  namespace: kube-system
data:
  coredns-vip: 10.96.0.3
  enable-vpc-dns: "true"
  nad-name: ovn-nad
  nad-provider: ovn-nad.default.ovn
```

- `enable-vpc-dns` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `true` ⬚
- `coredns-image` ⬚dns ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ coredns ⬚⬚⬚⬚⬚
- `coredns-vip` ⬚⬚ coredns ⬚⬚ lb ⬚⬚⬚ vip⬚
- `nad-name` ⬚⬚⬚⬚ `network-attachment-definitions` ⬚⬚⬚⬚⬚
- `nad-provider` ⬚⬚⬚⬚ provider ⬚⬚⬚
- `k8s-service-host` ⬚⬚⬚ coredns ⬚⬚ k8s apiserver ⬚⬚⬚ ip⬚⬚⬚⬚⬚⬚⬚ apiserver ⬚⬚⬚
- `k8s-service-port` ⬚⬚⬚ coredns ⬚⬚ k8s apiserver ⬚⬚⬚ port⬚⬚⬚⬚⬚⬚⬚ apiserver ⬚⬚⬚

## 5.7.4 ⬚⬚ vpc-dns

⬚⬚ vpc-dns yaml⬚

```
kind: VpcDns
apiVersion: kubeovn.io/v1
metadata:
  name: test-cjh1
spec:
  vpc: cjh-vpc-1
  subnet: cjh-subnet-1
  replicas: 2
```

- `vpc` ⬚ ⬚⬚⬚⬚ dns ⬚⬚⬚ vpc ⬚⬚⬚
- `subnet` ⬚⬚⬚⬚⬚ dns ⬚⬚⬚⬚⬚⬚⬚⬚
- `replicas` : vpc dns deployment replicas

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
# kubectl get vpc-dns
NAME        ACTIVE   VPC         SUBNET
test-cjh1   false    cjh-vpc-1   cjh-subnet-1
test-cjh2   true     cjh-vpc-1   cjh-subnet-2
```

`ACTIVE` : `true` ⬚⬚⬚⬚⬚⬚ dns ⬚⬚⬚ `false` ⬚⬚⬚⬚

⬚⬚⬚⬚⬚ VPC ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ DNS ⬚⬚⬚

- ⬚⬚⬚ VPC ⬚⬚⬚⬚⬚ vpc-dns ⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚⬚ subnet⬚⬚⬚⬚⬚⬚ vpc-dns ⬚⬚⬚⬚ `true` ⬚⬚⬚⬚ `false` ⬚
- ⬚ `true` ⬚ vpc-dns ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `false` ⬚ vpc-dns ⬚⬚⬚⬚⬚

## 5.7.5 ⬚⬚⬚⬚⬚⬚⬚

⬚⬚ vpc-dns Pod ⬚⬚⬚⬚⬚ label `app=vpc-dns` ⬚⬚⬚⬚⬚⬚⬚⬚ vpc-dns Pod ⬚⬚⬚

```
# kubectl -n kube-system get pods -l app=vpc-dns
NAME                                READY   STATUS    RESTARTS   AGE
vpc-dns-test-cjh1-7b878d96b4-g5979   1/1     Running   0          28s
vpc-dns-test-cjh1-7b878d96b4-ltmf9   1/1     Running   0          28s
```

⬚⬚ slr ⬚⬚⬚⬚⬚

```
# kubectl -n kube-system get slr
NAME               VIP          PORT(S)                 SERVICE                              AGE
vpc-dns-test-cjh1   10.96.0.3   53/UDP,53/TCP,9153/TCP   kube-system/slr-vpc-dns-test-cjh1   113s
```

⬚⬚⬚ VPC ⬚⬚ Pod⬚⬚⬚ dns ⬚⬚:

```
nslookup kubernetes.default.svc.cluster.local 10.96.0.3
```

⬚ VPC ⬚⬚ switch lb rule ⬚⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚⬚⬚⬚⬚⬚ Pod ⬚⬚⬚⬚⬚⬚

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## 5.7.6 ⬚⬚

# 5.8 SecurityGroup ⬚⬚

Kube-OVN ⬚⬚⬚⬚⬚⬚⬚⬚⬚ Pod ⬚⬚⬚⬚⬚⬚⬚

> **Warning**
>
> Kube-OVN ⬚⬚⬚⬚ NetworkPolicy⬚Network Policy API⬚Subnet ACL ⬚ Security Group ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ OVN ACL ⬚⬚⬚⬚⬚⬚ NetworkPolicy ⬚ NetworkPolicy API ⬚⬚⬚⬚ ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
    remoteType: address
  - ipVersion: ipv4
    policy: deny
    priority: 2
    protocol: all
    remoteAddress: 0.0.0.0/0
    remoteType: address
---
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-tier1
spec:
  tier: 1
  allowSameGroupTraffic: true
  ingressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: tcp
    remoteAddress: 10.16.0.0/16
    remoteType: address
    portRangeMin: 80
    portRangeMax: 443
```

[illegible] `sg-tier0` [illegible] `10.16.0.0/16` [illegible] TCP [illegible] tier 1[illegible] `sg-tier1` [illegible] 80-443 [illegible] TCP [illegible]

[illegible] Pod[illegible] annotation [illegible]

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: web
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-tier0,sg-tier1'
  name: multi-tier-pod
  namespace: default
spec:
  nodeName: kube-ovn-worker
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: nginx
```

## 5.8.3 [illegible]

[illegible] `localAddress` [illegible]

- **`localAddress`** [illegible] IP [illegible] CIDR[illegible]
- **`sourcePortRangeMin`** / **`sourcePortRangeMax`** [illegible]1-65535[illegible] TCP [illegible] UDP [illegible]

[illegible]

```
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-local-filter
spec:
  allowSameGroupTraffic: true
  ingressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: tcp
    remoteAddress: 10.16.0.0/16
    remoteType: address
    portRangeMin: 8080
    portRangeMax: 8080
    localAddress: 10.16.0.100
    sourcePortRangeMin: 1024
    sourcePortRangeMax: 65535
```

[illegible] `10.16.0.0/16` [illegible] `10.16.0.100` [illegible] 8080 [illegible] TCP [illegible] 1024-65535[illegible]

## 5.8.4 [illegible]

- [illegible] ACL [illegible]OVN [illegible] ACL [illegible] ACL [illegible]



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- ███████ `priority` ███████ 1-16384████████████████████ ACL ████████████████████ ACL ████████████████ `ACL = 18484 -` ████████████████ ACL ████

- `tier` ██████ `0` ▯ `1` ▯ `policy: pass` ████ tier `0` ██████ tier `1` ████████████

- ████████████████████████████████Kube-OVN ▯▯ CNI▯▯▯ Pod ████ Pod ████████████████████████████ Pod ████ ContainerCreating ██████████ Running ▯▯▯

## 5.8.5 ████

████ YAML ▯▯ Pod▯▯ annotation ████████████████

```yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: static
  annotations:
    ovn.kubernetes.io/security_groups: 'sg-example'
  name: sg-test-pod
  namespace: default
spec:
  nodeName: kube-ovn-worker
  containers:
  - image: docker.io/library/nginx:alpine
    imagePullPolicy: IfNotPresent
    name: qatest
```

████████████

```
# kubectl get pod -o wide
NAME                   READY   STATUS              RESTARTS   AGE     IP           NODE                   NOMINATED NODE   READINESS GATES
sg-test-pod            0/1     ContainerCreating   0          5h32m   <none>       kube-ovn-worker        <none>           <none>
test-99fff7f86-52h9r   1/1     Running             0          5h41m   10.16.0.14   kube-ovn-control-plane   <none>           <none>
test-99fff7f86-qcgjw   1/1     Running             0          5h43m   10.16.0.13   kube-ovn-worker        <none>           <none>
```

▯▯ kubectl describe pod ▯▯ Pod ████████████████

```
# kubectl describe pod sg-test-pod
Name:         sg-test-pod
Namespace:    default
Priority:     0
Node:         kube-ovn-worker/172.18.0.2
Start Time:   Tue, 28 Feb 2023 10:29:36 +0800
Labels:       app=static
Annotations:  ovn.kubernetes.io/allocated: true
              ovn.kubernetes.io/cidr: 10.16.0.0/16
              ovn.kubernetes.io/gateway: 10.16.0.1
              ovn.kubernetes.io/ip_address: 10.16.0.15
              ovn.kubernetes.io/logical_router: ovn-cluster
              ovn.kubernetes.io/logical_switch: ovn-default
              ovn.kubernetes.io/mac_address: 00:00:00:FA:17:97
              ovn.kubernetes.io/pod_nic_type: veth-pair
              ovn.kubernetes.io/port_security: true
              ovn.kubernetes.io/routed: true
              ovn.kubernetes.io/security_groups: sg-allow-reject
Status:       Pending
IP:
IPs:          <none>
.
.
.
Events:
  Type     Reason                  Age                   From     Message
  ----     ------                  ----                  ----     -------
  Warning  FailedCreatePodSandBox  5m3s (x70 over 4h59m)  kubelet  (combined from similar events): Failed to create pod sandbox: rpc error: code = Unknown
desc = failed to setup network for sandbox "40636e0c7f1ade5500fa958486163d74f2e2300051a71522a9afd7ba0538afb6": plugin type="kube-ovn" failed (add): RPC
failed; request ip return 500 configure nic failed 10.16.0.15 network not ready after 200 ping 10.16.0.1
```

████████████████████████████████

```yaml
apiVersion: kubeovn.io/v1
kind: SecurityGroup
metadata:
  name: sg-gw-both
spec:
  allowSameGroupTraffic: true
  egressRules:
  - ipVersion: ipv4
    policy: allow
    priority: 2
    protocol: all
    remoteAddress: 10.16.0.13
    remoteType: address
```



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```
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: all
    remoteAddress: 10.16.0.1
    remoteType: address
  ingressRules:
  - ipVersion: ipv4
    policy: deny
    priority: 2
    protocol: icmp
    remoteAddress: 10.16.0.14
    remoteType: address
  - ipVersion: ipv4
    policy: allow
    priority: 1
    protocol: icmp
    remoteAddress: 10.16.0.1
    remoteType: address
```

▯▯▯

# 5.9 OVN EIP FIP SNAT DNAT ▯▯

**Note**

▯▯▯ VPC ▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯▯▯▯VPC NAT ▯▯▯OVN ▯▯▯ Egress Gateway▯▯▯ VPC NAT ▯▯▯ Kube-OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ VPC NAT ▯▯▯▯▯▯▯▯▯ Pod▯▯▯▯▯▯▯▯▯▯▯ VPC ▯▯▯▯▯▯▯▯▯▯▯ Macvlan ▯▯▯▯▯▯▯▯▯▯▯▯ Pod ▯▯ iptables ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
###               vlan id                  underlay
```

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

- ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ provider network⬚vlan⬚subnet ⬚⬚⬚⬚
- ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ VPC enable_eip_snat ⬚⬚⬚⬚⬚⬚⬚⬚ vlan⬚subnet ⬚⬚⬚⬚⬚⬚⬚⬚⬚ ip ⬚ ipam⬚
- ⬚⬚⬚⬚⬚⬚⬚ VPC ⬚ enable_eip_snat ⬚⬚, ⬚⬚⬚⬚⬚⬚⬚⬚ pod annotation ⬚ fip snat⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚
- ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ VPC ⬚ enable_eip_snat ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ vlan⬚subnet ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚ eip snat ⬚⬚⬚⬚

### 1.1 ⬚⬚ **underlay** ⬚⬚⬚⬚

```
#    provider-network  vlan  subnet
# cat 01-provider-network.yaml

apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: external204
spec:
  defaultInterface: vlan

# cat 02-vlan.yaml

apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan204
spec:
  id: 204
  provider: external204

# cat 03-vlan-subnet.yaml

apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: external204
spec:
  protocol: IPv4
  cidrBlock: 10.5.204.0/24
  gateway: 10.5.204.254
  vlan: vlan204
  excludeIps:
  - 10.5.204.1..10.5.204.100
```

### 1.2 ⬚⬚ **VPC** ⬚⬚ **eip_snat**

```
#      VPC     underlay    provider subnet
# cat 00-centralized-external-gw-no-ip.yaml

apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-external-gw-config
  namespace: kube-system
data:
  enable-external-gw: "true"
  external-gw-nodes: "pc-node-1,pc-node-2,pc-node-3"
  type: "centralized"
  external-gw-nic: "vlan" #      ovs
  external-gw-addr: "10.5.204.254/24" # underlay       ip
```

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ logical router port (lrp) ip ⬚ mac⬚⬚⬚⬚⬚ underlay ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ lrp ⬚⬚⬚ ovn eip ⬚⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ ip ⬚⬚⬚⬚⬚⬚⬚ lrp ⬚⬚⬚ ovn-eip⬚ ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ lrp ⬚⬚⬚ ovn eip⬚

### 1.3 ⬚⬚⬚ **VPC** ⬚⬚ **eip snat fip** ⬚⬚

```
# cat 00-ns.yml

apiVersion: v1
kind: Namespace
metadata:
  name: vpc1

# cat 01-vpc-external.yml
```

```
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc1
spec:
  namespaces:
  - vpc1
  enableExternal: true
  staticRoutes:
    - cidr: 0.0.0.0/0
      nextHopIP: 10.5.204.254
      policy: policyDst

# VPC    enableExternal      lrp

# cat 02-subnet.yml

apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vpc1-subnet1
spec:
  cidrBlock: 192.168.0.0/24
  default: false
  disableGatewayCheck: false
  disableInterConnection: true
  gatewayNode: ""
  gatewayType: distributed
  #gatewayType: centralized
  natOutgoing: false
  private: false
  protocol: IPv4
  provider: ovn
  vpc: vpc1
  namespaces:
  - vpc1

#                  subnet
```

▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

```
# kubectl ko nbctl show vpc1

router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
    port vpc1-vpc1-subnet1
        mac: "00:00:00:ED:8E:C7"
        networks: ["192.168.0.1/24"]
    port vpc1-external204
        mac: "00:00:00:EF:05:C7"
        networks: ["10.5.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
    nat 21d853b0-f7b4-40bd-9a53-31d2e2745739
        external ip: "10.5.204.115"
        logical ip: "192.168.0.0/24"
        type: "snat"
```

```
# kubectl ko nbctl lr-route-list vpc1

IPv4 Routes
Route Table <main>:
               0.0.0.0/0             10.5.204.254 dst-ip
#    VPC CRD
```

> ▒▒▒▒▒▒ enableExternal ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

```
# cat 03-extra-vlan-subnet.yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: extra
spec:
  protocol: IPv4
  cidrBlock: 10.10.204.0/24
  gateway: 10.10.204.254
  vlan: vlan0
  excludeIps:
  - 10.10.204.1..10.10.204.100
```

#### 1.4.2 ▯▯▯ VPC ▯▯

```
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  name: vpc1
spec:
  namespaces:
  - vpc1
  enableExternal: true  #    enableExternal   VPC               external   ls
  extraExternalSubnets: #    extraExternalSubnets
  - extra
```

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
    port vpc1-vpc1-subnet1
        mac: "00:00:00:ED:8E:C7"
        networks: ["192.168.0.1/24"]
    port vpc1-external204
        mac: "00:00:00:EF:05:C7"
        networks: ["10.5.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
    port vpc1-extra
        mac: "00:00:00:EF:6A:C7"
        networks: ["10.10.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
```

## 5.9.3 2. ovn-eip

▯▯▯▯ iptables-eip ▯▯▯▯▯▯▯▯▯▯▯▯▯ovn-eip ▯▯▯▯▯ type

- nat: ▯▯ ovn dnat▯fip, snat ▯▯▯ nat ▯▯▯▯
- lrp: ▯▯▯▯▯▯▯▯▯ underlay ▯▯▯▯▯▯ lrp ▯▯▯ ip ▯▯▯▯▯ dnat snat ▯▯
- lsp: ▯▯▯▯▯▯▯▯▯▯▯LSP▯▯ EIP

```
---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  externalSubnet: external204
  type: nat

#       eip           fip
```

externalSubnet ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
metadata:
  name: eip-static
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  ovnEip: eip-static
  ipName: vpc-1-busybox01.vpc1  #       ip crd
  type: "centralized"           # centralized    distributed

--
#         VPC       ip

kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-static
spec:
  ovnEip: eip-static
  vpc: vpc1
  v4Ip: 192.168.0.2
  type: "centralized"           # centralized    distributed
```

```
# kubectl get ofip
NAME          VPC    V4EIP          V4IP          READY   IPTYPE   IPNAME
eip-for-vip   vpc1   10.5.204.106   192.168.0.3   true    vip      test-fip-vip
eip-static    vpc1   10.5.204.101   192.168.0.2   true             vpc-1-busybox01.vpc1
# kubectl get ofip eip-static
NAME          VPC    V4EIP          V4IP          READY   IPTYPE   IPNAME
eip-static    vpc1   10.5.204.101   192.168.0.2   true             vpc-1-busybox01.vpc1

[root@pc-node-1 03-cust-vpc]# ping 10.5.204.101
PING 10.5.204.101 (10.5.204.101) 56(84) bytes of data.
64 bytes from 10.5.204.101: icmp_seq=2 ttl=62 time=1.21 ms
64 bytes from 10.5.204.101: icmp_seq=3 ttl=62 time=0.624 ms
64 bytes from 10.5.204.101: icmp_seq=4 ttl=62 time=0.368 ms
^C
--- 10.5.204.101 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3049ms
rtt min/avg/max/mdev = 0.368/0.734/1.210/0.352 ms
[root@pc-node-1 03-cust-vpc]#

#       node ping    VPC    pod     ip
```

```
#     ip
# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
    port vpc1-vpc1-subnet1
        mac: "00:00:00:ED:8E:C7"
        networks: ["192.168.0.1/24"]
    port vpc1-external204
        mac: "00:00:00:EF:05:C7"
        networks: ["10.5.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
    nat 813523e7-c68c-408f-bd8c-cba30cb2e4f4
        external ip: "10.5.204.101"
        logical ip: "192.168.0.2"
        type: "dnat_and_snat"
```

### 2.2 ovn-fip  vip  fip

 vip  kubevirt  vip  keepalived kube-vip 

 fip  VPC  vip  vip 

```
#     vip eip    eip     vip
# cat vip.yaml

apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: test-fip-vip
spec:
  subnet: vpc1-subnet1

# cat 04-fip.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
```

```
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  ovnEip: eip-for-vip
  ipType: vip         #       fip     pod ip              vip
  ipName: test-fip-vip

---
#         VPC       ip

kind: OvnFip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-for-vip
spec:
  ovnEip: eip-for-vip
  ipType: vip         #       fip     pod ip              vip
  vpc: vpc1
  v4Ip: 192.168.0.3
```

```
# kubectl get ofip
NAME          VPC    V4EIP          V4IP          READY   IPTYPE   IPNAME
eip-for-vip   vpc1   10.5.204.106   192.168.0.3   true    vip      test-fip-vip


[root@pc-node-1 fip-vip]# ping  10.5.204.106
PING 10.5.204.106 (10.5.204.106) 56(84) bytes of data.
64 bytes from 10.5.204.106: icmp_seq=1 ttl=62 time=0.694 ms
64 bytes from 10.5.204.106: icmp_seq=2 ttl=62 time=0.436 ms

#   node    ping


# pod     ip

[root@pc-node-1 fip-vip]# kubectl -n vpc1 exec -it vpc-1-busybox03 -- bash
[root@vpc-1-busybox03 /]#
[root@vpc-1-busybox03 /]#
[root@vpc-1-busybox03 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
1568: eth0@if1569: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:56:40:e5 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.5/24 brd 192.168.0.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet 192.168.0.3/24 scope global secondary eth0  #      vip
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe56:40e5/64 scope link
       valid_lft forever preferred_lft forever

[root@vpc-1-busybox03 /]# tcpdump -i eth0 host  192.168.0.3 -netvv
tcpdump: listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
00:00:00:ed:8e:c7 > 00:00:00:56:40:e5, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 62, id 44830, offset 0, flags [DF], proto ICMP (1), length 84)
    10.5.32.51 > 192.168.0.3: ICMP echo request, id 177, seq 1, length 64
00:00:00:56:40:e5 > 00:00:00:ed:8e:c7, ethertype IPv4 (0x0800), length 98: (tos 0x0, ttl 64, id 43962, offset 0, flags [none], proto ICMP (1), length 84)
    192.168.0.3 > 10.5.32.51: ICMP echo reply, id 177, seq 1, length 64

# pod        fip     icmp
```

## 5.9.4 3. ovn-snat

### 3.1 ovn-snat [][][][] subnet [] cidr

[][][][] iptables-snat [][][][][][][][][][][][]

```
# cat 03-subnet-snat.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnSnatRule
```

```
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpcSubnet: vpc1-subnet1 # eip

---
#         VPC       subnet cidr

kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpc: vpc1
  v4IpCidr: 192.168.0.0/24 #        cidr      ip
```

□□□□□□□□□□□□□□□□□ externalSubnet □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ extra□

### 3.2 ovn-snat □□□□□ pod ip

□□□□ iptables-snat □□□□□□□□□□□□

```
# cat 03-pod-snat.yaml

---
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-pod-vpc-ip
spec:
  externalSubnet: external204
  type: nat

---
kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat01
spec:
  ovnEip: snat-for-pod-vpc-ip
  ipName: vpc-1-busybox02.vpc1 # eip      pod ip

---
#         VPC       ip

kind: OvnSnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: snat-for-subnet-in-vpc
spec:
  ovnEip: snat-for-subnet-in-vpc
  vpc: vpc1
  v4IpCidr: 192.168.0.4
```

□□□□□□□□□□□□□□□□□ externalSubnet □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ extra□

□□□□□□□□□□□□□ snat □□□□□□□□□□□□□

```
# kubectl ko nbctl show vpc1
router 87ad06fd-71d5-4ff8-a1f0-54fa3bba1a7f (vpc1)
    port vpc1-vpc1-subnet1
        mac: "00:00:00:ED:8E:C7"
        networks: ["192.168.0.1/24"]
    port vpc1-external204
        mac: "00:00:00:EF:05:C7"
        networks: ["10.5.204.105/24"]
        gateway chassis: [7cedd14f-265b-42e5-ac17-e03e7a1f2342 276baccb-fe9c-4476-b41d-05872a94976d fd9f140c-c45d-43db-a6c0-0d4f8ea298dd]
    nat 21d853b0-f7b4-40bd-9a53-31d2e2745739
        external ip: "10.5.204.115"
        logical ip: "192.168.0.0/24"
        type: "snat"
    nat da77a11f-c523-439c-b1d1-72c664196a0f
        external ip: "10.5.204.116"
        logical ip: "192.168.0.4"
        type: "snat"
```

```
[root@pc-node-1 03-cust-vpc]# kubectl get po -A -o wide  | grep busy
vpc1            vpc-1-busybox01                              1/1     Running   0               3d15h   192.168.0.2   pc-node-2   <none>           <none>
vpc1            vpc-1-busybox02                              1/1     Running   0               17h     192.168.0.4   pc-node-1   <none>           <none>
vpc1            vpc-1-busybox03                              1/1     Running   0               17h     192.168.0.5   pc-node-1   <none>           <none>
vpc1            vpc-1-busybox04                              1/1     Running   0               17h     192.168.0.6   pc-node-3   <none>           <none>
vpc1            vpc-1-busybox05                              1/1     Running   0               17h     192.168.0.7   pc-node-1   <none>           <none>
```

```
# kubectl exec -it -n vpc1            vpc-1-busybox04   bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] -- [COMMAND] instead.
[root@vpc-1-busybox04 /]#
[root@vpc-1-busybox04 /]#
[root@vpc-1-busybox04 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
17095: eth0@if17096: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:76:94:55 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.6/24 brd 192.168.0.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe76:9455/64 scope link
       valid_lft forever preferred_lft forever
[root@vpc-1-busybox04 /]# ping 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=1 ttl=114 time=22.2 ms
64 bytes from 223.5.5.5: icmp_seq=2 ttl=114 time=21.8 ms

[root@pc-node-1 03-cust-vpc]# kubectl exec -it -n vpc1            vpc-1-busybox02   bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] -- [COMMAND] instead.
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]#
[root@vpc-1-busybox02 /]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
1566: eth0@if1567: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 00:00:00:0b:e9:d0 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.0.4/24 brd 192.168.0.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fe0b:e9d0/64 scope link
       valid_lft forever preferred_lft forever
[root@vpc-1-busybox02 /]# ping 223.5.5.5
PING 223.5.5.5 (223.5.5.5) 56(84) bytes of data.
64 bytes from 223.5.5.5: icmp_seq=2 ttl=114 time=22.7 ms
64 bytes from 223.5.5.5: icmp_seq=3 ttl=114 time=22.6 ms
64 bytes from 223.5.5.5: icmp_seq=4 ttl=114 time=22.1 ms
^C
--- 223.5.5.5 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3064ms
rtt min/avg/max/mdev = 22.126/22.518/22.741/0.278 ms

#       pod          snat
```

## 5.9.5 4. ovn-dnat

### 4.1 ovn-dnat  pod  dnat

```yaml
kind: OvnEip
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  externalSubnet: underlay

---
kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ovnEip: eip-dnat
  ipName: vpc-1-busybox01.vpc1 #      pod ip crd
  protocol: tcp
  internalPort: "22"
  externalPort: "22"


---
#         VPC      ip

kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ovnEip: eip-dnat
  protocol: tcp
  internalPort: "22"
  externalPort: "22"
  vpc: vpc1
  v4Ip: 192.168.0.3
```

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ externalSubnet ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ extra⬚

OvnDnatRule ⬚⬚⬚⬚ IptablesDnatRule ⬚⬚

```
# kubectl get oeip eip-dnat
NAME       V4IP         V6IP   MAC                TYPE   READY
eip-dnat   10.5.49.4           00:00:00:4D:CE:49  dnat   true

# kubectl get odnat
NAME                EIP                 PROTOCOL   V4EIP       V4IP          INTERNALPORT   EXTERNALPORT   IPNAME                          READY
eip-dnat            eip-dnat            tcp        10.5.49.4   192.168.0.3   22             22             vpc-1-busybox01.vpc1            true
```

### 4.2 ovn-dnat ⬚ vip ⬚⬚⬚⬚ dnat

```
kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ipType: vip  #       dnat     pod ip              vip
  ovnEip: eip-dnat
  ipName: test-dnat-vip
  protocol: tcp
  internalPort: "22"
  externalPort: "22"

---
#          VPC       ip

kind: OvnDnatRule
apiVersion: kubeovn.io/v1
metadata:
  name: eip-dnat
spec:
  ipType: vip  #       dnat     pod ip              vip
  ovnEip: eip-dnat
  ipName: test-dnat-vip
  protocol: tcp
  internalPort: "22"
  externalPort: "22"
  vpc: vpc1
  v4Ip: 192.168.0.4
```

OvnDnatRule ⬚⬚⬚⬚ IptablesDnatRule ⬚⬚

```
# kubectl get vip test-dnat-vip
NAME            V4IP          PV4IP   MAC                PMAC   V6IP   PV6IP   SUBNET         READY
test-dnat-vip   192.168.0.4           00:00:00:D0:C0:B5                        vpc1-subnet1   true

# kubectl get oeip eip-dnat
NAME       V4IP         V6IP   MAC                TYPE   READY
eip-dnat   10.5.49.4           00:00:00:4D:CE:49  dnat   true

# kubectl get odnat eip-dnat
NAME       EIP        PROTOCOL   V4EIP       V4IP          INTERNALPORT   EXTERNALPORT   IPNAME          READY
eip-dnat   eip-dnat   tcp        10.5.49.4   192.168.0.4   22             22             test-dnat-vip   true
```

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# 5.10 VPC ⬚⬚

VPC ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ NAT ⬚⬚⬚

## 5.10.1 ⬚⬚⬚⬚

1. ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ VPC⬚
2. ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚⬚⬚⬚ CIDR ⬚⬚⬚⬚⬚
3. ⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

## 5.10.2 ⬚⬚⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ VPC⬚⬚⬚ VPC ⬚⬚⬚⬚⬚⬚ Subnet⬚Subnet ⬚ CIDR ⬚⬚⬚⬚⬚⬚

```yaml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-1
spec: {}
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net1
spec:
  vpc: vpc-1
  cidrBlock: 10.0.0.0/16
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-2
spec: {}
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: vpc-2
  cidrBlock: 172.31.0.0/16
```

⬚⬚⬚ VPC ⬚⬚⬚⬚⬚ `vpcPeerings` ⬚⬚⬚⬚⬚⬚⬚⬚⬚

```yaml
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-1
spec:
  vpcPeerings:
    - remoteVpc: vpc-2
      localConnectIP: 169.254.0.1/30
  staticRoutes:
    - cidr: 172.31.0.0/16
      nextHopIP: 169.254.0.2
      policy: policyDst
---
kind: Vpc
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-2
spec:
  vpcPeerings:
    - remoteVpc: vpc-1
      localConnectIP: 169.254.0.2/30
  staticRoutes:
    - cidr: 10.0.0.0/16
      nextHopIP: 169.254.0.1
      policy: policyDst
```

- `remoteVpc` : ⬚⬚⬚⬚⬚⬚ VPC ⬚⬚⬚⬚
- `localConnectIP` : ⬚⬚⬚⬚⬚⬚⬚ IP ⬚⬚⬚ CIDR⬚⬚⬚⬚⬚ IP ⬚⬚⬚⬚⬚ CIDR⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚
- `cidr` ⬚⬚⬚⬚ Subnet ⬚ CIDR⬚
- `nextHopIP` ⬚⬚⬚ VPC ⬚⬚⬚⬚ `localConnectIP` ⬚



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▯▯▯▯▯ Subnet ▯▯▯ Pod

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/logical_switch: net1
  name: vpc-1-pod
spec:
  containers:
    - name: vpc-1-pod
      image: docker.io/library/nginx:alpine
---
apiVersion: v1
kind: Pod
metadata:
  annotations:
    ovn.kubernetes.io/logical_switch: net2
  name: vpc-2-pod
spec:
  containers:
    - name: vpc-2-pod
      image: docker.io/library/nginx:alpine
```

▯▯▯▯▯▯▯

```
# kubectl exec -it vpc-1-pod -- ping $(kubectl get pod vpc-2-pod -o jsonpath='{.status.podIP}')
PING 172.31.0.2 (172.31.0.2): 56 data bytes
64 bytes from 172.31.0.2: seq=0 ttl=62 time=0.655 ms
64 bytes from 172.31.0.2: seq=1 ttl=62 time=0.086 ms
64 bytes from 172.31.0.2: seq=2 ttl=62 time=0.098 ms
^C
--- 172.31.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.086/0.279/0.655 ms
# kubectl exec -it vpc-2-pod -- ping $(kubectl get pod vpc-1-pod -o jsonpath='{.status.podIP}')
PING 10.0.0.2 (10.0.0.2): 56 data bytes
64 bytes from 10.0.0.2: seq=0 ttl=62 time=0.594 ms
64 bytes from 10.0.0.2: seq=1 ttl=62 time=0.093 ms
64 bytes from 10.0.0.2: seq=2 ttl=62 time=0.088 ms
^C
--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.088/0.258/0.594 ms
```

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5.10.3 ▯▯



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# 6. □□□□

## 6.1 kubectl □□□□

□□□□□□□□□□□□Kube-OVN □□□ kubectl □□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□ OVN □□□□□□□□□□OVN □□□ □□□□□□□OVS □□□□□□□□tcpdump □□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□

### 6.1.1 □□□□

Kube-OVN □□□□□□□□□□□□□□□□□□□□□ kubectl □□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□

□□ `kubectl-ko` □□□

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/kubectl-ko
```

□□□□□□□□ `$PATH` □□□□□

```
mv kubectl-ko /usr/local/bin/kubectl-ko
```

□□□□□□□□□

```
chmod +x /usr/local/bin/kubectl-ko
```

□□□□□□□□□□□□□□□

```
# kubectl plugin list
The following compatible plugins are available:

/usr/local/bin/kubectl-ko
```

### 6.1.2 □□□□

□□ `kubectl ko` □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

```
# kubectl ko
kubectl ko {subcommand} [option...]
Available Subcommands:
  [nb|sb] [status|kick|backup|dbstatus|restore]     ovn-db operations show cluster status, kick stale server, backup database, get db consistency status or
restore ovn nb db when met 'inconsistent data' error
  nbctl [ovn-nbctl options ...]    invoke ovn-nbctl
  sbctl [ovn-sbctl options ...]    invoke ovn-sbctl
  vsctl {nodeName} [ovs-vsctl options ...]   invoke ovs-vsctl on the specified node
  ofctl {nodeName} [ovs-ofctl options ...]   invoke ovs-ofctl on the specified node
  dpctl {nodeName} [ovs-dpctl options ...]   invoke ovs-dpctl on the specified node
  appctl {nodeName} [ovs-appctl options ...]   invoke ovs-appctl on the specified node
  tcpdump {namespace/podname} [tcpdump options ...]     capture pod traffic
  {trace|ovn-trace} ...    trace ovn microflow of specific packet
    {trace|ovn-trace} {namespace/podname} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]    trace ICMP/TCP/UDP
    {trace|ovn-trace} {namespace/podname} {target ip address} [target mac address] arp {request|reply}                     trace ARP request/reply
    {trace|ovn-trace} {node//nodename} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]       trace ICMP/TCP/UDP
    {trace|ovn-trace} {node//nodename} {target ip address} [target mac address] arp {request|reply}                         trace ARP request/reply
  diagnose {all|node|subnet|IPPorts} [nodename|subnetName|{proto1}-{IP1}-{Port1},{proto2}-{IP2}-{Port2}]    diagnose connectivity of all nodes or a specific
node or specify subnet's ds pod or IPPorts like 'tcp-172.18.0.2-53,udp-172.18.0.3-53'
  env-check    check the environment configuration
  reload    restart all kube-ovn components
  log {kube-ovn|ovn|ovs|linux|all}    save log to ./kubectl-ko-log/
  perf [image] performance test default image is docker.io/kubeovn/test:v1.13.0
  icnbctl [ovn-nbctl options ...]    invoke ovn-ic-nbctl
  icsbctl [ovn-sbctl options ...]    invoke ovn-ic-sbctl
```

□□□□□□□□□□□□□□□□□□□□

**[nb | sb] [status | kick | backup | dbstatus | restore]**

□□□□□□□□ OVN □□□□□□□□□□□□□□□□□□□□

### ██████

████████ OVN ████ leader ████ `ovs-appctl cluster/status` ██████:

```
# kubectl ko nb status
306b
Name: OVN_Northbound
Cluster ID: 9a87 (9a872522-3e7d-47ca-83a3-d74333e1a7ca)
Server ID: 306b (306b256b-b5e1-4eb0-be91-4ca96adf6bad)
Address: tcp:[172.18.0.2]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 280309 ms ago, reason: timeout
Last Election won: 280309 ms ago
Election timer: 5000
Log: [139, 139]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-8723 ->8723 <-85d6 ->85d6
Disconnections: 0
Servers:
    85d6 (85d6 at tcp:[172.18.0.4]:6643) next_index=139 match_index=138 last msg 763 ms ago
    8723 (8723 at tcp:[172.18.0.3]:6643) next_index=139 match_index=138 last msg 763 ms ago
    306b (306b at tcp:[172.18.0.2]:6643) (self) next_index=2 match_index=138
status: ok
```

█ `Server` ██ `match_index` ████████ `last msg` ████████ Server ██████████ ████████

### ███████

██████████ OVN ████████████████████████ ████████████████████████ `172.18.0.3` ██:

```
# kubectl ko nb kick 8723
started removal
```

████████████████████

```
# kubectl ko nb status
306b
Name: OVN_Northbound
Cluster ID: 9a87 (9a872522-3e7d-47ca-83a3-d74333e1a7ca)
Server ID: 306b (306b256b-b5e1-4eb0-be91-4ca96adf6bad)
Address: tcp:[172.18.0.2]:6643
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 324356 ms ago, reason: timeout
Last Election won: 324356 ms ago
Election timer: 5000
Log: [140, 140]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-85d6 ->85d6
Disconnections: 2
Servers:
    85d6 (85d6 at tcp:[172.18.0.4]:6643) next_index=140 match_index=139 last msg 848 ms ago
    306b (306b at tcp:[172.18.0.2]:6643) (self) next_index=2 match_index=139
status: ok
```

### ████

██████████ OVN ██████████████████

```
# kubectl ko nb backup
tar: Removing leading `/' from member names
backup ovn-nb db to /root/ovnnb_db.060223191654183154.backup
```

### ██████████

████████████████████

```
# kubectl ko nb dbstatus
status: ok
```

███████ `inconsistent data` ████████████



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...

... `inconsistent data` ...

```
# kubectl ko nb restore
deployment.apps/ovn-central scaled
ovn-central original replicas is 3
first nodeIP is 172.18.0.5
ovs-ovn pod on node 172.18.0.5 is ovs-ovn-8jxv9
ovs-ovn pod on node 172.18.0.3 is ovs-ovn-sjzb6
ovs-ovn pod on node 172.18.0.4 is ovs-ovn-t87zk
backup nb db file
restore nb db file, operate in pod ovs-ovn-8jxv9
deployment.apps/ovn-central scaled
finish restore nb db file and ovn-central replicas
recreate ovs-ovn pods
pod "ovs-ovn-8jxv9" deleted
pod "ovs-ovn-sjzb6" deleted
pod "ovs-ovn-t87zk" deleted
```

### [nbctl | sbctl] [options ...]

... OVN ... leader ... `ovn-nbctl` ... `ovn-sbctl` ... OVN ... ovn-nbctl(8) ... ovn-sbctl(8)...

```
# kubectl ko nbctl show
switch c7cd17e8-ceee-4a91-9bb3-e5a313fe1ece (snat)
    port snat-ovn-cluster
        type: router
        router-port: ovn-cluster-snat
switch 20e0c6d0-023a-4756-aec5-200e0c60f95d (join)
    port node-liumengxin-ovn3-192.168.137.178
        addresses: ["00:00:00:64:FF:A8 100.64.0.4"]
    port node-liumengxin-ovn1-192.168.137.176
        addresses: ["00:00:00:AF:98:62 100.64.0.2"]
    port node-liumengxin-ovn2-192.168.137.177
        addresses: ["00:00:00:D9:58:B8 100.64.0.3"]
    port join-ovn-cluster
        type: router
        router-port: ovn-cluster-join
switch 0191705c-f827-427b-9de3-3c3b7d971ba5 (central)
    port central-ovn-cluster
        type: router
        router-port: ovn-cluster-central
switch 2a45ff05-388d-4f85-9daf-e6fccd5833dc (ovn-default)
    port alertmanager-main-0.monitoring
        addresses: ["00:00:00:6C:DF:A3 10.16.0.19"]
    port kube-state-metrics-5d6885d89-4nf8h.monitoring
        addresses: ["00:00:00:6F:02:1C 10.16.0.15"]
    port fake-kubelet-67c55dfd89-pv86k.kube-system
        addresses: ["00:00:00:5C:12:E8 10.16.19.177"]
    port ovn-default-ovn-cluster
        type: router
        router-port: ovn-cluster-ovn-default
router 212f73dd-d63d-4d72-864b-a537e9afbee1 (ovn-cluster)
    port ovn-cluster-snat
        mac: "00:00:00:7A:82:8F"
        networks: ["172.22.0.1/16"]
    port ovn-cluster-join
        mac: "00:00:00:F8:18:5A"
        networks: ["100.64.0.1/16"]
    port ovn-cluster-central
        mac: "00:00:00:4D:8C:F5"
        networks: ["192.101.0.1/16"]
    port ovn-cluster-ovn-default
        mac: "00:00:00:A3:F8:18"
        networks: ["10.16.0.1/16"]
```

### vsctl {nodeName} [options ...]

... `nodeName` ... `ovs-ovn` ... `ovs-vsctl` ... `vswitchd` ... OVS ... ovs-vsctl(8)...

```
# kubectl ko vsctl kube-ovn-01 show
0d4c4675-c9cc-440a-8c1a-878e17f81b88
    Bridge br-int
        fail_mode: secure
        datapath_type: system
        Port a2c1a8a8b83a_h
            Interface a2c1a8a8b83a_h
        Port "4fa5c4cbb1a5_h"
            Interface "4fa5c4cbb1a5_h"
        Port ovn-eef07d-0
            Interface ovn-eef07d-0
                type: stt
                options: {csum="true", key=flow, remote_ip="192.168.137.178"}
        Port ovn0
```

```
            Interface ovn0
                type: internal
        Port mirror0
            Interface mirror0
                type: internal
        Port ovn-efa253-0
            Interface ovn-efa253-0
                type: stt
                options: {csum="true", key=flow, remote_ip="192.168.137.177"}
        Port br-int
            Interface br-int
                type: internal
    ovs_version: "2.17.2"
```

### ofctl {nodeName} [options ...]

... `nodeName` ... `ovs-ovn` ... `ovs-ofctl` ... OpenFlow ... OVS ... ovs-ofctl(8)。

```
# kubectl ko ofctl kube-ovn-01 dump-flows br-int
NXST_FLOW reply (xid=0x4): flags=[more]
 cookie=0xcf3429e6, duration=671791.432s, table=0, n_packets=0, n_bytes=0, idle_age=65534, hard_age=65534, priority=100,in_port=2 actions=load:0x4-
>NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x1->NXM_NX_REG14[],resubmit(,8)
 cookie=0xc91413c6, duration=671791.431s, table=0, n_packets=907489, n_bytes=99978275, idle_age=0, hard_age=65534, priority=100,in_port=7 actions=load:0x1-
>NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x4->NXM_NX_REG14[],resubmit(,8)
 cookie=0xf180459, duration=671791.431s, table=0, n_packets=17348582, n_bytes=2667811214, idle_age=0, hard_age=65534, priority=100,in_port=6317 actions=load:
0xa->NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x9->NXM_NX_REG14[],resubmit(,8)
 cookie=0x7806dd90, duration=671791.431s, table=0, n_packets=3235428, n_bytes=833821312, idle_age=0, hard_age=65534, priority=100,in_port=1 actions=load:0xd-
>NXM_NX_REG13[],load:0x9->NXM_NX_REG11[],load:0xb->NXM_NX_REG12[],load:0x4->OXM_OF_METADATA[],load:0x3->NXM_NX_REG14[],resubmit(,8)
...
```

### dpctl {nodeName} [options ...]

... `nodeName` ... `ovs-ovn` ... `ovs-dpctl` ... OVS datapath ... OVS ... ovs-dpctl(8)。

```
# kubectl ko dpctl kube-ovn-01 show
system@ovs-system:
  lookups: hit:350805055 missed:21983648 lost:73
  flows: 105
  masks: hit:1970748791 total:22 hit/pkt:5.29
  port 0: ovs-system (internal)
  port 1: ovn0 (internal)
  port 2: mirror0 (internal)
  port 3: br-int (internal)
  port 4: stt_sys_7471 (stt: packet_type=ptap)
  port 5: eeb4d9e51b5d_h
  port 6: a2c1a8a8b83a_h
  port 7: 4fa5c4cbb1a5_h
```

### appctl {nodeName} [options ...]

... `nodeName` ... `ovs-ovn` ... `ovs-appctl` ... daemon ... OVS ... ovs-appctl(8)。

```
# kubectl ko appctl kube-ovn-01 vlog/list
                console    syslog    file
                -------    ------    ------
backtrace         OFF        ERR       INFO
bfd               OFF        ERR       INFO
bond              OFF        ERR       INFO
bridge            OFF        ERR       INFO
bundle            OFF        ERR       INFO
bundles           OFF        ERR       INFO
...
```

### tcpdump {namespace/podname} [tcpdump options ...]

... `namespace/podname` ... `kube-ovn-cni` ... `tcpdump` ... veth ...

```
# kubectl ko tcpdump default/ds1-l6n7p icmp
+ kubectl exec -it kube-ovn-cni-wlg4s -n kube-ovn -- tcpdump -nn -i d7176fe7b4e0_h icmp
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on d7176fe7b4e0_h, link-type EN10MB (Ethernet), capture size 262144 bytes
06:52:36.619688 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 1, length 64
06:52:36.619746 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 1, length 64
06:52:37.619588 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 2, length 64
06:52:37.619630 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 2, length 64
06:52:38.619933 IP 100.64.0.3 > 10.16.0.4: ICMP echo request, id 2, seq 3, length 64
06:52:38.619973 IP 10.16.0.4 > 100.64.0.3: ICMP echo reply, id 2, seq 3, length 64
```

### trace [arguments ...]

... Pod ... OVN ... Openflow ...

...

```
kubectl ko trace {namespace/podname} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]
kubectl ko trace {namespace/podname} {target ip address} [target mac address] arp {request|reply}
kubectl ko trace {node//nodename} {target ip address} [target mac address] {icmp|tcp|udp} [target tcp/udp port]
kubectl ko trace {node//nodename} {target ip address} [target mac address] arp {request|reply}
```

...

```
# kubectl ko trace default/ds1-l6n7p 8.8.8.8 icmp
+ kubectl exec ovn-central-5bc494cb5-np9hm -n kube-ovn -- ovn-trace --ct=new ovn-default 'inport == "ds1-l6n7p.default" && ip.ttl == 64 && icmp && eth.src ==
0a:00:00:10:00:05 && ip4.src == 10.16.0.4 && eth.dst == 00:00:00:B8:CA:43 && ip4.dst == 8.8.8.8'
# icmp,reg14=0xf,vlan_tci=0x0000,dl_src=0a:00:00:10:00:05,dl_dst=00:00:00:b8:ca:
43,nw_src=10.16.0.4,nw_dst=8.8.8.8,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=0,icmp_code=0

ingress(dp="ovn-default", inport="ds1-l6n7p.default")
-----------------------------------------------------
 0. ls_in_port_sec_l2 (ovn-northd.c:4143): inport == "ds1-l6n7p.default" && eth.src == {0a:00:00:10:00:05}, priority 50, uuid 39453393
    next;
 1. ls_in_port_sec_ip (ovn-northd.c:2898): inport == "ds1-l6n7p.default" && eth.src == 0a:00:00:10:00:05 && ip4.src == {10.16.0.4}, priority 90, uuid 81bcd485
    next;
 3. ls_in_pre_acl (ovn-northd.c:3269): ip, priority 100, uuid 7b4f4971
    reg0[0] = 1;
    next;
 5. ls_in_pre_stateful (ovn-northd.c:3396): reg0[0] == 1, priority 100, uuid 36cdd577
    ct_next;

ct_next(ct_state=new|trk)
-------------------------
 6. ls_in_acl (ovn-northd.c:3759): ip && (!ct.est || (ct.est && ct_label.blocked == 1)), priority 1, uuid 7608af5b
    reg0[1] = 1;
    next;
10. ls_in_stateful (ovn-northd.c:3995): reg0[1] == 1, priority 100, uuid 2aba1b90
    ct_commit(ct_label=0/0x1);
    next;
16. ls_in_l2_lkup (ovn-northd.c:4470): eth.dst == 00:00:00:b8:ca:43, priority 50, uuid 5c9c3c9f
    outport = "ovn-default-ovn-cluster";
    output;

...
```

... trace ... Underlay ... Mac ...

```
kubectl ko trace default/virt-handler-7lvml 8.8.8.8 82:7c:9f:83:8c:01 icmp
```

### diagnose {all|node|subnet|IPPorts} [nodename|subnetName|{proto1}-{IP1}-{Port1},{proto2}-{IP2}-{Port2}]

... `kube-ovn-pinger` ...

```
# kubectl ko diagnose all
switch c7cd17e8-ceee-4a91-9bb3-e5a313fe1ece (snat)
    port snat-ovn-cluster
        type: router
        router-port: ovn-cluster-snat
switch 20e0c6d0-023a-4756-aec5-200e0c60f95d (join)
    port node-liumengxin-ovn3-192.168.137.178
        addresses: ["00:00:00:64:FF:A8 100.64.0.4"]
    port node-liumengxin-ovn1-192.168.137.176
        addresses: ["00:00:00:AF:98:62 100.64.0.2"]
    port join-ovn-cluster
        type: router
        router-port: ovn-cluster-join
switch 0191705c-f827-427b-9de3-3c3b7d971ba5 (central)
    port central-ovn-cluster
        type: router
        router-port: ovn-cluster-central
switch 2a45ff05-388d-4f85-9daf-e6fccd5833dc (ovn-default)
    port ovn-default-ovn-cluster
        type: router
        router-port: ovn-cluster-ovn-default
    port prometheus-k8s-1.monitoring
        addresses: ["00:00:00:AA:37:DF 10.16.0.23"]
router 212f73dd-d63d-4d72-864b-a537e9afbee1 (ovn-cluster)
    port ovn-cluster-snat
        mac: "00:00:00:7A:82:8F"
        networks: ["172.22.0.1/16"]
    port ovn-cluster-join
        mac: "00:00:00:F8:18:5A"
        networks: ["100.64.0.1/16"]
    port ovn-cluster-central
```

```
        mac: "00:00:00:4D:8C:F5"
        networks: ["192.101.0.1/16"]
    port ovn-cluster-ovn-default
        mac: "00:00:00:A3:F8:18"
        networks: ["10.16.0.1/16"]
Routing Policies
     31000                           ip4.dst == 10.16.0.0/16           allow
     31000                          ip4.dst == 100.64.0.0/16           allow
     30000                        ip4.dst == 192.168.137.177         reroute               100.64.0.3
     30000                        ip4.dst == 192.168.137.178         reroute               100.64.0.4
     29000                 ip4.src == $ovn.default.fake.6_ip4         reroute              100.64.0.22
     29000                 ip4.src == $ovn.default.fake.7_ip4         reroute              100.64.0.21
     29000                 ip4.src == $ovn.default.fake.8_ip4         reroute              100.64.0.23
     29000 ip4.src == $ovn.default.liumengxin.ovn3.192.168.137.178_ip4         reroute               100.64.0.4
     20000 ip4.src == $ovn.default.liumengxin.ovn1.192.168.137.176_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPv4         reroute               100.
64.0.2
     20000 ip4.src == $ovn.default.liumengxin.ovn2.192.168.137.177_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPv4         reroute               100.
64.0.3
     20000 ip4.src == $ovn.default.liumengxin.ovn3.192.168.137.178_ip4 && ip4.dst != $ovn.cluster.overlay.subnets.IPv4         reroute               100.
64.0.4
IPv4 Routes
Route Table <main>:
               0.0.0.0/0               100.64.0.1 dst-ip
UUID                                   LB                  PROTO      VIP                    IPs
e9bcfd9d-793e-4431-9073-6dec96b75d71   cluster-tcp-load    tcp        10.100.209.132:10660   192.168.137.176:10660
                                                           tcp        10.101.239.192:6641    192.168.137.177:6641
                                                           tcp        10.101.240.101:3000    10.16.0.7:3000
                                                           tcp        10.103.184.186:6642    192.168.137.177:6642
35d2b7a5-e3a7-485a-a4b7-b4970eb0e63b   cluster-tcp-sess    tcp        10.100.158.128:8080    10.16.0.10:8080,10.16.0.5:8080,10.16.63.30:8080
                                                           tcp        10.107.26.215:8080     10.16.0.19:8080,10.16.0.20:8080,10.16.0.21:8080
                                                           tcp        10.107.26.215:9093     10.16.0.19:9093,10.16.0.20:9093,10.16.0.21:9093
                                                           tcp        10.98.187.99:8080      10.16.0.22:8080,10.16.0.23:8080
                                                           tcp        10.98.187.99:9090      10.16.0.22:9090,10.16.0.23:9090
f43303e4-89aa-4d3e-a3dc-278a552fe27b   cluster-udp-load    udp        10.96.0.10:53          10.16.0.4:53,10.16.0.9:53
_uuid               : 06776304-5a96-43ed-90c4-c4854c251699
addresses           : []
external_ids        : {vendor=kube-ovn}
name                : node_liumengxin_ovn2_192.168.137.177_underlay_v6

_uuid               : 62690625-87d5-491c-8675-9fd83b1f433c
addresses           : []
external_ids        : {vendor=kube-ovn}
name                : node_liumengxin_ovn1_192.168.137.176_underlay_v6

_uuid               : b03a9bae-94d5-4562-b34c-b5f6198e180b
addresses           : ["10.16.0.0/16", "100.64.0.0/16", "172.22.0.0/16", "192.101.0.0/16"]
external_ids        : {vendor=kube-ovn}
name                : ovn.cluster.overlay.subnets.IPv4

_uuid               : e1056f3a-24cc-4666-8a91-75ee6c3c2426
addresses           : []
external_ids        : {vendor=kube-ovn}
name                : ovn.cluster.overlay.subnets.IPv6

_uuid               : 3e5d5fff-e670-47b2-a2f5-a39f4698a8c5
addresses           : []
external_ids        : {vendor=kube-ovn}
name                : node_liumengxin_ovn3_192.168.137.178_underlay_v6
_uuid               : 2d85dbdc-d0db-4abe-b19e-cc806d32b492
action              : drop
direction           : from-lport
external_ids        : {}
label               : 0
log                 : false
match               : "inport==@ovn.sg.kubeovn_deny_all && ip"
meter               : []
name                : []
options             : {}
priority            : 2003
severity            : []

_uuid               : de790cc8-f155-405f-bb32-5a51f30c545f
action              : drop
direction           : to-lport
external_ids        : {}
label               : 0
log                 : false
match               : "outport==@ovn.sg.kubeovn_deny_all && ip"
meter               : []
name                : []
options             : {}
priority            : 2003
severity            : []
Chassis "e15ed4d4-1780-4d50-b09e-ea8372ed48b8"
    hostname: liumengxin-ovn1-192.168.137.176
    Encap stt
        ip: "192.168.137.176"
        options: {csum="true"}
    Port_Binding node-liumengxin-ovn1-192.168.137.176
    Port_Binding perf-6vxkn.default
    Port_Binding kube-state-metrics-5d6885d89-4nf8h.monitoring
    Port_Binding alertmanager-main-0.monitoring
    Port_Binding kube-ovn-pinger-6ftdf.kube-system
    Port_Binding fake-kubelet-67c55dfd89-pv86k.kube-system
```

```
    Port_Binding prometheus-k8s-0.monitoring
Chassis "eef07da1-f8ad-4775-b14d-bd6a3b4eb0d5"
    hostname: liumengxin-ovn3-192.168.137.178
    Encap stt
        ip: "192.168.137.178"
        options: {csum="true"}
    Port_Binding kube-ovn-pinger-7twb4.kube-system
    Port_Binding prometheus-adapter-86df476d87-rl88g.monitoring
    Port_Binding prometheus-k8s-1.monitoring
    Port_Binding node-liumengxin-ovn3-192.168.137.178
    Port_Binding perf-ff475.default
    Port_Binding alertmanager-main-1.monitoring
    Port_Binding blackbox-exporter-676d976865-tvsjd.monitoring
Chassis "efa253c9-494d-4719-83ae-b48ab0f11c03"
    hostname: liumengxin-ovn2-192.168.137.177
    Encap stt
        ip: "192.168.137.177"
        options: {csum="true"}
    Port_Binding grafana-6c4c6b8fb7-pzd2c.monitoring
    Port_Binding node-liumengxin-ovn2-192.168.137.177
    Port_Binding alertmanager-main-2.monitoring
    Port_Binding coredns-6789c94dd8-9jqsz.kube-system
    Port_Binding coredns-6789c94dd8-25d4r.kube-system
    Port_Binding prometheus-operator-7bbc99fc8b-wgjm4.monitoring
    Port_Binding prometheus-adapter-86df476d87-gdxmc.monitoring
    Port_Binding perf-fjnws.default
    Port_Binding kube-ovn-pinger-vh2xg.kube-system
ds kube-proxy ready
kube-proxy ready
deployment ovn-central ready
deployment kube-ovn-controller ready
ds kube-ovn-cni ready
ds ovs-ovn ready
deployment coredns ready
ovn-nb leader check ok
ovn-sb leader check ok
ovn-northd leader check ok
### kube-ovn-controller recent log

### start to diagnose node liumengxin-ovn1-192.168.137.176
#### ovn-controller log:
2022-06-03T00:56:44.897Z|16722|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:06:44.912Z|16723|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:16:44.925Z|16724|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:26:44.936Z|16725|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:36:44.959Z|16726|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:46:44.974Z|16727|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T01:56:44.988Z|16728|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:06:45.001Z|16729|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:16:45.025Z|16730|inc_proc_eng|INFO|User triggered force recompute.
2022-06-03T02:26:45.040Z|16731|inc_proc_eng|INFO|User triggered force recompute.

#### ovs-vswitchd log:
2022-06-02T23:03:00.137Z|00079|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:f8:07:c8,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:f9d1
 with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-02T23:23:31.840Z|00080|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:f8:07:c8,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:15b2
 with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T00:09:15.659Z|00081|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:dc:e3:63,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.63.30,nw_dst=10.
16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0 icmp_csum:e5a5
 with metadata skb_priority(0),tunnel(tun_id=0x150017000004,src=192.168.137.178,dst=192.168.137.176,ttl=64,tp_src=9239,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.63.30,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T00:30:13.409Z|00064|dpif(handler2)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:f8:07:c8,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:6b4a
 with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0
2022-06-03T02:02:33.832Z|00082|dpif(handler1)|WARN|system@ovs-system: execute ct(commit,zone=14,label=0/0x1,nat(src)),8 failed (Invalid argument) on packet
icmp,vlan_tci=0x0000,dl_src=00:00:00:f8:07:c8,dl_dst=00:00:00:fa:1e:50,nw_src=10.16.0.5,nw_dst=10.16.0.10,nw_tos=0,nw_ecn=0,nw_ttl=64,icmp_type=8,icmp_code=0
icmp_csum:a819
 with metadata skb_priority(0),tunnel(tun_id=0x160017000004,src=192.168.137.177,dst=192.168.137.176,ttl=64,tp_src=38881,tp_dst=7471,flags(csum|
key)),skb_mark(0),ct_state(0x21),ct_zone(0xe),ct_tuple4(src=10.16.0.5,dst=10.16.0.10,proto=1,tp_src=8,tp_dst=0),in_port(4) mtu 0

#### ovs-vsctl show results:
0d4c4675-c9cc-440a-8c1a-878e17f81b88
    Bridge br-int
        fail_mode: secure
        datapath_type: system
        Port a2c1a8a8b83a_h
            Interface a2c1a8a8b83a_h
        Port "4fa5c4cbb1a5_h"
            Interface "4fa5c4cbb1a5_h"
        Port ovn-eef07d-0
            Interface ovn-eef07d-0
                type: stt
                options: {csum="true", key=flow, remote_ip="192.168.137.178"}
        Port ovn0
            Interface ovn0
```

```
                type: internal
        Port "04d03360e9a0_h"
            Interface "04d03360e9a0_h"
        Port eeb4d9e51b5d_h
            Interface eeb4d9e51b5d_h
        Port mirror0
            Interface mirror0
                type: internal
        Port "8e5d887ccd80_h"
            Interface "8e5d887ccd80_h"
        Port ovn-efa253-0
            Interface ovn-efa253-0
                type: stt
                options: {csum="true", key=flow, remote_ip="192.168.137.177"}
        Port "17512d5be1f1_h"
            Interface "17512d5be1f1_h"
        Port br-int
            Interface br-int
                type: internal
    ovs_version: "2.17.2"

#### pinger diagnose results:
I0603 10:35:04.349404   17619 pinger.go:19]
--------------------------------------------------------------------------------
Kube-OVN:
  Version:       v1.17.0
  Build:         2022-04-24_08:02:50
  Commit:        git-73f9d15
  Go Version:    go1.17.8
  Arch:          amd64
--------------------------------------------------------------------------------
I0603 10:35:04.376797   17619 config.go:166] pinger config is &{KubeConfigFile: KubeClient:0xc000493380 Port:8080 DaemonSetNamespace:kube-system
DaemonSetName:kube-ovn-pinger Interval:5 Mode:job ExitCode:0 InternalDNS:kubernetes.default ExternalDNS: NodeName:liumengxin-ovn1-192.168.137.176 HostIP:
192.168.137.176 PodName:kube-ovn-pinger-6ftdf PodIP:10.16.0.10 PodProtocols:[IPv4] ExternalAddress: NetworkMode:kube-ovn PollTimeout:2 PollInterval:15
SystemRunDir:/var/run/openvswitch DatabaseVswitchName:Open_vSwitch DatabaseVswitchSocketRemote:unix:/var/run/openvswitch/db.sock DatabaseVswitchFileDataPath:/
etc/openvswitch/conf.db DatabaseVswitchFileLogPath:/var/log/openvswitch/ovsdb-server.log DatabaseVswitchFilePidPath:/var/run/openvswitch/ovsdb-server.pid
DatabaseVswitchFileSystemIDPath:/etc/openvswitch/system-id.conf ServiceVswitchdFileLogPath:/var/log/openvswitch/ovs-vswitchd.log ServiceVswitchdFilePidPath:/
var/run/openvswitch/ovs-vswitchd.pid ServiceOvnControllerFileLogPath:/var/log/ovn/ovn-controller.log ServiceOvnControllerFilePidPath:/var/run/ovn/ovn-
controller.pid}
I0603 10:35:04.449166   17619 exporter.go:75] liumengxin-ovn1-192.168.137.176: exporter connect successfully
I0603 10:35:04.554011   17619 ovn.go:21] ovs-vswitchd and ovsdb are up
I0603 10:35:04.651293   17619 ovn.go:33] ovn_controller is up
I0603 10:35:04.651342   17619 ovn.go:39] start to check port binding
I0603 10:35:04.749613   17619 ovn.go:135] chassis id is 1d7f3d6c-eec5-4b3c-adca-2969d9cdfd80
I0603 10:35:04.763487   17619 ovn.go:49] port in sb is [node-liumengxin-ovn1-192.168.137.176 perf-6vxkn.default kube-state-metrics-5d6885d89-4nf8h.monitoring
alertmanager-main-0.monitoring kube-ovn-pinger-6ftdf.kube-system fake-kubelet-67c55dfd89-pv86k.kube-system prometheus-k8s-0.monitoring]
I0603 10:35:04.763583   17619 ovn.go:61] ovs and ovn-sb binding check passed
I0603 10:35:05.049309   17619 ping.go:259] start to check apiserver connectivity
I0603 10:35:05.053666   17619 ping.go:268] connect to apiserver success in 4.27ms
I0603 10:35:05.053786   17619 ping.go:129] start to check pod connectivity
I0603 10:35:05.249590   17619 ping.go:159] ping pod: kube-ovn-pinger-6ftdf 10.16.0.10, count: 3, loss count 0, average rtt 16.30ms
I0603 10:35:05.354135   17619 ping.go:159] ping pod: kube-ovn-pinger-7twb4 10.16.63.30, count: 3, loss count 0, average rtt 1.81ms
I0603 10:35:05.458460   17619 ping.go:159] ping pod: kube-ovn-pinger-vh2xg 10.16.0.5, count: 3, loss count 0, average rtt 1.92ms
I0603 10:35:05.458523   17619 ping.go:83] start to check node connectivity
```

██ diagnose ███████ subnet ██████ subnet ███ daemonset██ `kube-ovn-pinger` ██████ daemonset ███ pod ████████████████████ daemonset█

██ diagnose ███████ IPPorts ████████ `kube-ovn-pinger` pod ██████████IP█Port ██████

### reload

████████ Kube-OVN ██████

```
# kubectl ko reload
pod "ovn-central-8684dd94bd-vzgcr" deleted
Waiting for deployment "ovn-central" rollout to finish: 0 of 1 updated replicas are available...
deployment "ovn-central" successfully rolled out
pod "ovs-ovn-bsnvz" deleted
pod "ovs-ovn-m9b98" deleted
pod "kube-ovn-controller-8459db5ff4-64c62" deleted
Waiting for deployment "kube-ovn-controller" rollout to finish: 0 of 1 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
pod "kube-ovn-cni-2klnh" deleted
pod "kube-ovn-cni-t2jz4" deleted
Waiting for daemon set "kube-ovn-cni" rollout to finish: 0 of 2 updated pods are available...
Waiting for daemon set "kube-ovn-cni" rollout to finish: 1 of 2 updated pods are available...
daemon set "kube-ovn-cni" successfully rolled out
pod "kube-ovn-pinger-ln72z" deleted
pod "kube-ovn-pinger-w8lrk" deleted
Waiting for daemon set "kube-ovn-pinger" rollout to finish: 0 of 2 updated pods are available...
Waiting for daemon set "kube-ovn-pinger" rollout to finish: 1 of 2 updated pods are available...
daemon set "kube-ovn-pinger" successfully rolled out
pod "kube-ovn-monitor-7fb67d5488-7q6zb" deleted
Waiting for deployment "kube-ovn-monitor" rollout to finish: 0 of 1 updated replicas are available...
deployment "kube-ovn-monitor" successfully rolled out
```

### log

... kube-ovn ... Kube-OVN、OVN、Open vSwitch ... log ... linux ... debug ...

```
# kubectl ko log all
Collecting kube-ovn logging files
Collecting ovn logging files
Collecting openvswitch logging files
Collecting linux dmesg files
Collecting linux iptables-legacy files
Collecting linux iptables-nft files
Collecting linux route files
Collecting linux link files
Collecting linux neigh files
Collecting linux memory files
Collecting linux top files
Collecting linux sysctl files
Collecting linux netstat files
Collecting linux addr files
Collecting linux ipset files
Collecting linux tcp files
Collected files have been saved in the directory /root/kubectl-ko-log
```

...

```
# tree kubectl-ko-log/
kubectl-ko-log/
|-- kube-ovn-control-plane
|   |-- kube-ovn
|   |   |-- kube-ovn-cni.log
|   |   |-- kube-ovn-monitor.log
|   |   `-- kube-ovn-pinger.log
|   |-- linux
|   |   |-- addr.log
|   |   |-- dmesg.log
|   |   |-- ipset.log
|   |   |-- iptables-legacy.log
|   |   |-- iptables-nft.log
|   |   |-- link.log
|   |   |-- memory.log
|   |   |-- neigh.log
|   |   |-- netstat.log
|   |   |-- route.log
|   |   |-- sysctl.log
|   |   |-- tcp.log
|   |   `-- top.log
|   |-- openvswitch
|   |   |-- ovs-vswitchd.log
|   |   `-- ovsdb-server.log
|   `-- ovn
|       |-- ovn-controller.log
|       |-- ovn-northd.log
|       |-- ovsdb-server-nb.log
|       `-- ovsdb-server-sb.log
```

### perf [image]

... Kube-OVN ...

1. ...
2. Hostnetwork ...
3. ...
4. OVN-NB, OVN-SB, OVN-Northd leader ...

... image ... Pod ... `docker.io/kubeovn/test:v1.13.0` ...

```
# kubectl ko perf
============================== Prepareing Performance Test Resources ==============================
pod/test-client created
pod/test-host-client created
pod/test-server created
pod/test-host-server created
service/test-server created
pod/test-client condition met
pod/test-host-client condition met
pod/test-host-server condition met
pod/test-server condition met
===================================================================================================
============================ Start Pod Network Unicast Performance Test ============================
Size            TCP Latency     TCP Bandwidth   UDP Latency     UDP Lost Rate   UDP Bandwidth
64              82.8 us         97.7 Mbits/sec  67.6 us         (0%)            8.42 Mbits/sec
128             85.4 us         167 Mbits/sec   67.2 us         (0%)            17.2 Mbits/sec
```



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```
512             85.8 us         440 Mbits/sec   68.7 us         (0%)            68.4 Mbits/sec
1k              85.1 us         567 Mbits/sec   68.7 us         (0%)            134 Mbits/sec
4k              138 us          826 Mbits/sec   78.1 us         (1.4%)          503 Mbits/sec
============================================================================================
============================== Start Host Network Performance Test ==============================
Size            TCP Latency     TCP Bandwidth   UDP Latency     UDP Lost Rate   UDP Bandwidth
64              49.7 us         120 Mbits/sec   37.9 us         (0%)            18.6 Mbits/sec
128             49.7 us         200 Mbits/sec   38.1 us         (0%)            35.5 Mbits/sec
512             51.9 us         588 Mbits/sec   38.9 us         (0%)            142 Mbits/sec
1k              51.7 us         944 Mbits/sec   37.2 us         (0%)            279 Mbits/sec
4k              74.9 us         1.66 Gbits/sec  39.9 us         (0%)            1.20 Gbits/sec
============================================================================================
============================ Start Service Network Performance Test ============================
Size            TCP Latency     TCP Bandwidth   UDP Latency     UDP Lost Rate   UDP Bandwidth
64              111 us          96.3 Mbits/sec  88.4 us         (0%)            7.59 Mbits/sec
128             83.7 us         150 Mbits/sec   69.2 us         (0%)            16.9 Mbits/sec
512             87.4 us         374 Mbits/sec   75.8 us         (0%)            60.9 Mbits/sec
1k              88.2 us         521 Mbits/sec   73.1 us         (0%)            123 Mbits/sec
4k              148 us          813 Mbits/sec   77.6 us         (0.0044%)       451 Mbits/sec
============================================================================================
========================== Start Pod Multicast Network Performance Test ==========================
Size            UDP Latency     UDP Lost Rate   UDP Bandwidth
64              0.014 ms        (0.17%)         5.80 Mbits/sec
128             0.012 ms        (0%)            11.4 Mbits/sec
512             0.016 ms        (0%)            46.1 Mbits/sec
1k              0.023 ms        (0.073%)        89.8 Mbits/sec
4k              0.035 ms        (1.3%)          126 Mbits/sec
============================================================================================
========================== Start Host Multicast Network Performance ==========================
Size            UDP Latency     UDP Lost Rate   UDP Bandwidth
64              0.007 ms        (0%)            9.95 Mbits/sec
128             0.005 ms        (0%)            21.8 Mbits/sec
512             0.008 ms        (0%)            86.8 Mbits/sec
1k              0.013 ms        (0.045%)        168 Mbits/sec
4k              0.010 ms        (0.31%)         242 Mbits/sec
============================================================================================
================================ Start Leader Recover Time Test ================================
Delete ovn central nb pod
pod "ovn-central-5cb9c67d75-tlz9w" deleted
Waiting for ovn central nb pod running
============================ OVN nb Recovery takes 3.305236803 s ============================
Delete ovn central sb pod
pod "ovn-central-5cb9c67d75-szx4c" deleted
Waiting for ovn central sb pod running
============================ OVN sb Recovery takes 3.462698535 s ============================
Delete ovn central northd pod
pod "ovn-central-5cb9c67d75-zqmqv" deleted
Waiting for ovn central northd pod running
========================== OVN northd Recovery takes 2.691291403 s ==========================
============================================================================================
============================== Remove Performance Test Resource ==============================
rm -f unicast-test-client.log
rm -f unicast-test-host-client.log
rm -f unicast-test-client.log
kubectl ko nbctl lb-del test-server
rm -f multicast-test-server.log
kubectl exec ovs-ovn-gxdrf -n kube-system -- ip maddr del 01:00:5e:00:00:64 dev eth0
kubectl exec ovs-ovn-h57bf -n kube-system -- ip maddr del 01:00:5e:00:00:64 dev eth0
rm -f multicast-test-host-server.log
pod "test-client" deleted
pod "test-host-client" deleted
pod "test-host-server" deleted
pod "test-server" deleted
service "test-server" deleted
============================================================================================
```

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## 6.1.3 ⁇



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# 6.2 ▒▒▒▒▒▒▒

▒▒▒▒▒▒▒ Kubernetes ▒▒▒▒▒▒▒▒▒▒▒▒ `ovs-ovn` ▒▒▒▒▒ `ovn-controller` ▒▒▒▒▒▒▒▒▒▒▒▒ `ovn-central` ▒▒▒▒▒▒▒▒▒▒ ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ ▒▒▒▒▒ Kubernetes ▒▒▒▒▒▒▒▒▒ ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ `ovn-central` ▒▒▒▒▒▒▒▒▒▒▒ ovn-central ▒▒▒

## 6.2.1 ▒▒▒▒▒▒▒▒▒▒▒

```
 # kubectl drain kube-ovn-worker --ignore-daemonsets --force
 node/kube-ovn-worker cordoned
 WARNING: ignoring DaemonSet-managed Pods: kube-system/kube-ovn-cni-zt74b, kube-system/kube-ovn-pinger-5rxfs, kube-system/kube-proxy-jpmnm, kube-system/ovs-
ovn-v2kll
 evicting pod kube-system/coredns-64897985d-qsgpt
 evicting pod local-path-storage/local-path-provisioner-5ddd94ff66-llss6
 evicting pod kube-system/kube-ovn-controller-8459db5ff4-94lxb
 pod/kube-ovn-controller-8459db5ff4-94lxb evicted
 pod/coredns-64897985d-qsgpt evicted
 pod/local-path-provisioner-5ddd94ff66-llss6 evicted
 node/kube-ovn-worker drained
```

## 6.2.2 ▒▒ kubelet ▒ docker

▒▒▒▒▒▒ `ovs-ovn` ▒▒▒▒▒▒▒ `ovn-central` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

```
systemctl stop kubelet
systemctl stop docker
```

**Containerd**

> ▒▒ containerd ▒▒ CRI ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ Pod/▒▒▒▒▒▒▒▒▒▒ `ovs-ovn` ▒

▒▒ containerd ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ `ovs-ovn` ▒▒▒▒

1. ▒▒ kubelet

   ```
   systemctl stop kubelet
   ```

2. ▒▒ ovs-ovn Pod

   ```
   crictl rm -f $(crictl ps | grep openvswitch | awk '{print $1}')
   ```

3. ▒▒ containerd

   ```
   systemctl stop containerd
   ```

## 6.2.3 ▒▒ Node ▒▒▒▒▒▒▒

```
rm -rf /var/run/openvswitch
rm -rf /var/run/ovn
rm -rf /etc/origin/openvswitch/
rm -rf /etc/origin/ovn/
rm -rf /etc/cni/net.d/00-kube-ovn.conflist
rm -rf /etc/cni/net.d/01-kube-ovn.conflist
rm -rf /var/log/openvswitch
rm -rf /var/log/ovn
```

## 6.2.4 ▒▒ kubectl ▒▒▒▒

```
kubectl delete no kube-ovn-01
```



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## 6.2.5 □□□□□□□□□□ ovn-sb □□□

□□□□□□ kube-ovn-worker □□□□□□□□

```
# kubectl ko sbctl show
Chassis "b0564934-5a0d-4804-a4c0-476c93596a17"
  hostname: kube-ovn-worker
  Encap geneve
      ip: "172.18.0.2"
      options: {csum="true"}
  Port_Binding kube-ovn-pinger-5rxfs.kube-system
Chassis "6a29de7e-d731-4eaf-bacd-2f239ee52b28"
  hostname: kube-ovn-control-plane
  Encap geneve
      ip: "172.18.0.3"
      options: {csum="true"}
  Port_Binding coredns-64897985d-nbfln.kube-system
  Port_Binding node-kube-ovn-control-plane
  Port_Binding local-path-provisioner-5ddd94ff66-h4tn9.local-path-storage
  Port_Binding kube-ovn-pinger-hf2p6.kube-system
  Port_Binding coredns-64897985d-fhwlw.kube-system
```

## 6.2.6 □□□□□□ chassis □□□□□□□□□□□□

uuid □□□□□□□□□□ Chassis □□ id□

```
# kubectl ko sbctl chassis-del b0564934-5a0d-4804-a4c0-476c93596a17
# kubectl ko sbctl show
Chassis "6a29de7e-d731-4eaf-bacd-2f239ee52b28"
  hostname: kube-ovn-control-plane
  Encap geneve
      ip: "172.18.0.3"
      options: {csum="true"}
  Port_Binding coredns-64897985d-nbfln.kube-system
  Port_Binding node-kube-ovn-control-plane
  Port_Binding local-path-provisioner-5ddd94ff66-h4tn9.local-path-storage
  Port_Binding kube-ovn-pinger-hf2p6.kube-system
  Port_Binding coredns-64897985d-fhwlw.kube-system
```

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## 6.2.7 □□

# 6.3 ▯▯ ovn-central ▯▯

▯▯ `ovn-central` ▯▯ `ovn-nb` ▯ `ovn-sb` ▯▯▯▯▯▯▯ etcd ▯ Raft ▯▯▯▯▯▯▯ `ovn-central` ▯▯▯▯▯▯▯

**** **ovn-sb** ********

******** ovn-sb ********** ID**********

```
kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2395317 ms ago, reason: timeout
Last Election won: 2395316 ms ago
Election timer: 5000
Log: [130, 130]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-e9f7 ->e9f7 <-6e84 ->6e84
Disconnections: 0
Servers:
    e9f7 (e9f7 at tcp:[172.18.0.5]:6644) next_index=130 match_index=129 last msg 1006 ms ago
    6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=130 match_index=129 last msg 1004 ms ago
    3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=2 match_index=129
status: ok
```

`kube-ovn-control-plane2` **** IP **** `172.18.0.5` ******** ID **** `e9f7` ****** ovn-sb **********

```
# kubectl ko sb kick e9f7
started removal
```

**********

```
# kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 1
Leader: self
Vote: self

Last Election started 2481636 ms ago, reason: timeout
Last Election won: 2481635 ms ago
Election timer: 5000
Log: [131, 131]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: <-6e84 ->6e84
Disconnections: 2
Servers:
    6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=131 match_index=130 last msg 642 ms ago
    3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=2 match_index=130
status: ok
```

********** **ovn-central**

**** ovn-central **** `NODE_IPS` ****************

```
kubectl label node kube-ovn-control-plane2 kube-ovn/role-
kubectl scale deployment -n kube-system ovn-central --replicas=2
kubectl set env deployment/ovn-central -n kube-system NODE_IPS="172.18.0.3,172.18.0.4"
kubectl rollout status deployment/ovn-central -n kube-system
```

********* **ovn-central** **

** `ovs-ovn` ********************

```
# kubectl set env daemonset/ovs-ovn -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4"
daemonset.apps/ovs-ovn env updated
# kubectl delete pod -n kube-system -lapp=ovs
pod "ovs-ovn-4f6jc" deleted
```

```
pod "ovs-ovn-csn2w" deleted
pod "ovs-ovn-mpbmb" deleted
```

... `kube-ovn-controller` ...

```
# kubectl set env deployment/kube-ovn-controller -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4"
deployment.apps/kube-ovn-controller env updated

# kubectl rollout status deployment/kube-ovn-controller -n kube-system
Waiting for deployment "kube-ovn-controller" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "kube-ovn-controller" rollout to finish: 2 of 3 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
```

...

... `kube-ovn-control-plane2` ...

```
rm -rf /etc/origin/ovn
```

... Kubernetes ...

## 6.3.2 ovn-central ...

... Kubernetes ... `ovn-central` ...

...

... `/etc/origin/ovn` ... `ovnnb_db.db` ... `ovnsb_db.db` ...

```
rm -rf /etc/origin/ovn
```

... **ovn-central** ...

... `ovn-central` ...

```
# kubectl ko nb status
1b9a
Name: OVN_Northbound
Cluster ID: 32ca (32ca07fb-739b-4257-b510-12fa18e7cce8)
Server ID: 1b9a (1b9a5d76-e69b-410c-8085-39943d0cd38c)
Address: tcp:[172.18.0.3]:6643
Status: cluster member
Role: leader
Term: 44
Leader: self
Vote: self

Last Election started 1855739 ms ago, reason: timeout
Last Election won: 1855729 ms ago
Election timer: 5000
Log: [147, 147]
Entries not yet committed: 0
Entries not yet applied: 0
Connections: ->4984 <-4984
Disconnections: 0
Servers:
    4984 (4984 at tcp:[172.18.0.4]:6643) next_index=147 match_index=146 last msg 367 ms ago
    1b9a (1b9a at tcp:[172.18.0.3]:6643) (self) next_index=140 match_index=146
status: ok

# kubectl ko sb status
3722
Name: OVN_Southbound
Cluster ID: d4bd (d4bd37a4-0400-499f-b4df-b4fd389780f0)
Server ID: 3722 (3722d5ae-2ced-4820-a6b2-8b744d11fb3e)
Address: tcp:[172.18.0.3]:6644
Status: cluster member
Role: leader
Term: 33
Leader: self
Vote: self

Last Election started 1868589 ms ago, reason: timeout
Last Election won: 1868579 ms ago
Election timer: 5000
Log: [142, 142]
Entries not yet committed: 0
```

```
Entries not yet applied: 0
Connections: ->6e84 <-6e84
Disconnections: 0
Servers:
    6e84 (6e84 at tcp:[172.18.0.4]:6644) next_index=142 match_index=141 last msg 728 ms ago
    3722 (3722 at tcp:[172.18.0.3]:6644) (self) next_index=134 match_index=141
status: ok
```

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

⬚⬚⬚⬚ ovn-central ⬚⬚⬚⬚ `NODE_IPS` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
kubectl label node kube-ovn-control-plane2 kube-ovn/role=master
kubectl scale deployment -n kube-system ovn-central --replicas=3
kubectl set env deployment/ovn-central -n kube-system NODE_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
kubectl rollout status deployment/ovn-central -n kube-system
```

⬚⬚⬚⬚⬚⬚⬚⬚ **ovn-central** ⬚⬚

⬚⬚ `ovs-ovn` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
# kubectl set env daemonset/ovs-ovn -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
daemonset.apps/ovs-ovn env updated
# kubectl delete pod -n kube-system -lapp=ovs
pod "ovs-ovn-4f6jc" deleted
pod "ovs-ovn-csn2w" deleted
pod "ovs-ovn-mpbmb" deleted
```

⬚⬚ `kube-ovn-controller` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
# kubectl set env deployment/kube-ovn-controller -n kube-system OVN_DB_IPS="172.18.0.3,172.18.0.4,172.18.0.5"
deployment.apps/kube-ovn-controller env updated

# kubectl rollout status deployment/kube-ovn-controller -n kube-system
Waiting for deployment "kube-ovn-controller" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "kube-ovn-controller" rollout to finish: 2 of 3 updated replicas are available...
deployment "kube-ovn-controller" successfully rolled out
```

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6.3.3 ⬚⬚

# 6.4 OVN □□□□□□□□

□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

## 6.4.1 □□□□□

□□ kubectl □□□ backup □□□□□□□□□□□□□□□□□□□□□□□□□□□□

```
# kubectl ko nb backup
tar: Removing leading `/' from member names
backup ovn-nb db to /root/ovnnb_db.060223191654183154.backup

# kubectl ko sb backup
tar: Removing leading `/' from member names
backup ovn-sb db to /root/ovnsb_db.060223191654183154.backup
```

## 6.4.2 □□□□□□□□□

□□□□□□□□□□□□□□□□□□□

## 6.4.3 [illegible]

[illegible] leader[illegible]

### [illegible] ovn-central

[illegible] `ovn-central` [illegible] `ovn-central` [illegible]

```
kubectl scale deployment -n kube-system ovn-central --replicas=0
```

[illegible]

[illegible] [illegible] `ovsdb-tool cluster-to-standalone` [illegible]

[illegible] `ovn-central` [illegible] `NODE_IPS` [illegible] `/etc/origin/ovn` [illegible]

[illegible] `docker`

```
docker run -it -v /etc/origin/ovn:/etc/ovn kubeovn/kube-ovn:v1.17.0 bash
cd /etc/ovn/
ovsdb-tool cluster-to-standalone ovnnb_db_standalone.db ovnnb_db.db
ovsdb-tool cluster-to-standalone ovnsb_db_standalone.db ovnsb_db.db
```

[illegible] containerd[illegible] docker[illegible] `ctr`

```
ctr -n k8s.io image pull docker.io/kubeovn/kube-ovn:v1.17.0
ctr -n k8s.io run --rm -t \
  --mount type=bind,src=/etc/origin/ovn,dst=/etc/ovn,options=rbind:rw \
  docker.io/kubeovn/kube-ovn:v1.17.0 ovn-recover bash
cd /etc/ovn/
ovsdb-tool cluster-to-standalone ovnnb_db_standalone.db ovnnb_db.db
ovsdb-tool cluster-to-standalone ovnsb_db_standalone.db ovnsb_db.db
exit
```

[illegible] `ovn-central` Pod [illegible] `ovsdb-tool cluster-to-standalone` [illegible] `kubectl cp` [illegible]

### [illegible] ovn-central [illegible]

[illegible]

```
mv /etc/origin/ovn/ovnnb_db.db /tmp
mv /etc/origin/ovn/ovnsb_db.db /tmp
```

[illegible]

[illegible] `ovnnb_db.db` [illegible] `ovnsb_db.db` [illegible] `ovn-central` [illegible] `NODE_IPS` [illegible] `/etc/origin/ovn/` [illegible]

```
mv /etc/origin/ovn/ovnnb_db_standalone.db /etc/origin/ovn/ovnnb_db.db
mv /etc/origin/ovn/ovnsb_db_standalone.db /etc/origin/ovn/ovnsb_db.db
```

[illegible] `ovn-central` [illegible]

```
kubectl scale deployment -n kube-system ovn-central --replicas=3
kubectl rollout status deployment/ovn-central -n kube-system
```

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## 6.4.4 ⊠⊠

# 6.5 ▯▯▯▯ CIDR

▯▯▯▯▯▯▯ CIDR ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

> ▯▯▯▯ CIDR ▯▯▯▯▯▯▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ CIDR ▯▯▯▯▯▯▯▯▯▯ ▯▯ Join ▯▯ CIDR ▯▯▯▯▯▯ Join ▯▯ CIDR▯

## 6.5.1 ▯▯▯▯

▯▯ `kubectl edit` ▯▯▯▯ `cidrBlock` ▯ `gateway` ▯ `excludeIps` ▯

```
kubectl edit subnet test-subnet
```

## 6.5.2 ▯▯▯▯▯▯▯▯▯ Namespace ▯▯▯ Pod

▯▯▯▯▯ `test` Namespace ▯▯▯

```
for pod in $(kubectl get pod --no-headers -n test --field-selector spec.restartPolicy=Always -o custom-columns=NAME:.metadata.name,HOST:spec.hostNetwork |
awk '{if ($2!="true") print $1}'); do
  kubectl delete pod "$pod" -n test --ignore-not-found
done
```

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ host ▯▯▯▯▯ Pod▯

```
for ns in $(kubectl get ns --no-headers -o custom-columns=NAME:.metadata.name); do
  for pod in $(kubectl get pod --no-headers -n "$ns" --field-selector spec.restartPolicy=Always -o custom-columns=NAME:.metadata.name,HOST:spec.hostNetwork |
awk '{if ($2!="true") print $1}'); do
    kubectl delete pod "$pod" -n "$ns" --ignore-not-found
  done
done
```

## 6.5.3 ▯▯▯▯▯▯▯▯▯

▯▯▯▯▯▯▯▯▯▯ CIDR ▯▯▯▯▯ `kube-ovn-controller` Deployment ▯▯▯▯▯▯

```
args:
- --default-cidr=10.17.0.0/16
- --default-gateway=10.17.0.1
- --default-exclude-ips=10.17.0.1
```

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## 6.5.4 ▯▯

# 6.6 □□ Join □□ CIDR

□□□□□□ Join □□ CIDR □□□□□□□□□□□□□□□□□□□□□□□□

> □□ Join □□ CIDR □□□□□□□□ Pod □□□□□□□□□□□□□□□□□□□□□□□□ □□□□□□□□□□□

## 6.6.1 □□ Join □□

```
kubectl patch subnet join --type='json' -p '[{"op": "replace", "path": "/metadata/finalizers", "value": []}]'
kubectl delete subnet join
```

## 6.6.2 □□□□□□□□□

```
kubectl annotate node ovn.kubernetes.io/allocated=false --all --overwrite
```

## 6.6.3 □□ Join □□□□□□□

□□ `kube-ovn-controller` □ Join □□□□□□□

```
kubectl edit deployment -n kube-system kube-ovn-controller
```

□□□□□□□

```
args:
- --node-switch-cidr=100.51.0.0/16
```

□□ `kube-ovn-controller` □□ `join` □□□

```
kubectl delete pod -n kube-system -lapp=kube-ovn-controller
```

□□□□ Join □□□□□

```
# kubectl get subnet
NAME          PROVIDER   VPC           PROTOCOL   CIDR            PRIVATE   NAT     DEFAULT   GATEWAYTYPE   V4USED   V4AVAILABLE   V6USED   V6AVAILABLE
EXCLUDEIPS
join          ovn        ovn-cluster   IPv4       100.51.0.0/16   false     false   false     distributed   2        65531         0        0
["100.51.0.1"]
ovn-default   ovn        ovn-cluster   IPv4       10.17.0.0/16    false     true    true      distributed   5        65528         0        0
["10.17.0.1"]
```

## 6.6.4 □□□□ ovn0 □□□□

□□□□□ `ovn0` □□□□□□□□□□□□□□□□ `kube-ovn-cni` □□□□

```
kubectl delete pod -n kube-system -l app=kube-ovn-cni
```

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## 6.6.5 □□



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## 6.7 □□□□□□□

□□ `kube-ovn.yaml` □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

```
vi kube-ovn.yaml
# ...
        - name: kube-ovn-controller
          image: "docker.io/kubeovn/kube-ovn:v1.17.0"
          imagePullPolicy: IfNotPresent
          args:
          - /kube-ovn/start-controller.sh
          - --v=3
# ...
#
```

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### 6.7.1 □□

# 6.8 ■■■■■■■

■■■■■■■■■■■■■■■■ ■■■■■ ■■ Kube-OVN ■■■■■■■■■■■■■■■

## 6.8.1 ■■■■■■■

1. ■■■■■■■■■■
    - ■■ Pod ■ Pod ■■■■■■■
    - ■■ Pod ■■■■■■■■■■■■
    - ■■ Pod ■■ Service ■■■■■
    - ■■ Pod ■■■■■■■■■
    - ■■ Pod ■■■■■■■■■■■■
    - ■■■■■■■■ `kube-ovn-pinger` ■■■■■■■■■■■■■■■■■■■■■■■
2. ■■■■■■■■■■■■■■■■■■■■■■■■■
    - ■■ Pod ■■■■■■■■■■■■■■■■■■■■■■
    - ■■■■■ kube-ovn-cni ■■■■■■ CNI ■■■■■■■■■
    - ■■■■■ ovs-ovn ■■■■■■■■■■ OVS ■■■■■
    - ■■ kube-ovn-controller ■■■■■■■■■■■■■■■■■■■
    - ■■ ovn-central ■■■■■■■■■■ OVN ■■■■■
    - ■■■■■ dmesg ■■■■■■■■■■■■■■■■■■
    - ■■■■■ `netstat -s` ■■■■■■■■■■■■■■■■
3. ■■■■■■■■■■■■■■■■■■■■ CPU, ■■■■ IO ■■■■■■■■

■■■■■■■■■■■■■ `kubectl ko logs` ■■■■■■■

## 6.8.2 Pod ■■■■■■■■■■ IP ■■

■■

Pod ■■■■■ Running ■■■kubectl describe Pod ■■■■■ `duplicate IPv4 address <ip> found on logical switch port <port>`

■■■■■

1. ■■■■■ Pod ■■■■■ IP Pod■■■■■■ IP ■■■■ IP ■■■■■■■■■■ IP ■■■■■■■ IP Pod
2. ■■ kube-ovn-controller ■■■■■■■■ IP ■■■■■■■■■
3. ■■■■■■■ kube-ovn-controller ■■■■■■■■■■■■ IP ■■■■■
4. `kubectl ko nbctl show` ■■ OVN ■■■■■■■■■■■■■ IP ■■
5. ■ OVN ■■■■■■■■■■ Kubernetes ■■■■■■■■■■■■ IP ■■■■■■■■■■ `kubectl ko nbctl lsp-del <port>` ■■■■■

## 6.8.3 Pod ■■■■■■■■■■ ping gateway failed

■■

Pod ■■■■■ Running ■■■kubectl describe Pod ■■■■■ `network <ip> with gateway <gw ip> is not ready for interface eth0 after 30 checks`

■■■■■

1. `kubectl ko sbctl show` ■■ Pod ■■■■■■■■■■■■
2. ■■■■■■■■■■ ovn-central ■ ovs-ovn ■■■■■■■■■■■■■■■■■■■■ ovn-central ■ ovs-ovn ■■■■■



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3. □□□□□□□□□□□□□□□□□□□□ Pod □□□□□
4. □□ Underlay □□□□□□□□□□□□□□□□□□□□□□□□□□□□ Underlay □□□□ □□

## 6.8.4 Pod □□□□□□□□□

□□

□□ VPC □ Pod □□□□□□□□□□□□□□□□□□□□□□□□

□□□□

1. □□ `kubectl ko trace` □□ OVN □□□□□□□□□□□□ ACL □□
2. □□□ ACL □□□□□□ ACL □□
3. □□□□ ACL □□□□□□ Subnet stats □□□□□□□□□□□□□□□□□□□□□□□□□□□□
4. □□□□□□□□□□□□□□□□□□□□□□□□□□□□ Subnet Spec □□□□□□□□□□
5. □□ `kube-ovn-controller` □□□□□□□□□□□□□□□□□□□□□□

## 6.8.5 Pod IP □□□□□□ CIDR □

□□

Pod □□□□□□□□ IP □□□□□□ CIDR □□

□□□□

1. □□□□□□□□ `/etc/cni/net.d/` □□□□□□□□□□□□ Kube-OVN □ CNI □□□□
2. □□□□□□□□□□□□□□□□□□□□□□
3. □□ kubelet □□□□□□□□□□□□ Pod

## 6.8.6 Debug Pod □□□□□□□

□□

kubectl debug □□□ Pod □□□□□ ContainerCreating □□□Pod Event □□ network not ready □ no address allocated

□□□□

□□ debug Pod □□□□□□ yaml □□□□□□ yaml □□ Annotation□

```
ovn.kubernetes.io/ip_address
ovn.kubernetes.io/mac_address
ovn.kubernetes.io/allocated
ovn.kubernetes.io/routed
```

□□ debug Pod□□□□□□□□ yaml □□□□ debug Pod□

## 6.8.7 ARM □□□□□□□□□□□□□□□□

□□

ARM □□□□□□□□ Offload □□□□□□□□□□□□□□□□□□□□□□□□

□□ `netstat` □□□□□□

```
# netstat -us
IcmpMsg:
    InType0: 22
    InType3: 24
    InType8: 117852
    OutType0: 117852
```

```
    OutType3: 29
    OutType8: 22
Udp:
    3040636 packets received
    0 packets to unknown port received.
    4 packet receive errors
    602 packets sent
    0 receive buffer errors
    0 send buffer errors
    InCsumErrors: 4
UdpLite:
IpExt:
    InBcastPkts: 10244
    InOctets: 4446320361
    OutOctets: 1496815600
    InBcastOctets: 3095950
    InNoECTPkts: 7683903
```

□□□ `InCsumErrors` □□□□□□□□□□□□□□□□□□□□□□

□□□□

□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ □□□ `tx offload` □□□□□ TCP □□□□□□□□□

```
ethtool -K eth0 tx off
```

□□□□□□□□CentOS 7 □□□□□□□□□□□□□□□□□□□□□□□□□□□□□ `4.19.90-25.16.v2101` □□□□□□□□□□□□

## 6.8.8 Pod □□ Service □□

□□

Pod □□□□□ Service □□□□□□ `dmesg` □□□□□

```
netlink: Unknown conntrack attr (type=6, max=5)
openvswitch: netlink: Flow actions may not be safe on all matching packets.
```

□□□□□□□□ OVS □□□□□□□□□□ NAT □□□

□□□□

1. □□□□□□□□□□□□ OVS □□□□□
2. □□□□ Overlay □□□□□□□ `kube-ovn-controller` □□□□□□□ `--enable-lb=false` □□ OVN LB □□ kube-proxy □□ Service □□□

## 6.8.9 ovn-central □□□□□□□

□□

□ v1.11.x □□□□□1 □□ Pod □□□□□□□□ OVN NB □□ SB □□□□□□□□□□□□□□□□□ Kube-OVN □□□□□ ovsdb-server/compact □□□□□□□□□□□□

□□□□

□□□□□□□□ v1.13 □ `ENABLE_COMPACT` □□□□□ `false` □□□ `dist/images/install.sh` □ `start-db.sh` □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ ovn-central □□□□□□□□□□□□□□□ `true` □

```
- name: ENABLE_COMPACT
  value: "false"
```

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6.8.10 □□

# 7. □□□□

## 7.1 □□□□□

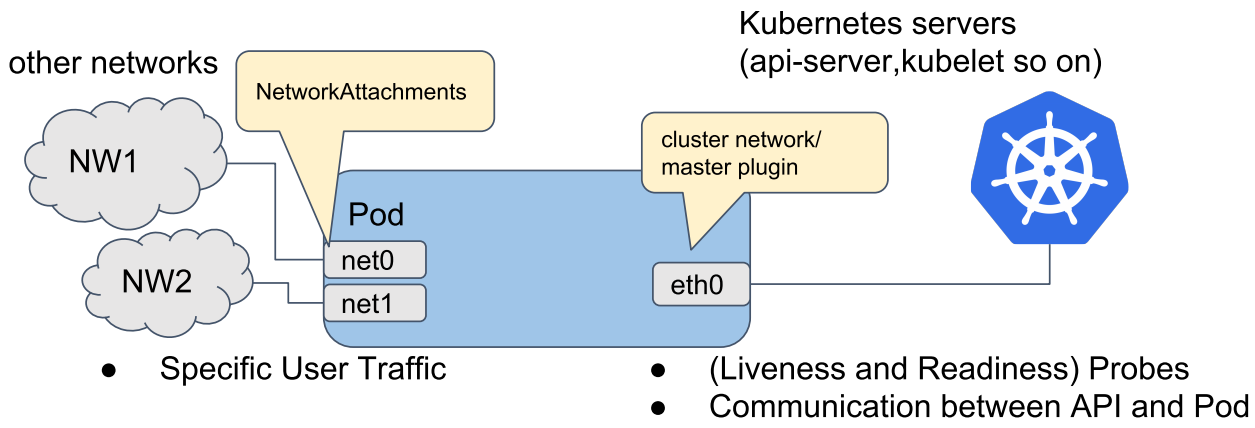
Kube-OVN □□□□□ CNI □□□□□□ macvlan□vlan□host-device □□□□□□□□□ IPAM □□□ □□□□□□□□□□□ Kube-OVN □□□□□□ IP □□□

Kube-OVN □□□□□□□□□□ Kube-OVN □□□□□□□□□□

### 7.1.1 □□□□

□□□□□□

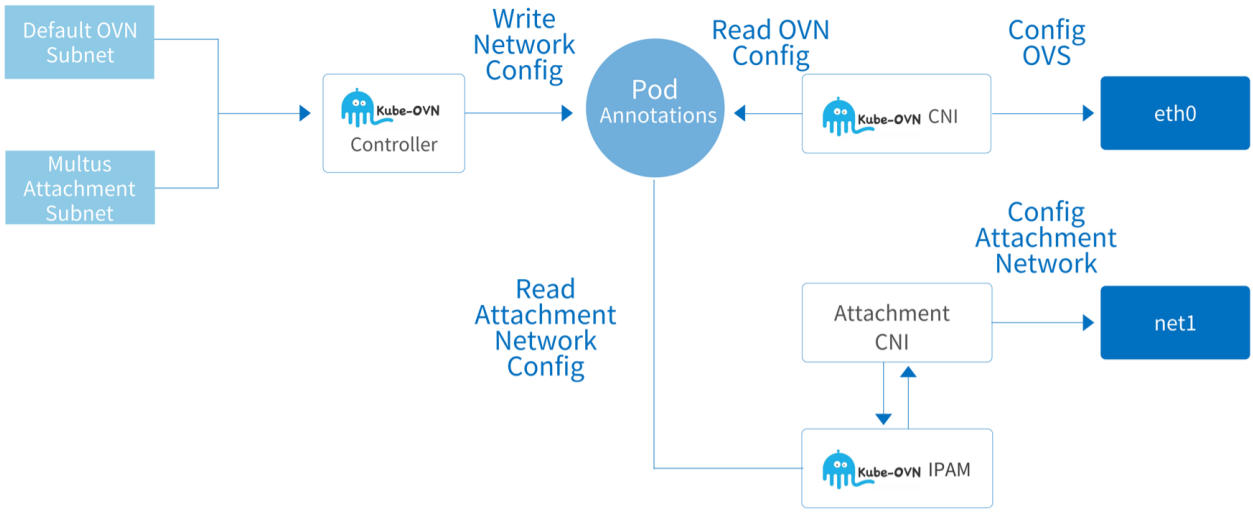
□□□□ Multus CNI □□□□□□ pod □□□□□□□□□□□□□ pod □□□□□□□eth0□net0 □ net1□ eth0 □□ kubernetes □□□□□□□□□ kubernetes □□□□□ kubernetes api-server□kubelet □□□ net0 □ net1 □□□□□□□□□□□□□□□ CNI □□□□ vlan/vxlan/ptp□□□□□□□□




IPAM□

□□□□ Multus CNI□□□□□□□□ Pod □□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□ IP □□□□□□□□□□□□ Kube-OVN □□□□□□□□□□□ Subnet □ IP □ CRD □□□ IP □□□□□□□ □□□□□□□□IP □□□□□□□ □□□□□□□□□□□□□□□□□□□□

⬚⬚⬚⬚




⬚⬚⬚⬚⬚⬚⬚⬚ Kube-OVN ⬚⬚⬚⬚⬚⬚⬚⬚⬚ IP ⬚⬚⬚⬚⬚⬚⬚ eth0 ⬚⬚⬚⬚ OVN ⬚⬚⬚net1 ⬚⬚⬚⬚⬚⬚ CNI ⬚⬚⬚ net1 ⬚⬚⬚⬚⬚⬚⬚⬚⬚ multus-cni ⬚⬚ NetworkAttachmentDefinition ⬚⬚⬚⬚⬚

⬚ Pod ⬚⬚⬚⬚ `kube-ovn-controller` ⬚⬚⬚⬚ Pod ⬚⬚⬚⬚⬚⬚⬚ Pod ⬚⬚ annotation ⬚⬚⬚⬚⬚⬚⬚ Subnet ⬚⬚⬚⬚⬚ IP ⬚⬚⬚⬚⬚⬚⬚ ⬚⬚ Pod ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ Pod annotation ⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚⬚CNI ⬚⬚⬚⬚⬚⬚⬚⬚⬚ `kube-ovn-cni` ⬚⬚⬚⬚ ipam ⬚⬚⬚⬚ `kube-ovn-cni` ⬚⬚⬚⬚⬚ Pod annotation ⬚⬚⬚⬚⬚⬚⬚⬚⬚ CNI ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ CNI ⬚⬚⬚⬚

## 7.1.2 ⬚⬚⬚⬚⬚⬚

`NetworkAttachmentDefinition` ⬚⬚ `spec` ⬚⬚⬚multus ⬚⬚⬚⬚⬚ `defaultConfDir` ⬚⬚⬚⬚⬚⬚ CNI ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: macvlan-conf-2
```

⬚⬚ `kube-ovn-controller` ⬚⬚⬚⬚⬚⬚⬚⬚⬚ `NetworkAttachmentDefinition` ⬚⬚ `provider` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: macvlan
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "macvlan",
      "master": "eth0",
      "mode": "bridge",
      "ipam": {
        "type": "kube-ovn",
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "macvlan.default"
      }
    }'
```

- `spec.config.ipam.type` : ▯▯▯▯▯ `kube-ovn` ▯▯▯ kube-ovn ▯▯▯▯▯▯▯▯▯▯
- `server_socket` : Kube-OVN ▯▯▯▯▯ socket ▯▯▯▯▯▯▯▯ `/run/openvswitch/kube-ovn-daemon.sock` ▯
- `provider` : ▯▯ NetworkAttachmentDefinition ▯ `<name>.<namespace>` ▯Kube-OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯ Subnet ▯▯▯
- `master` : ▯▯▯▯▯▯▯▯▯

> **Info**
>
> ▯▯▯ `provider` ▯ Underlay ▯▯ `ProviderNetwork` ▯▯▯▯▯▯▯▯▯▯▯▯▯

```
    macvlan.default.kubernetes.io/ip_address: 172.17.0.100
    macvlan.default.kubernetes.io/mac_address: 00:00:00:53:6B:BB
spec:
  containers:
  - name: static-ip
    image: docker.io/library/nginx:alpine
```

#### ... IP ...

... ippool ... `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/ip_pool` annotation...

```
apiVersion: apps/v1
kind: Deployment
metadata:
  namespace: default
  name: static-workload
  labels:
    app: static-workload
spec:
  replicas: 2
  selector:
    matchLabels:
      app: static-workload
  template:
    metadata:
      labels:
        app: static-workload
      annotations:
        k8s.v1.cni.cncf.io/networks: default/macvlan
        ovn.kubernetes.io/ip_pool: 10.16.0.15,10.16.0.16,10.16.0.17
        macvlan.default.kubernetes.io/ip_pool: 172.17.0.200,172.17.0.201,172.17.0.202
    spec:
      containers:
      - name: static-workload
        image: docker.io/library/nginx:alpine
```

#### ... macvlan ... Pod

... macvlan ... Pod... Pod ... annotation... `default-route` ...

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod-route
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{
      "name": "macvlan",
      "namespace": "default",
      "default-route": ["172.17.0.1"]
    }]'
spec:
  containers:
  - name: samplepod-route
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

#### ... macvlan ... Pod

... macvlan ... Pod... annotation `v1.multus-cni.io/default-network` ... NetworkAttachmentDefinition ... `<namespace>/<name>` ...

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod-macvlan
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: default/macvlan
spec:
  containers:
  - name: samplepod-macvlan
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

#### ... KUBE-OVN SUBNET...PROVIDER OVN...

... `provider` ... ovn ... subnet ... IP ... `provider` ... ovn ... Kube-OVN Subnet... `cidrBlock` ... `exclude_ips` ... Subnet ...

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: macvlan
spec:
  protocol: IPv4
  provider: ovn
```



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```
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

#### ... Pod

... `provider` ... ovn ... subnet ... IP ... Pod... annotation `k8s.v1.cni.cncf.io/networks` ... `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/logical_switch` ...

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/macvlan
    macvlan.default.kubernetes.io/logical_switch: macvlan
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

- `k8s.v1.cni.cncf.io/networks` : ... NetworkAttachmentDefinition ... `<namespace>/<name>`
- `macvlan.default.kubernetes.io/logical_switch` : ...

> ...
>
> - ... `<networkAttachmentName>.<networkAttachmentNamespace>.kubernetes.io/logical_switch` ... provider ...
> - ... ovn ... subnet ... ipam ... IP ... Pod... IP ... macvlan ... Pod...
> - ... macvlan ... Pod...

### ... Kube-OVN ...

... Kube-OVN ...

#### ... NETWORKATTACHMENTDEFINITION

... `provider` ... `ovn` ...

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "attachnet.default.ovn"
    }'
```

- `spec.config.type` : ... `kube-ovn` ... CNI ... Kube-OVN ...
- `server_socket` : Kube-OVN ... socket ... `/run/openvswitch/kube-ovn-daemon.sock` ...
- `provider` : ... NetworkAttachmentDefinition ... `<name>.<namespace>.ovn` ...Kube-OVN ... Subnet ... ovn...

#### ... KUBE-OVN SUBNET

... Kube-OVN ... `provider` ... NetworkAttachmentDefinition ... `<name>.<namespace>.ovn` ... `ovn` ... ... Kube-OVN ... Subnet ...

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: attachnet.default.ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```



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### ... Pod

... Pod... annotation `k8s.v1.cni.cncf.io/networks` ... NetworkAttachmentDefinition ... `<namespace>/<name>` ...

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

### ... IP/MAC

... NetworkAttachmentDefinition ... IP/MAC... `k8s.v1.cni.cncf.io/networks` ... `interface` ... annotation... `<nadName>.<nadNamespace>.kubernetes.io/ip_address.<interfaceName>` ... `<nadName>.<nadNamespace>.kubernetes.io/mac_address.<interfaceName>` ... - ... IP/MAC...

### ...

... Kube-OVN ... Pod... `<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/routes` annotation ...

```
apiVersion: v1
kind: Pod
metadata:
  name: custom-routes-attach
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
    attachnet.default.ovn.kubernetes.io/routes: |
      [{
        "dst": "192.168.0.101/24",
        "gw": "172.17.0.254"
      }, {
        "gw": "172.17.0.254"
      }]
spec:
  containers:
  - name: custom-routes-attach
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

> `dst` ...

... Deployment...DaemonSet ... StatefulSet... Annotation ... `.spec.template.metadata.annotations` ...

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: custom-routes-attach
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
      annotations:
        k8s.v1.cni.cncf.io/networks: default/attachnet
        attachnet.default.ovn.kubernetes.io/routes: |
          [{
            "dst": "192.168.0.101/24",
            "gw": "172.17.0.254"
          }, {
            "gw": "172.17.0.254"
          }]
    spec:
      containers:
      - name: nginx
        image: docker.io/library/nginx:alpine
```

### ... KUBE-OVN SUBNET...PROVIDER OVN...

... `provider` ... ovn ... subnet ... IP ... `provider` ... ovn ... Kube-OVN Subnet... `cidrBlock` ... `exclude_ips` ... Subnet ...



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```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: attachnet
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 172.17.0.0/16
  gateway: 172.17.0.1
  excludeIps:
  - 172.17.0.0..172.17.0.10
```

▯▯▯▯▯▯▯▯ **Pod**

▯▯▯▯▯ `provider` ▯▯▯ ovn ▯ subnet ▯▯▯ IP ▯ Pod▯▯▯ annotation `k8s.v1.cni.cncf.io/networks` ▯ `<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/logical_switch` ▯▯▯▯▯

```
apiVersion: v1
kind: Pod
metadata:
  name: samplepod
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
    attachnet.default.ovn.kubernetes.io/logical_switch: attachnet
spec:
  containers:
  - name: samplepod
    command: ["/bin/ash", "-c", "trap : TERM INT; sleep infinity & wait"]
    image: docker.io/library/alpine:edge
```

- `k8s.v1.cni.cncf.io/networks` : ▯▯▯▯▯▯ NetworkAttachmentDefinition ▯ `<namespace>/<name>`
- `attachnet.default.ovn.kubernetes.io/logical_switch` : ▯▯▯▯▯▯

> ▯▯▯
>
> - ▯▯ `<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/logical_switch` ▯▯▯▯▯▯▯▯▯▯▯▯ provider ▯▯▯▯▯
> - ▯▯▯▯▯▯▯ Kube-OVN ▯▯▯ Pod▯▯▯▯▯▯▯ IP ▯ Pod▯▯▯▯▯▯▯ IP ▯▯▯▯▯▯▯▯▯▯▯▯▯ macvlan ▯ Pod▯▯▯▯▯▯▯▯▯▯▯▯▯ Kube-OVN ▯▯▯ Pod▯▯▯▯▯▯▯▯▯▯▯▯▯
> - ▯▯▯▯▯▯▯ Kube-OVN ▯▯▯ Pod▯▯▯▯▯▯▯▯ `<networkAttachmentName>.<networkAttachmentNamespace>.ovn.kubernetes.io/routes` annotation ▯▯▯▯▯ ▯▯▯

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7.1.4 ▯▯



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# 7.2 ████

████████████████Kube-OVN ████████████████████████████████████ ████████████████████

████████████████████████████████████████████████████████

████████████████████████████Kube-OVN ████████

## 7.2.1 ██████████

████████████████████████████████████████████████████████████████████████████████████████████████████████

1. CNI ████████████████████████████████████ CNI ████████████████████
2. ████████████████████████ 10ns ████████████████ 20ns████████████████████████████████████
3. ████████████████████ CPU███████ CPU ████████████████
4. ████████████████ Macvlan ▯ SR-IOV ████████████████████

## 7.2.2 ████

> ████████████████████████████████████████████████████ ████████████████████████████████

**Overlay** ████████

████

- Kubernetes: 1.22.0
- OS: CentOS 7
- Kube-OVN: 1.8.0 *Overlay* ▯▯
- CPU: Intel(R) Xeon(R) E-2278G
- Network: 2*10Gbps, xmit_hash_policy=layer3+4

████ `qperf -t 60 <server ip> -ub -oo msg_size:1 -vu tcp_lat tcp_bw udp_lat udp_bw` ▯▯ 1 ████ tcp/udp ████████████████████████████

| Type | tcp_lat (us) | udp_lat (us) | tcp_bw (Mb/s) | udp_bw(Mb/s) |
|---|---|---|---|---|
| Kube-OVN Default | 25.7 | 22.9 | 27.1 | 1.59 |
| Kube-OVN Optimized | 13.9 | 12.9 | 27.6 | 5.57 |
| HOST Network | 13.1 | 12.4 | 28.2 | 6.02 |

**Overlay▯ Underlay** ██████

██████████ Kube-OVN ████████ Overlay ▯ Underlay ████████████

*Environment*:

- Kubernetes: 1.22.0
- OS: CentOS 7
- Kube-OVN: 1.8.0
- CPU: AMD EPYC 7402P 24-Core Processor
- Network: Intel Corporation Ethernet Controller XXV710 for 25GbE SFP28



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`qperf -t 60 <server ip> -ub -oo msg_size:1 -vu tcp_lat tcp_bw udp_lat udp_bw`

| Type | tcp_lat (us) | udp_lat (us) | tcp_bw (Mb/s) | udp_bw(Mb/s) |
|---|---|---|---|---|
| Kube-OVN Overlay | 15.2 | 14.6 | 23.6 | 2.65 |
| Kube-OVN Underlay | 14.3 | 13.8 | 24.2 | 3.46 |
| HOST Network | 16.6 | 15.4 | 24.8 | 2.64 |

`qperf -t 60 <server ip> -ub -oo msg_size:1K -vu tcp_lat tcp_bw udp_lat udp_bw`

| Type | tcp_lat (us) | udp_lat (us) | tcp_bw (Gb/s) | udp_bw(Gb/s) |
|---|---|---|---|---|
| Kube-OVN Overlay | 16.5 | 15.8 | 10.2 | 2.77 |
| Kube-OVN Underlay | 15.9 | 14.5 | 9.6 | 3.22 |
| HOST Network | 18.1 | 16.6 | 9.32 | 2.66 |

`qperf -t 60 <server ip> -ub -oo msg_size:4K -vu tcp_lat tcp_bw udp_lat udp_bw`

| Type | tcp_lat (us) | udp_lat (us) | tcp_bw (Gb/s) | udp_bw(Gb/s) |
|---|---|---|---|---|
| Kube-OVN Overlay | 34.7 | 41.6 | 16.0 | 9.23 |
| Kube-OVN Underlay | 32.6 | 44 | 15.1 | 6.71 |
| HOST Network | 35.9 | 45.9 | 14.6 | 5.59 |

> [illegible] netfilter [illegible] `kube-proxy` [illegible] netfilter[illegible] [illegible]

## 7.2.3 [illegible]

[illegible]

**CPU** [illegible]

[illegible] CPU [illegible] CPU [illegible]

```
cpupower frequency-set -g performance
```

[illegible]

[illegible]

[illegible]

```
# ethtool -g eno1
 Ring parameters for eno1:
 Pre-set maximums:
 RX:             4096
 RX Mini:        0
 RX Jumbo:       0
 TX:             4096
 Current hardware settings:
 RX:             255
 RX Mini:        0
 RX Jumbo:       0
 TX:             255
```

[illegible]

```
ethtool -G eno1 rx 4096
ethtool -G eno1 tx 4096
```

[illegible] **tuned** [illegible]

tuned [illegible] profile [illegible]

[illegible]

```
tuned-adm profile network-latency
```

[illegible]

```
tuned-adm profile network-throughput
```

[illegible]

[illegible] `irqbalance` [illegible] CPU [illegible] CPU [illegible]

[illegible] **OVN LB**

OVN [illegible] L2 LB [illegible] `conntrack` [illegible] recirculate [illegible] CPU [illegible] 20% [illegible] CPU [illegible] [illegible] Overlay [illegible] `kube-proxy` [illegible] Service [illegible] [illegible] Pod-to-Pod [illegible] `kube-ovn-controller` [illegible]

```
command:
- /kube-ovn/start-controller.sh
args:
...
- --enable-lb=false
...
```

> Underlay [illegible] `kube-proxy` [illegible] iptables [illegible] ipvs [illegible] LB [illegible] Service [illegible]

[illegible] **conntrack** [illegible]

[illegible] OVN LB [illegible] Service [illegible] Service [illegible] NetworkPolicy [illegible] Subnet A [illegible] Pod [illegible] Subnet B [illegible] Service[illegible] [illegible] kube-ovn-controller [illegible] `--skip-conntrack-dst-cidrs` [illegible] conntrack [illegible]

```
    --skip-conntrack-dst-cidrs="10.17.0.0/16,169.254.169.245/32"
```

[illegible] **FastPath** [illegible]

[illegible] network ns[illegible] netfilter [illegible] 20% [illegible] CPU [illegible] [illegible] netfilter [illegible] `FastPath` [illegible] netfilter [illegible] CPU [illegible] [illegible]

> [illegible] netfilter [illegible] iptables、ipvs、nftables [illegible]

[illegible]

[illegible] FastPath [illegible]

[illegible] `insmod kube_ovn_fastpath.ko` [illegible] `FastPath` [illegible] `dmesg` [illegible]

```
# dmesg
...
[619631.323788] init_module,kube_ovn_fastpath_local_out
[619631.323798] init_module,kube_ovn_fastpath_post_routing
[619631.323800] init_module,kube_ovn_fastpath_pre_routing
[619631.323801] init_module,kube_ovn_fastpath_local_in
...
```

**OVS** [illegible]

OVS [illegible] flow [illegible] 10% [illegible] CPU [illegible] x86 CPU [illegible] `popcnt` [illegible] `sse4.2` [illegible] [illegible]flow [illegible] CPU [illegible] [illegible] 5% [illegible]

[illegible] `FastPath` [illegible]



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[illegible] CPU [illegible]

```
cat /proc/cpuinfo  | grep popcnt
cat /proc/cpuinfo  | grep sse4_2
```

CENTOS [illegible]

[illegible]

```
yum install -y gcc kernel-devel-$(uname -r) python3 autoconf automake libtool rpm-build openssl-devel
```

[illegible] OVS [illegible] RPM [illegible]:

```
git clone -b branch-3.5 --depth=1 https://github.com/openvswitch/ovs.git
cd ovs
curl -s  https://github.com/kubeovn/ovs/commit/2d2c83c26d4217446918f39d5cd5838e9ac27b32.patch |  git apply
./boot.sh
./configure --with-linux=/lib/modules/$(uname -r)/build CFLAGS="-g -O2 -mpopcnt -msse4.2"
make rpm-fedora-kmod
cd rpm/rpmbuild/RPMS/x86_64/
```

[illegible] RPM [illegible]

```
rpm -i openvswitch-kmod-3.5.1-1.el7.x86_64.rpm
```

[illegible] Kube-OVN[illegible] OVS [illegible]

UBUNTU [illegible]

[illegible]

```
apt install -y autoconf automake libtool gcc build-essential libssl-dev
```

[illegible] OVS [illegible]

```
apt install -y autoconf automake libtool gcc build-essential libssl-dev

git clone -b branch-3.5 --depth=1 https://github.com/openvswitch/ovs.git
cd ovs
curl -s  https://github.com/kubeovn/ovs/commit/2d2c83c26d4217446918f39d5cd5838e9ac27b32.patch |  git apply
./boot.sh
./configure --prefix=/usr/ --localstatedir=/var --enable-ssl --with-linux=/lib/modules/$(uname -r)/build
make -j `nproc`
make install
make modules_install

cat > /etc/depmod.d/openvswitch.conf << EOF
override openvswitch * extra
override vport-* * extra
EOF

depmod -a
cp debian/openvswitch-switch.init /etc/init.d/openvswitch-switch
/etc/init.d/openvswitch-switch force-reload-kmod
```

[illegible] Kube-OVN[illegible] OVS [illegible]

## [illegible] STT [illegible]

> **Warning**
>
> OpenVswitch [illegible] 3.6 [illegible] STT [illegible] Tunnel [illegible]

[illegible] Geneve [illegible] Vxlan [illegible] UDP [illegible] UDP [illegible] TCP [illegible] TCP [illegible] offload [illegible] TCP [illegible] CPU [illegible] TCP [illegible]

STT [illegible] TCP [illegible] TCP [illegible] TCP [illegible] TCP [illegible] TCP [illegible]

STT [illegible] OVS [illegible]OVS [illegible]



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STT ⬚⬚⬚⬚⬚

```
kubectl set env daemonset/ovs-ovn -n kube-system TUNNEL_TYPE=stt

kubectl delete pod -n kube-system -lapp=ovs
```

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2022⬚5⬚24⬚

GitHub







## 7.2.4 ⬚⬚

## 7.3 ▯▯▯▯ FastPath ▯▯

▯▯▯▯▯▯▯▯▯ Profile▯ `Netfilter` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ 20% ▯▯▯ CPU ▯▯▯FastPath ▯▯▯▯▯▯▯ `Netfilter` ▯▯ ▯▯ CPU ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

2026⬜2⬜11⬜

2022⬜5⬜24⬜



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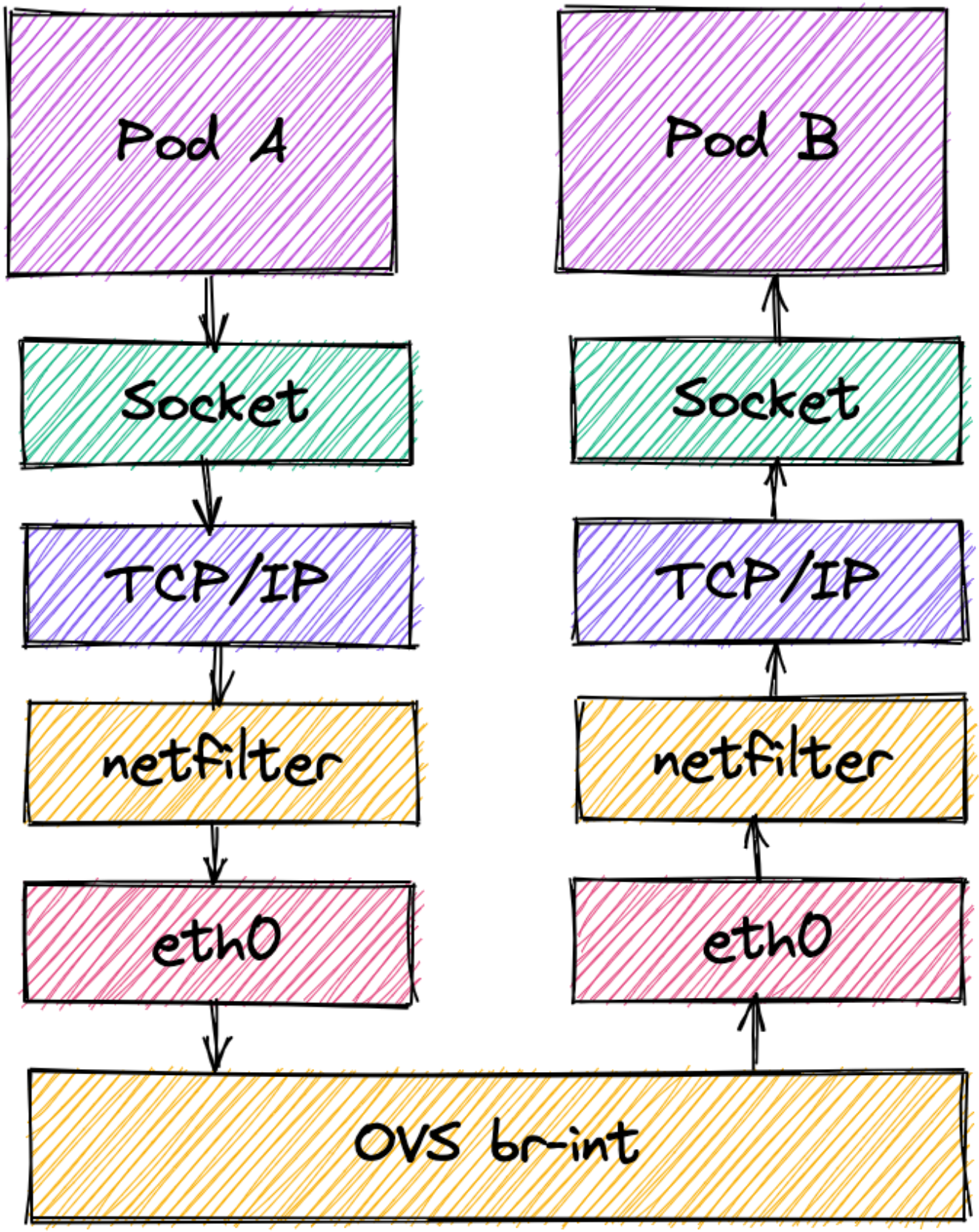
## 7.3.5 ⬜⬜

## 7.4 □□ eBPF □□□□□ TCP □□

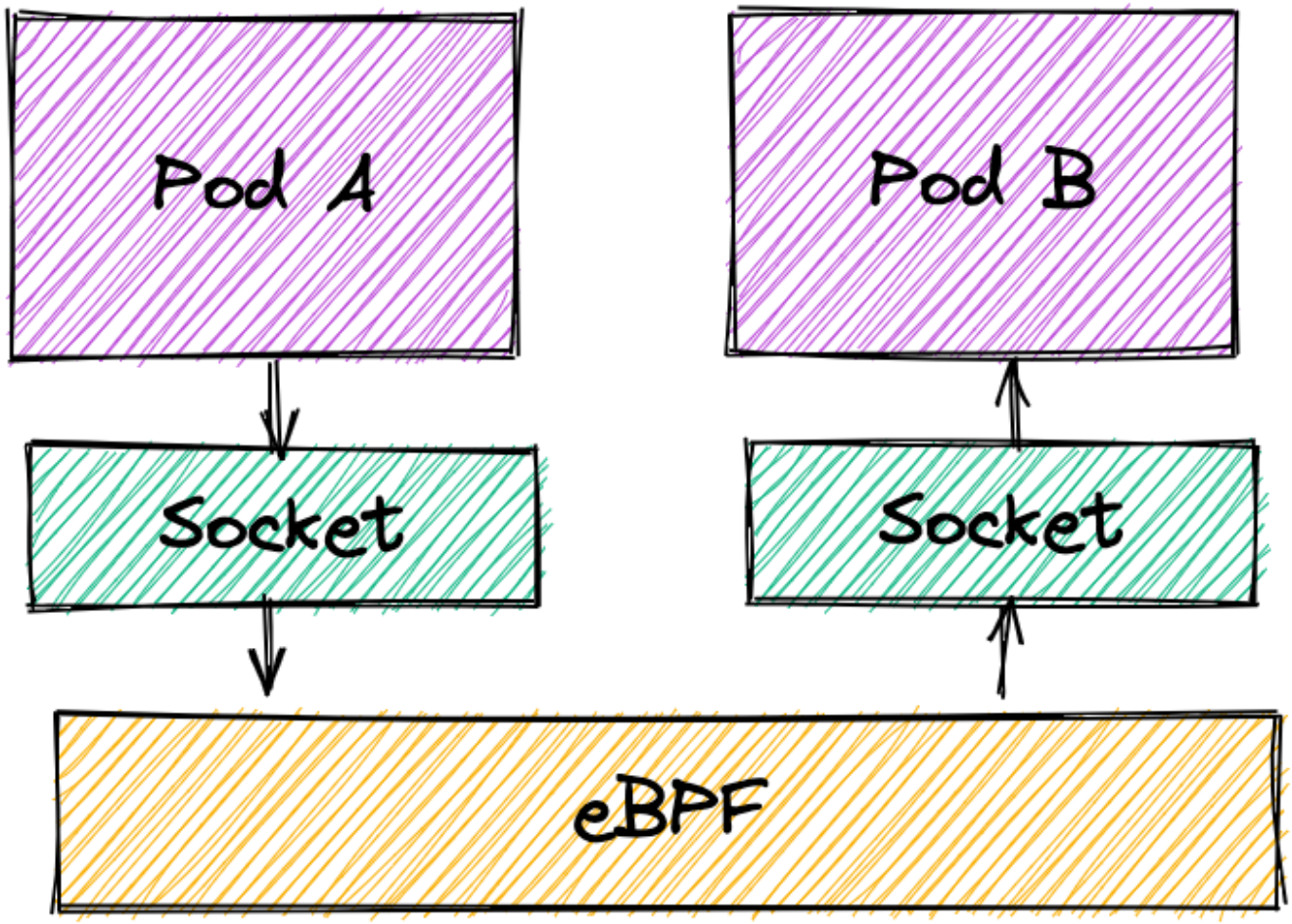
□□□□□□ 5G □□□□□□□□□□ Pod □□□□□□□□ TCP □□□□□□□ Intel □□□ istio-tcpip-bypass □□□Pod □□□□ eBPF □□□□□□□□ TCP/IP □□□□□□□□ socket □□□□□□□□□□□□□□□□□□□□

### 7.4.1 □□□□

□□□□□□□□□□□□□□□ Pod □□ TCP □□□□□□□□□□□□□□□□ TCP/IP, netfilter□OVS □□□□□□□




istio-tcpip-bypass ■■■■■■■■■■■■■■■■■■ TCP ■■■■■■■■■■■■■■■■■■ socket ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■




▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ TCP ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ Service Mesh ▒▒▒▒▒▒▒▒▒▒▒▒▒ Service Mesh ▒▒▒▒▒▒▒

▒▒▒▒▒▒▒▒▒▒▒▒▒ Tanzu Service Mesh Acceleration using eBPF▒

## 7.4.2 ▒▒▒▒

eBPF ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ Ubuntu 20.04 ▒ Linux 5.4.0-74-generic ▒▒▒▒▒▒▒▒▒▒

## 7.4.3 ▒▒▒▒

▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ Pod▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ `nodeSelector` ▒

```
# kubectl create deployment perf --image=kubeovn/perf:dev --replicas=2
deployment.apps/perf created
# kubectl get pod -o wide
NAME                    READY   STATUS    RESTARTS   AGE   IP           NODE     NOMINATED NODE   READINESS GATES
perf-7697bc6ddf-b2cpv   1/1     Running   0          28s   100.64.0.3   sealos   <none>           <none>
perf-7697bc6ddf-p2xpt   1/1     Running   0          28s   100.64.0.2   sealos   <none>           <none>
```

▒▒▒▒▒▒ Pod ▒▒ qperf server▒▒▒▒▒▒ Pod ▒▒▒ qperf client ▒▒▒▒▒▒▒▒

```
# kubectl exec -it perf-7697bc6ddf-b2cpv sh
/ # qperf

# kubectl exec -it perf-7697bc6ddf-p2xpt sh
/ # qperf -t 60 100.64.0.3 -ub -oo msg_size:1:16K:*4 -vu tcp_lat tcp_bw
```

▒▒ istio-tcpip-bypass ▒▒▒

```
kubectl apply -f https://raw.githubusercontent.com/intel/istio-tcpip-bypass/main/bypass-tcpip-daemonset.yaml
```

▯▯▯▯ perf client ▯▯▯▯▯▯▯▯▯▯

```
# kubectl exec -it perf-7697bc6ddf-p2xpt sh
/ # qperf -t 60 100.64.0.3 -ub -oo msg_size:1:16K:*4 -vu tcp_lat tcp_bw
```

## 7.4.4 ▯▯▯▯

▯▯▯▯▯▯ TCP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ 40% ~ 60% ▯▯▯▯▯▯▯▯▯▯▯▯▯ 1024 ▯▯▯▯▯▯▯▯▯ 40% ~ 80% ▯▯▯

| Packet Size (byte) | eBPF tcp_lat (us) | Default tcp_lat (us) | eBPF tcp_bw (Mb/s) | Default tcp_bw(Mb/s) |
|---|---|---|---|---|
| 1 | 20.2 | 44.5 | 1.36 | 4.27 |
| 4 | 20.2 | 48.7 | 5.48 | 16.7 |
| 16 | 19.6 | 41.6 | 21.7 | 63.5 |
| 64 | 18.8 | 41.3 | 96.8 | 201 |
| 256 | 19.2 | 36 | 395 | 539 |
| 1024 | 18.3 | 42.4 | 1360 | 846 |
| 4096 | 16.5 | 62.6 | 4460 | 2430 |
| 16384 | 20.2 | 58.8 | 9600 | 6900 |

> ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ 512 ▯▯▯▯▯▯▯ eBPF ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ TCP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯ eBPF ▯▯▯▯▯▯▯▯▯▯▯▯ eBPF TCP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

## 7.4.5 ▯▯▯▯

1. istio-tcpip-bypass
2. Deep Dive TCP/IP Bypass with eBPF in Service Mesh
3. Tanzu Service Mesh Acceleration using eBPF

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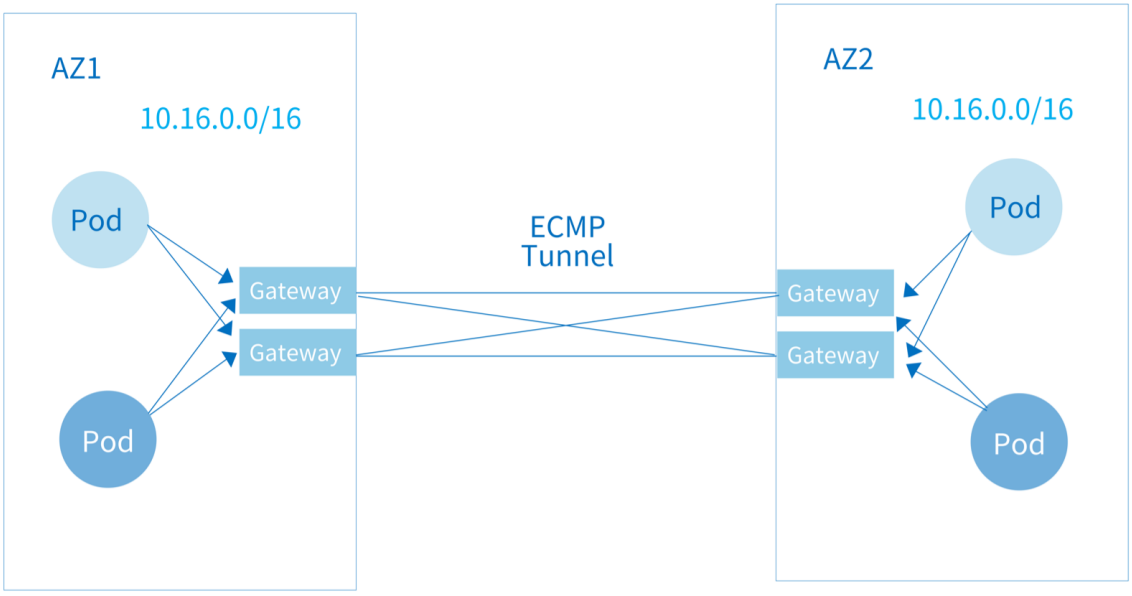


## 7.4.6 ▯▯

# 7.5 ▒▒ OVN-IC ▒▒▒▒▒▒▒

Kube-OVN ▒▒▒▒ OVN-IC ▒▒▒ Kubernetes ▒▒ Pod ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ Pod ▒▒▒▒ Pod IP ▒▒▒▒▒▒▒ Kube-OVN ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ IP ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

> ▒▒▒▒▒▒▒▒▒▒▒ Overlay ▒▒▒▒▒Underlay ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒




> **Limitation**
>
> OVN-IC ▒▒▒▒▒▒▒ Pod IP ▒▒▒▒▒▒▒▒▒▒▒▒ Service▒DNS ▒ NetworkPolicy ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ Istio ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

## 7.5.1 ▒▒▒▒

1. 1.11.16 ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

▒▒▒▒▒ `install-ovn-ic.sh` ▒▒▒▒▒▒▒▒▒

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install-ic-server.sh
```

▒▒▒▒▒▒▒▒▒▒ `TS_NUM` ▒▒▒▒▒▒▒▒ ECMP Path ▒▒▒

```
sed 's/VERSION=.*/VERSION=v1.17.0/' dist/images/install-ic-server.sh | TS_NUM=3 bash
```

▒▒▒▒▒▒▒▒▒▒

```
deployment.apps/ovn-ic-server created
Waiting for deployment spec update to be observed...
Waiting for deployment "ovn-ic-server" rollout to finish: 0 out of 3 new replicas have been updated...
Waiting for deployment "ovn-ic-server" rollout to finish: 0 of 3 updated replicas are available...
Waiting for deployment "ovn-ic-server" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "ovn-ic-server" rollout to finish: 2 of 3 updated replicas are available...
deployment "ovn-ic-server" successfully rolled out
OVN IC Server installed Successfully
```

▒▒ `kubectl ko icsbctl show` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

```
kubectl ko icsbctl show
availability-zone az0
    gateway 059b5c54-c540-4d77-b009-02d65f181a02
        hostname: kube-ovn-worker
        type: geneve
            ip: 172.18.0.3
        port ts-az0
            transit switch: ts
            address: ["00:00:00:B4:8E:BE 169.254.100.97/24"]
    gateway 74ee4b9a-ba48-4a07-861e-1a8e4b9f905f
        hostname: kube-ovn-worker2
        type: geneve
            ip: 172.18.0.2
        port ts1-az0
            transit switch: ts1
            address: ["00:00:00:19:2E:F7 169.254.101.90/24"]
    gateway 7e2428b6-344c-4dd5-a0d5-972c1ccec581
        hostname: kube-ovn-control-plane
        type: geneve
            ip: 172.18.0.4
        port ts2-az0
            transit switch: ts2
            address: ["00:00:00:EA:32:BA 169.254.102.103/24"]
availability-zone az1
    gateway 034da7cb-3826-4318-81ce-6a877a9bf285
        hostname: kube-ovn1-worker
        type: geneve
            ip: 172.18.0.6
        port ts-az1
            transit switch: ts
            address: ["00:00:00:25:3A:B9 169.254.100.51/24"]
    gateway 2531a683-283e-4fb8-a619-bdbcb33539b8
        hostname: kube-ovn1-worker2
        type: geneve
            ip: 172.18.0.5
        port ts1-az1
            transit switch: ts1
            address: ["00:00:00:52:87:F4 169.254.101.118/24"]
    gateway b0efb0be-e5a7-4323-ad4b-317637a757c4
        hostname: kube-ovn1-control-plane
        type: geneve
            ip: 172.18.0.8
        port ts2-az1
            transit switch: ts2
            address: ["00:00:00:F6:93:1A 169.254.102.17/24"]
```

▒▒▒▒▒▒▒ **2**

▒▒▒▒▒ `kube-ovn-controller` ▒▒▒ IP ▒▒▒▒▒▒▒▒▒ `OVN-IC` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

▒▒ `docker` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ `OVN-IC` ▒▒▒▒▒

```
docker run --name=ovn-ic-db -d --env "ENABLE_OVN_LEADER_CHECK="false"" --network=host --privileged  -v /etc/ovn/:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /
var/log/ovn:/var/log/ovn kubeovn/kube-ovn:v1.17.0 bash start-ic-db.sh
```

▒▒▒▒▒ `containerd` ▒▒ `docker` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒



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```
ctr -n k8s.io run -d --env "ENABLE_OVN_LEADER_CHECK="false"" --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --
mount="type=bind,src=/var/run/ovn,dst=/var/run/ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw" docker.io/kubeovn/
kube-ovn:v1.17.0 ovn-ic-db bash start-ic-db.sh
```

## 7.5.3 ■■■■■■■

■■■■■■■■■■■■■■■■■■■■■■ VPC ■ Subnet ■ CIDR ■■■■■■ `OVN-IC` ■■■■■■■■■■■■■ Subnet CIDR ■■■■■■■

■ `kube-system` Namespace ■■■ `ovn-ic-config` ConfigMap■

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

- `enable-ic`：■■■■■■■■■■
- `az-name`：■■■■■■■■■■■■■■■■■■■■■■■■■■■■
- `ic-db-host`：■■ `OVN-IC` ■■■■■■■■■■
- `ic-nb-port`：`OVN-IC` ■■■■■■■■■■■■■ 6645■
- `ic-sb-port`：`OVN-IC` ■■■■■■■■■■■■■ 6646■
- `gw-nodes`：■■■■■■■■■■■■■■■■■■■■■■■■■■■■
- `auto-route`：■■■■■■■■■■■■■■■■■

■■■ ■■■■■■■■■■■■■ `ovn-ic-config` ■■ ConfigMap ■■■■■■■■■■■■■■■■■■■■■■ ConfigMap■■■■■■■■■ ConfigMap■

■ `ovn-ic` ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ `ts` ■

```
# ovn-ic-sbctl show
availability-zone az1
    gateway deee03e0-af16-4f45-91e9-b50c3960f809
        hostname: az1-gw
        type: geneve
            ip: 192.168.42.145
        port ts-az1
            transit switch: ts
            address: ["00:00:00:50:AC:8C 169.254.100.45/24"]
availability-zone az2
    gateway e94cc831-8143-40e3-a478-90352773327b
        hostname: az2-gw
        type: geneve
            ip: 192.168.42.149
        port ts-az2
            transit switch: ts
            address: ["00:00:00:07:4A:59 169.254.100.63/24"]
```

■■■■■■■■■■■■■■■■■■■■■■■■■■■■

```
# kubectl ko nbctl lr-route-list ovn-cluster
IPv4 Routes
                10.42.1.1            169.254.100.45 dst-ip (learned)
                10.42.1.3                100.64.0.2 dst-ip
                10.16.0.2                100.64.0.2 src-ip
                10.16.0.3                100.64.0.2 src-ip
                10.16.0.4                100.64.0.2 src-ip
                10.16.0.6                100.64.0.2 src-ip
             10.17.0.0/16            169.254.100.45 dst-ip (learned)
            100.65.0.0/16            169.254.100.45 dst-ip (learned)
```

■■■■■■■■■■■ 1 ■■■■ Pod ■■■ `ping` ■■ 2 ■■■■ Pod IP ■■■■■■■■■■

■■■■■■■■■■■■■■■■■■■■■■■■■■■■ Subnet ■■ `disableInterConnection` ■■■■■■■■■

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
```



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```
    name: no-advertise
  spec:
    cidrBlock: 10.199.0.0/16
    disableInterConnection: true
```

## 7.5.4 ⬚⬚⬚⬚⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚ CIDR ⬚⬚⬚⬚⬚⬚⬚⬚

```
docker run --name=ovn-ic-db -d --env "ENABLE_OVN_LEADER_CHECK="false"" --network=host --privileged -v /etc/ovn/:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/
log/ovn:/var/log/ovn -e LOCAL_IP="192.168.65.3"  -e NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1"   kubeovn/kube-ovn:v1.17.0 bash start-ic-db.sh
```

 `containerd`

```
ctr -n k8s.io run -d --env "ENABLE_OVN_LEADER_CHECK="false"" --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --
mount="type=bind,src=/var/run/ovn,dst=/var/run/ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw"  --
env="NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1"" --env="LOCAL_IP="192.168.65.3"" docker.io/kubeovn/kube-ovn:v1.17.0 ovn-ic-db bash start-ic-db.sh
```

- `LOCAL_IP` IP
- `NODE_IPS`  `OVN-IC`  IP

 `OVN-IC`  follower

 `docker`

```
docker run --name=ovn-ic-db -d --network=host --privileged -v /etc/ovn/:/etc/ovn -v /var/run/ovn:/var/run/ovn -v /var/log/ovn:/var/log/ovn -e
LOCAL_IP="192.168.65.2"  -e NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1" -e LEADER_IP="192.168.65.3"  kubeovn/kube-ovn:v1.17.0 bash start-ic-db.sh
```

 `containerd`

```
ctr -n k8s.io run -d --net-host --privileged --mount="type=bind,src=/etc/ovn/,dst=/etc/ovn,options=rbind:rw" --mount="type=bind,src=/var/run/ovn,dst=/var/run/
ovn,options=rbind:rw" --mount="type=bind,src=/var/log/ovn,dst=/var/log/ovn,options=rbind:rw"  --env="NODE_IPS="192.168.65.3,192.168.65.2,192.168.65.1"" --
env="LOCAL_IP="192.168.65.2"" --env="LEADER_IP="192.168.65.3"" docker.io/kubeovn/kube-ovn:v1.17.0 ovn-ic-db bash start-ic-db.sh
```

- `LOCAL_IP` IP
- `NODE_IPS`  `OVN-IC`  IP
- `LEADER_IP` :  `OVN-IC`  leader  IP

 `ovn-ic-config`  `OVN-IC`

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-ic-config
  namespace: kube-system
data:
  enable-ic: "true"
  az-name: "az1"
  ic-db-host: "192.168.65.3,192.168.65.2,192.168.65.1"
  ic-nb-port: "6645"
  ic-sb-port: "6646"
  gw-nodes: "az1-gw"
  auto-route: "true"
```

## 7.5.6 ECMP

1

ECMPECMP path  3 ECMP path

```
kubectl edit deployment ovn-ic-server -n kube-system
```

 'TS_NUM'  `TS_NUM`  ECMP Path

## 7.5.7

Configmap

 `ovn-ic-config` Configmap

```
kubectl -n kube-system delete cm ovn-ic-config
```

 `ts`

```
kubectl ko nbctl ls-del ts
```

[illegible]

## 7.5.8 az-name

[illegible] `kubectl edit` [illegible] `ovn-ic-config` configmap `az-name` [illegible] ovn-cni pod [illegible] 10 [illegible]

```
ovn-appctl -t ovn-controller inc-engine/recompute
```

## 7.5.9 [illegible]

[illegible] `ovn-ic-config` Configmap

```
kubectl -n kube-system delete cm ovn-ic-config
```

[illegible] `ts` [illegible]

```
kubectl ko nbctl ls-del ts
```

[illegible] OVN-IC [illegible]

[illegible] `docker` [illegible]

```
docker stop ovn-ic-db
docker rm ovn-ic-db
```

[illegible] `containerd` [illegible]

```
ctr -n k8s.io task kill ovn-ic-db
ctr -n k8s.io containers rm ovn-ic-db
```

[illegible] deployment `ovn-ic-server` [illegible]

```
kubectl delete deployment ovn-ic-server -n kube-system
```

[illegible] master [illegible] DB[illegible]

```
rm -f /etc/origin/ovn/ovn_ic_nb_db.db
rm -f /etc/origin/ovn/ovn_ic_sb_db.db
```

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## 7.5.10 [illegible]



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# 7.6 ▯▯ Submariner ▯▯▯▯▯▯▯

Submariner ▯▯▯▯▯▯▯ Kubernetes ▯▯ Pod ▯ Service ▯▯▯▯▯▯▯▯▯▯▯▯▯ Kube-OVN ▯▯▯▯▯▯▯

▯▯▯▯ OVN-IC ▯▯▯▯▯▯▯▯▯▯▯▯Submariner ▯▯▯▯ Kube-OVN ▯▯ Kube-OVN ▯▯▯▯▯▯▯ ▯▯▯ Service ▯▯▯▯▯▯▯▯▯ Submariner ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

## 7.6.1 ▯▯▯▯

- ▯▯▯▯▯ Service CIDR ▯▯▯▯▯▯ CIDR ▯▯▯▯▯

## 7.6.2 ▯▯ Submariner

> **Note**
>
> Submariner ▯ v0.22.0 ▯▯▯ nftables ▯▯▯▯▯▯▯▯▯▯▯▯ Kube-OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Submariner▯
>
> ```
> kubectl create namespace submariner-operator
> kubectl create configmap submariner-global --namespace=submariner-operator --from-literal=use-nftables=false
> ```

▯▯ `subctl` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
curl -Ls https://get.submariner.io | bash
export PATH=$PATH:~/.local/bin
echo export PATH=\$PATH:~/.local/bin >> ~/.profile
```

▯▯ `kubeconfig` ▯▯▯▯▯ `submariner-broker` ▯▯▯▯▯▯▯▯

```
subctl deploy-broker
```

▯▯▯▯▯ `cluster0` ▯▯▯▯▯ CIDR ▯ `10.16.0.0/16` ▯join ▯▯ CIDR ▯ `100.64.0.0/16` ▯ `cluster1` ▯▯▯▯▯ CIDR ▯ `11.16.0.0/16` ▯join ▯▯ CIDR ▯ `100.68.0.0/16` ▯

▯▯ `kubeconfig` ▯ `cluster0` ▯▯▯▯▯ broker▯▯▯▯▯▯▯▯:

```
subctl  join broker-info.subm --clusterid  cluster0 --clustercidr 100.64.0.0/16,10.16.0.0/16  --natt=false --cable-driver vxlan --health-check=false
kubectl label nodes cluster0 submariner.io/gateway=true
```

▯▯ `kubeconfig` ▯ `cluster1` ▯▯▯▯▯ broker▯▯▯▯▯▯▯▯:

```
subctl  join broker-info.subm --clusterid  cluster1 --clustercidr 100.68.0.0/16,11.16.0.0/16  --natt=false --cable-driver vxlan --health-check=false
kubectl label nodes cluster1 submariner.io/gateway=true
```

▯▯▯▯ `join` ▯▯▯▯▯▯▯▯ `gateway, routeagent` pod ▯▯▯▯, ▯▯ `submariner-operator` ▯▯ `clusterrole` ▯▯▯▯▯▯:

```
- apiGroups:
  - "apps"
  resources:
  - daemonsets
  verbs:
  - create
  - get
  - list
  - watch
  - update
```

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `subnet` `ovn-default` ▯▯▯▯▯▯▯▯ `centralized` ▯▯▯▯▯▯▯ `gatewayNode` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: ovn-default
spec:
  gatewayType: centralized
  gatewayNode: "node1,node2"   #   submariner     gateway
```

▯ submariner ▯▯▯ `gateway` ▯▯▯▯▯ `gatewayNode` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯



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⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ Pod ⬚⬚⬚⬚⬚ IP ⬚⬚⬚⬚⬚⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `subctl` ⬚⬚⬚⬚⬚⬚⬚

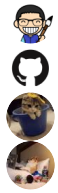
```
subctl show all
subctl diagnose all
```

⬚⬚ Submariner ⬚⬚⬚⬚⬚⬚⬚ Submariner ⬚⬚⬚⬚⬚

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## 7.6.3 ⬚⬚

# 7.7 Overlay

Underlay Pod IP

> VPC Overlay Pod IP IP

## 7.7.1

- ip_forward
- iptables forward Drop
- ct INVALID

## 7.7.2

natOutgoing false nat Pod IP

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: routed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: distributed
  natOutgoing: false
```

Pod

Kubernetes

```
ip route add 10.166.0.0/16 via 192.168.2.10 dev eth0
```

10.166.0.0/16 192.168.2.10 Kubernetes

Keepalived VIP VIP

Subnet gatewayType centralized gatewayNode IP

```yaml
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: routed
spec:
  protocol: IPv4
  cidrBlock: 10.166.0.0/16
  default: false
  excludeIps:
  - 10.166.0.1
  gateway: 10.166.0.1
  gatewayType: centralized
  gatewayNode: "node1"
  natOutgoing: false
```

nat VPC NAT

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## 7.7.3 ⊠⊠



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# 7.8 Overlay □□□□□□□□

□□□□□□□□□□□□□□ Overlay □□□□□□□□□□□□□□□ tunnel □□□□□□□□□

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## 7.8.1 □□□□□□□□□

□□□□□□□□□□□□□□□Kube-OVN □□□□□ Kubernetes Node IP □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

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```
IFACE=eth1
```

□□□□□□□□□□□□□□□□□□□□□□ `ens[a-z0-9]*,eth[a-z0-9]*` □

□□□□□□□□□□ `kube-ovn-cni` DaemonSet □□□□□□□□□

```
args:
- --iface=eth1
```

□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ annotation `ovn.kubernetes.io/tunnel_interface` □□□□□□□□□□□□□□□□□ annotation □□□□□ `iface` □□□□□□□□ annotation□

```
kubectl annotate node no1 ovn.kubernetes.io/tunnel_interface=ethx
```

## 7.8.2 □□□□□□□□□□□□□

□□□□□□□□□□□□□□□□□Kube-OVN □□□□□□□□□□□□□□□□□□□□□□□□ IP □□ tunnel □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

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- □□□□□□□ Overlay □□□□□□

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1. □□□□ Node □□□□□□□□□□□□□□□□□□□□□□□□ IP□
2. □ Subnet □□□□ `nodeNetwork` □□□□□□□□□□□□□□□□□□□
3. kube-ovn-daemon □□□□□□□□□□□□□□□□ IP □□□□ OVS□
4. □ Pod □□□□□□□□□□□□□□□□□ `nodeNetwork` □□□□□□ Pod □ OVS □□□□□□□□□□ IP□

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□□□□□□□□□□□□□□□□□□□□□□ `ovn.kubernetes.io/node_networks` □□□□□□□□ JSON □□□key □□□□□□□value □□□□□□□□□□ IP□

```
kubectl annotate node <node-name> ovn.kubernetes.io/node_networks='{"storage": "192.168.100.10", "app": "172.16.0.10"}'
```

□□□□□□□□□□□□□□□□□

- `storage` □□□□ IP `192.168.100.10` □□□□□ IP □□□□□□□□□□□□□
- `app` □□□□ IP `172.16.0.10` □□□□□ IP □□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□

■■■■

■■■■■■■■■ `spec.nodeNetwork` ■■■■■■■■■■■■■■■■■■

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: storage-subnet
spec:
  protocol: IPv4
  cidrBlock: 10.100.0.0/16
  gateway: 10.100.0.1
  nodeNetwork: storage
```

■■■■■■ `storage-subnet` ■■■■ Pod ■■■ `storage` ■■■■■■■ tunnel ■■■

■■■■■■ `nodeNetwork` ■■■■■■■■■■■■■■■ IP■■ `IFACE` ■■■■■■■■ IP■■

■■■■

■■ OVS ■■ IP ■■

■■■■■■■■■■■■■■■ OVS ■■■■■■ IP■

```
ovs-vsctl get open . external-ids:ovn-encap-ip
```

■■■■■■■■■■■■■■■ IP■■■■■

```
"192.168.1.10,192.168.100.10,172.16.0.10"
```

■■■■■■■ IP■

```
ovs-vsctl get open . external-ids:ovn-encap-ip-default
```

■■ POD ■■■■■■ IP

■■ Pod ■■■■■■■ Pod ■■ OVS ■■■■■■ IP ■■■■

```
ovs-vsctl --columns=external_ids find interface external-ids:iface-id="<pod-name>.<namespace>"
```

■■■■■■■■ `nodeNetwork` ■■■■■■■■ `encap-ip` ■■■■

```
external_ids        : {encap-ip="192.168.100.10", iface-id="test-pod.default", ...}
```

■■■■

■■■■■■■■■■■■■■■■■■■■

1. ■■■■■■■■■

■■■■■■■■■■■■■■■■■■■■■■■■■

- `eth0` ■■■■■■■IP ■■■■ `192.168.1.10` ■ `192.168.1.11`
- `eth1` ■■■■■■■IP ■■■■ `10.10.10.10` ■ `10.10.10.11`

```
kubectl annotate node node1 ovn.kubernetes.io/node_networks='{"storage": "10.10.10.10"}'
kubectl annotate node node2 ovn.kubernetes.io/node_networks='{"storage": "10.10.10.11"}'
```

2. ■■■■■■■

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: storage-net
spec:
  protocol: IPv4
  cidrBlock: 10.200.0.0/16
  gateway: 10.200.0.1
  nodeNetwork: storage
```



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```
  namespaces:
  - storage-namespace
```

3. ▯▯▯▯▯▯▯▯▯ POD

```
apiVersion: v1
kind: Pod
metadata:
  name: storage-pod
  namespace: storage-namespace
spec:
  containers:
  - name: app
    image: docker.io/library/nginx:alpine
```

▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `eth1` ▯▯▯▯▯▯▯

▯▯▯▯

1. ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯
2. ▯▯▯▯▯▯▯▯▯ IP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ IP ▯▯▯▯
3. ▯▯▯▯▯▯▯▯ Overlay ▯▯▯▯▯▯Underlay ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Underlay ▯▯▯▯▯▯▯ Underlay ▯▯▯▯▯▯
4. ▯▯▯▯▯▯▯▯ `nodeNetwork` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

# 7.9 BGP ▯▯

Kube-OVN ▯▯▯ Pods▯Subnets▯Services ▯ EIPs ▯ IP ▯▯▯▯ BGP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ IP ▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `kube-ovn-speaker` ▯▯▯▯▯▯▯▯▯▯ Pod ▯ Subnet ▯▯▯▯▯ annotation▯

▯▯▯▯ EIP ▯▯▯ BGP▯▯▯▯▯▯▯▯▯▯▯▯ VPC NAT Gateway▯▯▯▯▯▯▯▯▯▯▯▯▯▯ EIPs▯

## 7.9.1 ▯▯ kube-ovn-speaker

`kube-ovn-speaker` ▯▯▯ GoBGP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

▯▯▯▯ `kube-ovn-speaker` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
kubectl label nodes speaker-node-1 ovn.kubernetes.io/bgp=true
kubectl label nodes speaker-node-2 ovn.kubernetes.io/bgp=true
```

> ▯▯▯▯▯ kube-ovn-speaker ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ECMP▯

▯▯▯▯ yaml:

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/yamls/speaker.yaml
```

▯▯ yaml ▯▯▯▯▯▯

▯▯▯▯▯▯▯▯▯▯▯▯

```
- --neighbor-address=10.32.32.254
- --neighbor-ipv6-address=2409:AB00:AB00:2000::AFB:8AFE
- --neighbor-as=65030
- --cluster-as=65000
#       IP
# - --allowed-source-addresses=10.32.32.2,10.32.32.3,10.32.32.4,10.32.32.5
```

▯▯▯▯▯▯▯▯▯▯▯

```
- --neighbor-address=10.32.32.252,10.32.32.253
- --neighbor-ipv6-address=2409:AB00:AB00:2000::AFB:8AFC,2409:AB00:AB00:2000::AFB:8AFD
- --neighbor-as=65030
- --cluster-as=65000
#       IP
# - --allowed-source-addresses=10.32.32.2,10.32.32.3,10.32.32.4,10.32.32.5
```

- `neighbor-address` : BGP Peer ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯
- `neighbor-as` : BGP Peer ▯ AS ▯▯
- `cluster-as` : ▯▯▯▯▯ AS ▯▯
- `allowed-source-addresses` : Speaker ▯▯▯▯▯▯▯▯ IP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯Speaker ▯▯▯ `ip route get` ▯▯▯▯ BGP Peer ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ IP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Speaker ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ECMP ▯▯▯▯▯▯▯▯▯▯ IP ▯▯▯▯▯▯▯▯▯▯ IP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

▯▯ yaml:

```
kubectl apply -f speaker.yaml
```

## 7.9.2 ▯▯ Pod/Subnet ▯▯

▯▯▯▯ BGP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Subnet ▯ `natOutgoing` ▯▯▯ `false` ▯▯▯ Pod IP ▯▯▯▯▯▯▯▯▯▯▯▯

▯▯ annotation ▯▯▯▯▯▯

```
kubectl annotate pod sample ovn.kubernetes.io/bgp=true
kubectl annotate subnet ovn-default ovn.kubernetes.io/bgp=true
```

▯▯ annotation ▯▯▯▯▯▯



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```
kubectl annotate pod sample ovn.kubernetes.io/bgp-
kubectl annotate subnet ovn-default ovn.kubernetes.io/bgp-
```

BGP

### 7.9.3 `ClusterIP` Service

Service ClusterIP `kube-ovn-speaker` `--announce-cluster-ip` `true` BGP

annotation

```
kubectl annotate service sample ovn.kubernetes.io/bgp=true
```

annotation

```
kubectl annotate service sample ovn.kubernetes.io/bgp-
```

### 7.9.4 EIPs

EIPs VPC NAT Gateway `VpcNatGateway` BGP BGP Sidecar

VPC NAT Gateway BGP BGP Speaker Sidecar `NetworkAttachmentDefinition` NAD VPC Subnet Sidecar Kubernetes API EIPs VPC CoreDNS NAD

`NetworkAttachmentDefinition` `Subnet` `provider` `{nadName}.{nadNamespace}.ovn`

```yaml
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: api-ovn-nad
  namespace: default
spec:
  config: '{
      "cniVersion": "0.3.0",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "api-ovn-nad.default.ovn"
    }'
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vpc-apiserver-subnet
spec:
  protocol: IPv4
  cidrBlock: 100.100.100.0/24
  provider: api-ovn-nad.default.ovn
```

`ovn-vpc-nat-config` ConfigMap `apiNadProvider` BGP Speaker:

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: ovn-vpc-nat-config
  namespace: kube-system
data:
  apiNadProvider: api-ovn-nad.default.ovn             # What NetworkAttachmentDefinition provider to use so that the sidecar
                                                      # can access the K8S API, as it can't by default due to VPC segmentation
  bgpSpeakerImage: docker.io/kubeovn/kube-ovn:v1.13.0  # Sets the BGP speaker image used
  image: docker.io/kubeovn/vpc-nat-gateway:v1.13.0
```

`ovn-default` `provider`

```yaml
provider: api-ovn-nad.default.ovn
```

VPC NAT Gateway BGP

```yaml
kind: VpcNatGateway
apiVersion: kubeovn.io/v1
metadata:
  name: vpc-natgw
spec:
  vpc: vpc1
  subnet: net1
  lanIp: 10.0.1.10
```

```
  bgpSpeaker:
    enabled: true
    asn: 65500
    remoteAsn: 65000
    neighbors:
      - 100.127.4.161
      - fd:01::1
    enableGracefulRestart: true # Optional
    routerId: 1.1.1.1           # Optional
    holdTime: 1m                # Optional
    password: "password123"     # Optional
    extraArgs:                  # Optional, passed directly to the BGP speaker
      - -v5                     # Enables verbose debugging of the BGP speaker sidecar
  selector:
    - "kubernetes.io/os: linux"
  externalSubnets:
  - ovn-vpc-external-network # Network on which we'll speak BGP and receive/send traffic to the outside world
                             # BGP neighbors need to be on that network
```

████████████████ BGP ████ EIP█

```
kubectl annotate eip sample ovn.kubernetes.io/bgp=true
```

## 7.9.5 ████

`kube-ovn-speaker` ████████████:

- **Cluster**: ██████████ speaker ███████ Pod IPs/Subnet CIDRs████████████████ IP ██████████ CIDR █ Pod████████████████ speaker ████████████████████ Pod██ ████████████████████ Pod █ Subnet ██████████
- **Local**: ████████Pod IPs ███████ Pod ████████████████ Cluster ████████████ Pod ████████████

> ██: ████ `Local` ████████████████ `kube-ovn-speaker` ███ Pod ████████ speaker ████████████████

████████ `Cluster` ██████ Pod/Subnet █ annotation `ovn.kubernetes.io/bgp` ███████

- `ovn.kubernetes.io/bgp=cluster` █ `ovn.kubernetes.io/bgp=true` ████ `Cluster` ███
- `ovn.kubernetes.io/bgp=local` ████ `Local` ███

> ██████ Service ████████ `kube-proxy` ██████ `ClusterIP` ██ Service ██████████ `Cluster` ███

## 7.9.6 BGP ████

`kube-ovn-speaker` ████ BGP ████████████████████████████

- `announce-cluster-ip` : ███████ Service ███████ `false` █
- `auth-password` : BGP peer ███████
- `holdtime` : BGP ████████████████████████████████████ 90 ██
- `graceful-restart` : ████ BGP Graceful Restart█
- `graceful-restart-time` : BGP Graceful restart time ███ RFC4724 3█
- `graceful-restart-deferral-time` : BGP Graceful restart deferral time ███ RFC4724 4.1█
- `passivemode` : Speaker ███ passive ████████ peer█
- `ebgp-multihop` : ebgp ttl ████ 1█
- `allowed-source-addresses` : █ IP ████████████████Speaker ██████ `ip route get` ████ BGP Peer ████████████ IP ████████████████████ ECMP ██████████ █ IP ██████
- `enable-bfd` : ██ `kube-ovn-speaker` ██ BFD████████████████████ `false` ██████“████ `kube-ovn-speaker` ██ BFD”█
- `bfd-min-tx` : `kube-ovn-speaker` ██ BFD ████████████████ `1000` ██1 ███
- `bfd-min-rx` : `kube-ovn-speaker` ██ BFD ████████████████ `1000` ██1 ███
- `bfd-detection-multiplier` : `kube-ovn-speaker` ██ BFD ████████ `3` ███████ = `bfd-min-rx * bfd-detection-multiplier` █



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## 7.9.7 BFD ⬚⬚⬚⬚⬚⬚

⬚ BGP ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ BFD⬚Bidirectional Forwarding Detection⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ BGP ECMP ⬚⬚⬚⬚⬚⬚⬚⬚⬚

⬚ Kube-OVN ⬚⬚⬚⬚BFD ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

- ⬚⬚⬚⬚⬚⬚⬚ **host network** ⬚⬚⬚⬚⬚⬚ Kubernetes Node ⬚ host network ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ BFD ⬚⬚⬚⬚⬚⬚ BGP ECMP ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚
- ⬚⬚⬚⬚**VPC** ⬚⬚⬚⬚⬚⬚ `OVN logical router BFD` ⬚⬚ `VPC` CRD ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ kube-ovn controller ⬚⬚⬚⬚⬚⬚⬚⬚ VPC ⬚⬚⬚⬚⬚⬚⬚⬚ BFD⬚

⬚⬚⬚⬚⬚⬚

```
flowchart TD
    A[BFD     ] --> B[   :     host network]
    A --> C[   : VPC   ]
    B --> D[   : kube-ovn-speaker    BFD]
    B --> E[   : openbfdd     ]
    C --> F[OVN logical router BFD]
```

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
    "port": 3784,
    "desired_minimum_tx_interval": 1000000,
    "required_minimum_receive": 1000000,
    "detection_multiplier": 3
  },
  "bfd_state": {
    "session_state": 1,
    "bfd_async": {
      "transmitted_packets": 100,
      "received_packets": 99
    }
  }
}
```

 `session_state`  `1` = UP `2` = DOWN `3` = ADMIN_DOWN `4` = INIT

#### 

 BFD GoBGP BGP hard reset `BFD is down`  BGP 

### openbfdd

 BFD  `kube-ovn-speaker`  openbfdd  BFD DaemonSet node host network  BGP  BFD  `BFD_MULTI * max(BFD_MIN_TX, BFD_MIN_RX) = 3 * 1000ms = 3` 

DaemonSet  `hostNetwork: true`  Pod

- **init-peer**:  `ip route get`  IP Volume
- **bfdd**: openbfdd  init  IP  BFD  `start-bfdd.sh`  `bfdd-prestart.sh`  startupProbe
- **reconcile**:  5  BFD  peer

 yaml

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/yamls/bfdd-daemonset.yaml
```

 `GATEWAY_ADDRESS`  `ALLOWED_SOURCE_ADDRESSES`  Speaker  `--allowed-source-addresses`

```
env:
  - name: GATEWAY_ADDRESS
    value: "10.32.32.1"                          #
  - name: ALLOWED_SOURCE_ADDRESSES
    value: "10.32.32.2,10.32.32.3,10.32.32.4,10.32.32.5"  #   IP
  - name: BFD_MIN_TX
    value: "1000"                                #
  - name: BFD_MIN_RX
    value: "1000"                                #
  - name: BFD_MULTI
    value: "3"                                   #
```

 DaemonSet

```
kubectl apply -f bfdd-daemonset.yaml
```

#### BFD 

```
#    BFD daemon
bfdd-control status

#      BFD    remote=  , local=    IP
bfdd-control status remote <gateway-ip> local <local-ip>

#    BFD
bfdd-control allow <gateway-ip>

#
bfdd-control session new set mintx <ms> ms   #
bfdd-control session new set minrx <ms> ms   #
bfdd-control session new set multi <n>       #

#
bfdd-control log type command no

#    init        IP
cat /bfdd-peer/local-ip

#    reconcile sidecar
kubectl logs -n kube-system ds/openbfdd -c reconcile
```

```
#    init
kubectl logs -n kube-system ds/openbfdd -c init-peer
```

**OVN  BFD **

OVN  BFD kube-ovn controller 

```
#    OVN BFD
kubectl ko nbctl list bfd

#           BFD
kubectl ko nbctl find bfd logical_port=<lrp-name>

#    OVN BFD
kubectl ko nbctl destroy bfd <uuid>
```

## 7.9.8 BGP routes debug

```
# show peer neighbor
gobgp neighbor

# show announced routes to one peer
gobgp neighbor 10.32.32.254 adj-out
```

PDF Slack Support

2026-7-3

2022-6-14



GitHub 



## 7.9.9 

# 7.10 □□ MetalLB □ Kube-OVN Underlay □□

MetalLB □□□□ Kubernetes □□□□□□□□□□□□□□□□□□□□□□ Kubernetes □□□□ Service □□□□□□□□□□□□□□□

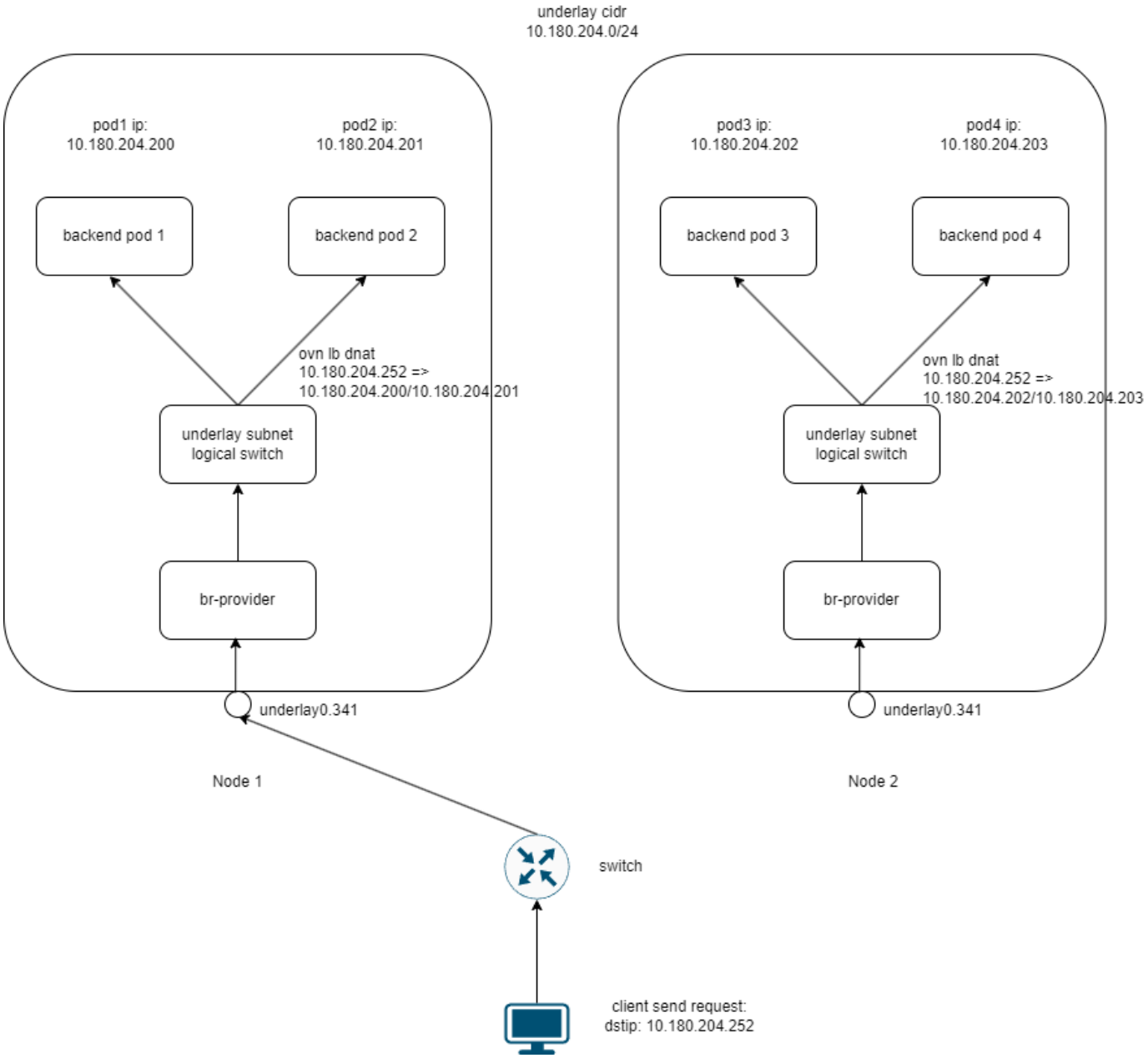
□□□□□□□□ MetalLB □ Kube-OVN □ Underlay □□□□□□□□

## 7.10.1 □□□□

Kube-OVN □□□ 1.14.0 □□□□ MetalLB □ Underlay □□□□□□□□□□□□□□□□□□

- □□□□□□□□□□□ MetalLB □□□ IP □□□□□□□□□□□□
- □□□□□ Pod □ MetalLB VIP □□□□ Underlay □□□
- □□□□□□ IP □□□□□□□□□□ SNAT

## 7.10.2 □□□□






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□ *1□MetalLB VIP □ Kube-OVN Underlay □□□□□□□□□*

MetalLB □ Kube-OVN Underlay □□□□□□□□□□

1. □□□□□□□□□□□□ VIP□□□ 10.180.204.252□□□ IP □ MetalLB □□ L2 □□□□□□□□□□□□ Node1 □□ metallb □ VIP
2. □□□□□□□□□□□□ VIP □□□□□□□□□□ `underlay0.341` □□□□
3. □□□□□□□□ `br-provider` □□□□□ Underlay □□□□□□
4. `br-provider` □□ OpenFlow □□□□□□□□□□ OVN □□□□□
5. □□□□ `underlay subnet` □□□□□□□ OVN □□□□□□ `ovn lb dnat` □□□
6. OVN □□□□□□□□□□□□□□□□ Pod□

□□□□□□□ 10.180.204.0/24□□□ VIP □□□ Pod □ IP □□□□□□□□□□

## 7.10.3 □□□□

- Kube-OVN □□□□□ `--enable-ovn-lb-prefer-local=true` □□
- Underlay □□□□ `enableExternalLBAddress=true`
- □ Underlay □□□ `excludeIps` □□□ MetalLB □□□□□□ IP □□

## 7.10.4 □□□□

**1. □□ Kube-OVN**

□□□□□□□□ Kube-OVN□□□ Kube-OVN □□□□□ `--enable-ovn-lb-prefer-local=true` □ `--ls-ct-skip-dst-lport-ips=false` □□□

```
#   kube-ovn-controller   Deployment
kubectl edit deployment -n kube-system kube-ovn-controller
```

□□□□□□□□□□□

```
--enable-ovn-lb-prefer-local=true
--ls-ct-skip-dst-lport-ips=false
```

**2. □□ Underlay □□**

□□□□□ Underlay □□□□□□□ LoadBalancer □□□□□□□ excludeIps □□□ MetalLB □□□□ IP □□□

```
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: underlay-subnet
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 10.180.204.0/24  #
  gateway: 10.180.204.1
  excludeIps:
  - 10.180.204.250
  - 10.180.204.251
  - 10.180.204.252  # MetalLB            VIP 10.180.204.252
  natOutgoing: false
  enableExternalLBAddress: true   #     subnet   cidr    ip      metallb vip
```

**3. □□ MetalLB**

□□ MetalLB □□□□ □□ MetalLB□

```
kubectl apply -f https://raw.githubusercontent.com/metallb/metallb/v0.13.7/config/manifests/metallb-native.yaml
```

□□ MetalLB □□□□□ L2 □□□□□

```
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
```

```
  name: underlay-pool
  namespace: metallb-system
spec:
  addresses:
  - 10.180.204.250-10.180.204.254  #      VIP 10.180.204.252
---
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  name: l2-advert
  namespace: metallb-system
spec:
  ipAddressPools:
  - underlay-pool
```

**4. □□ LoadBalancer □□□ Service**

□□□□ LoadBalancer □□□ Service□□□□□□ Underlay □□□□ Pod□

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: nginx
  name: deploy-169402624
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      annotations:
        ovn.kubernetes.io/logical_switch: underlay-subnet
      labels:
        app: nginx
    spec:
      containers:
      - args:
        - netexec
        - --http-port
        - "80"
        image: kubeovn/agnhost:2.47
        imagePullPolicy: IfNotPresent
        name: nginx
---
apiVersion: v1
kind: Service
metadata:
  name: nginx-lb
spec:
  externalTrafficPolicy: Local
  ipFamilies:
  - IPv4
  ipFamilyPolicy: PreferDualStack
  ports:
  - port: 80
    protocol: TCP
    targetPort: 80
  selector:
    app: nginx
  type: LoadBalancer
```

## 7.10.5 □□□□□

1. □□ Service □□□□ MetalLB □□□ IP □□□

```
kubectl get svc nginx-lb
```

□□□□ EXTERNAL-IP □□□□□□ IP □□□□ 10.180.204.252□□

1. □□□□□ Service □ IP □□□

```
curl http://10.180.204.252
```

1. □□□□□□□□□□□□□□□ Pod□

□□□□□□ Service □ endpoints □ Pod □□□□□□□□□□□□□□□□□□□□

```
#    Service   endpoints
kubectl get endpoints nginx-lb
```



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```
#    Pod
kubectl get pods -l app=nginx -o wide
```

1. □□□□□ IP □□□□□

□ nginx Pod □□□□□□□□□□□□□□□□□□□ IP □□□□□□□□□ IP□□□□□□ SNAT □□ IP□

```
kubectl exec -it $(kubectl get pods -l app=nginx -o name | head -n1) -- cat /var/log/nginx/access.log
```

## 7.10.6 □□□□

**IP □□□□□**

MetalLB □□□□□□□□□ Underlay □□ CIDR □□□□□□□□□ Underlay □□ `excludeIps` □□□□□□□□□□□ IP □□□□□

**□□□□□□**

MetalLB □□□□□ Kube-OVN Underlay □□□□□□□□□□□□□□□ `underlay0.341` □□□□□□□□□ VLAN □□□□□□□□□□□□□□□□ VLAN □□□ ARP □□□□□□□□□ MetalLB VIP□

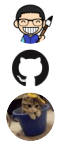
**□□□□□□**

□□□□□□□□□□□□□□□□□□□□ - Kube-OVN □□□□ `--enable-ovn-lb-prefer-local=true` □□ - Service □□□□□ `externalTrafficPolicy: Local`

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## 7.10.7 □□



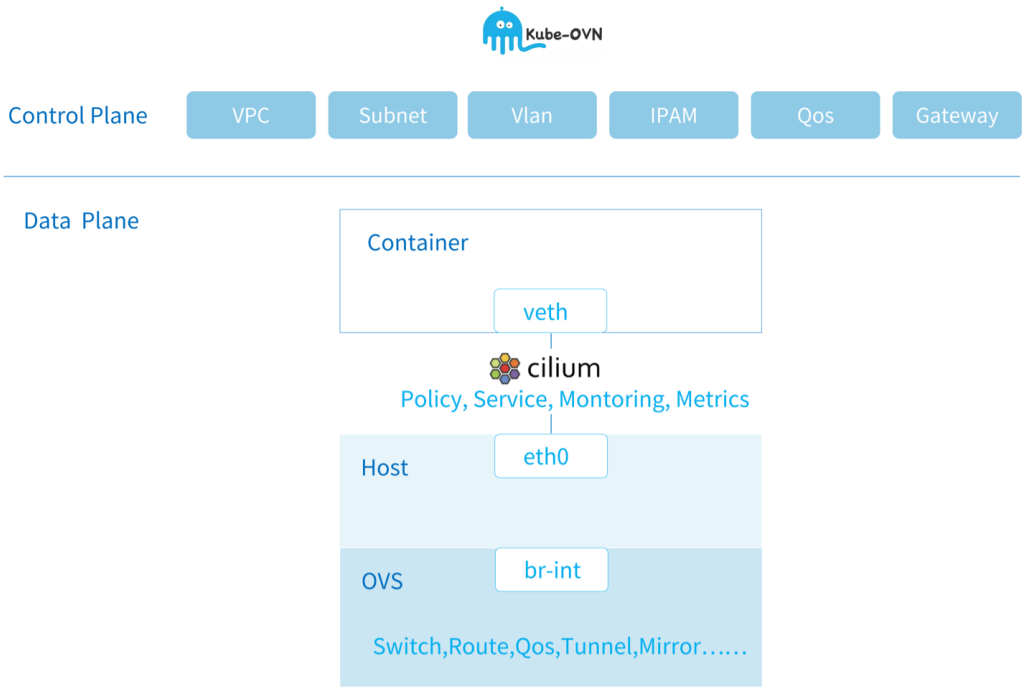
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# 7.11 Cilium ▯▯

Cilium ▯▯▯▯▯ eBPF ▯▯▯▯▯▯▯▯▯Kube-OVN ▯▯▯▯▯ CNI Chaining ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯ Kube-OVN ▯▯▯▯▯▯▯▯▯▯▯ eBPF ▯▯▯▯▯▯▯▯▯▯▯▯▯

▯▯▯▯ Cilium▯Kube-OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯

- ▯▯▯▯▯▯▯▯▯▯▯▯▯
- ▯▯ Hubble ▯▯▯▯▯▯▯




## 7.11.1 ▯▯▯▯▯

1. Linux ▯▯▯▯▯▯▯ 4.19 ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ eBPF ▯▯▯▯▯▯
2. ▯▯▯▯▯ Helm ▯▯▯▯ Cilium ▯▯▯▯▯▯▯ Helm ▯▯▯▯ Installing Helm▯

## 7.11.2 ▯▯ Kube-OVN

▯▯▯▯▯▯▯ Cilium ▯▯▯▯▯▯▯▯▯▯▯▯ Kube-OVN ▯▯ `networkpolicy` ▯▯▯▯▯▯▯ CNI ▯▯▯▯▯▯▯

▯ `install.sh` ▯▯▯▯▯▯▯▯▯▯▯▯

```
ENABLE_NP=false
CNI_CONFIG_PRIORITY=10
```

▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `kube-ovn-controller` ▯▯▯▯▯▯▯▯▯▯ `networkpolicy` ▯

```
args:
- --enable-np=false
```

▯▯ `kube-ovn-cni` ▯▯▯▯▯▯▯ CNI ▯▯▯▯▯▯▯

```
args:
- --cni-conf-name=10-kube-ovn.conflist
```



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▯▯▯▯▯▯▯ Kube-OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Cilium ▯▯▯▯▯▯

```
mv /etc/cni/net.d/01-kube-ovn.conflist /etc/cni/net.d/10-kube-ovn.conflist
```

## 7.11.3 ▯▯ Cilium

▯▯ `chaining.yaml` ▯▯▯▯▯▯▯▯ Cilium ▯ generic-veth ▯▯▯

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: cni-configuration
  namespace: kube-system
data:
  cni-config: |-
    {
      "name": "generic-veth",
      "cniVersion": "0.3.1",
      "plugins": [
        {
          "type": "kube-ovn",
          "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
          "ipam": {
              "type": "kube-ovn",
              "server_socket": "/run/openvswitch/kube-ovn-daemon.sock"
          }
        },
        {
          "type": "portmap",
          "snat": true,
          "capabilities": {"portMappings": true}
        },
        {
          "type": "cilium-cni"
        }
      ]
    }
```

▯▯▯▯▯▯▯▯

```
kubectl apply -f chaining.yaml
```

▯▯ Helm ▯▯ Cilium▯

```
helm repo add cilium https://helm.cilium.io/
helm install cilium cilium/cilium --version 1.11.6 \
    --namespace kube-system \
    --set cni.chainingMode=generic-veth \
    --set cni.customConf=true \
    --set cni.configMap=cni-configuration \
    --set tunnel=disabled \
    --set enableIPv4Masquerade=false \
    --set devices="eth+ ovn0 genev_sys_6081 vxlan_sys_4789" \
    --set enableIdentityMark=false
```

▯▯ Cilium ▯▯▯▯▯

```
# cilium  status
    /¯¯\
 /¯¯\__/¯¯\    Cilium:         OK
 \__/¯¯\__/    Operator:       OK
 /¯¯\__/¯¯\    Hubble:         disabled
 \__/¯¯\__/    ClusterMesh:    disabled
    \__/

DaemonSet         cilium             Desired: 2, Ready: 2/2, Available: 2/2
Deployment        cilium-operator    Desired: 2, Ready: 2/2, Available: 2/2
Containers:       cilium             Running: 2
                  cilium-operator    Running: 2
Cluster Pods:     8/11 managed by Cilium
Image versions    cilium             quay.io/cilium/cilium:v1.11.6: 2
                  cilium-operator    quay.io/cilium/operator-generic:v1.11.6: 2
```

## 7.11.4 ▯ Cilium ▯▯▯▯▯ Kube-OVN NAT ▯▯

Cilium ▯▯▯▯▯▯▯▯▯▯▯▯▯ NAT ▯▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯ NAT ▯▯▯▯▯ VPC ▯ Pod ▯▯▯▯▯ SNAT/DNAT ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ IP ▯▯ Pod ▯▯▯ IP▯

Cilium ▯▯ **SIP**▯Source IP▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯——▯▯▯▯▯▯▯ Pod IP ▯▯▯▯▯▯▯▯▯▯▯▯ Hubble ▯▯▯ **DROPPED**▯▯



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▒▒▒▒▒▒▒▒▒▒▒▒ IP ▒▒▒▒▒▒▒▒ NAT ▒▒▒▒▒▒▒▒

▒▒▒▒▒▒**/Pod ▒▒▒▒▒▒▒▒▒Cilium ≥ 1.20**▒

Cilium 1.20 ▒▒cilium/cilium#43505▒▒ ▒▒▒▒▒▒▒ **annotation** ▒ Pod ▒▒▒▒▒ SIP ▒▒▒▒

1. ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ annotation▒

```
kubectl annotate namespace <nat-gateway-namespace> \
  config.cilium.io/delegate-source-ip-verification=true
```

2. NAT ▒▒ Pod ▒▒▒▒▒ SIP ▒▒▒▒ annotation▒

```
apiVersion: v1
kind: Pod
metadata:
  annotations:
    config.cilium.io/disable-source-ip-verification: "true"
```

▒▒▒▒▒▒▒▒▒▒ NAT ▒▒ Pod ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

▒▒▒▒▒▒**Cilium < 1.20**▒

▒▒▒▒▒▒ Cilium▒▒▒ Helm Chart ▒▒▒▒▒▒ `enableSourceIPVerification` ▒ `false` ▒▒▒▒▒ SIP ▒▒▒▒

```
enableSourceIPVerification: false
```

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GitHub

# 7.12 Cilium NetworkPolicy ⬚⬚

Kube-OVN ⬚⬚⬚⬚⬚⬚ Cilium ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ Cilium ⬚⬚⬚

⬚⬚⬚ Cilium ⬚⬚⬚⬚⬚⬚⬚ Cilium ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ Cilium L3 ⬚ L4 ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

## 7.12.1 ⬚⬚⬚⬚

⬚⬚⬚⬚ **Pod**

⬚⬚ namespace `test` ⬚⬚⬚⬚⬚ yaml⬚⬚ test namespace ⬚⬚⬚⬚⬚ label `app=test` ⬚ Pod⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ Pod⬚

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: test
  name: test
  namespace: test
spec:
  replicas: 1
  selector:
    matchLabels:
      app: test
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      labels:
        app: test
    spec:
      containers:
      - image: docker.io/library/nginx:alpine
        imagePullPolicy: IfNotPresent
        name: nginx
```

⬚⬚⬚⬚⬚⬚ yaml⬚⬚ default namespace ⬚⬚⬚⬚⬚ label `app=dynamic` ⬚ Pod ⬚⬚⬚⬚⬚⬚⬚⬚ Pod⬚

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: dynamic
  name: dynamic
  namespace: default
spec:
  replicas: 2
  selector:
    matchLabels:
      app: dynamic
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: dynamic
    spec:
      containers:
      - image: docker.io/library/nginx:alpine
        imagePullPolicy: IfNotPresent
        name: nginx
```

⬚⬚⬚⬚ Pod ⬚⬚ Label ⬚⬚:

```
# kubectl get pod -o wide --show-labels
NAME                       READY   STATUS    RESTARTS   AGE   IP           NODE                   NOMINATED NODE   READINESS GATES   LABELS
dynamic-7d8d7874f5-9v5c4   1/1     Running   0          28h   10.16.0.35   kube-ovn-worker        <none>           <none>            app=dynamic,pod-
template-hash=7d8d7874f5
dynamic-7d8d7874f5-s8z2n   1/1     Running   0          28h   10.16.0.36   kube-ovn-control-plane   <none>           <none>            app=dynamic,pod-
template-hash=7d8d7874f5
# kubectl get pod -o wide -n test --show-labels
NAME                       READY   STATUS    RESTARTS   AGE     IP           NODE                   NOMINATED NODE   READINESS GATES   LABELS
dynamic-7d8d7874f5-6dsg6   1/1     Running   0          7h20m   10.16.0.2    kube-ovn-control-plane   <none>           <none>            app=dynamic,pod-
template-hash=7d8d7874f5
```



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```
dynamic-7d8d7874f5-tjgtp       1/1     Running   0          7h46m   10.16.0.42   kube-ovn-worker          <none>           <none>           app=dynamic,pod-
template-hash=7d8d7874f5
label-test1-77b6764857-swq4k   1/1     Running   0          3h43m   10.16.0.12   kube-ovn-worker          <none>           <none>           app=test1,pod-
template-hash=77b6764857

#          Pod
test-54c98bc466-mft5s          1/1     Running   0          8h      10.16.0.41   kube-ovn-worker          <none>           <none>           app=test,pod-
template-hash=54c98bc466
```

**L3 □□□□□□**

□□□□ yaml□□□ `CiliumNetworkPolicy` □□:

```yaml
apiVersion: "cilium.io/v2"
kind: CiliumNetworkPolicy
metadata:
  name: "l3-rule"
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
  - fromEndpoints:
    - matchLabels:
        app: dynamic
```

□ default namespace □□□□ Pod □□□□□□□ Pod □□□□□□□□□□□ □□□ test namespace □□□□□□□ Pod □□□□□□□□□

default namespace □□□□□:

```
# kubectl exec -it dynamic-7d8d7874f5-9v5c4 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
```

test namespace □ Pod □□□□□□□□:

```
# kubectl exec -it -n test dynamic-7d8d7874f5-6dsg6 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes
64 bytes from 10.16.0.41: seq=0 ttl=64 time=2.558 ms
64 bytes from 10.16.0.41: seq=1 ttl=64 time=0.223 ms
64 bytes from 10.16.0.41: seq=2 ttl=64 time=0.304 ms

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.223/1.028/2.558 ms
```

□□ Cilium □□□□□□□ `CiliumNetworkPolicy` □□□□□□□□□ `Namespace` □□□□□□□□□□□ Cilium □□□

□□□□□□□□□□□□□□□□□□□□□ **Namespace** □ Pod □□□□□□□□□□□□□□□□□□□□□ **Namespace** □ Pod □□□□□

□□□□□□ Namespace □□□□□□□□□□□□□□□ Namespace □□□

□□□□□□□ `CiliumNetworkPolicy` □□□□□ namespace □□:

```yaml
  ingress:
  - fromEndpoints:
    - matchLabels:
        app: dynamic
        k8s:io.kubernetes.pod.namespace: default    #      Namespace    Pod
```

□□□□□□ `CiliumNetworkPolicy` □□□□:

```
# kubectl get cnp -n test  -o yaml l3-rule
apiVersion: cilium.io/v2
kind: CiliumNetworkPolicy
metadata:
  name: l3-rule
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
  - fromEndpoints:
    - matchLabels:
```

```
        app: dynamic
    - matchLabels:
        app: dynamic
        k8s:io.kubernetes.pod.namespace: default
```

... default namespace ... Pod ... Pod ...:

```
# kubectl exec -it dynamic-7d8d7874f5-9v5c4 -n test -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes
64 bytes from 10.16.0.41: seq=0 ttl=64 time=2.383 ms
64 bytes from 10.16.0.41: seq=1 ttl=64 time=0.115 ms
64 bytes from 10.16.0.41: seq=2 ttl=64 time=0.142 ms

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.115/0.880/2.383 ms
```

... Kubernetes ... networkpolicy... Cilium ... Namespace ... Namespace ...

... Kube-OVN ...Kube-OVN ... k8s ... Namespace ... **Pod**... Pod... Namespace ... Namespace ... Pod... Pod ...

### L4 ...

... yaml... L4 ...:

```
apiVersion: "cilium.io/v2"
kind: CiliumNetworkPolicy
metadata:
  name: "l4-rule"
  namespace: test
spec:
  endpointSelector:
    matchLabels:
      app: test
  ingress:
  - fromEndpoints:
    - matchLabels:
        app: dynamic
    toPorts:
    - ports:
      - port: "80"
        protocol: TCP
```

... Namespace ... Pod ...

```
# kubectl exec -it -n test dynamic-7d8d7874f5-6dsg6 -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
bash-5.0#
bash-5.0# curl 10.16.0.41:80
<html>
<head>
        <title>Hello World!</title>
        <link href='//fonts.googleapis.com/css?family=Open+Sans:400,700' rel='stylesheet' type='text/css'>
        <style>
        body {
                background-color: white;
                text-align: center;
                padding: 50px;
                font-family: "Open Sans","Helvetica Neue",Helvetica,Arial,sans-serif;
        }
        #logo {
                margin-bottom: 40px;
        }
        </style>
</head>
<body>
                <h1>Hello World!</h1>
                                <h3>Links found</h3>
        <h3>I am on  test-54c98bc466-mft5s</h3>
        <h3>Cookie                  =</h3>
                                    <b>KUBERNETES</b> listening in 443 available at tcp://10.96.0.1:443<br />
                                            <h3>my name is hanhouchao!</h3>
                        <h3> RequestURI='/'</h3>
</body>
</html>
```

▯▯ Namespace ▯▯▯▯▯▯▯▯▯▯▯▯ Pod ▯▯▯▯

```
# kubectl exec -it -n test label-test1-77b6764857-swq4k -- bash
bash-5.0# ping -c 3 10.16.0.41
PING 10.16.0.41 (10.16.0.41): 56 data bytes

--- 10.16.0.41 ping statistics ---
3 packets transmitted, 0 packets received, 100% packet loss
bash-5.0#
bash-5.0# curl -v 10.16.0.41:80 --connect-timeout 10
*   Trying 10.16.0.41:80...
* After 10000ms connect time, move on!
* connect to 10.16.0.41 port 80 failed: Operation timed out
* Connection timeout after 10001 ms
* Closing connection 0
curl: (28) Connection timeout after 10001 ms
```

▯▯▯▯▯▯▯▯▯ Namespace ▯▯▯▯▯▯▯▯▯▯▯▯▯ L3 ▯▯▯▯▯▯▯▯▯▯▯▯

▯ L4 ▯▯▯▯▯▯▯▯▯ping ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ TCP ▯▯▯▯▯▯▯▯▯▯▯▯▯

▯▯ ICMP ▯▯▯▯▯▯▯▯▯▯▯▯▯ L4 ▯▯▯▯▯

**L7 ▯▯▯▯▯▯▯**

chaining ▯▯▯▯L7 ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Cilium ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Generic Veth Chaining▯

▯▯▯▯▯▯ issue 12454 ▯▯▯▯▯▯▯▯▯▯▯▯

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GitHub



# 7.13 Cilium ■■■■■■

Kube-OVN ■■■■■■■ Cilium ■■■■■■■■■■■■ Cilium ■■■

Cilium ■■■■■■■■■■■■■■■■■■■■■■■■■■■■ Hubble ■■■■Hubble ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

## 7.13.1 ■■ Hubble

■■■ Cilium ■■■■■■■■■■■■ Hubble ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ Hubble■

■■■■■■■■■■ helm ■■ Hubble■

```
helm upgrade cilium cilium/cilium --version 1.11.6 \
   --namespace kube-system \
   --reuse-values \
   --set hubble.relay.enabled=true \
   --set hubble.ui.enabled=true
```

■■■■ Hubble ■■■■■■ `cilium status` ■■■■■■■■■■ Hubble ■■■■■■

```
# cilium status
    /¯¯\
 /¯¯\__/¯¯\    Cilium:         OK
 \__/¯¯\__/    Operator:       OK
 /¯¯\__/¯¯\    Hubble:         OK
 \__/¯¯\__/    ClusterMesh:    disabled
    \__/

Deployment        hubble-relay       Desired: 1, Ready: 1/1, Available: 1/1
Deployment        cilium-operator    Desired: 2, Ready: 2/2, Available: 2/2
DaemonSet         cilium             Desired: 2, Ready: 2/2, Available: 2/2
Deployment        hubble-ui          Desired: 1, Ready: 1/1, Available: 1/1
Containers:       cilium             Running: 2
                  hubble-ui          Running: 1
                  hubble-relay       Running: 1
                  cilium-operator    Running: 2
Cluster Pods:     16/17 managed by Cilium
Image versions    hubble-relay       quay.io/cilium/hubble-relay:v1.11.6@sha256:fd9034a2d04d5b973f1e8ed44f230ea195b89c37955ff32e34e5aa68f3ed675a: 1
                  cilium-operator    quay.io/cilium/operator-generic:v1.11.6@sha256:9f6063c7bcaede801a39315ec7c166309f6a6783e98665f6693939cf1701bc17: 2
                  cilium             quay.io/cilium/cilium:v1.11.6@sha256:f7f93c26739b6641a3fa3d76b1e1605b15989f25d06625260099e01c8243f54c: 2
                  hubble-ui          quay.io/cilium/hubble-ui:v0.9.0@sha256:0ef04e9a29212925da6bdfd0ba5b581765e41a01f1cc30563cef9b30b457fea0: 1
                  hubble-ui          quay.io/cilium/hubble-ui-backend:v0.9.0@sha256:000df6b76719f607a9edefb9af94dfd1811a6f1b6a8a9c537cba90bf12df474b: 1
apple@bogon cilium %
```

■■ Hubble ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■ Hubble CLI :

```
curl -L --fail --remote-name-all https://github.com/cilium/hubble/releases/download/v0.10.0/hubble-linux-amd64.tar.gz
sudo tar xzvfC hubble-linux-amd64.tar.gz /usr/local/bin
```

## 7.13.2 ■■■■■■■

Cilium ■■■■■■■■■■■■■■■■■■■

```
replicaset.apps/client-7df6cfbf7b           1         1         1       21s
replicaset.apps/client2-547996d7d8          1         1         1       21s
replicaset.apps/echo-other-node-d79544ccf   1         1         1       21s
replicaset.apps/echo-same-node-5d466d5444   1         1         1       21s
```

## 7.13.3 □□□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□□□□ Cilium □□□□□□□□□□ □□□ `kube-system namespace` □□ Cilium □□ pod □□□ `hubble observe` □□□□□□□□□□□□□□□□□

```
# kubectl get pod -n kube-system -o wide
NAME                                      READY   STATUS    RESTARTS   AGE     IP           NODE                   NOMINATED NODE   READINESS GATES
cilium-d6h56                              1/1     Running   0          2d20h   172.18.0.2   kube-ovn-worker        <none>           <none>
cilium-operator-5887f78bbb-c7sb2          1/1     Running   0          2d20h   172.18.0.2   kube-ovn-worker        <none>           <none>
cilium-operator-5887f78bbb-wj8gt          1/1     Running   0          2d20h   172.18.0.3   kube-ovn-control-plane <none>           <none>
cilium-tq5xb                              1/1     Running   0          2d20h   172.18.0.3   kube-ovn-control-plane <none>           <none>
kube-ovn-pinger-7lgk8                     1/1     Running   0          21h     10.16.0.19   kube-ovn-control-plane <none>           <none>
kube-ovn-pinger-msvcn                     1/1     Running   0          21h     10.16.0.18   kube-ovn-worker        <none>           <none>

# kubectl exec -it -n kube-system cilium-d6h56 -- bash
root@kube-ovn-worker:/home/cilium# hubble observe --from-namespace kube-system
Jul 29 03:24:25.551: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.561: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: RST)
Jul 29 03:24:25.561: kube-system/kube-ovn-pinger-msvcn:35576 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, RST)
Jul 29 03:24:25.572: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: SYN)
Jul 29 03:24:25.572: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.651: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.661: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: RST)
Jul 29 03:24:25.661: kube-system/kube-ovn-pinger-msvcn:35578 -> 172.18.0.3:6642 to-stack FORWARDED (TCP Flags: ACK, RST)
Jul 29 03:24:25.761: kube-system/kube-ovn-pinger-msvcn:52004 -> 172.18.0.3:6443 to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 29 03:24:25.779: kube-system/kube-ovn-pinger-msvcn -> kube-system/kube-ovn-pinger-7lgk8 to-stack FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:25.779: kube-system/kube-ovn-pinger-msvcn <- kube-system/kube-ovn-pinger-7lgk8 to-endpoint FORWARDED (ICMPv4 EchoReply)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 <- kube-system/hubble-relay-959988db5-zc5vv:4245 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 <- kube-system/hubble-relay-959988db5-zc5vv:80 to-endpoint FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 -> kube-system/hubble-relay-959988db5-zc5vv:4245 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.866: kube-system/hubble-ui-7596f7ff6f-7j6f2:55836 -> kube-system/hubble-relay-959988db5-zc5vv:4245 to-endpoint FORWARDED (TCP Flags: ACK)
Jul 29 03:24:25.975: kube-system/kube-ovn-pinger-7lgk8 -> kube-system/kube-ovn-pinger-msvcn to-endpoint FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:25.975: kube-system/kube-ovn-pinger-7lgk8 <- kube-system/kube-ovn-pinger-msvcn to-stack FORWARDED (ICMPv4 EchoReply)
Jul 29 03:24:25.979: kube-system/kube-ovn-pinger-msvcn -> 172.18.0.3 to-stack FORWARDED (ICMPv4 EchoRequest)
Jul 29 03:24:26.037: kube-system/coredns-6d4b75cb6d-lbgjg:36430 -> 172.18.0.3:6443 to-stack FORWARDED (TCP Flags: ACK)
Jul 29 03:24:26.282: kube-system/kube-ovn-pinger-msvcn -> 172.18.0.2 to-stack FORWARDED (ICMPv4 EchoRequest)
```

□□ Hubble Relay □□Hubble □□□□□□□□□□□□□□□□□□□□□□

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□□□□□□□ Hubble API□□□□□□□□□□□□□□□□□□□□ Hubble Service□□□□□ `kubectl port-forward deployment/hubble-relay -n kube-system 4245:4245` □□□□□□□□□□□□□□□□

□□□□□□□□□□□ □□□□□

`kubectl port-forward` □□□□□□□□□□□□□□□□□□□□□□□□□□

□□□□□□□□□□□□□□□□□ `hubble status` □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

```
# hubble status
Healthcheck (via localhost:4245): Ok
Current/Max Flows: 8,190/8,190 (100.00%)
Flows/s: 22.86
Connected Nodes: 2/2
```

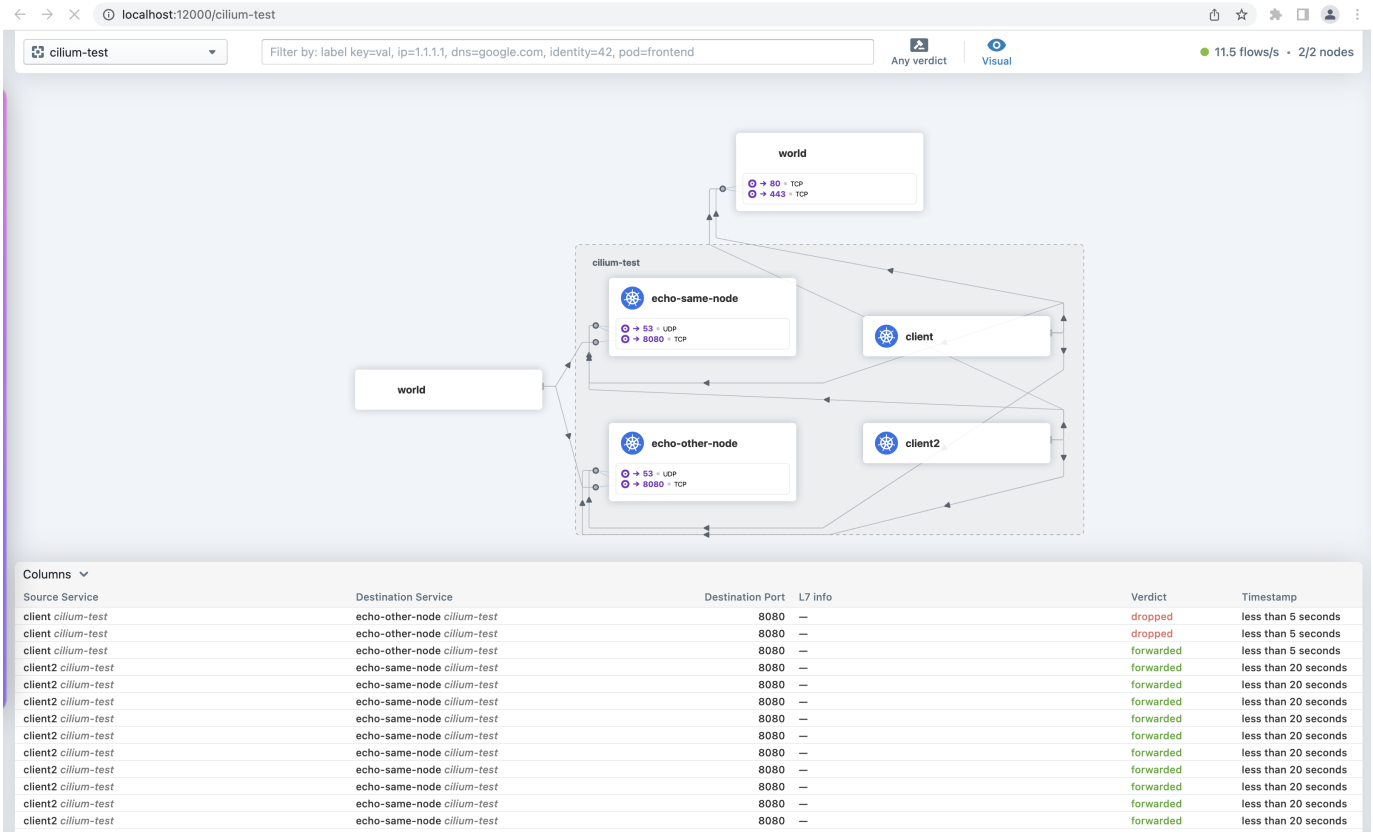
▒▒▒▒▒▒

▒▒▒▒▒▒▒ `hubble observe` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ ▒▒▒▒▒ cilium-test ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

```
apple@bogon ovn-test % hubble observe
Jul 28 08:00:07.033: cilium-test/client-7df6cfbf7b-qn7q6:56906 (ID:15432) -> kube-system/coredns-6d4b75cb6d-b444j:53 (ID:11307) to-endpoint FORWARDED (UDP)
Jul 28 08:00:07.033: cilium-test/client-7df6cfbf7b-qn7q6:56906 (ID:15432) <- kube-system/coredns-6d4b75cb6d-b444j:53 (ID:11307) to-stack FORWARDED (UDP)
Jul 28 08:00:07.095: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.095: 100.64.0.3:43037 (world) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.096: 100.64.0.3:43037 (world) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.097: 100.64.0.3:43037 (world) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.249: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.249: 100.64.0.2:33419 (host) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.250: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.251: 100.64.0.2:33419 (host) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.323: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) <- 172.18.0.3:31514 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.323: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) <- cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) <- 172.18.0.3:31514 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.324: cilium-test/client2-547996d7d8-jgblc:34927 (ID:14804) -> 172.18.0.3:31514 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.324: 100.64.0.2:34927 (world) -> cilium-test/echo-same-node-5d466d5444-4vllm:8080 (ID:15976) to-endpoint FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) -> 172.18.0.2:4244 (host) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:44656 (ID:53347) -> 172.18.0.3:4244 (kube-apiserver) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.340: kube-system/hubble-relay-959988db5-zc5vv:46938 (ID:53347) <- 172.18.0.2:4244 (host) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.341: kube-system/hubble-relay-959988db5-zc5vv:44656 (ID:53347) <- 172.18.0.3:4244 (kube-apiserver) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN)
Jul 28 08:00:07.409: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: SYN, ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK)
Jul 28 08:00:07.410: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.411: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.411: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, PSH)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) -> cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-stack FORWARDED (TCP Flags: ACK, FIN)
Jul 28 08:00:07.412: cilium-test/client-7df6cfbf7b-qn7q6:57326 (ID:15432) <- cilium-test/echo-other-node-d79544ccf-r688b:8080 (ID:50956) to-endpoint FORWARDED (TCP Flags: ACK, FIN)
apple@bogon ovn-test %
```

▒▒▒▒▒▒▒▒ `hubble observe` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

... `cilium-test` namespace... Cilium ...




## 7.13.5 Hubble ...

Hubble ... Pod ... Hubble ...

... `hubble.metrics.enabled` ...:

```
helm upgrade cilium cilium/cilium --version 1.11.6 \
   --namespace kube-system \
   --reuse-values \
   --set hubble.relay.enabled=true \
   --set hubble.ui.enabled=true \
   --set hubble.metrics.enabled="{dns,drop,tcp,flow,icmp,http}"
```

... kube-system namespace ... `hubble-metrics` ... Endpoints ... Hubble ...:

```
# curl 172.18.0.2:9091/metrics
# HELP hubble_drop_total Number of drops
# TYPE hubble_drop_total counter
hubble_drop_total{protocol="ICMPv6",reason="Unsupported L3 protocol"} 2
# HELP hubble_flows_processed_total Total number of flows processed
# TYPE hubble_flows_processed_total counter
hubble_flows_processed_total{protocol="ICMPv4",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 335
hubble_flows_processed_total{protocol="ICMPv4",subtype="to-stack",type="Trace",verdict="FORWARDED"} 335
hubble_flows_processed_total{protocol="ICMPv6",subtype="",type="Drop",verdict="DROPPED"} 2
hubble_flows_processed_total{protocol="TCP",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 8282
hubble_flows_processed_total{protocol="TCP",subtype="to-stack",type="Trace",verdict="FORWARDED"} 6767
hubble_flows_processed_total{protocol="UDP",subtype="to-endpoint",type="Trace",verdict="FORWARDED"} 1642
hubble_flows_processed_total{protocol="UDP",subtype="to-stack",type="Trace",verdict="FORWARDED"} 1642
# HELP hubble_icmp_total Number of ICMP messages
# TYPE hubble_icmp_total counter
hubble_icmp_total{family="IPv4",type="EchoReply"} 335
hubble_icmp_total{family="IPv4",type="EchoRequest"} 335
hubble_icmp_total{family="IPv4",type="RouterSolicitation"} 2
# HELP hubble_tcp_flags_total TCP flag occurrences
# TYPE hubble_tcp_flags_total counter
hubble_tcp_flags_total{family="IPv4",flag="FIN"} 2043
hubble_tcp_flags_total{family="IPv4",flag="RST"} 301
hubble_tcp_flags_total{family="IPv4",flag="SYN"} 1169
hubble_tcp_flags_total{family="IPv4",flag="SYN-ACK"} 1169
```

PDF Slack Support

2023⬚3⬚26⬚

2022⬚8⬚2⬚

GitHub 

7.13.6 ⬚⬚



2025 Kube-OVN Team

# 7.14 ▯▯▯▯▯▯▯

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## 7.14.1 ▯▯▯▯

```yaml
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: external
spec:
  cidrBlock: 172.31.0.0/16
  gatewayType: centralized
  natOutgoing: false
  externalEgressGateway: 192.168.0.1
  policyRoutingTableID: 1000
  policyRoutingPriority: 1500
```

- `natOutgoing`：▯▯▯▯▯ `false` ▯
- `externalEgressGateway` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯
- `policyRoutingTableID` ▯▯▯▯▯▯▯▯▯▯▯▯▯ TableID ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯
- `policyRoutingPriority` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

# 7.15 VIP ▯▯ IP

VIP ▯▯▯ IP▯▯▯▯▯ IP ▯▯▯▯▯▯▯▯ VIP ▯▯▯ kube-ovn ▯ IP ▯ POD ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ IP ▯▯▯▯▯▯ IP ▯▯▯▯ VIP ▯▯▯▯▯▯▯ Openstack neutron Allowed-Address-Pairs ▯AAP▯ ▯▯▯▯▯▯▯▯ Openstack octavia ▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯POD▯IP▯▯▯▯▯▯▯▯ aliyun terway ▯▯▯ ▯▯▯▯▯▯ neutron ▯▯▯ IP ▯▯▯▯▯▯▯▯ VIP ▯▯▯▯▯▯▯▯▯▯▯▯ VIP ▯▯▯▯▯▯▯▯ POD ▯▯ IP▯▯▯▯▯▯▯▯▯▯▯▯ VIP ▯ IP ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
# kubectl get vip
NAME             V4IP         PV4IP   MAC                PMAC   V6IP   PV6IP   SUBNET        READY
static-vip01   10.16.0.121           00:00:00:F0:DB:26                       ovn-default   true
```

[illegible] VIP [illegible] IP [illegible]

### 1.3 Pod [illegible] VIP [illegible] AAP

Pod [illegible] annotation [illegible] VIP [illegible] AAP [illegible]Pod [illegible] labels [illegible] VIP [illegible] `selector` [illegible]

Pod annotation [illegible] VIP[illegible]ovn.kubernetes.io/aaps: vip-aap,vip-aap2,vip-aap3

AAP [illegible][illegible] Pod [illegible]AAP [illegible] Pod [illegible] VIP [illegible] subnet [illegible] Port [illegible]

#### 1.3.1 [illegible] VIP [illegible] AAP

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: vip-aap
spec:
  subnet: ovn-default
  namespace: default
  selector:
    - "app: aap1"
```

VIP [illegible]

- `namespace` : AAP [illegible]VIP [illegible]VIP [illegible] AAP [illegible]
- `selector` : AAP [illegible] VIP [illegible] Pod [illegible] `namespace` [illegible] Pod[illegible] `key: value` [illegible] AND [illegible]

[illegible] VIP [illegible] Port[illegible]

```
# kubectl ko nbctl show ovn-default
switch e32e1d3b-c539-45f4-ab19-be4e33a061f6 (ovn-default)
    port aap-vip
        type: virtual
```

```
apiVersion: v1
kind: Pod
metadata:
  name: busybox
  annotations:
    ovn.kubernetes.io/aaps: vip-aap
  labels:
    app: aap1
spec:
  containers:
    - name: busybox
      image: busybox
      command: ["sleep", "3600"]
      securityContext:
        capabilities:
          add:
            - NET_ADMIN
```

[illegible] AAP [illegible]

```
# kubectl ko nbctl list logical_switch_port aap-vip
_uuid               : cd930750-0533-4f06-a6c0-217ddac73272
addresses           : []
dhcpv4_options      : []
dhcpv6_options      : []
dynamic_addresses   : []
enabled             : []
external_ids        : {ls=ovn-default, vendor=kube-ovn}
ha_chassis_group    : []
mirror_rules        : []
name                : aap-vip
options             : {virtual-ip="10.16.0.100", virtual-parents="busybox.default"}
parent_name         : []
port_security       : []
tag                 : []
tag_request         : []
type                : virtual
up                  : false
```

virtual-ip [illegible] VIP [illegible] IP[illegible]virtual-parents [illegible] AAP [illegible] Pod [illegible] Port[illegible]



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□□□□□□□□ Pod □□□□□□

```
# kubectl exec -it busybox -- ip addr add 10.16.0.100/16 dev eth0
# kubectl exec -it busybox01 -- ip addr show eth0
35: eth0@if36: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 1400 qdisc noqueue
    link/ether 00:00:00:e2:ab:0c brd ff:ff:ff:ff:ff:ff
    inet 10.16.0.7/16 brd 10.16.255.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet 10.16.0.100/16 scope global secondary eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:fee2:ab0c/64 scope link
       valid_lft forever preferred_lft forever
```

□ Pod □□□□□□□□ IP□VIP □ IP □□□□□□□□□□□ subnet □□□□ Pod □□□□□ IP □□□□□

## 7.15.2 2. Switch LB rule vip

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: slr-01
spec:
  subnet: ovn-default
  type: switch_lb_vip
```

- `subnet` : □□□ Subnet □□□ IP□
- `type` : □□□□□□□□□□□□□□□□□□ ipam ip □□□ `switch_lb_vip` □□□ vip □□□ switch lb □□ vip □□□ ip □□□□□□□□□

## 7.15.3 3. Pod □□ VIP □□□ IP

> □□□□ v1.12 □□□□□□□□ v1.17 □□□□□□□□□□□ `ovn.kubernetes.io/vip` □□ Pod IP □□□□□□□ VIP □ AAP □ SwitchLB □□□

□□□□□□□□□□□□□□□□□□ IP □□□□□□□□□□ IPPool □□ Pod annotation□

```
apiVersion: kubeovn.io/v1
kind: Vip
metadata:
  name: pod-use-vip
spec:
  subnet: ovn-default
  type: ""
```

□□□□ annotation □□□ VIP □□□□□ Pod□

```
apiVersion: v1
kind: Pod
metadata:
  name: static-ip
  annotations:
    ovn.kubernetes.io/vip: pod-use-vip #    vip
  namespace: default
spec:
  containers:
    - name: static-ip
      image: docker.io/library/nginx:alpine
```

## 7.15.4 4. StatefulSet □ Kubevirt VM □□ IP

StatefulSet □ VM □ IP □□□□□□□ VIP □□□

□□ `StatefulSet` □ `VM` □□ Pod □□□□□□□□□□□□□□□□□ IP□

VM □□ IP □□□□ `kube-ovn-controller` □ `keep-vm-ip` □□□ `true` □□□□ Kubevirt VM □□□□□□□□□

PDF Slack Support

2026⊠10⊠20⊠

2022⊠5⊠24⊠



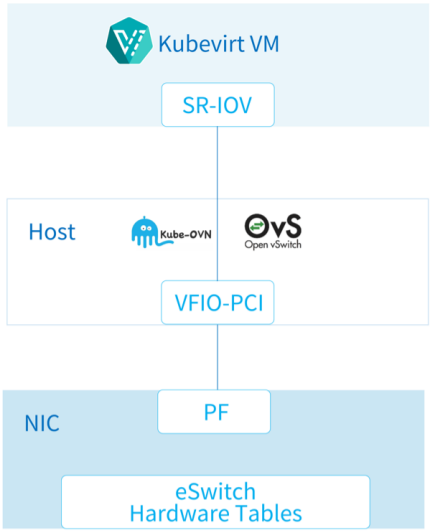
GitHub 

+2

## 7.15.5 ⊠⊠

# 7.16 Mellanox □□ Offload □□

Kube-OVN □□□□□□□□□□ OVS □□□□□□□□□□□□□□□□□□□□□□□□□□□□ CPU □□□□□□□□□□□□□□□□ CPU □□□□□□ □□□□□□□□□□□□Mellanox □ Accelerated Switching And Packet Processing (ASAP²) □□□□□□ OVS □□□□□□□□□□□□□□□□ eSwitch □□□□□□□□□□□□□□□ OVS □□□□□□□□□□□□□□□□□□□□□□□□□□□□ CPU □□□□□□□□□□□□□□□□□□□□□□□□□□




**Note**

□□□□□□□□ 2022 □□□□□□□□□□□□□□□□□□□□

▯▯▯▯ ID ▯▯▯▯▯▯▯

```
# ls -l /sys/class/net/ | grep 84:00.0
lrwxrwxrwx 1 root root 0 Feb 4 16:16 enp132s0f0np0 -> ../../devices/pci0000:80/0000:80:08.0/0000:84:00.0/net/enp132s0f0np0
# ls -l /sys/class/net/ | grep 84:00.1
lrwxrwxrwx 1 root root 0 Feb 4 16:16 enp132s0f1np1 -> ../../devices/pci0000:80/0000:80:08.0/0000:84:00.1/net/enp132s0f1np1
```

▯▯▯▯▯▯▯▯ bond▯

> ▯▯▯▯▯▯ enp132s0f0np0 ▯ enp132s0f1np1 ▯▯ bond1

```
# ip link show enp132s0f0np0 | grep bond
160: enp132s0f0np0: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master bond1 state UP mode DEFAULT group default qlen 1000
# ip link show enp132s0f1np1 | grep bond
169: enp132s0f1np1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master bond1 state UP mode DEFAULT group default qlen 1000
```

▯▯ bond ▯▯▯▯ VF▯

```
ifenslave -d bond1 enp132s0f0np0
ifenslave -d bond1 enp132s0f1np1
echo 0 > /sys/class/net/enp132s0f0np0/device/sriov_numvfs
echo 0 > /sys/class/net/enp132s0f1np1/device/sriov_numvfs
ip link set enp132s0f0np0 down
ip link set enp132s0f1np1 down
```

▯▯▯▯▯▯▯▯▯

> OVS ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯SMFS ▯ DMFS

- SMFS (software-managed flow steering)▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
84:00.5 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.6 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:00.7 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:01.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
84:01.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

▯ VF ▯▯▯▯▯▯▯

```
echo 0000:84:00.2 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.3 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.4 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.5 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.6 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:00.7 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:01.0 > /sys/bus/pci/drivers/mlx5_core/unbind
echo 0000:84:01.1 > /sys/bus/pci/drivers/mlx5_core/unbind
```

▯▯ eSwitch ▯▯▯▯▯▯▯▯▯▯▯▯

```
devlink dev eswitch set pci/0000:84:00.0 mode switchdev
devlink dev eswitch set pci/0000:84:00.1 mode switchdev
ethtool -K enp132s0f0np0 hw-tc-offload on
ethtool -K enp132s0f1np1 hw-tc-offload on
```

SR-IOV VF ▯▯▯▯▯▯▯

SR-IOV VF LAG ▯▯▯▯▯ PF ▯▯ OVS ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ e-switch ▯▯▯▯▯ bond ▯▯▯▯▯

- Active-backup
- XOR
- LACP

> SR-IOV VF LAG ▯▯ LAG ▯▯▯▯▯▯▯▯▯▯bond ▯▯▯▯▯▯▯▯ bond PF ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ VF▯▯▯ VF ▯▯▯▯▯▯ bond ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ PF ▯▯▯▯▯ ▯▯▯▯▯▯ VF ▯▯▯▯▯▯▯▯ PF▯▯ XOR ▯ LACP ▯▯▯▯▯▯▯▯ PF ▯▯▯▯▯▯▯▯▯▯▯▯ VF ▯▯▯▯▯▯▯▯▯▯ PF ▯▯▯▯▯

▯▯▯▯▯▯▯ LACP ▯▯▯▯▯▯▯▯▯▯▯▯

```
modprobe bonding mode=802.3ad
ip link set enp132s0f0np0 master bond1
ip link set enp132s0f1np1 master bond1
ip link set enp132s0f0np0 up
ip link set enp132s0f1np1 up
ip link set bond1 up
```

> ▯▯▯▯▯▯▯▯▯▯ bond▯▯▯▯▯▯▯▯▯▯

▯▯▯▯▯▯▯▯▯▯ VF ▯▯▯

```
echo 0000:84:00.2 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.3 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.4 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.5 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.6 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:00.7 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:01.0 > /sys/bus/pci/drivers/mlx5_core/bind
echo 0000:84:01.1 > /sys/bus/pci/drivers/mlx5_core/bind
```

`NetworkManager` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `NetworkManager` ▯▯▯▯▯▯▯

```
systemctl stop NetworkManager
systemctl disable NetworkManager
```

#### ▯▯ DEVICE PLUGIN

▯▯▯▯▯▯▯ VF ▯▯▯▯▯▯▯▯▯▯▯▯▯ Pod ▯▯▯ VF ▯▯▯▯▯▯▯▯▯▯ SR-IOV Device Plugin ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯

▯▯ SR-IOV ▯▯ Configmap▯

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
```

```
    {
      "resourceList": [{
          "resourcePrefix": "mellanox.com",
          "resourceName": "cx5_sriov_switchdev",
          "selectors": {
                  "vendors": ["15b3"],
                  "devices": ["1018"],
                  "drivers": ["mlx5_core"]
              }
      }
      ]
    }
```

SR-IOV Device Plugin ▯▯▯▯▯▯ ConfigMap ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ConfigMap ▯ name ▯ sriovdp-config▯

- `selectors` : VF ▯▯▯
- `vendors` : ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯
- `devices` : ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯
- `drivers` : ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

`selectors` ▯▯▯▯▯ `pciAddresses` ▯ `acpiIndexes` ▯▯▯▯▯ VF ▯▯▯▯▯▯▯▯▯▯▯▯▯▯SR-IOV ConfigMap ▯▯

▯▯ SR-IOV ▯▯▯▯▯▯

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/v3.6.2/deployments/sriovdp-daemonset.yaml
```

▯▯ SR-IOV ▯▯▯▯▯▯▯▯▯ Kubernetes Node ▯▯

```
kubectl describe node kube-ovn-01  | grep mellanox

mellanox.com/cx5_sriov_switchdev:  8
mellanox.com/cx5_sriov_switchdev:  8
mellanox.com/cx5_sriov_switchdev  0           0
```

### ▯▯ sriov-network-operator ▯▯ SR-IOV ▯ Device Plugin

▯▯ node-feature-discovery▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
kubectl apply -k https://github.com/kubernetes-sigs/node-feature-discovery/deployment/overlays/default?ref=v0.11.3
```

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ offload ▯▯▯▯▯▯▯▯ annotation:

```
kubectl label nodes [offloadNicNode] feature.node.kubernetes.io/network-sriov.capable=true
```

▯▯▯▯▯▯▯▯▯▯ Operator▯

```
git clone --depth=1 https://github.com/kubeovn/sriov-network-operator.git
kubectl apply -k sriov-network-operator/deploy
```

▯▯ Operator ▯▯▯▯▯▯▯▯▯▯

```
# kubectl get -n kube-system all | grep sriov
NAME                                            READY   STATUS    RESTARTS   AGE
pod/sriov-network-config-daemon-bf9nt           1/1     Running   0          8s
pod/sriov-network-operator-54d7545f65-296gb     1/1     Running   0          10s

NAME                                          DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE
SELECTOR                                                  AGE
daemonset.apps/sriov-network-config-daemon    1         1         1       1            1           beta.kubernetes.io/os=linux,feature.node.kubernetes.io/
network-sriov.capable=true   8s

NAME                                       READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/sriov-network-operator     1/1     1            1           10s

NAME                                                  DESIRED   CURRENT   READY   AGE
replicaset.apps/sriov-network-operator-54d7545f65     1         1         1       10s
```

▯▯ `SriovNetworkNodeState` ▯▯▯▯ `node1` ▯▯▯▯▯▯▯▯▯▯▯▯ Mellanox ▯▯▯

```
# kubectl get sriovnetworknodestates.sriovnetwork.openshift.io -n kube-system node1 -o yaml
apiVersion: sriovnetwork.openshift.io/v1
kind: SriovNetworkNodeState
spec: ...
```

```
status:
  interfaces:
  - deviceID: "1017"
    driver: mlx5_core
    mtu: 1500
    pciAddress: "0000:5f:00.0"
    totalvfs: 8
    vendor: "15b3"
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    name: ens41f0np0
  - deviceID: "1017"
    driver: mlx5_core
    mtu: 1500
    pciAddress: "0000:5f:00.1"
    totalvfs: 8
    vendor: "15b3"
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    name: ens41f1np1
```

 `SriovNetworkNodePolicy` `nicSelector`

```
apiVersion: sriovnetwork.openshift.io/v1
kind: SriovNetworkNodePolicy
metadata:
  name: policy
  namespace: kube-system
spec:
  nodeSelector:
    feature.node.kubernetes.io/network-sriov.capable: "true"
  eSwitchMode: switchdev
  numVfs: 3
  nicSelector:
    pfNames:
    - ens41f0np0
    - ens41f1np1
  resourceName: cx_sriov_switchdev
```

 `SriovNetworkNodeState`  `status`

```
# kubectl get sriovnetworknodestates.sriovnetwork.openshift.io -n kube-system node1 -o yaml

...
spec:
  interfaces:
  - eSwitchMode: switchdev
    name: ens41f0np0
    numVfs: 3
    pciAddress: 0000:5f:00.0
    vfGroups:
    - policyName: policy
      vfRange: 0-2
      resourceName: cx_sriov_switchdev
  - eSwitchMode: switchdev
    name: ens41f1np1
    numVfs: 3
    pciAddress: 0000:5f:00.1
    vfGroups:
    - policyName: policy
      vfRange: 0-2
      resourceName: cx_sriov_switchdev
status:
  interfaces
  - Vfs:
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.2
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.3
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.4
      vendor: "15b3"
    deviceID: "1017"
    driver: mlx5_core
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:ab
    mtu: 1500
    name: ens41f0np0
    numVfs: 3
    pciAddress: 0000:5f:00.0
    totalvfs: 3
    vendor: "15b3"
  - Vfs:
```

```
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.5
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.6
      vendor: "15b3"
    - deviceID: 1018
      driver: mlx5_core
      pciAddress: 0000:5f:00.7
      vendor: "15b3"
    deviceID: "1017"
    driver: mlx5_core
    linkSeed: 25000Mb/s
    linkType: ETH
    mac: 08:c0:eb:f4:85:bb
    mtu: 1500
    name: ens41f1np1
    numVfs: 3
    pciAddress: 0000:5f:00.1
    totalvfs: 3
    vendor: "15b3"
```

▯▯ VF ▯▯▯▯

```
# lspci -nn | grep ConnectX
5f:00.0 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
5f:00.1 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5] [15b3:1017]
5f:00.2 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.3 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.4 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.5 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.6 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
5f:00.7 Ethernet controller [0200]: Mellanox Technologies MT27800 Family [ConnectX-5 Virtual Function] [15b3:1018]
```

▯▯ PF ▯▯▯▯▯

```
# cat /sys/class/net/ens41f0np0/compat/devlink/mode
switchdev
```

## 7.16.3 ▯▯ Multus-CNI

SR-IOV Device Plugin ▯▯▯▯▯▯▯▯ ID ▯▯▯▯ Multus-CNI ▯▯▯ Kube-OVN▯▯▯▯▯▯▯ Multus-CNI ▯▯▯▯▯▯▯▯▯

▯▯ Multus-CNI ▯▯▯▯▯▯▯

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/v4.0.2/deployments/multus-daemonset-thick.yml
```

> ▯▯▯multus ▯▯▯ Thin ▯ Thick ▯▯▯▯▯▯▯▯▯▯ SR-IOV ▯▯▯▯▯ Thick ▯▯▯

▯▯ `NetworkAttachmentDefinition` ▯

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: sriov
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: mellanox.com/cx5_sriov_switchdev
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
        {
            "type":"kube-ovn",
            "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
            "provider": "sriov.default.ovn"
        },
        {
            "type":"portmap",
            "capabilities":{
                "portMappings":true
            }
        }
    ]
}'
```

- `provider`：▯▯▯▯▯ `NetworkAttachmentDefinition` ▯ {name}.{namespace}.ovn▯

## 7.16.4 Overlay ⬚⬚

**Kube-OVN ⬚⬚⬚⬚⬚⬚⬚**

⬚⬚⬚⬚⬚⬚⬚

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

⬚⬚⬚⬚⬚⬚⬚ `IFACE` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ IP⬚

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="bond1"
#      SR-IOV   Device Plugin           bond    IFACE     bond1      bond     IFACE     enp132s0f0np0   enp132s0f1np1
```

⬚⬚ Kube-OVN⬚

```
bash install.sh
```

⬚⬚⬚⬚ **VF ⬚⬚⬚ Pod**

⬚⬚⬚⬚⬚⬚ yaml ⬚⬚⬚⬚⬚⬚ VF ⬚⬚⬚⬚⬚⬚⬚⬚⬚ Pod:

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-overlay
  annotations:
    v1.multus-cni.io/default-network: default/sriov
    sriov.default.ovn.kubernetes.io/logical_switch: ovn-default
spec:
  containers:
  - name: nginx-overlay
    image: docker.io/library/nginx:alpine
    resources:
      requests:
        mellanox.com/cx5_sriov_switchdev: '1'
      limits:
        mellanox.com/cx5_sriov_switchdev: '1'
```

- `v1.multus-cni.io/default-network`：⬚⬚⬚⬚⬚⬚ `NetworkAttachmentDefinition` ⬚ {namespace}/{name}⬚
- `sriov.default.ovn.kubernetes.io/logical_switch`：⬚⬚ Pod ⬚⬚⬚⬚⬚⬚⬚⬚ Pod ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

## 7.16.5 Underlay ⬚⬚

**Kube-OVN ⬚⬚⬚⬚⬚⬚⬚**

⬚⬚⬚⬚⬚⬚⬚

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

⬚⬚⬚⬚⬚⬚⬚ `IFACE` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ IP⬚

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE=""
#     Underlay    IFACE         PF      IFACE         K8s                 PF
```

⬚⬚ Kube-OVN⬚

```
bash install.sh
```

⬚⬚⬚⬚ **VF ⬚⬚⬚ Pod**

⬚⬚⬚⬚⬚⬚ yaml ⬚⬚⬚⬚⬚⬚ VF ⬚⬚⬚⬚⬚⬚⬚⬚⬚ Pod:



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```yaml
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: underlay-offload
spec:
  defaultInterface: bond1

---
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan0
spec:
  id: 0
  provider: underlay-offload

---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: vlan0
spec:
  protocol: IPv4
  provider: ovn
  cidrBlock: 10.10.204.0/24
  gateway: 10.10.204.254
  vlan: vlan0
  excludeIps:
  - 10.10.204.1..10.10.10.204.100

---
apiVersion: v1
kind: Pod
metadata:
  name: nginx-underlay
  annotations:
    k8s.v1.cni.cncf.io/networks: '[{
      "name": "sriov",
      "namespace": "default",
      "default-route": ["10.10.204.254"]
    }]'
    sriov.default.ovn.kubernetes.io/logical_switch: vlan0
spec:
  containers:
  - name: nginx-underlay
    image: docker.io/library/nginx:alpine
    resources:
      requests:
        mellanox.com/cx5_sriov_switchdev: '1'
      limits:
        mellanox.com/cx5_sriov_switchdev: '1'
```

- `v1.multus-cni.io/default-network` : ... `NetworkAttachmentDefinition` ... {namespace}/{name}...

> ... multus ... VF ... Pod... VF ... Pod ... VF ... Pod ... multus ...

... yaml ... VF ... Pod:

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-underlay-noVF
  annotations:
    ovn.kubernetes.io/logical_switch: vlan0
spec:
  containers:
  - name: nginx-underlay-noVF
    image: docker.io/library/nginx:alpine
```

... VF ... Pod... ovs-kernel ... e-switch ...

## 7.16.6 ...

... Pod ... `ovs-ovn` ...

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:91cc45de-e7e9-4935-8f82-1890430b0f66, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/
0),in_port(5b45c61b307e_h),packet_type(ns=0/0,id=0/0),eth(src=00:00:00:c5:6d:4e,dst=00:00:00:e7:16:ce),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.
0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no), packets:941539, bytes:62142230, used:0.260s, offloaded:yes, dp:tc, actions:54235e5753b8_h
ufid:e00768d7-e652-4d79-8182-3291d852b791, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/
0),in_port(54235e5753b8_h),packet_type(ns=0/0,id=0/0),eth(src=00:00:00:e7:16:ce,dst=00:00:00:c5:6d:4e),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.
0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no), packets:82386659, bytes:115944854173, used:0.260s, offloaded:yes, dp:tc, actions:5b45c61b307e_h
```

... `offloaded:yes, dp:tc` ...

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2025⬚9⬚10⬚

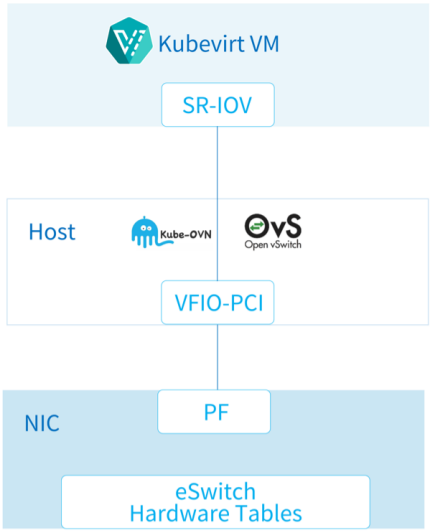
2022⬚5⬚24⬚

GitHub 

7.16.7 ⬚⬚

# 7.17 □□□□□ Offload □□

Kube-OVN □□□□□□□□□□ OVS □□□□□□□□□□□□□□□□□□□□□□□□□□□□ CPU □□□□□□□□□□□□□□□□ CPU □□□□□□ □□□□□□□□□□□□□□□□□ Agilio CX □□□□□□□□□□ OVS □□□□□□□□□□□□□□□□□□□ □□□□□□□ □□□ OVS □□□□□□□□□□□□□□□□□□□□□□□□□□□□ CPU □□□□□□□□□□□□□□□□□□□□□□□□□□□




**Note**

□□□□□□□□ 2022 □□□□□□□□□□□□□□□□□□□□

```
    echo ASSY: defaults to ${DEFAULT_ASSY}
    echo APP: defaults to flower. flower-next is supported if updated
    echo      firmware has been installed.
    exit 1
fi

# It is recommended that the assembly be determined by inspection
# The following code determines the value via the debug interface
if [ "${ASSY}x" = "scanx" ]; then
    ethtool -W ${DEVICE} 0
    DEBUG=$(ethtool -w ${DEVICE} data /dev/stdout | strings)
    SERIAL=$(echo "${DEBUG}" | grep "^SN:")
    ASSY=$(echo ${SERIAL} | grep -oE AMDA[0-9]{4})
fi

PCIADDR=$(basename $(readlink -e /sys/class/net/${DEVICE}/device))
FWDIR="/lib/firmware/netronome"

# AMDA0081 and AMDA0097 uses the same firmware
if [ "${ASSY}" = "AMDA0081" ]; then
    if [ ! -e ${FWDIR}/${APP}/nic_AMDA0081.nffw ]; then
       ln -sf nic_AMDA0097.nffw ${FWDIR}/${APP}/nic_AMDA0081.nffw
   fi
fi

FW="${FWDIR}/pci-${PCIADDR}.nffw"
ln -sf "${APP}/nic_${ASSY}.nffw" "${FW}"

# insert distro-specific initramfs section here...
```

□□□□□□□□□□□□□

```
./agilio-tc-fw-select.sh ens47np0 scan
rmmod nfp
modprobe nfp
```

□□□□ VF □□□□□□ VF□

```
# cat /sys/class/net/ens3/device/sriov_totalvfs
65

# echo 4 > /sys/class/net/ens47/device/sriov_numvfs
```

## 7.17.3 □□ SR-IOV Device Plugin

□□□□□□□ VF □□□□□□□□□□□□ Pod □□□ VF □□□□□□□□□ SR-IOV Device Plugin □□□□□□□□□□□□□□□□□□□□ □□□□□□□

□□ SR-IOV □□ Configmap□

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
      "resourceList": [{
          "resourcePrefix": "corigine.com",
          "resourceName": "agilio_sriov",
          "selectors": {
                  "vendors": ["19ee"],
                  "devices": ["6003"],
                  "drivers": ["nfp_netvf"]
              }
      }
      ]
    }
```

□□ SR-IOV Network Device Plugin □□□□□□□□□□ intel/sriov-network-device-plugin □□□ k8snetworkplumbingwg□□

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/master/deployments/sriovdp-daemonset.yaml
```

□□ SR-IOV □□□□□□□□□ Kubernetes Node □□

```
kubectl describe no containerserver  | grep corigine

corigine.com/agilio_sriov:  4
corigine.com/agilio_sriov:  4
corigine.com/agilio_sriov  0           0
```

## 7.17.4 □□ Multus-CNI

SR-IOV Device Plugin □□□□□□□□ ID □□□□ Multus-CNI □□□ Kube-OVN□□□□□□□ Multus-CNI □□□□□□□□□

□□ Multus-CNI □□□□□□□

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/master/deployments/multus-daemonset.yml
```

□□ `NetworkAttachmentDefinition` □

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: default
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: corigine.com/agilio_sriov
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
        {
            "type":"kube-ovn",
            "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
            "provider": "default.default.ovn"
        },
        {
            "type":"portmap",
            "capabilities":{
                "portMappings":true
            }
        }
    ]
}'
```

- `provider` : □□□□□ `NetworkAttachmentDefinition` □ {name}.{namespace}.ovn□

## 7.17.5 Kube-OVN □□□□□□□

□□□□□□□

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

□□□□□□□ `IFACE` □□□□□□□□□□□□□□□□□□□□ IP□

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="ensp01"
```

□□ Kube-OVN□

```
bash install.sh
```

## 7.17.6 □□□□ VF □□□ Pod

□□□□□□ yaml □□□□□□ VF □□□□□□□□□ Pod:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: default/default
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          corigine.com/agilio_sriov: '1'
```



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```
        limits:
          corigine.com/agilio_sriov: '1'
```

- `v1.multus-cni.io/default-network`：⬚⬚⬚⬚⬚⬚ `NetworkAttachmentDefinition` ⬚ {namespace}/{name}⬚

⬚⬚⬚⬚ Pod ⬚⬚⬚⬚⬚ `ovs-ovn` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:91cc45de-e7e9-4935-8f82-1890430b0f66, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/
0),in_port(5b45c61b307e_h),packet_type(ns=0/0,id=0/0),eth(src=00:00:00:c5:6d:4e,dst=00:00:00:e7:16:ce),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.
0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no), packets:941539, bytes:62142230, used:0.260s, offloaded:yes, dp:tc, actions:54235e5753b8_h
ufid:e00768d7-e652-4d79-8182-3291d852b791, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0x1),recirc_id(0),dp_hash(0/
0),in_port(54235e5753b8_h),packet_type(ns=0/0,id=0/0),eth(src=00:00:00:e7:16:ce,dst=00:00:00:c5:6d:4e),eth_type(0x0800),ipv4(src=0.0.0.0/0.0.0.0,dst=0.
0.0.0/0.0.0.0,proto=0/0,tos=0/0,ttl=0/0,frag=no), packets:82386659, bytes:115944854173, used:0.260s, offloaded:yes, dp:tc, actions:5b45c61b307e_h
```

⬚⬚⬚ `offloaded:yes, dp:tc` ⬚⬚⬚⬚⬚⬚⬚⬚

PDF Slack Support

2026⬚4⬚28⬚

2022⬚5⬚24⬚

GitHub



7.17.7 ⬚⬚

# 7.18 ▯▯▯▯▯▯ Offload ▯▯

Kube-OVN ▯▯▯▯▯▯▯▯▯▯ OVS ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ CPU ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ CPU ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ metaScale ▯▯▯▯▯▯▯▯▯ OVS ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯ OVS ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ CPU ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

> **Note**
>
> 1. ▯▯▯▯▯▯▯▯ 2024 ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
ls -l /sys/class/net/ | grep b3:00.0
lrwxrwxrwx 1 root root 0 May  7 16:30 p3p1 -> ../../devices/pci0000:b2/0000:b2:00.0/0000:b3:00.0/net/p3p1
```

1. ▯▯▯▯▯ VF ▯▯▯

```
cat /sys/class/net/p3p1/device/sriov_totalvfs
512
```

1. ▯▯ VF▯

```
echo '10' > /sys/class/net/p3p1/device/sriov_numvfs
```

1. ▯▯ VF ▯▯▯▯▯

```
lspci -d 1f67:
b3:00.0 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.1 Ethernet controller: Device 1f67:1111 (rev 02)
b3:00.2 Ethernet controller: Device 1f67:1112
b3:00.3 Ethernet controller: Device 1f67:1112
b3:00.4 Ethernet controller: Device 1f67:1112
b3:00.5 Ethernet controller: Device 1f67:1112
b3:00.6 Ethernet controller: Device 1f67:1112
b3:00.7 Ethernet controller: Device 1f67:1112
b3:01.0 Ethernet controller: Device 1f67:1112
b3:01.1 Ethernet controller: Device 1f67:1112
b3:01.2 Ethernet controller: Device 1f67:1112
b3:01.3 Ethernet controller: Device 1f67:1112
```

1. ▯▯ switchdev ▯▯▯

```
devlink dev eswitch set pci/0000:b3:00.0 mode switchdev
```

### ▯▯ SR-IOV Device Plugin

1. ▯▯ SR-IOV ▯▯ ConfigMap▯

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
        "resourceList": [{
                "resourceName": "xsc_sriov",
                "resourcePrefix": "yunsilicon.com",
                "selectors": {
                    "vendors": ["1f67"],
                    "devices": ["1012", "1112"]
                }}
        ]
    }
```

1. ▯▯ SR-IOV Device Plugin ▯▯ DevicePlugin▯
2. ▯▯▯▯▯▯▯▯▯▯ SR-IOV ▯▯▯

```
# kubectl describe node <node name> | grep yunsilicon.com/xsc_sriov
  yunsilicon.com/xsc_sriov:  10
  yunsilicon.com/xsc_sriov:  10
  yunsilicon.com/xsc_sriov  0            0
```

### ▯▯ Multus-CNI

1. ▯▯ Multus-CNI ▯▯▯ Multus-CNI

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/master/deployments/multus-daemonset.yml
```

1. ▯▯ NetworkAttachmentDefinition

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
```



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```
metadata:
  name: sriov-net1
  namespace: default
  annotations:
    k8s.v1.cni.cncf.io/resourceName: yunsilicon.com/xsc_sriov
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
        {
            "type":"kube-ovn",
            "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
            "provider": "sriov-net1.default.ovn"
        },
        {
            "type":"portmap",
            "capabilities":{
                "portMappings":true
            }
        }
    ]
}'
```

⬚⬚⬚⬚ **SR-IOV ⬚ Pod**

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  annotations:
    v1.multus-cni.io/default-network: default/sriov-net1
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          yunsilicon.com/xsc_sriov: '1'
        limits:
          yunsilicon.com/xsc_sriov: '1'
```

**Offload ⬚⬚**

⬚⬚⬚⬚ Pod ⬚⬚⬚⬚⬚ `ovs-ovn` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
ovs-appctl dpctl/dump-flows type=offloaded
flow-dump from pmd on cpu core: 9
ct_state(-new+est-rel+rpl+trk),ct_mark(0/0x3),recirc_id(0x2d277),in_port(15),packet_type(ns=0,id=0),eth(src=00:00:00:9d:fb:1a,dst=00:
00:00:ce:cf:b9),eth_type(0x0800),ipv4(dst=10.16.0.14,frag=no), packets:6, bytes:588, used:7.276s, actions:ct(zone=4,nat),recirc(0x2d278)
ct_state(-new+est-rel-rpl+trk),ct_mark(0/0x3),recirc_id(0x2d275),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9,dst=00:00:00:9d:fb:
1a),eth_type(0x0800),ipv4(dst=10.16.0.18,frag=no), packets:5, bytes:490, used:7.434s, actions:ct(zone=6,nat),recirc(0x2d276)
ct_state(-new+est-rel-rpl+trk),ct_mark(0/0x1),recirc_id(0x2d276),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9,dst=00:00:00:9d:fb:1a/
01:00:00:00:00:00),eth_type(0x0800),ipv4(frag=no), packets:5, bytes:490, used:7.434s, actions:15
recirc_id(0),in_port(15),packet_type(ns=0,id=0),eth(src=00:00:00:9d:fb:1a/01:00:00:00:00:00,dst=00:00:00:ce:cf:b9),eth_type(0x0800),ipv4(dst=10.
16.0.14/255.192.0.0,frag=no), packets:6, bytes:588, used:7.277s, actions:ct(zone=6,nat),recirc(0x2d277)
recirc_id(0),in_port(8),packet_type(ns=0,id=0),eth(src=00:00:00:ce:cf:b9/01:00:00:00:00:00,dst=00:00:00:9d:fb:1a),eth_type(0x0800),ipv4(dst=10.
16.0.18/255.192.0.0,frag=no), packets:6, bytes:588, used:7.434s, actions:ct(zone=4,nat),recirc(0x2d275)
ct_state(-new+est-rel+rpl+trk),ct_mark(0/0x1),recirc_id(0x2d278),in_port(15),packet_type(ns=0,id=0),eth(dst=00:
00:00:ce:cf:b9/01:00:00:00:00:00),eth_type(0x0800),ipv4(frag=no), packets:6, bytes:588, used:7.277s, actions:8
```

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7.18.3 ⬚⬚



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# 7.19 □□□□□□ Offload □□

> **Note**
>
> □□□□□□□ 2024 □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

## 7.19.1 □□□□

- □□□□ 2200E □□□□□□
- □□□□ HADOS □□□□□□□□□□
- BIOS □□ SR-IOV

## 7.19.2 □□□□

**□□ SR-IOV**

1. □□□□□□ 2200E □□□ vendor ID□1f47□□□□□□□□□□□□□□□□ ID□□□□□00:0a.0□□□00:0b.0□□□□□□ 2200E □□□□□□□□□□□□□□□□□□□□□□□□□□

```
lspci | grep 1f47
00:0a.0 Ethernet controller: Device 1f47:1001 (rev 10)
00:0b.0 Ethernet controller: Device 1f47:1001 (rev 10)
```

1. □□□□□□□□□□□□□□□□□□□□ VF □□□

```
cat /sys/bus/pci/devices/0000\:00\:0a.0/sriov_totalvfs
256
```

1. □□□□ VF□□□□□□□□□□□□□□ VF □□□

```
echo 7 > /sys/bus/pci/devices/0000\:00\:0a.0/sriov_numvfs
```

1. □□ VF □□□□

```
lspci | grep 1f47
00:0a.0 Ethernet controller: Device 1f47:1001 (rev 10)
00:0a.1 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.2 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.3 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.4 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.5 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.6 Ethernet controller: Device 1f47:110f (rev 10)
00:0a.7 Ethernet controller: Device 1f47:110f (rev 10)
00:0b.0 Ethernet controller: Device 1f47:1001 (rev 10)
```

**□□□□□ SR-IOV Device Plugin**

1. □□ SR-IOV □□ Configmap□□□□□□□□ SR-IOV Device Plugin □□□□□□□□□□□□ VF □□□□□□ Pod □□□

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: sriovdp-config
  namespace: kube-system
data:
  config.json: |
    {
        "resourceList": [{
                "resourceName": "sriov_dpu",
                "resourcePrefix": "yusur.tech",
                "selectors": {
                    "vendors": ["1f47"],
                    "devices": ["110f"]
                }}
```



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```
        ]
    }
```

1. □□□□□ SR-IOV Device Plugin

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/sriov-network-device-plugin/v3.6.2/deployments/sriovdp-daemonset.yaml
```

1. □□ SR-IOV □□□□□□□□□□ kubernetes Node □□□□□□

```
kubectl describe node node1 | grep yusur
  yusur.tech/sriov_dpu:  7
  yusur.tech/sriov_dpu:  7
  yusur.tech/sriov_dpu  0           0
```

## 7.19.3 □□ Multus-CNI

□□ Multus-CNI□□□□ Kube-OVN □□□□ SRIOV □□□ Device ID□

```
kubectl apply -f https://raw.githubusercontent.com/k8snetworkplumbingwg/multus-cni/v4.0.2/deployments/multus-daemonset-thick.yml
```

□□ `NetworkAttachmentDefinition` □

```
apiVersion:
  "k8s.cni.cncf.io/v1"
kind:
  NetworkAttachmentDefinition
metadata:
  name: test
  namespace: kube-system
  annotations:
    k8s.v1.cni.cncf.io/resourceName: yusur.tech/sriov_dpu
spec:
  config: '{
    "cniVersion": "0.3.1",
    "name": "kube-ovn",
    "plugins":[
        {
            "type":"kube-ovn",
            "server_socket":"/run/openvswitch/kube-ovn-daemon.sock",
            "provider": "test.kube-system.ovn"
        },
        {
            "type":"portmap",
            "capabilities":{
                "portMappings":true
            }
        }
    ]
}'
```

- `provider` : □□□□□ NetworkAttachmentDefinition □ `{name}.{namespace}.ovn` □

## 7.19.4 Kube-OVN □□□□□□□

1. □□ □□□□

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.12/dist/images/install.sh
```

1. □□□□□□□IFACE □□□□□□□□□□□□□□□□□□□ IP□

```
ENABLE_MIRROR=${ENABLE_MIRROR:-false}
HW_OFFLOAD=${HW_OFFLOAD:-true}
ENABLE_LB=${ENABLE_LB:-false}
IFACE="p0"
```

1. □□ kube-ovn

```
bash install.sh
```

□□□□ **VF □□□ pod**

□□□□□□ yaml □□□□□□ VF □□□□□□□□□ Pod□



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```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: default
  annotations:
    v1.multus-cni.io/default-network: kube-system/test
spec:
  containers:
    - name: nginx
      image: docker.io/library/nginx:alpine
      resources:
        requests:
          yusur.tech/sriov_dpu: '1'
        limits:
          yusur.tech/sriov_dpu: '1'
```

- `v1.multus-cni.io/default-network`：▯▯▯▯▯▯ `NetworkAttachmentDefinition` ▯ `{namespace}/{name}` ▯

**Offload ▯▯**

▯▯▯▯ Pod ▯▯▯▯▯ `ovs-ovn` ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
# ovs-appctl dpctl/dump-flows -m type=offloaded
ufid:67c2e10f-92d4-4574-be70-d072815ff166, skb_priority(0/0),skb_mark(0/0),ct_state(0/0x23),ct_zone(0/0),ct_mark(0/0),ct_label(0/0),recirc_id(0),dp_hash(0/
0),in_port(d85b161b6840_h),packet_type(ns=0/0,id=0/0),eth(src=0a:c9:1c:70:01:09,dst=8a:18:a4:22:b7:7d),eth_type(0x0800),ipv4(src=10.0.1.10,dst=10.
0.1.6,proto=6,tos=0/0x3,ttl=0/0,frag=no),tcp(src=60774,dst=9001), packets:75021, bytes:109521630, offload_packets:75019, offload_bytes:109521498, used:
3.990s,offloaded:yes,dp:tc, actions:set(tunnel(tun_id=0x5,dst=192.168.201.12,ttl=64,tp_dst=6081,geneve({class=0x102,type=0x80,len=4,0xa0006}),flags(csum|
key))),genev_sys_6081
ufid:7940666e-a0bd-42a5-8116-1e84e81bb338, skb_priority(0/0),tunnel(tun_id=0x5,src=192.168.201.12,dst=192.168.201.11,ttl=0/
0,tp_dst=6081,geneve({class=0x102,type=0x80,len=4,0x6000a}),flags(+key)),skb_mark(0/0),ct_state(0/0),ct_zone(0/0),ct_mark(0/0),ct_label(0/
0),recirc_id(0),dp_hash(0/0),in_port(genev_sys_6081),packet_type(ns=0/0,id=0/0),eth(src=8a:18:a4:22:b7:7d,dst=0a:c9:1c:70:01:09),eth_type(0x0800),ipv4(src=10.
0.1.6,dst=10.0.1.10,proto=6,tos=0/0,ttl=0/0,frag=no),tcp(src=9001,dst=60774), packets:6946, bytes:459664, offload_packets:6944, offload_bytes:459532, used:
4.170s, dp:tc,offloaded:yes,actions:d85b161b6840_h
```

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7.19.5 ▯▯

# 7.20 DPDK ⬚⬚

⬚⬚⬚⬚ Kube-OVN ⬚⬚⬚ OVS-DPDK ⬚⬚⬚⬚ KubeVirt ⬚⬚⬚⬚ DPDK ⬚⬚⬚⬚⬚⬚⬚

> ⬚⬚⬚ KubeVirt ⬚⬚⬚⬚⬚⬚ OVS-DPDK⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ patch Vhostuser implementation ⬚⬚ KubeVirt ⬚ KVM Device Plugin ⬚⬚⬚ OVS-DPDK⬚

## 7.20.1 ⬚⬚⬚⬚

- ⬚⬚⬚⬚⬚⬚⬚ DPDK ⬚⬚⬚⬚⬚⬚⬚⬚
- ⬚⬚⬚⬚⬚ Hugepages⬚

## 7.20.2 ⬚⬚⬚⬚ DPDK ⬚⬚

⬚⬚⬚⬚⬚⬚ `driverctl` ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ DPDK ⬚⬚⬚⬚⬚⬚

```
driverctl set-override 0000:00:0b.0 uio_pci_generic
```

## 7.20.3 ⬚⬚⬚⬚

⬚⬚⬚ OVS-DPDK ⬚⬚⬚⬚⬚⬚⬚⬚⬚ Kube-OVN ⬚⬚⬚⬚⬚⬚⬚⬚

```
kubectl label nodes <node> ovn.kubernetes.io/ovs_dp_type="userspace"
```

⬚⬚⬚ OVS-DPDK ⬚⬚⬚ `/opt/ovs-config` ⬚⬚⬚⬚⬚⬚⬚⬚⬚ `ovs-dpdk-config` ⬚

```
ENCAP_IP=192.168.122.193/24
DPDK_DEV=0000:00:0b.0
```

- `ENCAP_IP` : ⬚⬚⬚⬚⬚⬚⬚⬚
- `DPDK_DEV` : ⬚⬚⬚ PCI ID⬚

## 7.20.4 ⬚⬚ Kube-OVN

⬚⬚⬚⬚⬚⬚⬚⬚

```
wget https://raw.githubusercontent.com/kubeovn/kube-ovn/release-1.16/dist/images/install.sh
```

⬚⬚ DPDK ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
bash install.sh --with-hybrid-dpdk
```

## 7.20.5 ⬚⬚⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ vhostuser ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ OVS-DPDK ⬚⬚⬚

⬚⬚ KVM Device Plugin ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ KVM Device Plugin⬚

```
kubectl apply -f https://raw.githubusercontent.com/kubevirt/kubernetes-device-plugins/master/manifests/kvm-ds.yml
```

⬚⬚ NetworkAttachmentDefinition⬚

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: ovn-dpdk
  namespace: default
spec:
  config: >-
    {
        "cniVersion": "0.3.0",
        "type": "kube-ovn",
```

```
        "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
        "provider": "ovn-dpdk.default.ovn",
        "vhost_user_socket_volume_name": "vhostuser-sockets",
        "vhost_user_socket_name": "sock"
    }
```

□□□□□ Dockerfile □□ VM □□□

```
FROM quay.io/kubevirt/virt-launcher:v0.46.1

# wget http://cloud.centos.org/centos/7/images/CentOS-7-x86_64-GenericCloud.qcow2
COPY CentOS-7-x86_64-GenericCloud.qcow2 /var/lib/libvirt/images/CentOS-7-x86_64-GenericCloud.qcow2
```

□□□□□□

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: vm-config
data:
  start.sh: |
    chmod u+w /etc/libvirt/qemu.conf
    echo "hugetlbfs_mount = \"/dev/hugepages\"" >> /etc/libvirt/qemu.conf
    virtlogd &
    libvirtd &

    mkdir /var/lock

    sleep 5

    virsh define /root/vm/vm.xml
    virsh start vm

    tail -f /dev/null
  vm.xml: |
    <domain type='kvm'>
      <name>vm</name>
      <uuid>4a9b3f53-fa2a-47f3-a757-dd87720d9d1d</uuid>
      <memory unit='KiB'>2097152</memory>
      <currentMemory unit='KiB'>2097152</currentMemory>
      <memoryBacking>
        <hugepages>
          <page size='2' unit='M' nodeset='0'/>
        </hugepages>
      </memoryBacking>
      <vcpu placement='static'>2</vcpu>
      <cputune>
        <shares>4096</shares>
        <vcpupin vcpu='0' cpuset='4'/>
        <vcpupin vcpu='1' cpuset='5'/>
        <emulatorpin cpuset='1,3'/>
      </cputune>
      <os>
        <type arch='x86_64' machine='pc'>hvm</type>
        <boot dev='hd'/>
      </os>
      <features>
        <acpi/>
        <apic/>
      </features>
      <cpu mode='host-model'>
        <model fallback='allow'/>
        <topology sockets='1' cores='2' threads='1'/>
        <numa>
          <cell id='0' cpus='0-1' memory='2097152' unit='KiB' memAccess='shared'/>
        </numa>
      </cpu>
      <on_reboot>restart</on_reboot>
      <devices>
        <emulator>/usr/libexec/qemu-kvm</emulator>
        <disk type='file' device='disk'>
          <driver name='qemu' type='qcow2' cache='none'/>
          <source file='/var/lib/libvirt/images/CentOS-7-x86_64-GenericCloud.qcow2'/>
          <target dev='vda' bus='virtio'/>
        </disk>

        <interface type='vhostuser'>
          <mac address='00:00:00:0A:30:89'/>
          <source type='unix' path='/var/run/vm/sock' mode='server'/>
           <model type='virtio'/>
          <driver queues='2'>
            <host mrg_rxbuf='off'/>
          </driver>
        </interface>
        <serial type='pty'>
          <target type='isa-serial' port='0'>
            <model name='isa-serial'/>
          </target>
        </serial>
        <console type='pty'>
          <target type='serial' port='0'/>
```

```
        </console>
        <channel type='unix'>
          <source mode='bind' path='/var/lib/libvirt/qemu/channel/target/domain-1-vm/org.qemu.guest_agent.0'/>
          <target type='virtio' name='org.qemu.guest_agent.0' state='connected'/>
          <alias name='channel0'/>
          <address type='virtio-serial' controller='0' bus='0' port='1'/>
        </channel>

      </devices>
    </domain>
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: vm-deployment
  labels:
    app: vm
spec:
  replicas: 1
  selector:
    matchLabels:
      app: vm
  template:
    metadata:
      labels:
        app: vm
      annotations:
        k8s.v1.cni.cncf.io/networks: default/ovn-dpdk
        ovn-dpdk.default.ovn.kubernetes.io/ip_address: 10.16.0.96
        ovn-dpdk.default.ovn.kubernetes.io/mac_address: 00:00:00:0A:30:89
    spec:
      nodeSelector:
        ovn.kubernetes.io/ovs_dp_type: userspace
      securityContext:
        runAsUser: 0
      volumes:
        - name: vhostuser-sockets
          emptyDir: {}
        - name: xml
          configMap:
            name: vm-config
        - name: hugepage
          emptyDir:
            medium: HugePages-2Mi
        - name: libvirt-runtime
          emptyDir: {}
      containers:
        - name: vm
          image: vm-vhostuser:latest
          command: ["bash", "/root/vm/start.sh"]
          securityContext:
            capabilities:
              add:
                - NET_BIND_SERVICE
                - SYS_NICE
                - NET_RAW
                - NET_ADMIN
            privileged: false
            runAsUser: 0
          resources:
            limits:
              cpu: '2'
              devices.kubevirt.io/kvm: '1'
              memory: '8784969729'
              hugepages-2Mi: 2Gi
            requests:
              cpu: 666m
              devices.kubevirt.io/kvm: '1'
              ephemeral-storage: 50M
              memory: '4490002433'
          volumeMounts:
            - name: vhostuser-sockets
              mountPath: /var/run/vm
            - name: xml
              mountPath: /root/vm/
            - mountPath: /dev/hugepages
              name: hugepage
            - name: libvirt-runtime
              mountPath: /var/run/libvirt
```

... Pod ...

```
# virsh set-user-password vm root 12345
Password set successfully for root in vm

# virsh console vm
Connected to domain 'vm'
Escape character is ^] (Ctrl + ])

CentOS Linux 7 (Core)
Kernel 3.10.0-1127.el7.x86_64 on an x86_64
```

```
localhost login: root
Password:
Last login: Fri Feb 25 09:52:54 on ttyS0
```

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

```
ip link set eth0 mtu 1400
ip addr add 10.16.0.96/16 dev eth0
ip ro add default via 10.16.0.1
ping 114.114.114.114
```

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## 7.20.6 ▯▯



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# 7.21 OpenStack ▯▯

▯▯▯▯▯▯▯▯▯▯▯▯▯ OpenStack ▯▯▯▯▯▯▯▯ Kubernetes ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ OpenStack Neutron ▯▯▯▯▯▯ OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯ Kube-OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ OVN ▯▯▯▯▯▯▯ OpenStack ▯ Kubernetes ▯▯▯▯▯

## 7.21.1 ▯▯▯▯▯

▯▯▯▯▯▯ OVN-IC ▯▯▯▯▯▯▯▯▯▯▯▯▯ Kubernetes ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

▒ OpenStack ▒▒ OVN ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

```
ovn-nbctl set NB_Global . name=op-az
```

▒▒▒▒ `OVN-IC` ▒▒▒▒▒▒▒▒▒ `OVN-IC` ▒▒▒▒

```
/usr/share/ovn/scripts/ovn-ctl --ovn-ic-nb-db=tcp:192.168.65.3:6645 \
  --ovn-ic-sb-db=tcp:192.168.65.3:6646 \
  --ovn-northd-nb-db=unix:/run/ovn/ovnnb_db.sock \
  --ovn-northd-sb-db=unix:/run/ovn/ovnsb_db.sock \
  start_ic
```

- `ovn-ic-nb-db` ▒ `ovn-ic-sb-db` : OVN-IC ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒
- `ovn-northd-nb-db` ▒ `ovn-northd-sb-db` : ▒▒▒▒ OVN ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

▒▒▒▒▒▒▒▒▒▒

```
ovs-vsctl set open_vswitch . external_ids:ovn-is-interconn=true
```

▒▒▒▒▒▒ OpenStack ▒ OVN ▒▒▒▒▒▒▒▒▒▒▒▒▒

▒▒ `ts` ▒▒▒▒▒▒▒ `router0` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

```
ovn-nbctl lrp-add router0 lrp-router0-ts 00:02:ef:11:39:4f 169.254.100.73/24
ovn-nbctl lsp-add ts lsp-ts-router0 -- lsp-set-addresses lsp-ts-router0 router \
  -- lsp-set-type lsp-ts-router0 router \
  -- lsp-set-options lsp-ts-router0  router-port=lrp-router0-ts
ovn-nbctl lrp-set-gateway-chassis lrp-router0-ts {gateway chassis} 1000
ovn-nbctl set NB_Global . options:ic-route-adv=true options:ic-route-learn=true
```

▒▒▒▒▒▒ Kubernetes ▒▒▒▒▒▒

```
# ovn-nbctl lr-route-list router0
IPv4 Routes
               10.0.0.22            169.254.100.34 dst-ip (learned)
            10.16.0.0/16            169.254.100.34 dst-ip (learned)
```

▒▒▒▒▒▒ `router0` ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ Kubernetes ▒ Pod ▒▒▒

## 7.21.2 ▒▒▒▒ OVN

▒▒▒▒▒▒OpenStack ▒ Kubernetes ▒▒▒▒▒▒▒▒ OVN▒▒▒▒▒▒▒▒▒▒ VPC ▒ Subnet ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

▒▒▒▒▒▒▒▒▒▒▒▒ Kube-OVN ▒▒ OVN▒OpenStack ▒▒ Neutron ▒▒▒▒▒▒▒▒▒▒ OVN ▒▒▒▒OpenStack ▒▒▒ networking-ovn ▒▒ Neutron ▒▒▒▒▒▒

**Neutron ▒▒▒▒**

▒▒ Neutron ▒▒▒▒ `/etc/neutron/plugins/ml2/ml2_conf.ini` ▒

```
[ovn]
...
ovn_nb_connection = tcp:[192.168.137.176]:6641,tcp:[192.168.137.177]:6641,tcp:[192.168.137.178]:6641
ovn_sb_connection = tcp:[192.168.137.176]:6642,tcp:[192.168.137.177]:6642,tcp:[192.168.137.178]:6642
ovn_l3_scheduler = OVN_L3_SCHEDULER
```

- `ovn_nb_connection` ▒ `ovn_sb_connection` : ▒▒▒▒▒▒▒ Kube-OVN ▒▒ `ovn-central` ▒▒▒▒▒▒▒

▒▒▒▒▒▒▒▒ OVS ▒▒▒

```
ovs-vsctl set open . external-ids:ovn-remote=tcp:[192.168.137.176]:6642,tcp:[192.168.137.177]:6642,tcp:[192.168.137.178]:6642
ovs-vsctl set open . external-ids:ovn-encap-type=geneve
ovs-vsctl set open . external-ids:ovn-encap-ip=192.168.137.200
```

- `external-ids:ovn-remote` : ▒▒▒▒▒▒▒ Kube-OVN ▒▒ `ovn-central` ▒▒▒▒▒▒▒
- `ovn-encap-ip` : ▒▒▒▒▒▒▒▒▒ IP ▒▒▒



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▒ **Kubernetes ▒▒▒ OpenStack ▒▒▒**

▒▒▒▒▒▒▒▒ Kubernetes ▒▒▒ OpenStack ▒▒▒▒▒▒▒ OpenStack ▒▒▒▒▒▒ Pod▒

> **Note**
>
> ▒▒▒▒▒▒▒▒ `kube-ovn-controller` ▒ args ▒▒▒▒▒▒▒ `--enable-external-vpc=true` ▒

▒▒ OpenStack ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒

```
# openstack router list
+--------------------------------------+---------+--------+-------+----------------------------------+
| ID                                   | Name    | Status | State | Project                          |
+--------------------------------------+---------+--------+-------+----------------------------------+
| 22040ed5-0598-4f77-bffd-e7fd4db47e93 | router0 | ACTIVE | UP    | 62381a21d569404aa236a5dd8712449c |
+--------------------------------------+---------+--------+-------+----------------------------------+
# openstack network list
+--------------------------------------+----------+--------------------------------------+
| ID                                   | Name     | Subnets                              |
+--------------------------------------+----------+--------------------------------------+
| cd59e36a-37db-4c27-b709-d35379a7920f | provider | 01d73d9f-fdaa-426c-9b60-aa34abbfacae |
+--------------------------------------+----------+--------------------------------------+
# openstack subnet list
+--------------------------------------+-------------+--------------------------------------+----------------+
| ID                                   | Name        | Network                              | Subnet         |
+--------------------------------------+-------------+--------------------------------------+----------------+
| 01d73d9f-fdaa-426c-9b60-aa34abbfacae | provider-v4 | cd59e36a-37db-4c27-b709-d35379a7920f | 192.168.1.0/24 |
+--------------------------------------+-------------+--------------------------------------+----------------+
# openstack server list
+--------------------------------------+-------------------+--------+----------------------+--------+--------+
| ID                                   | Name              | Status | Networks             | Image  | Flavor |
+--------------------------------------+-------------------+--------+----------------------+--------+--------+
| 8433d622-a8d6-41a7-8b31-49abfd64f639 | provider-instance | ACTIVE | provider=192.168.1.61 | ubuntu | m1     |
+--------------------------------------+-------------------+--------+----------------------+--------+--------+
```

▒ Kubernetes ▒▒▒▒ VPC ▒▒▒

```
# kubectl get vpc
NAME                                    STANDBY   SUBNETS
neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93   true      ["neutron-cd59e36a-37db-4c27-b709-d35379a7920f"]
ovn-cluster                             true      ["join","ovn-default"]
```

`neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93` ▒▒ OpenStack ▒▒▒▒▒ VPC ▒▒▒

▒▒▒▒▒▒▒ Kube-OVN ▒▒▒ VPC ▒ Subnet ▒▒▒▒ Pod ▒▒▒▒

VPC, Subnet ▒▒ Namespace `net2` ▒▒▒▒ Pod:

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: net2
---
apiVersion: kubeovn.io/v1
kind: Vpc
metadata:
  creationTimestamp: "2021-06-20T13:34:11Z"
  generation: 2
  labels:
    ovn.kubernetes.io/vpc_external: "true"
  name: neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93
  resourceVersion: "583728"
  uid: 18d4c654-f511-4def-a3a0-a6434d237c1e
spec:
  namespaces:
  - net2
---
kind: Subnet
apiVersion: kubeovn.io/v1
metadata:
  name: net2
spec:
  vpc: neutron-22040ed5-0598-4f77-bffd-e7fd4db47e93
  namespaces:
    - net2
  cidrBlock: 12.0.1.0/24
  natOutgoing: false
---
apiVersion: v1
kind: Pod
metadata:
  name: ubuntu
```



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```
  namespace: net2
spec:
  containers:
    - image: docker.io/kubeovn/kube-ovn:v1.8.0
      command:
        - "sleep"
        - "604800"
      imagePullPolicy: IfNotPresent
      name: ubuntu
  restartPolicy: Always
```

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## 7.21.3 ⬚⬚

# 7.22 □□ IPsec □□□□□□□

Kube-OVN □ v1.13.0 □□□□□ OVN/OVS □□ IPsec □□□□□□□□□□Geneve/Vxlan/STT□□□□□□□□□

## 7.22.1 □□□□

- □□□□□□ UDP 500□IKE□□ UDP 4500□NAT-T□□□□
- □□□□ `xfrm` / `af_key` □□□□□□□□□ strongSwan/libreswan □□□□□

## 7.22.2 □□□□

`kube-ovn-cni` □□□□□□ CertificateSigningRequest□□ `kube-ovn-controller` □□ approve □□□□□□□□ `kube-ovn-cni` □□□□□□□ IPsec □□□□□ ipsec □□□

## 7.22.3 □□ IPsec

□ `kube-ovn-controller` Deployment □ `kube-ovn-cni` DaemonSet □□□□□ `--enable-ovn-ipsec=false` □□ `--enable-ovn-ipsec=true` □□□□□□□□□□□

```
ENABLE_OVN_IPSEC=true
```

## 7.22.4 □□ cert-manager □□□□□□□□

□□□□□□□ cert-manager□□□□□□□□ IPsec □□□□□□□□□□□□□□□ `--cert-manager-ipsec-cert=true` □□□□ kube-ovn-cni □□□ cert-manager □□□□□□□ IPsec □□□□□□ Kube-OVN □□□□□ CA□

```
args:
- --enable-ovn-ipsec=true
- --cert-manager-ipsec-cert=true
```

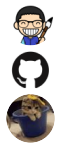
## 7.22.5 □□□□□

- □□□□□□ `kube-ovn-cni` □□□□□ `ovs-appctl ipsec/show` □□□□□□□□□□ IPsec □□□SA□SPI □□□□
- □□ `kubectl get csr` □□□□□ kube-ovn-cni □ CSR□□□□□□ IPsec □□□□□□□□ CSR □□ approve□□□□□□ controller □□□ `signer.go` □□□□□□
- □□ IPsec □□□□□□□□□□ `false` □□□□□□□□□□ IPsec □□□□□□□□□□□□ SA□□□ `kube-ovn-cni` Pod □□□

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## 7.22.6 □□



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# 7.23 OVN □□□□

□□□□□□□□ Pod□□□□□□□□□□□□ GRE/ERSPAN □□□□□□□□□□□□

> □□□□□ Kube-OVN □□□□□ v1.12□

## 7.23.1 □□ Multus-CNI

□□ Multus-CNI □□ □□ Multus□

## 7.23.2 □□□□□□□

□□□□□□□□□□□□□□□

```
apiVersion: "k8s.cni.cncf.io/v1"
kind: NetworkAttachmentDefinition
metadata:
  name: attachnet
  namespace: default
spec:
  config: |
    {
      "cniVersion": "0.3.1",
      "type": "kube-ovn",
      "server_socket": "/run/openvswitch/kube-ovn-daemon.sock",
      "provider": "attachnet.default.ovn"
    }
```

□□ `provider` □□□□□ `<NAME>.<NAMESPACE>.ovn` □

## 7.23.3 □□ Underlay □□

□□□□□□□□□□□□□□□□□□□□□□□□□□□MTU □□□□□□□□ LSP/Pod□□□□□□□ Underlay □□□□□□□

□□□□□□□□ Underlay □□□

```
apiVersion: kubeovn.io/v1
kind: ProviderNetwork
metadata:
  name: net1
spec:
  defaultInterface: eth1
---
apiVersion: kubeovn.io/v1
kind: Vlan
metadata:
  name: vlan1
spec:
  id: 0
  provider: net1
---
apiVersion: kubeovn.io/v1
kind: Subnet
metadata:
  name: subnet1
spec:
  protocol: IPv4
  cidrBlock: 172.19.0.0/16
  excludeIps:
  - 172.19.0.2..172.19.0.20
  gateway: 172.19.0.1
  vlan: vlan1
  provider: attachnet.default.ovn
```

□□□□□□ `provider` □□□□□□□□ `provider` □□□

## 7.23.4 □□□□□□ Pod

□□□□□□□□□□□□□□□□□□ Pod□

```
apiVersion: v1
kind: Pod
```



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```
metadata:
  name: pod1
  annotations:
    k8s.v1.cni.cncf.io/networks: default/attachnet
spec:
  containers:
  - name: bash
    image: docker.io/kubeovn/kube-ovn:v1.17.0
    args:
    - bash
    - -c
    - sleep infinity
    securityContext:
      privileged: true
```

▒▒▒▒▒▒▒▒ Pod ▒ IP ▒▒▒

```
$ kubectl get ips | grep pod1
pod1.default                        10.16.0.12   00:00:00:FF:34:24  kube-ovn-worker  ovn-default
pod1.default.attachnet.default.ovn  172.19.0.21  00:00:00:A0:30:68  kube-ovn-worker  subnet1
```

▒▒▒▒▒▒▒ IP ▒▒ `172.19.0.21` ▒

## 7.23.5 ▒▒ OVN ▒▒▒▒

▒▒▒▒▒▒▒▒ OVN ▒▒▒▒▒

```
kubectl ko nbctl mirror-add mirror1 gre 99 from-lport 172.19.0.21
kubectl ko nbctl lsp-attach-mirror coredns-787d4945fb-gpnkb.kube-system mirror1
```

▒▒ `coredns-787d4945fb-gpnkb.kube-system` ▒ OVN LSP ▒▒▒▒▒▒▒▒▒ `<POD_NAME>.<POD_NAMESPACE>` ▒

▒▒▒ OVN ▒▒▒▒▒▒▒▒▒

```
ovn-nbctl mirror-add <NAME> <TYPE> <INDEX> <FILTER> <IP>

NAME   - add a mirror with given name
TYPE   - specify TYPE 'gre' or 'erspan'
INDEX  - specify the tunnel INDEX value
         (indicates key if GRE, erpsan_idx if ERSPAN)
FILTER - specify FILTER for mirroring selection
         ('to-lport' / 'from-lport')
IP     - specify Sink / Destination i.e. Remote IP

ovn-nbctl mirror-del [NAME]         remove mirrors
ovn-nbctl mirror-list               print mirrors

ovn-nbctl lsp-attach-mirror PORT MIRROR   attach source PORT to MIRROR
ovn-nbctl lsp-detach-mirror PORT MIRROR   detach source PORT from MIRROR
```

## 7.23.6 ▒▒▒▒▒▒ Pod

▒▒▒▒▒▒ Pod ▒▒▒▒▒▒▒▒

```
root@pod1:/kube-ovn# ip link add mirror1 type gretap local 172.19.0.21 key 99 dev net1
root@pod1:/kube-ovn# ip link set mirror1 up
```

▒▒▒▒▒▒▒▒▒▒▒▒ Pod ▒▒▒▒▒▒▒▒

```
root@pod1:/kube-ovn# tcpdump -i mirror1 -nnve
tcpdump: listening on mirror1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
05:13:30.328808 00:00:00:a3:f5:e2 > 00:00:00:97:0f:6e, ethertype ARP (0x0806), length 42: Ethernet (len 6), IPv4 (len 4), Request who-has 10.16.0.7 tell 10.
16.0.4, length 28
05:13:30.559167 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 212: (tos 0x0, ttl 64, id 57364, offset 0, flags [DF], proto UDP (17),
length 198)
    10.16.0.4.53 > 10.16.0.6.50472: 34511 NXDomain*- 0/1/1 (170)
05:13:30.559343 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 212: (tos 0x0, ttl 64, id 57365, offset 0, flags [DF], proto UDP (17),
length 198)
    10.16.0.4.53 > 10.16.0.6.45177: 1659 NXDomain*- 0/1/1 (170)
05:13:30.560625 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 200: (tos 0x0, ttl 64, id 57367, offset 0, flags [DF], proto UDP (17),
length 186)
    10.16.0.4.53 > 10.16.0.6.43848: 2636*- 0/1/1 (158)
05:13:30.562774 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 191: (tos 0x0, ttl 64, id 57368, offset 0, flags [DF], proto UDP (17),
length 177)
    10.16.0.4.53 > 10.16.0.6.37755: 48737 NXDomain*- 0/1/1 (149)
05:13:30.563523 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 187: (tos 0x0, ttl 64, id 57369, offset 0, flags [DF], proto UDP (17),
length 173)
    10.16.0.4.53 > 10.16.0.6.53887: 45519 NXDomain*- 0/1/1 (145)
05:13:30.564940 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 201: (tos 0x0, ttl 64, id 57370, offset 0, flags [DF], proto UDP (17),
```

```
length 187)
    10.16.0.4.53 > 10.16.0.6.40846: 25745 NXDomain*- 0/1/1 (159)
05:13:30.565140 00:00:00:a3:f5:e2 > 00:00:00:89:d5:cc, ethertype IPv4 (0x0800), length 201: (tos 0x0, ttl 64, id 57371, offset 0, flags [DF], proto UDP (17),
length 187)
    10.16.0.4.53 > 10.16.0.6.45214: 61875 NXDomain*- 0/1/1 (159)
05:13:30.566023 00:00:00:a3:f5:e2 > 00:00:00:55:e4:4e, ethertype IPv4 (0x0800), length 80: (tos 0x0, ttl 64, id 45937, offset 0, flags [DF], proto UDP (17),
length 66)
    10.16.0.4.44116 > 172.18.0.1.53: 16025+ [1au] AAAA? kube-ovn.io. (38)
```

## 7.23.7 ▯▯▯▯

1. ▯▯▯▯ ERSPAN ▯▯▯▯▯▯▯OVN ▯▯▯▯▯▯▯▯ Linux ▯▯▯▯▯▯▯▯ 4.14▯▯▯▯ ERSPAN ▯▯▯▯▯▯▯▯▯ IPv6 ▯▯▯▯▯▯▯Linux ▯▯▯▯▯▯▯▯ 4.16▯
2. ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯▯

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## 7.23.8 ▯▯



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# 7.24 □□□□ DNS □□□ Kube-OVN □□

NodeLocal DNSCache □□□□□□□□□□ DaemonSet □□ DNS □□□□□□□□ DNS □□□□□□□□□□□ Kube-OVN □□□

## 7.24.1 □□□□ DNS □□□□

### □□ Kubernetes □□□□□ DNS □□

□□□□□ Kubernetes □□□□ Nodelocaldnscache□

□□□□□□□□□□

```
#!bin/bash

localdns=169.254.20.10
domain=cluster.local
kubedns=10.96.0.10

wget https://raw.githubusercontent.com/kubernetes/kubernetes/master/cluster/addons/dns/nodelocaldns/nodelocaldns.yaml
sed -i "s/__PILLAR__LOCAL__DNS__/$localdns/g; s/__PILLAR__DNS__DOMAIN__/$domain/g; s/,__PILLAR__DNS__SERVER__//g; s/__PILLAR__CLUSTER__DNS__/$kubedns/g"
nodelocaldns.yaml

kubectl apply -f nodelocaldns.yaml
```

□□□□□□□□ kubelet □□□□□□□ `/var/lib/kubelet/config.yaml` □□ clusterDNS □□□□□□□□ DNS IP 169.254.20.10□□□□□ kubelet □□□

### Kube-OVN □□ DNS □□

□□□ Kubernetes □ Nodelocal DNScache □□□□Kube-OVN □□□□□□□□□

#### UNDERLAY SUBNET □□ U2O □□

□□□ Underlay Subnet □□□□□□□ DNS □□□□□□□□ U2O □□□□□ `kubectl edit subnet {your subnet}` □□□□ `spec.u2oInterconnection = true` , □□□ Overlay Subnet □□□□□□□□□□

#### □ KUBE-OVN-CONTROLLER □□□□□□□□ DNS IP

```
kubectl edit deployment kube-ovn-controller -n kube-system
```

□ `spec.template.spec.containers.args` □□□□□ `--node-local-dns-ip=169.254.20.10`

#### □□□□□□□□ POD

□□□□□□ Pod □□□□ `/etc/resolv.conf` □ nameserver □□□□ DNS IP□□□□□□□□ Pod □ nameserver □□□□□□□□□ DNS ClusterIP□□□ u2o □□□□□□□□□□□□ Pod □□□□□ Pod □ □□

## 7.24.2 □□□□□□□ DNS □□□□

□□□□□□□□□□□□ Pod □□□□□□□□□□ Pod □ DNS □□□□□□□□□ 169.254.20.10□□□□□□□□□□

```
# kubectl exec -it pod1 -- nslookup github.com
Server:         169.254.20.10
Address:        169.254.20.10:53


Name:   github.com
Address: 20.205.243.166
```

□□□□□□□□□□□□□□□□□□□ DNS □□□□□□□ ovn0 □□□□□□□□ DNS □□□DNS □□□□□□□□□□

```
# tcpdump -i any port 53

06:20:00.441889 659246098c56_h P   ifindex 17 00:00:00:73:f1:06 ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1291+ A? baidu.com.
(27)
06:20:00.441889 ovn0  In  ifindex 7 00:00:00:50:32:cd ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1291+ A? baidu.com. (27)
06:20:00.441950 659246098c56_h P   ifindex 17 00:00:00:73:f1:06 ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1611+ AAAA?
baidu.com. (27)
06:20:00.441950 ovn0  In  ifindex 7 00:00:00:50:32:cd ethertype IPv4 (0x0800), length 75: 10.16.0.2.40230 > 169.254.20.10.53: 1611+ AAAA? baidu.com. (27)
06:20:00.442203 ovn0  Out ifindex 7 00:00:00:52:99:d8 ethertype IPv4 (0x0800), length 145: 169.254.20.10.53 > 10.16.0.2.40230: 1611* 0/1/0 (97)
```

```
06:20:00.442219 659246098c56_h Out ifindex 17 00:00:00:ea:b3:5e ethertype IPv4 (0x0800), length 145: 169.254.20.10.53 > 10.16.0.2.40230: 1611* 0/1/0 (97)
06:20:00.442273 ovn0  Out ifindex 7 00:00:00:52:99:d8 ethertype IPv4 (0x0800), length 125: 169.254.20.10.53 > 10.16.0.2.40230: 1291* 2/0/0 A 39.156.66.10, A
110.242.68.66 (77)
06:20:00.442278 659246098c56_h Out ifindex 17 00:00:00:ea:b3:5e ethertype IPv4 (0x0800), length 125: 169.254.20.10.53 > 10.16.0.2.40230: 1291* 2/0/0 A 39.
156.66.10, A 110.242.68.66 (77)
```

## 7.24.3 ████

⚠️ ███

████████ NetworkPolicy██████ NetworkPolicy ███████ DNS IP██ 169.254.20.10████ CIDR████ NetworkPolicy ██ DNS ██████████ Pod ██████████

**NetworkPolicy ████**

███████ Pod ████ DNS ████████ NetworkPolicy ████

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-local-dns-and-node-cidr
  namespace: default  #
spec:
  podSelector: {}  #       Pod
  policyTypes:
  - Ingress
  - Egress
  egress:
  #       DNS
  - to:
    - ipBlock:
        cidr: 169.254.20.10/32
  #       CIDR          CIDR
  - to:
    - ipBlock:
        cidr: 10.0.0.0/8  #      CIDR
  ingress:
  #       DNS
  - from:
    - ipBlock:
        cidr: 169.254.20.10/32
  #       CIDR              CIDR
  - from:
    - ipBlock:
        cidr: 10.0.0.0/8  #      CIDR
```

████

- `169.254.20.10/32` ███ DNS ███ IP ██
- `10.0.0.0/8` ████ CIDR████████████████████

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## 7.24.4 ██



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# 7.25 □□ VPC NAT □□□□

## 7.25.1 □□

□□ VPC □□ Overlay □□□□□ `natOutgoing` □□□□Subnet □□□□ Pod □□□□□□□□ SNAT □□□□ IP□□□□□□□□□□□□□□□□□□□□ Pod □□□□□□ SNAT□

□□ NAT □□□□□□□□□□□□□□□□□□□□□□□□□□□□ CIDR □□ IP □□□□□ SNAT□

## 7.25.2 □□□□

□ `subnet.Spec` □□□ `natOutgoing` □□□ □□□□□□ `natOutgoingPolicyRules` □□□

```
spec:
  natOutgoing: true
  natOutgoingPolicyRules:
    - action: forward
      match:
        srcIPs: 10.0.11.0/30,10.0.11.254
    - action: nat
      match:
        srcIPs: 10.0.11.128/26
        dstIPs: 114.114.114.114,8.8.8.8
```

□□□□□□□□□ NAT □□□□□

1. □ IP □ 10.0.11.0/30 □□ 10.0.11.254 □□□□□□□□□□□ SNAT□
2. □ IP □ 10.0.11.128/26 □□□□ IP □ 114.114.114.114 □□ 8.8.8.8 □□□□□□□□□□ SNAT□

□□□□□

`action` □□□ `match` □□□□□□□□□□□□□ action, action □□□□ `forward` □ `nat` □ `forward` □□□□□□□□□□ SNAT, `nat` □□□□□□□□ SNAT□ □□□□ natOutgoingPolicyRules □□□ □□□□□□□□□ SNAT□

`match` □□□□□□□□□□□□□□□□ `srcIPs` □ `dstIPs` □ □□□□□□□□□□□□□□□□□□□□□□ IP □ □□ IP□ `match.srcIPs` □ `match.dstIPs` □□□□ CIDR □ IP□□□□□□□□□□

□□□□□□ match □□□□□□□□□ `natOutgoingPolicyRules` □□□□□□□□□□□□□□□□ action □□□□□

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## 7.25.3 □□



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# 8. □□□□

## 8.1 □□□□

□□□□□□ Kube-OVN □□□□□□□□□□□□□□□□□□□□□□□□□□□□

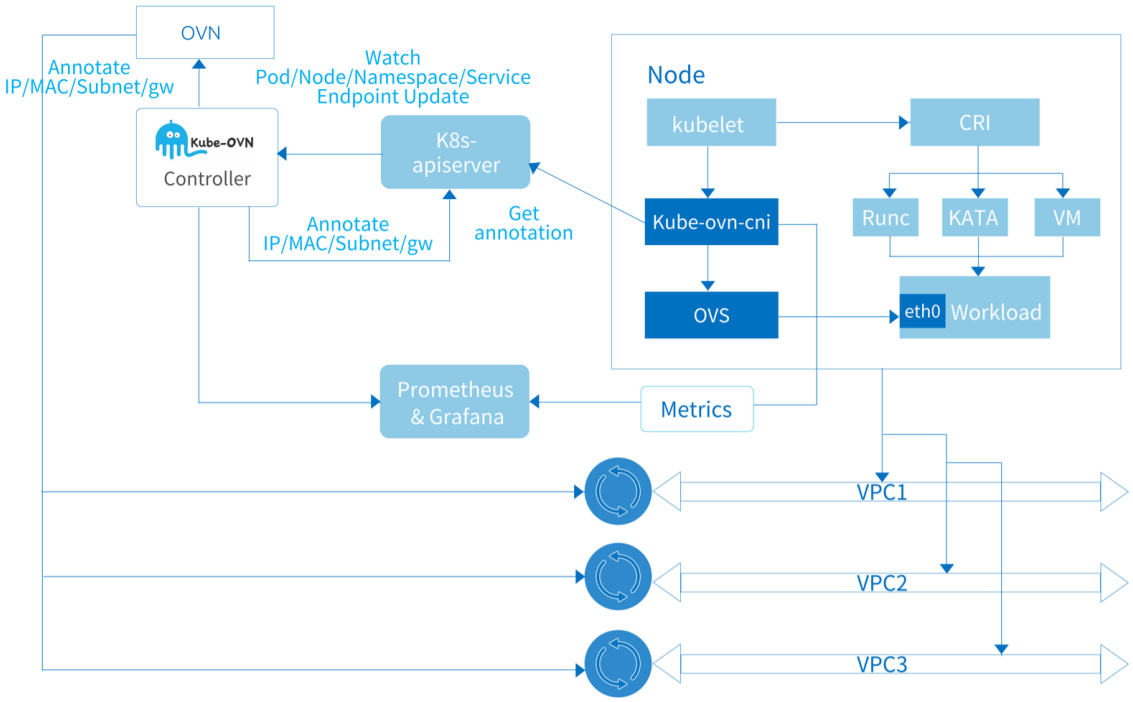
□□□□□Kube-OVN □□ Kubernetes □ OVN □□□□□□□□□□□□□ SDN □□□□□□□□□ □□□□□ Kube-OVN □□□□□ OVN □□□□ Kubernetes □□□□□□□□□□ CNI□Service □ Networkpolicy□□□□□ □□ SDN □□□□□□□□□□□□□□□□□□□□□□□□□□□□□VPC□□□□QoS□ACL □□□□□□□

□□ Kube-OVN □□□□□□□□□□□□□□□□□□□□□□□□□□□□ Cilium□Submariner□Prometheus□KubeVirt □□□

### 8.1.1 □□□□

Kube-OVN □□□□□□□□□□□□□□

- □□ OVN/OVS □□□
- □□□□□□□ Agent□
- □□□□□□□□□□□□□□□




□□ **OVN/OVS** □□

□□□□□□□ OVN/OVS □□□□□□ Kube-OVN □□□□□□□□□□□□□□OVN/OVS □□□□□□□□□□□□□□□□□□ SDN □□□□□□□□□□ □ Kube-OVN □□□□□□□□□□□□□□□ ovn-architecture(7) □□□□□□□ OVN □□ □□□□□□□□□□Kube-OVN □□ OVN □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ Kubernetes □□□

□□ OVN/OVS □□□□□□□□□□□□□□□□□□□ Kubernetes □□□□

#### OVN-CENTRAL

`ovn-central` Deployment □□ OVN □□□□□□□□□□□ `ovn-nb`, `ovn-sb`, □ `ovn-northd` □

- `ovn-nb` □□□□□□□□□□□□□□□ API □□□□□□□□□□ `kube-ovn-controller` □□□□□□ `ovn-nb` □□□□□□□□□□



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- `ovn-sb` ████ `ovn-nb` ████████████████████████████████
- `ovn-northd` ██ `ovn-nb` ████████ `ovn-sb` ███████

██ `ovn-central` ████ Raft ████████████

#### OVS-OVN

`ovs-ovn` ▯ DaemonSet ██████████ Pod ████ `openvswitch`, `ovsdb`, ▯ `ovn-controller` ███████ `ovn-central` ▯ Agent ██████████████

### ██████ Agent

████ Kube-OVN ████████ OVN ▯ Kubernetes ████████████████████████████ ████████████████

#### KUBE-OVN-CONTROLLER

██████ Deployment ████ Kubernetes ████ OVN ████████████ Kube-OVN ████████ `kube-ovn-controller` ████████████████████████████ OVN ████ ████████████ Pod▯Service▯Endpoint▯Node▯NetworkPolicy▯VPC▯Subnet▯Vlan▯ProviderNetwork▯

▯ Pod ████ `kube-ovn-controller` ███ Pod ████████████ IPAM ██████████ `ovn-central` ██ ████████████ ACL ██████ `kube-ovn-controller` ██████ ████████ CIDR████████████ Pod ▯ annotation ███ annotation ████ `kube-ovn-cni` ████████████

#### KUBE-OVN-CNI

██████ DaemonSet ██████████ CNI ██████████ OVS ███████

▯ DaemonSet ███ `kube-ovn` ████████████ `kubelet` ▯ `kube-ovn-cni` ██████████ CNI ██ ███ `kube-ovn-cni` ████████████████ `/opt/cni/bin` ████

`kube-ovn-cni` ████████████████████████

1. ██ `ovn-controller` ▯ `vswitchd` ▯
2. ██ CNI add/del ███
   a. ████ veth ██ OVS ████
   b. ██ OVS ████
   c. ██████ iptables/ipset/route ████
3. ██████ QoS.
4. ████ `ovn0` ████████████
5. ████████ Vlan/Underlay/EIP ████
6. ██████████

### ██████████

████████████████████ Kube-OVN ████████████████

#### KUBE-OVN-SPEAKER

██████ DaemonSet ████████████████████████████████ Pod IP ████

██████████ BGP ▯▯▯

#### KUBE-OVN-PINGER

██████ DaemonSet ██████████ OVS ████████████████████████████ Kube-OVN ▯▯▯▯▯

#### KUBE-OVN-MONITOR

██████ Deployment ▯▯ OVN ████████████ Kube-OVN ▯▯▯▯▯

#### KUBECTL-KO

████ kubectl ████████████████ kubectl ▯▯▯▯▯



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## 8.2 What's Next

This document lists the features merged into the master branch for the next minor release.

### 8.2.1 Post-v1.16.0

- QoS: support configurable ingress/egress burst limits via per-NIC annotations ( `ovn.kubernetes.io/ingress_burst` ). #6628
- VPC Egress Gateway: accept integer Mbps values and `M`, `Mi`, `G`, or `Gi` bit-rate quantities for per-replica bandwidth limits. #7115
- VPC NAT Gateway: allow placing the NAT gateway StatefulSet and its iptables-EIP custom resources in a user-specified namespace. #6595
- iptables-EIP: surface the bound external subnet in the `kubectl get iptables-eip` printer. #6672
- Helm Chart (kube-ovn-v2): ship PrometheusRule and Grafana dashboards out of the box. #6630

### 8.2.2 Post-v1.15.0

- NetworkPolicy now supports provider-scoped policies for multi-network pods using the `ovn.kubernetes.io/policy-for` annotation. #6223
- Support static IP/MAC for multiple interfaces on the same logical switch. #6060
- MetalLB underlay integration now supports IPv6 and dual-stack. #6159
- KubeVirt live-migration multi-chassis options now apply to all VM NICs, not just the primary one. #6241
- Add human-readable descriptions to all Kube-OVN CRD fields for better `kubectl explain` output. #6133 #6147
- Support BGP and EVPN (L3VPN) for VPC Egress Gateway, running FRR in the egress gateway Pod. #6224
- Support per-pod NIC granular DHCP control via annotations, overriding subnet-level DHCP settings. #6475
- Extend SecurityGroup API with tiers, larger priority range (1-16384), localAddress and port matches. #6330
- Use BigInt for SubnetStatus IP count fields to support large address ranges. #6403
- Improve hairpin logic to support FIP, SNAT from any CIDR within the VPC and LoadBalancers. #6445
- VPC NAT Gateway
- Support user-defined annotations on NAT gateway Pod template. #6256
- Allow SNAT EIP to FIP EIP traffic. #6329
- Allow any EIP to share external subnet gateway in one native VLAN. #6426
- VPC Egress Gateway
- Support custom resources and bandwidth limits. #6407
- Interconnection
- Add vendor ID to transit switches to avoid conflicts with other OVN controllers. #6186
- Reliability
- OpenFlow synchronization: detect and recover from stale or missing OVS flows automatically. #6117
- OVN DB: back up Raft header before rejoining cluster to improve recovery. #6106
- Performance
- Strip `managedFields` from informer cache to reduce memory usage. #6119
- Add field selectors to informer factory to reduce API server load. #6091
- Security
- Replace wildcard RBAC verbs with explicit verb lists. #6233
- Specify ephemeral storage limits for containers. #6259
- Helm Chart
- Make DaemonSet update strategy configurable via `values.yaml` . #6136
- Introduce `extraEnv` variable for all components. #6142



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- Add `affinity` and `nodeSelector` support for ovs-ovn and ovs-ovn-dpdk DaemonSets. #6308
- Add `external-gateway-config-ns` option for controller. #6211
- Introduce ServiceMonitor and labels to services. #6340

## 8.2.3 Post-v1.14.0

- ACL log supports ratelimiting. #5938
- Subnet with centralized gateway now supports nodeSelectors. #5956
- Overlay encapsulation NIC selection. #5946
- Performace: skip conntrack for specific dst CIDRs. #5821
- NetworkPolicy supports `lax` mode which only deny traffic type of TCP, UDP and SCTP. That means ARP, ICMP and DHCP traffic are alaways allowed. #5745
- Remove internal-port type interface code. #5794
- IPPool
- Multiple IPPools now can bind to the same Namespace. #5731
- Pods in a bound namespace will only get IPs from the bound pool(s), not other ranges in the subnet. #5731
- IPPool will create an AddressSet that can be work with VPC Policy Route and ACL. #5920
- `AdminNetworkPolicy` now supports specify egress peers using FQDNs. #5703
- Using ARP for IPv4 network ready check: now you don't need ACL allow rules for gateway to make Pod running. #5716
- Non-primary CNI mode: you can run Kube-OVN as the secondary only network, without annoying unused annotations and logical switch port allocations. #5618
- VPC NAT Gateway:
- No default EIP mode: the secondary interface can initialize without a default EIP to avoid the waste. #5605
- Custom routes: you can control the route rules within the vpc-nat-gateway Pods to control traffic paths. #5608
- Gratuitous ARP: VPC NAT Gateway automatically sends gratuitous ARP packets during initialization to accelerate network convergence. #5607
- Healthchecks for static endpoints in `SwitchLBRules`: SLR with both selector or endpoints key can support healthchecks. #5435
- Underlay
- Node Selectors for `ProviderNetwork`: instead of adding/removing nodes to the `ProviderNetwork` one by one, you can use node selectors to simplify the workflow. #5518
- Different `NetworkProvider`s can now share the same VLAN. #5471
- Auto create VLAN sub-interfaces. #5966
- Auto move VLAN sub-interfaces to OVS bridges. #5949
- Adding `pod_name` and `pod_namespace` labels to interface metrics. #5463
- IPSec
- Support `cert-manager` to issue certificates. #5365
- Request new certificate if current certificate is not trusted. #5710
- kubectl-ko
- Collect IPSec and xFRM information. #5472
- Replace `Endpoint` with `EndpointSlice`. #5425
- NetworkAttachment caching: reduce APIServer load in large-scale deployments with Multus. #5386
- Upgrade `OVS` to 3.5 and `OVN` to 25.03. #5537

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## 8.2.4 ⬚⬚

# 8.3 ████████

██████████████████Kube-OVN ██████ ████████████████████████████████████████████████████████████ ████████████████

## 8.3.1 ███████

Kube-OVN ██████████████████

1. `default` ██████ Pod ██ IP ████████████ CIDR ▯ `10.16.0.0/16` ████ `10.16.0.1` ▯
2. `join` ██████ Node ▯ Pod ████████████████, ██ CIDR ▯ `100.64.0.0/16` ████ `100.64.0.1` ▯

████████████████████████

```
POD_CIDR="10.16.0.0/16"
POD_GATEWAY="10.16.0.1"
JOIN_CIDR="100.64.0.0/16"
EXCLUDE_IPS=""
```

`EXCLUDE_IP` ███ `POD_CIDR` ██████████████████ 192.168.10.20..192.168.10.30 ▯

████ Overlay ████████████████████ Service CIDR ███

████████████████████████████████████ Join ███

## 8.3.2 Service ████

████ `kube-proxy` ███ iptables ███████ Kube-OVN ████████████ Kube-OVN ████ Service ▯ CIDR ████████████

████████████████

```
SVC_CIDR="10.96.0.0/12"
```

███████

████████████ `kube-ovn-controller` Deployment ████

```
args:
- --service-cluster-ip-range=10.96.0.0/12
```

███████

## 8.3.3 Overlay ████

██████████████████Kube-OVN ██████ Kubernetes Node IP ████████████████████████████████

████████████████████████████████████

```
IFACE=eth1
```

████████████████████ `ens[a-z0-9]*,eth[a-z0-9]*` ▯

██████████ `kube-ovn-cni` DaemonSet ██████████

```
args:
- --iface=eth1
```

████████████████████████████████ annotation `ovn.kubernetes.io/tunnel_interface` ██████████████████ annotation ██████ `iface` ████████ annotation▯

```
kubectl annotate node no1 ovn.kubernetes.io/tunnel_interface=ethx
```

████████████████████ Overlay ██████████▯



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### 8.3.4 MTU ⬚⬚

⬚⬚ Overlay ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚Kube-OVN ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ MTU ⬚⬚⬚⬚⬚⬚⬚ MTU ⬚⬚⬚ ⬚⬚⬚⬚⬚ Overlay ⬚⬚⬚ Pod ⬚⬚ MTU ⬚⬚⬚⬚⬚ MTU - 100⬚Underlay ⬚⬚⬚⬚Pod ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

```
ENABLE_EIP_SNAT=false
```

⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `kube-ovn-controller` Deployment ⬚⬚⬚⬚⬚⬚⬚⬚

```
args:
- --enable-eip-snat=false
```

EIP ⬚ SNAT ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

### 8.3.13 [illegible]

Kube-OVN [illegible] Overlay [illegible] Geneve[illegible] Vxlan [illegible] STT[illegible]

```
TUNNEL_TYPE="vxlan"
```

[illegible] `ovs-ovn` DaemonSet [illegible]

```
env:
- name: TUNNEL_TYPE
  value: "vxlan"
```

[illegible] STT [illegible] ovs [illegible]

[illegible]

### 8.3.14 SSL [illegible]

OVN DB [illegible] API [illegible] SSL [illegible]:

```
ENABLE_SSL=true
```

SSL [illegible]

### 8.3.15 [illegible] conntrack [illegible] CIDR [illegible]

[illegible] conntrack [illegible] NAT/[illegible] `kube-ovn-controller` [illegible] CIDR [illegible]

```
SKIP_CONNTRACK_DST_CIDRS="10.10.0.0/16,fd00:10::/64"
```

```
args:
- --skip-conntrack-dst-cidrs=10.10.0.0/16,fd00:10::/64
```

[illegible]

### 8.3.16 [illegible] OVN IPSec

[illegible] Kube-OVN [illegible] OVN [illegible] IPSec [illegible] Geneve/Vxlan/STT [illegible]

```
ENABLE_OVN_IPSEC=true
```

[illegible] daemon [illegible]

```
args:
- --enable-ovn-ipsec=true
```

[illegible] OVN IPSec[illegible]

### 8.3.17 AdminNetworkPolicy / DNS [illegible]

[illegible] AdminNetworkPolicy[illegible]ANP/CNP[illegible] DNS [illegible] `DNSNameResolver` CRD[illegible]

```
ENABLE_ANP=true
ENABLE_DNS_NAME_RESOLVER=true
```

```
args:
- --enable-anp=true
- --enable-dns-name-resolver=true
```

[illegible][illegible]



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### 8.3.18 VPC NAT Gateway ...

...... VPC NAT Gateway .. controller......

```
ENABLE_NAT_GW=true
```

.. `ovn-vpc-nat-gw-config` ConfigMap ... `enable-vpc-nat-gw` ..........

### 8.3.19 ovn-northd ...

.......... `ovn-northd` ..............

```
OVN_NORTHD_N_THREADS=4
```

.. 1..... CPU .....

### 8.3.20 NetworkPolicy ....

Kube-OVN .. `standard` / `lax` .. NetworkPolicy ......................

```
NP_ENFORCEMENT=standard
```

```
args:
- --np-enforcement=standard
```

..... NetworkPolicy ... `ovn.kubernetes.io/network_policy_enforcement` annotation ..... NetworkPolicy ...

### 8.3.21 VXLAN .. TX checksum offload

.... VXLAN ..... checksum offload .........

```
SET_VXLAN_TX_OFF=true
```

```
args:
- --set-vxlan-tx-off=true
```

### 8.3.22 KubeVirt .....

......... KubeVirt ..................................

```
ENABLE_LIVE_MIGRATION_OPTIMIZE=true
```

### 8.3.23 OVN LB .... endpoint

... OVN .. LoadBalancer ..... endpoint... `externalTrafficPolicy=Local` ...

```
ENABLE_OVN_LB_PREFER_LOCAL=true
```

```
args:
- --enable-ovn-lb-prefer-local=true
```

### 8.3.24 .... ConfigMap ....

......... ConfigMap.. `external-gw-config` .... `kube-system` ..................

```
EXTERNAL_GATEWAY_CONFIG_NS=kube-system
```



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## 8.3.25 Non-Primary CNI ⬚⬚

⬚⬚⬚⬚ Kube-OVN ⬚⬚⬚ attachment network ⬚⬚⬚⬚⬚⬚⬚ Pod ⬚⬚⬚ IP⬚⬚⬚⬚⬚⬚ CNI ⬚⬚⬚controller ⬚ daemon ⬚⬚⬚⬚⬚⬚⬚⬚

```
args:
- --non-primary-cni-mode=true
```

⬚⬚⬚ Helm `cni_conf.NON_PRIMARY_CNI=true` ⬚chart v1⬚/ `cni.nonPrimaryCNI=true` ⬚chart v2⬚⬚⬚⬚⬚ Non-Primary CNI ⬚⬚⬚

## 8.3.26 ⬚⬚⬚⬚ ip

kube-ovn-controller/kube-ovn-cni/kube-ovn-monitor ⬚⬚⬚⬚⬚⬚⬚⬚⬚ ip⬚⬚⬚⬚⬚⬚⬚⬚⬚

## 8.4 □□□□

Kube-OVN □□□□□□ Minor □□□ Patch □□□Minor □□□□□□□□□□□□OVN/OVS □□□□□□□□□□□□□API □□□Patch □□□□□□ Bug □□□□□□□□□□□□□□□□□□□□□□□□□ API□

### 8.4.1 □□□□

Kube-OVN □□□□□□□□□□□□□□□□□□□□□□□□□□ `master`, `release-1.12` □ `release-1.11` □□□□□□□□□□□□□ `release-1.12` □□□□□□□□□□□□□□□□□□ Bug □□□□□□□□□□□□□□□□□□□ backport □□□□□□□□□

□□□□□□□□□□□ `release-1.11` □□ backport □□□□□ Bug □□□□□□□□□□□□□□□□□□□

### 8.4.2 □□□□

Minor □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□Patch □□□□□□ Bug □□□□□□□□□□□□ Bug □□□□□□□□□□□□

### 8.4.3 Patch □□□□□□

□□ Patch □□□□□□□□□□□□□ hack/release.sh □□□□□□□□□□□□□□□□□□□□□□□□□□□□

1. □□□□□□ Build □□□□□□□□
2. □□□ tag □□□ Docker Hub□□□□□
3. □□□ tag □□□ Github□□□□□
4. □□□□□□□□□□□□□□□□□
5. □□□□□□□□□□□□□□□□□
6. □□ Release Note PR□□□□□
7. □□ Release Note (□□)
8. □□ Merge github action □□□ Release Note PR
9. □□ Github Release □□□□□□
10. □ Github Release □□□□□□□□ Release□□□□□□□□□□□□□□□□ `v1.12.12` □□□□□□□□□□□□ Release Note □ Release □□

### 8.4.4 Minor □□□□□□

□□ Minor □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

1. □ Github □□□□□□□□□□□□□□□□ `release-1.13` (□□)
2. □□□□□□ `VERSION`, `dist/images/install.sh`, `charts/kube-ovn/values.yaml` □ `charts/kube-ovn/Chart.yaml` □□□□□□□□□□□□ Minor □□□□□ `v1.14.0` (□□)
3. □□□ tag □□□ Docker Hub (□□)
4. □□□□□□□□□ tag □□□ Github (□□)
5. □□□□□□□□□□□□□□□□ `v1.13` □□□ `mkdocs.yml` □□□□ `version` □ `branch` □□ (□□)
6. □□ Release Note PR□□□□□
7. □□ Release Note (□□)
8. □□ Merge github action □□□ Release Note PR
9. □□ Github Release □□□□□□
10. □ Github Release □□□□□□□□ Release□□□□□□□□□□□□□□□□ `v1.13.0` □□□□□□□□□□□□ Release Note □ Release □□
11. □□□□□□□□ `VERSION` □□□□□□□□□□□ Patch □□□□□ `v1.13.1`

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GitHub 



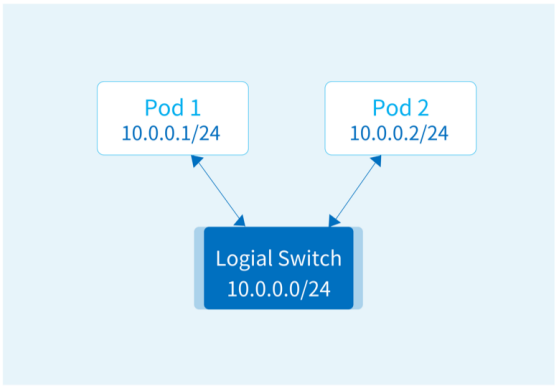
## 8.4.5 ⬜⬜

# 8.5 Underlay ▯▯▯▯

▯▯▯▯ Underlay ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯

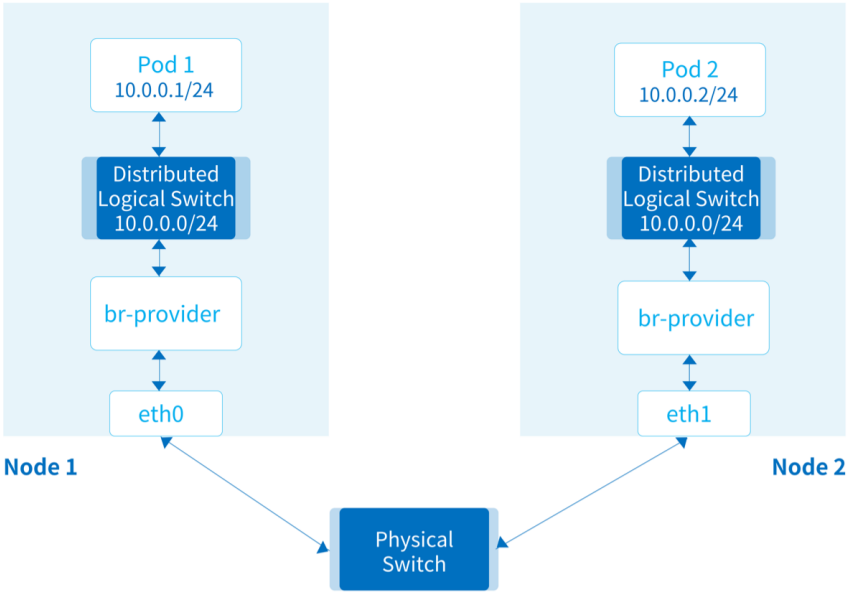
## 8.5.1 ▯▯▯▯▯▯▯

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯



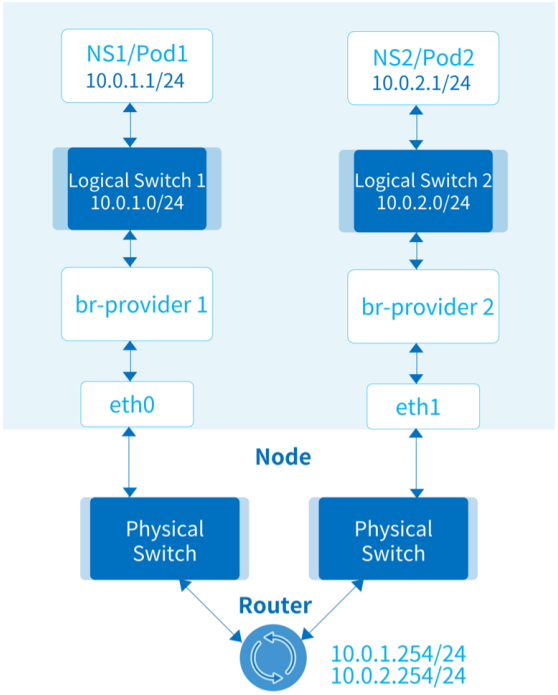

## 8.5.2 ▯▯▯▯▯▯▯

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯




## 8.5.3 ████████

████████████████████████████████████████████



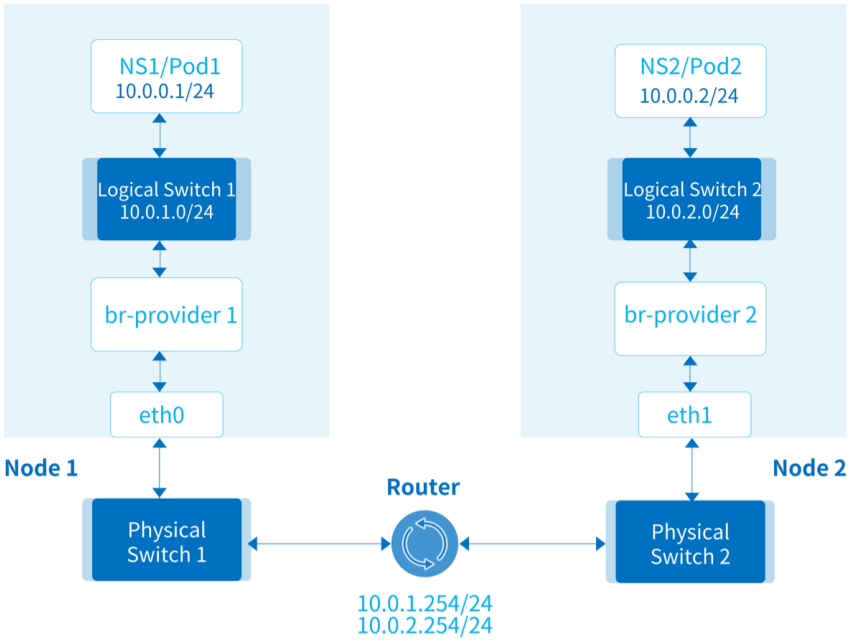

> ██ br-provider-1 ▯ br-provider-2 ██████ OVS ████████████████████ Provider Network▯



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## 8.5.4 ████████

████████████████████████████████████████████

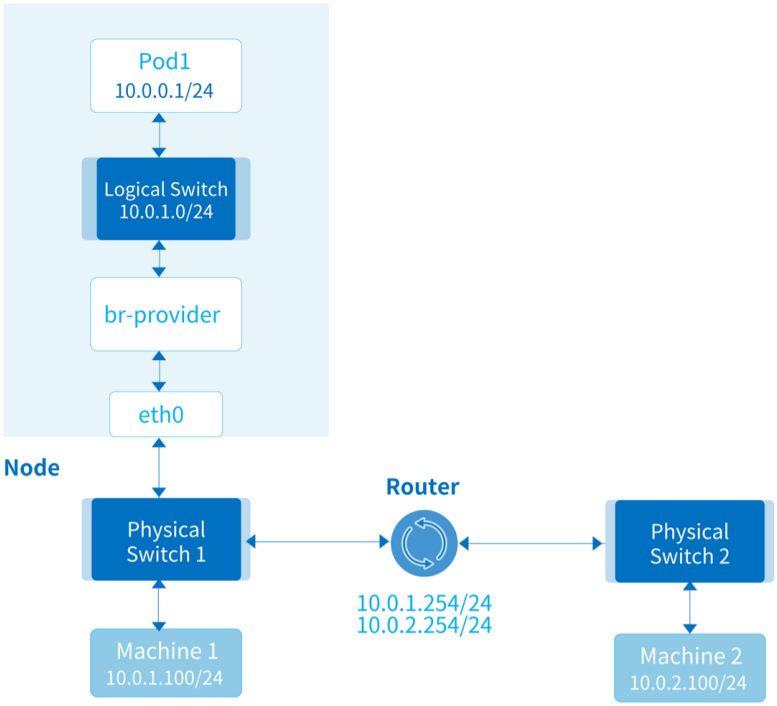



## 8.5.5 ████

████████████████████████████████████████████

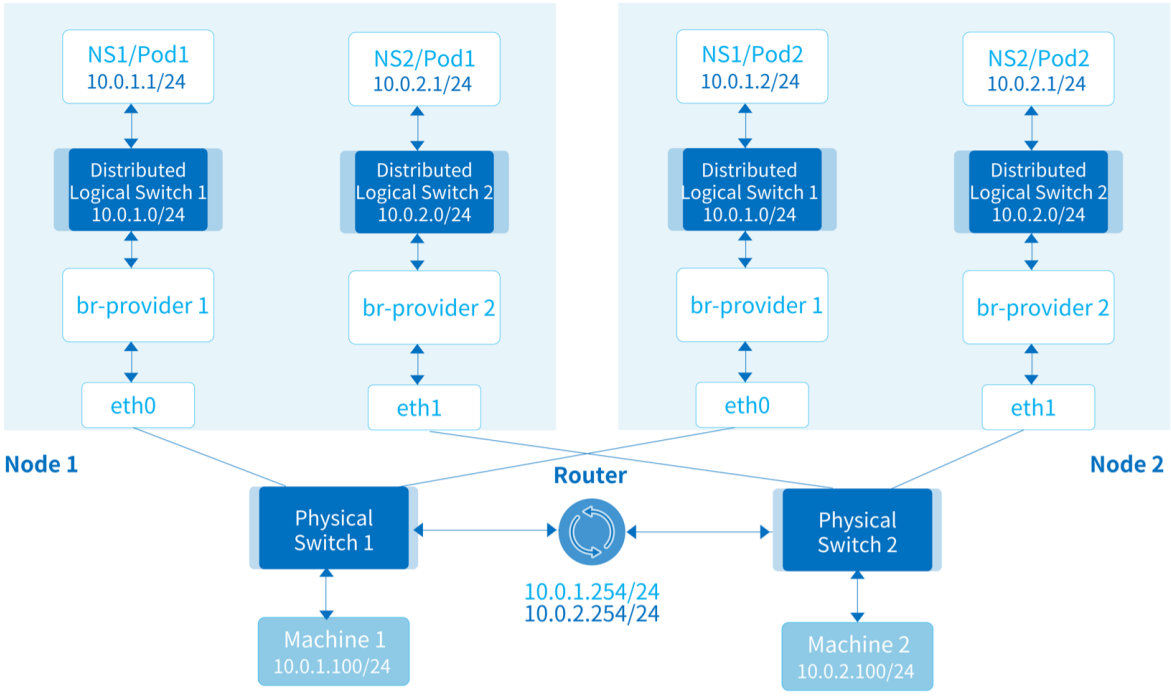


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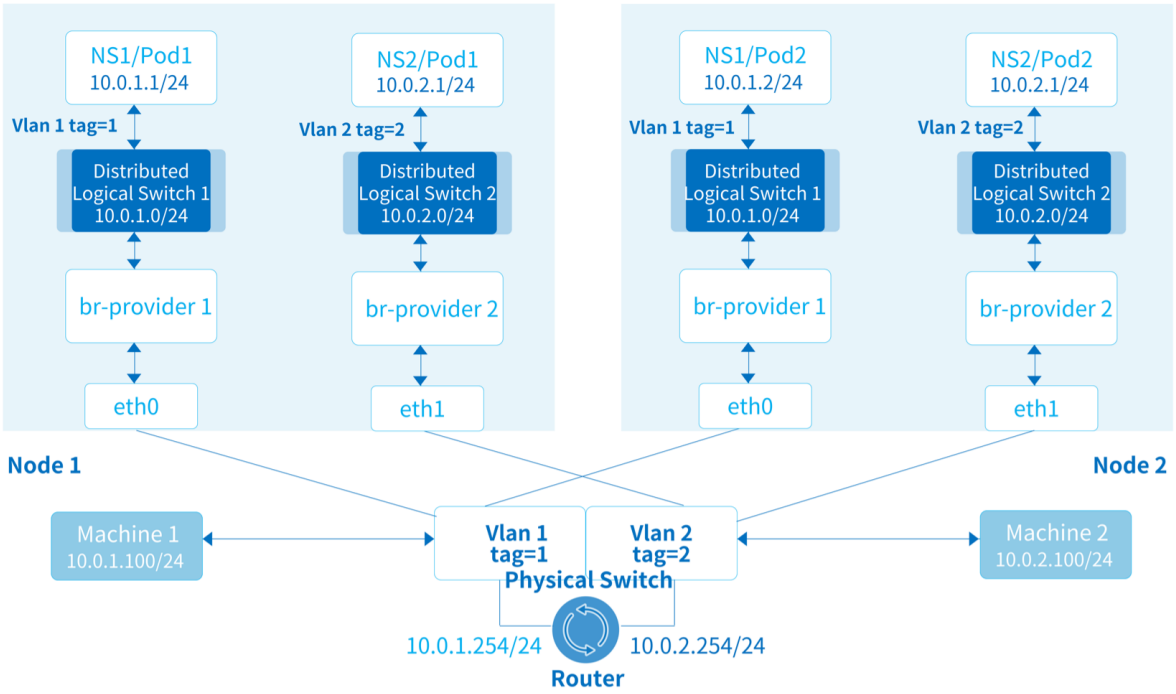



Pod

## 8.5.6 Vlan Tag



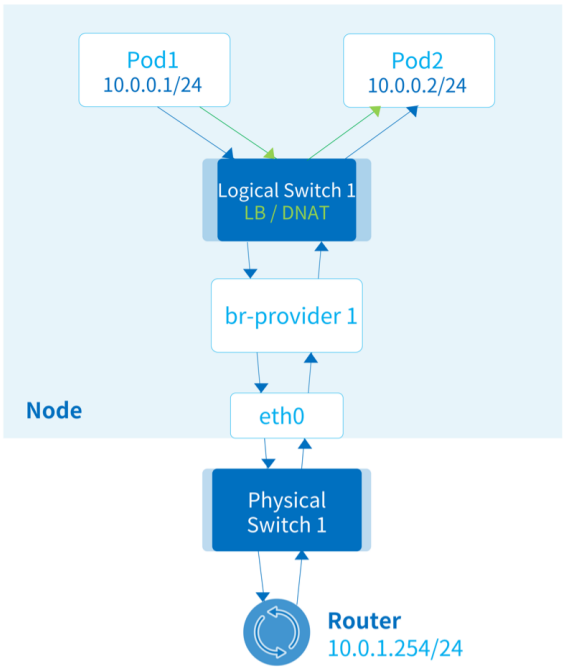
## 8.5.7 □ VLAN □□



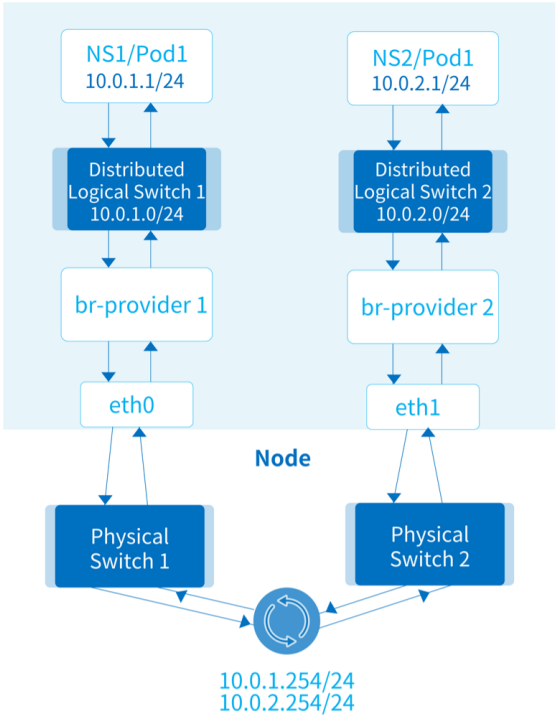

## 8.5.8 Pod □□ Service IP

Kube-OVN □□□ Kubernetes Service □□□□□□□□□□□□□□□□□□□□□□□ □ Pod □□□□ Service IP □□□□ Pod □□□□□□□□□□□□□ Service IP□□□ MAC □□□□□ MAC □□□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□□□ DNAT □□□□□□ IP □□□□□□ Service □□□□□ Endpoint □ IP □□□□ □□□□□□□□□□□□□□□□□□□ MAC □□□□□□□□□□□□□□□□□□□□

**Service ▯▯▯▯▯▯▯▯▯▯ Pod**




**Service ▯▯▯▯▯▯▯▯▯▯ Pod**




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## 8.5.9 □□



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## 8.6 Iptables ⬚⬚

Kube-OVN ⬚⬚ ipset ⬚ iptables ⬚⬚⬚⬚⬚ VPC ⬚⬚⬚⬚⬚Overlay⬚⬚⬚ NAT ⬚⬚⬚⬚

⬚⬚⬚ ipset ⬚⬚⬚⬚⬚⬚

| ⬚⬚⬚**IPv4/IPv6**⬚ | ⬚⬚ | ⬚⬚⬚⬚ |
|---|---|---|
| ovn40services/ovn60services | hash:net | Service ⬚⬚ |
| ovn40subnets/ovn60subnets | hash:net | Overlay ⬚⬚⬚⬚⬚⬚ NodeLocal DNS IP ⬚⬚ |
| ovn40subnets-nat/ovn60subnets-nat | hash:net | ⬚⬚ NatOutgoing ⬚ Overlay ⬚⬚⬚⬚ |
| ovn40subnets-distributed-gw/ovn60subnets-distributed-gw | hash:net | ⬚⬚⬚⬚⬚⬚⬚⬚ Overlay ⬚⬚⬚⬚ |
| ovn40other-node/ovn60other-node | hash:net | ⬚⬚⬚⬚⬚⬚⬚ IP ⬚⬚ |
| ovn40local-pod-ip-nat/ovn60local-pod-ip-nat | hash:ip | ⬚⬚⬚⬚ |
| ovn40subnets-nat-policy | hash:net | ⬚⬚⬚ natOutgoingPolicyRules ⬚⬚⬚⬚⬚⬚⬚ |
| ovn40natpr-418e79269dc5-dst | hash:net | natOutgoingPolicyRules ⬚ rule ⬚⬚⬚ dstIPs |
| ovn40natpr-418e79269dc5-src | hash:net | natOutgoingPolicyRules ⬚ rule ⬚⬚⬚ srcIPs |



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▯▯▯ iptables ▯▯▯IPv4▯▯▯▯▯▯▯

| □ | □ | □□ | □□ | □□ |
|---|---|---|---|---|
| filter | INPUT | -m set --match-set ovn40services src -j ACCEPT | □□ k8s Service □ Pod □□□□ □□ | -- |
| filter | INPUT | -m set --match-set ovn40services dst -j ACCEPT | □□ | -- |
| filter | INPUT | -m set --match-set ovn40subnets src -j ACCEPT | □□ | -- |
| filter | INPUT | -m set --match-set ovn40subnets dst -j ACCEPT | □□ | -- |
| filter | FORWARD | -m set --match-set ovn40services src -j ACCEPT | □□ | -- |
| filter | FORWARD | -m set --match-set ovn40services dst -j ACCEPT | □□ | -- |
| filter | FORWARD | -m set --match-set ovn40subnets src -j ACCEPT | □□ | -- |
| filter | FORWARD | -m set --match-set ovn40subnets dst -j ACCEPT | □□ | -- |
| filter | FORWARD | -s 10.16.0.0/16 -m comment --comment "ovn-subnet-gateway,ovn-default" | □□□□□ subnet □□□□□□□□□□ | 10.16.0.0/16 □ subnet □ cidr □comment □□□ □ ovn-subnet-gateway □□□□□ iptables □□□□ subnet □□□□□□□□□□□□□□ ovn-default □□ subnet □□ |
| filter | FORWARD | -d 10.16.0.0/16 -m comment --comment "ovn-subnet-gateway,ovn-default" | □□□□□□□□□□□□ subnet □□□ | □□ |
| filter | OUTPUT | -p udp -m udp --dport 6081 -j MARK --set-xmark 0x0 | □□□□□□□□□□□□ SNAT | UDP: bad checksum on VXLAN interface |
| nat | PREROUTING | -m comment --comment "kube-ovn prerouting rules" -j OVN-PREROUTING | □□ OVN-PREROUTING □□□ | -- |
| nat | POSTROUTING | -m comment --comment "kube-ovn postrouting rules" -j OVN-POSTROUTING | □□ OVN-POSTROUTING □ □□ | -- |
| nat | OVN-PREROUTING | -i ovn0 -m set --match-set ovn40subnets src -m set --match-set ovn40services dst -j MARK --set-xmark 0x4000/0x4000 | □ Pod □□ Service □□□□ masquerade □□ | □□□□□□□ LB □□□ |
| nat | OVN-PREROUTING | -p tcp -m addrtype --dst-type LOCAL -m set --match- | | □ kube-proxy □□ ipvs □□□□□ |



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| ⬚ | ⬚ | ⬚⬚ | ⬚⬚ | ⬚⬚ |
|---|---|---|---|---|
| | | set KUBE-NODE-PORT-LOCAL-TCP dst -j MARK --set-xmark 0x80000/0x80000 | ⬚ ExternalTrafficPolicy ⬚ Local ⬚ Service ⬚⬚⬚TCP⬚⬚ ⬚⬚⬚⬚⬚ | |
| nat | OVN-PREROUTING | -p udp -m addrtype --dst-type LOCAL -m set --match-set KUBE-NODE-PORT-LOCAL-UDP dst -j MARK --set-xmark 0x80000/0x80000 | ⬚ ExternalTrafficPolicy ⬚ Local ⬚ Service ⬚⬚⬚UDP⬚⬚ ⬚⬚⬚⬚⬚ | ⬚⬚ |
| nat | OVN-POSTROUTING | -m set --match-set ovn40services src -m set --match-set ovn40subnets dst -m mark --mark 0x4000/0x4000 -j SNAT --to-source | ⬚⬚⬚⬚ Service IP ⬚⬚ Overlay Pod ⬚⬚⬚⬚ IP ⬚⬚⬚ IP⬚ | ⬚ kube-proxy ⬚⬚ ipvs ⬚⬚⬚⬚ |
| nat | OVN-POSTROUTING | -m mark --mark 0x4000/0x4000 -j MASQUERADE | ⬚⬚⬚⬚⬚⬚⬚⬚⬚ SNAT | -- |
| nat | OVN-POSTROUTING | -m set --match-set ovn40subnets src -m set --match-set ovn40subnets dst -j MASQUERADE | ⬚⬚⬚⬚⬚ Pod ⬚⬚⬚ Service ⬚⬚ ⬚⬚ SNAT | -- |
| nat | OVN-POSTROUTING | -m mark --mark 0x80000/0x80000 -m set --match-set ovn40subnets-distributed-gw dst -j RETURN | ⬚⬚ ExternalTrafficPolicy ⬚ Local ⬚ Service ⬚⬚⬚⬚ Endpoint ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ SNAT | -- |
| nat | OVN-POSTROUTING | -m mark --mark 0x80000/0x80000 -j MASQUERADE | ⬚⬚ ExternalTrafficPolicy ⬚ Local ⬚ Service ⬚⬚⬚⬚ Endpoint ⬚⬚⬚⬚⬚⬚⬚⬚⬚ SNAT | -- |
| nat | OVN-POSTROUTING | -p tcp -m tcp --tcp-flags SYN NONE -m conntrack --ctstate NEW -j RETURN | Pod IP ⬚⬚⬚⬚⬚⬚⬚⬚ SNAT | -- |
| nat | OVN-POSTROUTING | -s 10.16.0.0/16 -m set ! --match-set ovn40subnets dst -j SNAT --to-source 192.168.0.101 | Pod ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ NatOutgoing ⬚⬚⬚⬚⬚ IP ⬚⬚⬚ ⬚⬚⬚⬚⬚⬚ SNAT | 10.16.0.0/16 ⬚⬚⬚⬚⬚⬚192.168.0.101 ⬚⬚⬚⬚⬚⬚⬚ IP |
| nat | OVN-POSTROUTING | -m set --match-set ovn40subnets-nat src -m set ! --match-set ovn40subnets dst -j MASQUERADE | Pod ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ NatOutgoing⬚⬚⬚ SNAT | -- |
| nat | OVN-POSTROUTING | -m set --match-set ovn40subnets-nat-policy src -m set ! --match-set ovn40subnets dst -j OVN-NAT-POLICY | Pod ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ natOutgoingPolicyRules⬚ ⬚⬚⬚⬚⬚⬚⬚⬚⬚ SNAT | ⬚⬚⬚ natOutgoingPolicyRules ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ OVN-NAT-POLICY ⬚ |
| nat | | | ⬚⬚ | |



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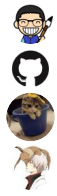
| ⬚ | ⬚ | ⬚⬚ | ⬚⬚ | ⬚⬚ |
|---|---|---|---|---|
| | OVN-POSTROUTING | -m mark --mark 0x90001/0x90001 -j MASQUERADE --random-fully | | ⬚ OVN-NAT-POLICY ⬚⬚⬚⬚⬚⬚⬚⬚ tag 0x90001/0x90001 ⬚⬚⬚ SNAT |
| nat | OVN-POSTROUTING | -m mark --mark 0x90002/0x90002 -j RETURN | ⬚⬚ | ⬚ OVN-NAT-POLICY ⬚⬚⬚, ⬚⬚⬚⬚⬚ tag 0x90002/0x90002 ⬚⬚⬚ SNAT |
| nat | OVN-NAT-POLICY | -s 10.0.11.0/24 -m comment --comment natPolicySubnet-net1 -j OVN-NAT-PSUBNET-aa98851157c5 | ⬚⬚ | 10.0.11.0/24 ⬚⬚⬚⬚ net1 ⬚ CIDR⬚ OVN-NAT-PSUBNET-aa98851157c5 ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ natOutgoingPolicyRules ⬚⬚ |
| nat | OVN-NAT-PSUBNET-xxxxxxxxxxxx | -m set --match-set ovn40natpr-418e79269dc5-src src -m set --match-set ovn40natpr-418e79269dc5-dst dst -j MARK --set-xmark 0x90002/0x90002 | ⬚⬚ | 418e79269dc5 ⬚⬚ natOutgoingPolicyRules ⬚⬚⬚⬚⬚⬚ ID⬚⬚⬚⬚⬚ status.natOutgoingPolicyRules[index].Rule ⬚⬚⬚⬚ ⬚⬚ srcIPs ⬚⬚ ovn40natpr-418e79269dc src⬚ dstIPS ⬚⬚ ovn40natpr-418e79269dc5-dst ⬚⬚⬚ tag 0x90002 |
| mangle | OVN-OUTPUT | -d 10.241.39.2/32 -p tcp -m tcp --dport 80 -j MARK --set-xmark 0x90003/0x90003 | ⬚ kubelet ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ ⬚ tproxy | |
| mangle | OVN-PREROUTING | -d 10.241.39.2/32 -p tcp -m tcp --dport 80 -j TPROXY --on-port 8102 --on-ip 172.18.0.3 --tproxy-mark 0x90004/0x90004 | ⬚ kubelet ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ ⬚ tproxy | |

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# 8.7 ████████

## 8.7.1 ████

Kube-OVN ████████████████████ Github ████████████/████████████ Github ███ Issue ▯ PR███ Maintainer ▯ Review ███ Github Action ████████████

## 8.7.2 ████

Kube-OVN ▯▯ Go ████ Go Modules ████ ███████ `GO111MODULE="on"` ▯

golangci-lint ████████████████████████████local-installation▯

████████████Kube-OVN ████ Docker buildx ███████ Docker ████ ███ buildx:

```
docker buildx create --use
```

## 8.7.3 ████

████████████████ Kube-OVN ████

```
git clone https://github.com/kubeovn/kube-ovn.git
cd kube-ovn
make release
```

████ ARM ████████████████████

```
make release-arm
```

## 8.7.4 ▯▯ base ▯▯

████████████████OVS/OVN ███████ base ██████████

base ████ Dockerfile ▯ `dist/images/Dockerfile.base` ▯

████

```
# build x86 base image
make base-amd64

# build arm base image
make base-arm64
```

## 8.7.5 ▯▯ E2E

Kube-OVN ▯▯:

- KIND ████ Kubernetes ███ `go install sigs.k8s.io/kind@latest`
- jinjanator ████: `pip install jinjanator`
- Ginkgo ███████ `go install github.com/onsi/ginkgo/v2/ginkgo; go get github.com/onsi/gomega/...`

> ██████████████████

███ E2E ███

```
make kind-init
make kind-install
make e2e
```

███ Underlay E2E ██████████



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```
make kind-init
make kind-install-underlay
make e2e-underlay-single-nic
```

▯▯▯▯ ovn vpc nat gw eip, fip, snat, dnat ▯▯▯▯▯▯▯▯▯▯

```
make kind-init
make kind-install
make ovn-vpc-nat-gw-conformance-e2e
```

▯▯▯▯ iptables vpc nat gw eip, fip, snat, dnat ▯▯▯▯▯▯▯▯▯▯

```
make kind-init
make kind-install-vpc-nat-gw
make iptables-vpc-nat-gw-conformance-e2e
```

▯▯▯▯ loadbalancer service ▯▯▯▯▯▯▯▯▯▯

```
make kind-init
make kind-install-lb-svc
make kube-ovn-lb-svc-conformance-e2e
```

▯▯▯▯▯▯▯▯▯▯▯▯▯

```
make kind-clean
```

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GitHub

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8.7.6 ▯▯

## 8.8 □□□ OVS/OVN □□

□□ OVN/OVS □□□□□□□□□ SDN □□□□□□□□□□□ Kubernetes □□□□□□□□□□□□□ □□ Kube-OVN □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□Kube-OVN □□□ OVN/OVS □□□□□□□□□□□□□ □□□□ OVN/OVS □□ Kube-OVN □□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□

□□□□□□□□□

- 4228eab1d7 □□ vswitchd ofport_usage □□□□□□□□□□□□□
- 54056ea65d □□□□ timer□□□□□□□□□□□□□□□
- 6b4dcb311f □□ fdb □□□□□□
- f627b7721e □□ hairpin □□□ fdb □□□□□□□□□
- 3f3e3a436f □ ovsdb-tool □ join-cluster □□□□□ Server ID □□□
- a6cb8215a8 □□ QoS □□□□□□□□
- d4d76ddb2e ovsdb-tool□□□ fix-cluster □□□
- ffd2328d4a netdev□□□□□□□□□□□ CPU □□□□
- d088c5d8c2 ovs-router□□□□□ kube-ipvs0 □□□□□
- 1b31f07dc6 □□□□□□□□□□□□□□□□□
- 54b7678229 □□ ovs-sandbox □□□ docker run□
- 9ee66bd91b □□□□□□□□□□
- e889d46924 □□ Underlay □□□□□ resubmit □□□□□□□
- f9e97031b5 ovn-controller□□□ Kube-OVN □□□ localnet □□□ GARP□
- 78cade0187 □□□□□□□□ conntrack □□□□
- 85aa6263ad northd□□□□□□□□ DNS IP □□□ conntrack□
- 34dc3e3fcf lflow□□□ lport □□□□□□□□□□□ conntrack□
- a297b840c2 □□□□□□□□□□□□□□ DNAT □□□□□□□□□ lsp□
- 03e35ed9c5 ovn-controller□□□□□□□□□□□□□□□□□□□
- e7d3ba53cd □□ ACL □□□□□□□□ DNS IP □ conntrack□
- 9286e1fd57 □□□□□□□□□□
- e5916eb53a □□□□□□□□□□□□□□□□ lr-lb DNAT □□□
- e4e6ea9c5f □□□□□□ BFD LRP□
- e76880e792 northd□□□ nb □□ version_compatibility□
- 477695a010 northd□□□□ localnet □□□ lrp □□□ arp/nd □□□
- 2209d9a335 □□ ovn-controller □□□□ LSP □□□□□□□□
- 52b727b331 □□□□□□□□□□□□ ovn-northd □□□□□□□□
- 53b9837f7f □□ LB □ ACL □□□□□□□□□□□□□
- 737d9f932e □□□ GW □□□□□□□□□□□□□
- sbrec-delete-null-check □□ ovn-controller □ sbrec □□□□□□□□□□□□□

□□□□□□□□□

- 20626ea909 □□□□□□□ LB □ ACL □□□□□□□□□□□□□□□□□□□□
- a2d9ff3ccd Deb □□□□□□□□□□□□□

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## 8.8.1 ▯▯



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## 8.9 ▯▯▯▯▯▯

Kube-OVN ▯▯ OVN/OVS ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `Geneve` ▯ `Vxlan` ▯ `STT` ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯

## 8.9.5 ⬚⬚



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# 8.10 Kube-OVN □□□□

□□□□□ Kube-OVN □□□□□□□□□□

## 8.10.1 ovn-monitor

OVN □□□□□□□□□□

| □□ | □□□ | □□ |
|---|---|---|
| Gauge | kube_ovn_ovn_status | OVN □□□□□ (2) □ follower□ (1) □ leader, (0) □□□□□□ |
| Gauge | kube_ovn_failed_req_count | OVN □□□□□□□ |
| Gauge | kube_ovn_log_file_size_bytes | OVN □□□□□□□□□ |
| Gauge | kube_ovn_db_file_size_bytes | OVN □□□□□□□□□□ |
| Gauge | kube_ovn_chassis_info | OVN chassis □□ (1) □□□□(0) □□□ |
| Gauge | kube_ovn_db_status | OVN □□□□□, (1) □□□□ (0) □□□□ |
| Gauge | kube_ovn_logical_switch_info | OVN logical switch □□□□□ (1)□□□□□□ logical switch □□□ |
| Gauge | kube_ovn_logical_switch_external_id | OVN logical switch external_id □□□□□ (1)□□□□□□ external-id □□□ |
| Gauge | kube_ovn_logical_switch_port_binding | OVN logical switch □ logical switch port □□□□□□□ (1)□□□□□□□□□□ |
| Gauge | kube_ovn_logical_switch_tunnel_key | □ OVN logical switch □□□ tunnel key □□□ |
| Gauge | kube_ovn_logical_switch_ports_num | OVN logical switch □ logical port □□□□ |
| Gauge | kube_ovn_logical_switch_port_info | OVN logical switch port □□□□□ (1)□□□□□□□□□□□ |
| Gauge | kube_ovn_logical_switch_port_tunnel_key | □ OVN logical switch port □□□ tunnel key □□□ |
| Gauge | kube_ovn_cluster_enabled | (1) OVN □□□□□□□□□ (0) OVN □□□□□□□□□□ |
| Gauge | kube_ovn_cluster_role | □□□□□□□□□□□□□ (1)□□□□□□□□□□□□□□□ |
| Gauge | kube_ovn_cluster_status | □□□□□□□□□□□□□ (1)□□□□□□□□□□□□□□□ |
| Gauge | kube_ovn_cluster_term | RAFT term □□□ |
| Gauge | kube_ovn_cluster_leader_self | □□□□□□□□□□ leader (1) □□ (0) □□□ |
| Gauge | kube_ovn_cluster_vote_self | □□□□□□□□□□□□□□□ leader (1) □□ (0) □□□ |
| Gauge | kube_ovn_cluster_election_timer | □□ election timer □□ |
| Gauge | kube_ovn_cluster_log_not_committed | □ commit □ RAFT □□□□□ |
| Gauge | kube_ovn_cluster_log_not_applied | □ apply □ RAFT □□□□□ |
| Gauge | kube_ovn_cluster_log_index_start | □□ RAFT □□□□□□□□□ |
| Gauge | kube_ovn_cluster_log_index_next | RAFT □□□□□□□□□□ |
| Gauge | kube_ovn_cluster_inbound_connections_total | □□□□□□□□□□□□ |
| Gauge | kube_ovn_cluster_outbound_connections_total | □□□□□□□□□□□□ |
| Gauge | kube_ovn_cluster_inbound_connections_error_total | □□□□□□□□□□□□□□□ |
| Gauge | kube_ovn_cluster_outbound_connections_error_total | □□□□□□□□□□□□□□□ |



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## 8.10.2 ovs-monitor

`ovsdb  vswitchd `

| □□ | □□□ | □□ |
|---|---|---|
| Gauge | ovs_status | OVS □□□□□ (1) □□□□(0) □□□□ |
| Gauge | ovs_info | OVS □□□□□□□ (1)□□□□□□□□□□□□ |
| Gauge | failed_req_count | OVS □□□□□□□ |
| Gauge | log_file_size_bytes | OVS □□□□□□□□□□□□□□ |
| Gauge | db_file_size_bytes | OVS □□□□□□□□□□□□□□□ |
| Gauge | datapath | Datapath □□□□□□□ (1)□□□□□□□□□□□□ |
| Gauge | dp_total | □□ OVS □ datapath □□□ |
| Gauge | dp_if | Datapath □□□□□□□□□ (1)□□□□□□□□□□□□ |
| Gauge | dp_if_total | □□ datapath □ port □□□ |
| Gauge | dp_flows_total | Datapath □ flow □□□ |
| Gauge | dp_flows_lookup_hit | Datapath □□□□□ flow □□□□□□ |
| Gauge | dp_flows_lookup_missed | Datapath □□□□□□ flow □□□□□□ |
| Gauge | dp_flows_lookup_lost | Datapath □□□□□□ userspace □□□□□□□□□ |
| Gauge | dp_masks_hit | Datapath □□□□□ mask □□□□□□ |
| Gauge | dp_masks_total | Datapath □ mask □□□□ |
| Gauge | dp_masks_hit_ratio | Datapath □ □□□□□ mask □□□□ |
| Gauge | interface | OVS □□□□□□□□□ (1)□□□□□□□□□□□□ |
| Gauge | interface_admin_state | □□□□□□□□ (0) □ down, (1) □ up, (2) □□□□□□ |
| Gauge | interface_link_state | □□□□□□□□ (0) □ down, (1) □ up, (2) □□□□□□ |
| Gauge | interface_mac_in_use | OVS Interface □□□ MAC □□ |
| Gauge | interface_mtu | OVS Interface □□□ MTU□ |
| Gauge | interface_of_port | OVS Interface □□□ OpenFlow Port ID□ |
| Gauge | interface_if_index | OVS Interface □□□ Index□ |
| Gauge | interface_tx_packets | OVS Interface □□□□□□ |
| Gauge | interface_tx_bytes | OVS Interface □□□□□□ |
| Gauge | interface_rx_packets | OVS Interface □□□□□□ |
| Gauge | interface_rx_bytes | OVS Interface □□□□□□ |
| Gauge | interface_rx_crc_err | OVS Interface □□□□□□□□□□□ |
| Gauge | interface_rx_dropped | OVS Interface □□□□□□□□ |
| Gauge | interface_rx_errors | OVS Interface □□□□□□□□ |
| Gauge | interface_rx_frame_err | OVS Interface □□□□□□□□ |
| Gauge | interface_rx_missed_err | OVS Interface □□□ miss □□□ |
| Gauge | interface_rx_over_err | OVS Interface □□□ overrun □□□ |
| Gauge | interface_tx_dropped | OVS Interface □□□□□□□□ |

| □□ | □□□ | □□ |
| --- | --- | --- |
| Gauge | interface_tx_errors | OVS Interface □□□□□□□□ |
| Gauge | interface_collisions | OVS interface □□□□□ |

## 8.10.3 kube-ovn-pinger

□□□□□□□□□□□□

| □□ | □□□ | □□ |
| --- | --- | --- |
| Gauge | pinger_ovs_up | □□ OVS □□□ |
| Gauge | pinger_ovs_down | □□ OVS □□□ |
| Gauge | pinger_ovn_controller_up | □□ ovn-controller □□□ |
| Gauge | pinger_ovn_controller_down | □□ ovn-controller □□□ |
| Gauge | pinger_inconsistent_port_binding | OVN-SB □ portbinding □□□□□ OVS interface □□□□□□□ |
| Gauge | pinger_apiserver_healthy | kube-ovn-pinger □□□□ apiserver□ |
| Gauge | pinger_apiserver_unhealthy | kube-ovn-pinger □□□□ apiserver□ |
| Histogram | pinger_apiserver_latency_ms | kube-ovn-pinger □□ apiserver □□□ |
| Gauge | pinger_internal_dns_healthy | kube-ovn-pinger □□□□□□□□□ |
| Gauge | pinger_internal_dns_unhealthy | kube-ovn-pinger □□□□□□□□□ |
| Histogram | pinger_internal_dns_latency_ms | kube-ovn-pinger □□□□□□□□□ |
| Gauge | pinger_external_dns_healthy | kube-ovn-pinger □□□□□□□□□ |
| Gauge | pinger_external_dns_unhealthy | kube-ovn-pinger □□□□□□□□□ |
| Histogram | pinger_external_dns_latency_ms | kube-ovn-pinger □□□□□□□□□ |
| Histogram | pinger_pod_ping_latency_ms | kube-ovn-pinger ping Pod □□□ |
| Gauge | pinger_pod_ping_lost_total | kube-ovn-pinger ping Pod □□□□□ |
| Gauge | pinger_pod_ping_count_total | kube-ovn-pinger ping Pod □□□ |
| Histogram | pinger_node_ping_latency_ms | kube-ovn-pinger ping Node □□□ |
| Gauge | pinger_node_ping_lost_total | kube-ovn-pinger ping Node □□□ |
| Gauge | pinger_node_ping_count_total | kube-ovn-pinger ping Node □□□ |
| Histogram | pinger_external_ping_latency_ms | kube-ovn-pinger ping □□□□ □□□ |
| Gauge | pinger_external_ping_lost_total | kube-ovn-pinger ping □□□□□□□ |



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### 8.10.4 kube-ovn-controller

`kube-ovn-controller` [illegible]

| [illegible] | [illegible] | [illegible] |
|---|---|---|
| Histogram | rest_client_request_latency_seconds | [illegible] apiserver [illegible] |
| Counter | rest_client_requests_total | [illegible] apiserver [illegible] |
| Counter | lists_total | API list [illegible] |
| Summary | list_duration_seconds | API list [illegible] |
| Summary | items_per_list | API list [illegible] |
| Counter | watches_total | API watch [illegible] |
| Counter | short_watches_total | [illegible] API watch [illegible] |
| Summary | watch_duration_seconds | API watch [illegible] |
| Summary | items_per_watch | API watch [illegible] |
| Gauge | last_resource_version | [illegible] resource version[illegible] |
| Histogram | ovs_client_request_latency_milliseconds | [illegible] OVN [illegible] |
| Gauge | subnet_available_ip_count | [illegible] IP [illegible] |
| Gauge | subnet_used_ip_count | [illegible] IP [illegible] |

### 8.10.5 kube-ovn-cni

`kube-ovn-cni` [illegible]

| [illegible] | [illegible] | [illegible] |
|---|---|---|
| Histogram | cni_op_latency_seconds | CNI [illegible] |
| Counter | cni_wait_address_seconds_total | CNI [illegible] |
| Counter | cni_wait_connectivity_seconds_total | CNI [illegible] |
| Counter | cni_wait_route_seconds_total | CNI [illegible] |
| Histogram | rest_client_request_latency_seconds | [illegible] apiserver [illegible] |
| Counter | rest_client_requests_total | [illegible] apiserver [illegible] |
| Counter | lists_total | API list [illegible] |
| Summary | list_duration_seconds | API list [illegible] |
| Summary | items_per_list | API list [illegible] |
| Counter | watches_total | API watch [illegible] |
| Counter | short_watches_total | [illegible] API watch [illegible] |
| Summary | watch_duration_seconds | API watch [illegible] |
| Summary | items_per_watch | API watch [illegible] |
| Gauge | last_resource_version | [illegible] resource version[illegible] |
| Histogram | ovs_client_request_latency_milliseconds | [illegible] OVN [illegible] |



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PDF Slack Support

2026⬚4⬚28⬚

2022⬚6⬚21⬚

GitHub 

8.10.6 ⬚⬚

# 8.11 Kube-OVN ⬚⬚⬚⬚

⬚⬚ Kube-OVN ⬚⬚⬚⬚⬚⬚⬚ Kube-OVN ⬚⬚⬚ CRD ⬚⬚⬚⬚⬚⬚ CRD ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

## 8.11.1 ⬚⬚⬚ Condition ⬚⬚

| ⬚⬚⬚⬚ | ⬚⬚ | ⬚⬚ |
| --- | --- | --- |
| type | String | ⬚⬚⬚⬚ |
| status | String | ⬚⬚⬚⬚⬚⬚⬚ `True` ⬚ `False` ⬚ `Unknown` |
| reason | String | ⬚⬚⬚⬚⬚⬚⬚ |
| message | String | ⬚⬚⬚⬚⬚⬚⬚⬚⬚ |
| observedGeneration | Int64 | ⬚⬚⬚⬚⬚⬚⬚⬚ |
| lastUpdateTime | Time | ⬚⬚⬚⬚⬚⬚⬚⬚ |
| lastTransitionTime | Time | ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ |

⬚⬚ CRD ⬚⬚⬚⬚⬚Status ⬚⬚ Condition ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚

## 8.11.2 ⬚⬚⬚⬚⬚⬚

**Subnet** ⬚⬚

SUBNET

| ⬚⬚⬚⬚ | ⬚⬚ | ⬚⬚ |
| --- | --- | --- |
| apiVersion | String | ⬚⬚ Kubernetes ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ kubeovn.io/v1 |
| kind | String | ⬚⬚ Kubernetes ⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚⬚ `Subnet` |
| metadata | ObjectMeta | ⬚⬚ Kubernetes ⬚⬚⬚⬚⬚⬚⬚ |
| spec | SubnetSpec | Subnet ⬚⬚⬚⬚⬚⬚⬚⬚ |
| status | SubnetStatus | Subnet ⬚⬚⬚⬚⬚⬚ |



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**SubnetSpec**

| ▯▯▯▯ | ▯▯ | ▯▯ |
|---|---|---|
| default | Bool | ▯▯▯▯▯▯▯▯▯▯ |
| vpc | String | ▯▯▯▯ VPC▯▯▯▯ ovn-cluster |
| protocol | String | IP ▯▯▯▯▯▯▯▯ `IPv4` ▯ `IPv6` ▯ `Dual` |
| namespaces | []String | ▯▯▯▯▯▯▯ namespace ▯▯ |
| cidrBlock | String | ▯▯▯▯▯▯▯▯▯ 10.16.0.0/16 |
| gateway | String | ▯▯▯▯▯▯▯▯▯▯▯▯ CIDRBlock ▯▯▯▯▯▯▯▯▯ |
| excludeIps | []String | ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ |
| provider | String | ▯▯▯ OVN▯▯▯▯▯▯▯▯▯▯▯▯▯▯ NetworkAttachmentDefinition ▯▯Kube-OVN ▯▯▯▯▯▯▯▯▯▯▯▯▯ Subnet ▯▯ |
| gatewayType | String | Overlay ▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `distributed` ▯ `centralized` |
| gatewayNode | String | ▯▯▯▯▯▯ centralized ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ |
| natOutgoing | Bool | ▯▯▯▯▯▯▯▯ NAT▯▯▯▯▯ `externalEgressGateway` ▯▯▯▯▯▯▯▯▯▯ |
| externalEgressGateway | String | ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `natOutgoing` ▯▯▯▯▯▯▯▯▯ |
| policyRoutingPriority | Uint32 | ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ |
| policyRoutingTableID | Uint32 | ▯▯▯▯▯▯▯▯▯▯▯▯ TableID▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ |
| mtu | Uint32 | ▯▯▯ MTU ▯▯ |
| private | Bool | ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ |
| allowSubnets | []String | ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ |
| vlan | String | ▯▯▯▯▯▯ Vlan ▯▯ |
| vips | []String | ▯▯▯ virtual ▯▯ lsp ▯ virtual-ip ▯▯▯▯ |
| logicalGateway | Bool | ▯▯▯▯▯▯▯▯▯ |
| disableGatewayCheck | Bool | ▯▯ Pod ▯▯▯▯▯▯▯▯▯▯▯▯▯ |
| disableInterConnection | Bool | ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ |
| enableDHCP | Bool | ▯▯▯▯▯▯▯▯▯▯ lsp ▯ dhcp ▯▯▯▯ |
| dhcpV4Options | String | ▯▯▯ lsp dhcpv4_options ▯▯▯ DHCP_Options ▯▯ |
| dhcpV6Options | String | ▯▯▯ lsp dhcpv6_options ▯▯▯ DHCP_Options ▯▯ |
| enableIPv6RA | Bool | ▯▯▯▯▯▯▯▯▯▯▯▯ lrp ▯▯▯▯▯▯▯▯ ipv6_ra_configs ▯▯ |
| ipv6RAConfigs | String | ▯▯▯▯▯▯▯▯▯ lrp ▯▯▯ipv6_ra_configs ▯▯▯▯▯▯▯ |
| acls | []Acl | ▯▯▯▯ logical-switch ▯▯▯ acls ▯▯ |
| allowEWTraffic | Bool | ▯▯▯▯▯▯▯▯▯▯ |
| natOutgoingPolicyRules | []NatOutgoingPolicyRule | NAT ▯▯▯▯▯▯▯ |
| u2oInterconnectionIP | String | Underlay/Overlay ▯▯▯▯▯▯ IP ▯▯ |
| u2oInterconnection | Bool | ▯▯▯▯ Overlay/Underlay ▯▯▯▯▯▯ |
| enableLb | *Bool | ▯▯▯▯▯▯▯▯ logical-switch ▯▯▯▯ load-balancer ▯▯ |
| enableEcmp | Bool | ▯▯▯▯▯▯▯▯▯▯▯▯ ECMP ▯▯ |



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| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| enableMulticastSnoop | Bool | [illegible] |
| enableExternalLBAddress | Bool | [illegible] |
| routeTable | String | [illegible] |
| namespaceSelectors | []LabelSelector | [illegible] |

Acl

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| direction | String | Acl [illegible] `from-lport` [illegible] `to-lport` |
| priority | Int | Acl [illegible] 0 [illegible] 32767 |
| match | String | Acl [illegible] |
| action | String | Acl [illegible] `allow-related` [illegible] `allow-stateless` [illegible] `allow` [illegible] `drop` [illegible] `reject` [illegible] |

NatOutgoingPolicyRule

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| match | NatOutGoingPolicyMatch | [illegible] |
| action | String | [illegible] |

NatOutGoingPolicyMatch

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| srcIPs | String | [illegible] IP [illegible] |
| dstIPs | String | [illegible] IP [illegible] |



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SubnetStatus

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| conditions | []SubnetCondition | [illegible] Condition [illegible] |
| v4availableIPs | BigInt | [illegible] IPv4 IP [illegible] IPv6 [illegible] float64 [illegible] |
| v4availableIPrange | String | [illegible] IPv4 [illegible] |
| v4usingIPs | BigInt | [illegible] IPv4 IP [illegible] |
| v4usingIPrange | String | [illegible] IPv4 [illegible] |
| v6availableIPs | BigInt | [illegible] IPv6 IP [illegible] |
| v6availableIPrange | String | [illegible] IPv6 [illegible] |
| v6usingIPs | BigInt | [illegible] IPv6 IP [illegible] |
| v6usingIPrange | String | [illegible] IPv6 [illegible] |
| activateGateway | String | [illegible] |
| dhcpV4OptionsUUID | String | [illegible] lsp dhcpv4_options [illegible] DHCP_Options [illegible] |
| dhcpV6OptionsUUID | String | [illegible] lsp dhcpv6_options [illegible] DHCP_Options [illegible] |
| u2oInterconnectionIP | String | [illegible] Overlay/Underlay [illegible] IP [illegible] |
| u2oInterconnectionMAC | String | [illegible] Overlay/Underlay [illegible] MAC [illegible] |
| u2oInterconnectionVPC | String | [illegible] Overlay/Underlay [illegible] VPC |
| natOutgoingPolicyRules | []NatOutgoingPolicyRuleStatus | NAT [illegible] |
| mcastQuerierIP | String | [illegible] IP [illegible] |
| mcastQuerierMAC | String | [illegible] MAC [illegible] |

**IP** [illegible]

IP

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| apiVersion | String | [illegible] Kubernetes [illegible] kubeovn.io/v1 |
| kind | String | [illegible] Kubernetes [illegible] `IP` |
| metadata | ObjectMeta | [illegible] Kubernetes [illegible] |
| spec | IPSpec | IP [illegible] |



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IPSpec

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| podName | String | [illegible] Pod [illegible] |
| namespace | String | [illegible] Pod [illegible] Namespace [illegible] |
| subnet | String | IP [illegible] Subnet |
| attachSubnets | []String | [illegible] IP [illegible] |
| nodeName | String | [illegible] Pod [illegible] |
| ipAddress | String | IP [illegible] `v4IP` `v6IP` [illegible] |
| v4IpAddress | String | IPv4 IP [illegible] |
| v6IpAddress | String | IPv6 IP [illegible] |
| attachIps | []String | [illegible] IP [illegible] IP [illegible] |
| macAddress | String | [illegible] Pod [illegible] MAC [illegible] |
| attachMacs | []String | [illegible] IP [illegible] MAC [illegible] |
| containerID | String | [illegible] Pod [illegible] Container ID |
| podType | String | [illegible] Pod[illegible] `StatefulSet` [illegible] `VirtualMachine` [illegible] |

**Vpc** [illegible]

VPC

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| apiVersion | String | [illegible] Kubernetes [illegible] kubeovn.io/v1 |
| kind | String | [illegible] Kubernetes [illegible] `Vpc` |
| metadata | ObjectMeta | [illegible] Kubernetes [illegible] |
| spec | VpcSpec | Vpc [illegible] |
| status | VpcStatus | Vpc [illegible] |

VpcSpec

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| defaultSubnet | String | [illegible] |
| namespaces | []String | Vpc [illegible] |
| staticRoutes | []StaticRoute | [illegible] |
| policyRoutes | []PolicyRoute | [illegible] |
| vpcPeerings | []VpcPeering | VPC [illegible] |
| enableExternal | Bool | [illegible] |
| extraExternalSubnets | []String | [illegible] |
| enableBfd | Bool | [illegible] BFD ([illegible]) |
| bfdPort | BFDPort | BFD [illegible] |



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StaticRoute

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| policy | String | [illegible] |
| cidr | String | [illegible] |
| nextHopIP | String | [illegible] IP [illegible] |
| ecmpMode | String | ECMP [illegible] |
| bfdId | String | BFD ID |
| routeTable | String | [illegible] |

PolicyRoute

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| priority | Int | [illegible] |
| match | String | [illegible] |
| action | String | [illegible] `allow` [illegible] `drop` [illegible] `reroute` |
| nextHopIP | String | [illegible] IP [illegible] action [illegible] reroute [illegible] |

VpcPeering

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| remoteVpc | String | [illegible] VPC [illegible] |
| localConnectIP | String | [illegible] IP [illegible] |

BFDPort

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| enabled | Bool | [illegible] BFD |
| ip | String | BFD [illegible] IP [illegible] |
| nodeSelector | LabelSelector | [illegible] BFD LRP [illegible] |



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VpcStatus

| ▯▯▯▯ | ▯▯ | ▯▯ |
|---|---|---|
| conditions | []VpcCondition | Vpc ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ Condition ▯▯ |
| standby | Bool | ▯▯▯▯▯ VPC |
| default | Bool | ▯▯▯▯▯ VPC |
| defaultLogicalSwitch | String | ▯▯▯▯▯▯▯▯▯▯ |
| router | String | ▯▯▯▯▯▯▯▯▯ |
| tcpLoadBalancer | String | TCP ▯▯▯▯▯▯▯▯ |
| udpLoadBalancer | String | UDP ▯▯▯▯▯▯▯▯ |
| sctpLoadBalancer | String | SCTP ▯▯▯▯▯▯▯▯ |
| tcpSessionLoadBalancer | String | TCP ▯▯▯▯▯▯▯▯▯▯ |
| udpSessionLoadBalancer | String | UDP ▯▯▯▯▯▯▯▯▯▯ |
| sctpSessionLoadBalancer | String | SCTP ▯▯▯▯▯▯▯▯▯▯ |
| subnets | []String | VPC ▯▯▯▯▯▯▯ |
| vpcPeerings | []String | VPC ▯▯▯▯▯▯▯ |
| enableExternal | Bool | ▯▯▯▯▯▯▯▯▯ |
| extraExternalSubnets | []String | ▯▯▯▯▯▯▯▯ |
| enableBfd | Bool | ▯▯▯▯ BFD |

## 8.11.3 Underlay ▯▯▯▯

**Vlan**

| ▯▯▯▯ | ▯▯ | ▯▯ |
|---|---|---|
| apiVersion | String | ▯▯ Kubernetes ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ kubeovn.io/v1 |
| kind | String | ▯▯ Kubernetes ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `Vlan` |
| metadata | ObjectMeta | ▯▯ Kubernetes ▯▯▯▯▯▯▯▯ |
| spec | VlanSpec | Vlan ▯▯▯▯▯▯▯▯▯ |
| status | VlanStatus | Vlan ▯▯▯▯▯▯▯ |

VLANSPEC

| ▯▯▯▯ | ▯▯ | ▯▯ |
|---|---|---|
| id | Int | Vlan tag ▯▯▯▯▯▯▯▯ 0~4096 |
| provider | String | Vlan ▯▯▯ ProviderNetwork ▯▯ |



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#### VLANSTATUS

| ... | ... | ... |
| --- | --- | --- |
| subnets | []String | Vlan ... |
| conflict | Bool | ... |
| conditions | []VlanCondition | Vlan ... Condition ... |

### ProviderNetwork

| ... | ... | ... |
| --- | --- | --- |
| apiVersion | String | ... Kubernetes ... kubeovn.io/v1 |
| kind | String | ... Kubernetes ... `ProviderNetwork` |
| metadata | ObjectMeta | ... Kubernetes ... |
| spec | ProviderNetworkSpec | ProviderNetwork ... |
| status | ProviderNetworkStatus | ProviderNetwork ... |

#### PROVIDERNETWORKSPEC

| ... | ... | ... |
| --- | --- | --- |
| defaultInterface | String | ... |
| customInterfaces | []CustomInterface | ... |
| nodeSelector | LabelSelector | ... OVS ... matchLabels ... matchExpressions... nodeSelector...excludeNodes ... |
| excludeNodes | []String | ... |
| exchangeLinkName | Bool | ... OVS ... |

#### CustomInterface

| ... | ... | ... |
| --- | --- | --- |
| interface | String | Underlay ... |
| nodes | []String | ... |

#### PROVIDERNETWORKSTATUS

| ... | ... | ... |
| --- | --- | --- |
| ready | Bool | ... |
| readyNodes | []String | ... |
| notReadyNodes | []String | ... |
| vlans | []String | ... Vlan ... |
| conditions | []ProviderNetworkCondition | ProviderNetwork ... Condition ... |



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## 8.11.4 [illegible]

**SecurityGroup**

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| apiVersion | String | [illegible] Kubernetes [illegible] kubeovn.io/v1 |
| kind | String | [illegible] Kubernetes [illegible] `SecurityGroup` |
| metadata | ObjectMeta | [illegible] Kubernetes [illegible] |
| spec | SecurityGroupSpec | SecurityGroup [illegible] |
| status | SecurityGroupStatus | SecurityGroup [illegible] |

SECURITYGROUPSPEC

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| ingressRules | []SecurityGroupRule | [illegible] |
| egressRules | []SecurityGroupRule | [illegible] |
| allowSameGroupTraffic | Bool | [illegible] |

SecurityGroupRule

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| ipVersion | String | IP [illegible] `ipv4` [illegible] `ipv6` |
| protocol | SgProtocol | [illegible] `all` [illegible] `icmp` [illegible] `tcp` [illegible] `udp` |
| priority | Int | [illegible] 1-200[illegible] |
| remoteType | SgRemoteType | [illegible] `address` [illegible] `securityGroup` |
| remoteAddress | String | [illegible] |
| remoteSecurityGroup | String | [illegible] |
| portRangeMin | Int | [illegible] 1 |
| portRangeMax | Int | [illegible] 65535 |
| policy | SgPolicy | [illegible] `allow` [illegible] `drop` |

SECURITYGROUPSTATUS

| [illegible] | [illegible] | [illegible] |
| --- | --- | --- |
| portGroup | String | [illegible] |
| allowSameGroupTraffic | Bool | [illegible] |
| ingressMd5 | String | [illegible] MD5 [illegible] |
| egressMd5 | String | [illegible] MD5 [illegible] |
| ingressLastSyncSuccess | Bool | [illegible] |
| egressLastSyncSuccess | Bool | [illegible] |



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## 8.11.5 □□□□□□□ IP

### Vip

| □□□□ | □□ | □□ |
| --- | --- | --- |
| apiVersion | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ kubeovn.io/v1 |
| kind | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ `Vip` |
| metadata | ObjectMeta | □□ Kubernetes □□□□□□□□ |
| spec | VipSpec | Vip □□□□□□□□□ |
| status | VipStatus | Vip □□□□□□□ |

VIPSPEC

| □□□□ | □□ | □□ |
| --- | --- | --- |
| namespace | String | VIP □□□□□□□□ |
| subnet | String | VIP □□□□□□ |
| type | String | VIP □□ |
| v4ip | String | IPv4 □□ |
| v6ip | String | IPv6 □□ |
| macAddress | String | MAC □□ |
| selector | []String | □□□□ |
| attachSubnets | []String | □□□□□□□□ |

VIPSTATUS

| □□□□ | □□ | □□ |
| --- | --- | --- |
| conditions | []VipCondition | VIP □□□□□□□□□□□□□□□□□□□ Condition □□ |
| type | String | VIP □□ |
| v4ip | String | IPv4 □□ |
| v6ip | String | IPv6 □□ |
| mac | String | MAC □□ |

### SwitchLBRule

| □□□□ | □□ | □□ |
| --- | --- | --- |
| apiVersion | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ kubeovn.io/v1 |
| kind | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ `SwitchLBRule` |
| metadata | ObjectMeta | □□ Kubernetes □□□□□□□□ |
| spec | SwitchLBRuleSpec | SwitchLBRule □□□□□□□□□ |
| status | SwitchLBRuleStatus | SwitchLBRule □□□□□□□ |



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#### SWITCHLBRULESPEC

| 字段名 | 类型 | 说明 |
| --- | --- | --- |
| vip | String | 虚拟 IP 地址 |
| namespace | String | 命名空间 |
| selector | []String | 选择器 |
| endpoints | []String | 端点列表 |
| sessionAffinity | String | 会话亲和性 |
| ports | []SwitchLBRulePort | 端口列表 |

#### SwitchLBRulePort

| 字段名 | 类型 | 说明 |
| --- | --- | --- |
| name | String | 端口名 |
| port | Int32 | 端口号 |
| targetPort | Int32 | 目标端口 |
| protocol | String | 协议类型 |

#### SWITCHLBRULESTATUS

| 字段名 | 类型 | 说明 |
| --- | --- | --- |
| conditions | []SwitchLBRuleCondition | SwitchLBRule 当前状态的详细条件 Condition 列表 |
| ports | String | SwitchLBRule 端口列表 |
| service | String | SwitchLBRule 关联的 service 名称 |

## 8.11.6 QoS 和 IP 管理

### QoSPolicy

| 字段名 | 类型 | 说明 |
| --- | --- | --- |
| apiVersion | String | 遵循 Kubernetes 的约定，值为 kubeovn.io/v1 |
| kind | String | 遵循 Kubernetes 的约定，值为 `QoSPolicy` |
| metadata | ObjectMeta | 遵循 Kubernetes 的约定 |
| spec | QoSPolicySpec | QoSPolicy 的规格定义 |

#### QOSPOLICYSPEC

| 字段名 | 类型 | 说明 |
| --- | --- | --- |
| bandwidthLimitRules | QoSPolicyBandwidthLimitRules | 带宽限制规则 |
| shared | Bool | 是否共享 |
| bindingType | QoSPolicyBindingType | 绑定类型 |



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**IPPool**

| □□□□ | □□ | □□ |
| --- | --- | --- |
| apiVersion | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ kubeovn.io/v1 |
| kind | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ `IPPool` |
| metadata | ObjectMeta | □□ Kubernetes □□□□□□□□ |
| spec | IPPoolSpec | IPPool □□□□□□□□□ |
| status | IPPoolStatus | IPPool □□□□□□□ |

IPPOOLSPEC

| □□□□ | □□ | □□ |
| --- | --- | --- |
| subnet | String | □□□□ |
| namespaces | []String | □□□□□□□□□□ |
| ips | []String | IP □□□□ |

IPPOOLSTATUS

| □□□□ | □□ | □□ |
| --- | --- | --- |
| v4AvailableIPs | BigInt | IPv4 □□ IP □□□□ |
| v4AvailableIPRange | String | IPv4 □□ IP □□□□ |
| v4UsingIPs | BigInt | IPv4 □□□ IP □□□□ |
| v4UsingIPRange | String | IPv4 □□□ IP □□□□ |
| v6AvailableIPs | BigInt | IPv6 □□ IP □□□□ |
| v6AvailableIPRange | String | IPv6 □□ IP □□□□ |
| v6UsingIPs | BigInt | IPv6 □□□ IP □□□□ |
| v6UsingIPRange | String | IPv6 □□□ IP □□□□ |
| conditions | []IPPoolCondition | IP □□□□□□□□□□□□□□□□□□□ Condition □□ |

## 8.11.7 NAT □□□ IP □□

**IptablesEIP**

| □□□□ | □□ | □□ |
| --- | --- | --- |
| apiVersion | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ kubeovn.io/v1 |
| kind | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ `IptablesEIP` |
| metadata | ObjectMeta | □□ Kubernetes □□□□□□□□ |
| spec | IptablesEIPSpec | IptablesEIP □□□□□□□□□ |
| status | IptablesEIPStatus | IptablesEIP □□□□□□□ |



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IPTABLESEIPSPEC

| ... | ... | ... |
|---|---|---|
| v4ip | String | IPv4 ... |
| v6ip | String | IPv6 ... |
| macAddress | String | MAC ... |
| natGwDp | String | NAT ... |
| qosPolicy | String | QoS ... |
| externalSubnet | String | ... |

IPTABLESEIPSTATUS

| ... | ... | ... |
|---|---|---|
| ready | Bool | IptablesEIP ... |
| ip | String | IptablesEIP ... IP ... IPv4 ... |
| redo | String | IptablesEIP CRD ... |
| nat | String | IptablesEIP ... `fip` ... `snat` ... `dnat` |
| qosPolicy | String | QoS ... |
| conditions | []IptablesEIPCondition | IptablesEIP ... Condition ... |

**OvnEip**

| ... | ... | ... |
|---|---|---|
| apiVersion | String | ... Kubernetes ... kubeovn.io/v1 |
| kind | String | ... Kubernetes ...`OvnEip` |
| metadata | ObjectMeta | ... Kubernetes ... |
| spec | OvnEipSpec | OvnEip ... |
| status | OvnEipStatus | OvnEip ... |

OVNEIPSPEC

| ... | ... | ... |
|---|---|---|
| externalSubnet | String | ... |
| v4Ip | String | IPv4 ... |
| v6Ip | String | IPv6 ... |
| macAddress | String | MAC ... |
| type | String | ... lrp、lsp ... nat |



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**IptablesFIPRule**

| □□□□ | □□ | □□ |
| --- | --- | --- |
| apiVersion | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ kubeovn.io/v1 |
| kind | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ `IptablesFIPRule` |
| metadata | ObjectMeta | □□ Kubernetes □□□□□□□□ |
| spec | IptablesFIPRuleSpec | IptablesFIPRule □□□□□□□□□ |

IPTABLESFIPRULESPEC

| □□□□ | □□ | □□ |
| --- | --- | --- |
| eip | String | □□ IP □□ |
| internalIP | String | □□ IP □□ |

**OvnFip**

| □□□□ | □□ | □□ |
| --- | --- | --- |
| apiVersion | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ kubeovn.io/v1 |
| kind | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ `OvnFip` |
| metadata | ObjectMeta | □□ Kubernetes □□□□□□□□ |
| spec | OvnFipSpec | OvnFip □□□□□□□□□ |
| status | OvnFipStatus | OvnFip □□□□□□□ |

OVNFIPSPEC

| □□□□ | □□ | □□ |
| --- | --- | --- |
| ovnEip | String | □□□ OVN EIP |
| ipType | String | IP □□□□□□ vip □ ip |
| ipName | String | IP □□ |
| vpc | String | □□ VPC |
| v4Ip | String | IPv4 □□ |
| v6Ip | String | IPv6 □□ |
| type | String | □□□□□□ distributed □ centralized |

**IptablesDnatRule**

| □□□□ | □□ | □□ |
| --- | --- | --- |
| apiVersion | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ kubeovn.io/v1 |
| kind | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ `IptablesDnatRule` |
| metadata | ObjectMeta | □□ Kubernetes □□□□□□□□ |
| spec | IptablesDnatRuleSpec | IptablesDnatRule □□□□□□□□□ |



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IPTABLESDNATRULESPEC

| □□□□ | □□ | □□ |
| --- | --- | --- |
| eip | String | □□ IP □□ |
| externalPort | String | □□□□ |
| protocol | String | □□□□ |
| internalIP | String | □□ IP □□ |
| internalPort | String | □□□□ |

**OvnDnatRule**

| □□□□ | □□ | □□ |
| --- | --- | --- |
| apiVersion | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ kubeovn.io/v1 |
| kind | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ `OvnDnatRule` |
| metadata | ObjectMeta | □□ Kubernetes □□□□□□□□ |
| spec | OvnDnatRuleSpec | OvnDnatRule □□□□□□□□□ |
| status | OvnDnatRuleStatus | OvnDnatRule □□□□□□□ |

OVNDNATRULESPEC

| □□□□ | □□ | □□ |
| --- | --- | --- |
| ovnEip | String | □□□ OVN EIP |
| ipType | String | IP □□□□□□ vip □ ip |
| ipName | String | IP □□ |
| internalPort | String | □□□□ |
| externalPort | String | □□□□ |
| protocol | String | □□□□ |
| vpc | String | □□ VPC |
| v4Ip | String | IPv4 □□ |
| v6Ip | String | IPv6 □□ |



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OVNDNATRULESTATUS

| □□□□ | □□ | □□ |
| --- | --- | --- |
| vpc | String | □□ VPC |
| v4Eip | String | IPv4 EIP □□ |
| v6Eip | String | IPv6 EIP □□ |
| externalPort | String | □□□□ |
| v4Ip | String | IPv4 □□ |
| v6Ip | String | IPv6 □□ |
| internalPort | String | □□□□ |
| protocol | String | □□□□ |
| ipName | String | IP □□ |
| ready | Bool | DNAT □□□□□□□□ |
| conditions | []OvnDnatRuleCondition | OVN DNAT □□□□□□□□□□□□□□□□□□□□ Condition □□ |

**IptablesSnatRule**

| □□□□ | □□ | □□ |
| --- | --- | --- |
| apiVersion | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ kubeovn.io/v1 |
| kind | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ `IptablesSnatRule` |
| metadata | ObjectMeta | □□ Kubernetes □□□□□□□□ |
| spec | IptablesSnatRuleSpec | IptablesSnatRule □□□□□□□□ |

IPTABLESSNATRULESPEC

| □□□□ | □□ | □□ |
| --- | --- | --- |
| eip | String | □□ IP □□ |
| internalCIDR | String | □□ CIDR □□ |

**OvnSnatRule**

| □□□□ | □□ | □□ |
| --- | --- | --- |
| apiVersion | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ kubeovn.io/v1 |
| kind | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ `OvnSnatRule` |
| metadata | ObjectMeta | □□ Kubernetes □□□□□□□□ |
| spec | OvnSnatRuleSpec | OvnSnatRule □□□□□□□□ |
| status | OvnSnatRuleStatus | OvnSnatRule □□□□□□ |



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#### OVNSNATRULESPEC

| □□□□ | □□ | □□ |
| --- | --- | --- |
| ovnEip | String | □□□ OVN EIP |
| vpcSubnet | String | VPC □□ |
| ipName | String | IP □□ |
| vpc | String | □□ VPC |
| v4IpCidr | String | IPv4 CIDR □□ |
| v6IpCidr | String | IPv6 CIDR □□ |

## 8.11.8 VPC □□□□

**VpcNatGateway**

| □□□□ | □□ | □□ |
| --- | --- | --- |
| apiVersion | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ kubeovn.io/v1 |
| kind | String | □□ Kubernetes □□□□□□□□□□□□□□□□□□□□ `VpcNatGateway` |
| metadata | ObjectMeta | □□ Kubernetes □□□□□□□□ |
| spec | VpcNatGatewaySpec | VpcNatGateway □□□□□□□□□ |
| status | VpcNatGatewayStatus | VpcNatGateway □□□□□□□ |

#### VPCNATGATEWAYSPEC

| □□□□ | □□ | □□ |
| --- | --- | --- |
| vpc | String | VPC □□ Pod □□□ VPC □□ |
| subnet | String | VPC □□ Pod □□□□□□□□ |
| externalSubnets | []String | □□□□□□□ |
| lanIp | String | VPC □□ Pod □□□□□ IP □□ |
| selector | []String | □□ Kubernetes Selector □□□□ |
| tolerations | []Toleration | □□ Kubernetes □□□□ |
| affinity | Affinity | □□ Kubernetes □□□□□ |
| qosPolicy | String | QoS □□□□ |
| bgpSpeaker | VpcBgpSpeaker | BGP speaker □□ |
| routes | []Route | □□□□□□□ |



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**VpcBgpSpeaker**

| ■■■■ | ■■ | ■■ |
|---|---|---|
| enabled | Bool | ■■■■ BGP speaker |
| asn | Uint32 | ■■■■■■■■ |
| remoteAsn | Uint32 | ■■■■■■■■ |
| neighbors | []String | BGP ■■■■ |
| holdTime | Duration | BGP ■■■■ |
| routerId | String | BGP ■■■ ID |
| password | String | BGP ■■■■ |
| enableGracefulRestart | Bool | ■■■■■■■■■ |
| extraArgs | []String | ■■■■■■■ |

**Route**

| ■■■■ | ■■ | ■■ |
|---|---|---|
| cidr | String | ■■■■ CIDR |
| nextHopIP | String | ■■■ IP ■■ |

**VPCNATGATEWAYSTATUS**

| ■■■■ | ■■ | ■■ |
|---|---|---|
| qosPolicy | String | QoS ■■■■ |
| externalSubnets | []String | ■■■■■■■ |
| selector | []String | ■■ Kubernetes Selector ■■■■ |
| tolerations | []Toleration | ■■ Kubernetes ■■■■ |
| affinity | Affinity | ■■ Kubernetes ■■■■■■ |

### VpcEgressGateway

| ■■■■ | ■■ | ■■ |
|---|---|---|
| apiVersion | String | ■■ Kubernetes ■■■■■■■■■■■■■■■■■■■■ kubeovn.io/v1 |
| kind | String | ■■ Kubernetes ■■■■■■■■■■■■■■■■■■■■ `VpcEgressGateway` |
| metadata | ObjectMeta | ■■ Kubernetes ■■■■■■■■ |
| spec | VpcEgressGatewaySpec | VpcEgressGateway ■■■■■■■■■ |
| status | VpcEgressGatewayStatus | VpcEgressGateway ■■■■■■■ |



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#### VPCEGRESSGATEWAYSPEC

| ... | ... | ... |
|---|---|---|
| vpc | String | ... VPC... VPC |
| bgpConf | String | ... `BgpConf` ... |
| evpnConf | String | ... `EvpnConf` ... EVPN ... |
| replicas | Int32 | ... 1 |
| prefix | String | ... `<prefix><veg-name>` |
| image | String | ... kube-ovn-controller ... |
| internalSubnet | String | ... VPC ... |
| externalSubnet | String | ... |
| internalIPs | []String | ... IP ... `v4,v6` ... |
| externalIPs | []String | ... IP ... `v4,v6` ... |
| internalIPPool | String | ... `IPPool` ... `internalSubnet` ... `internalIPs` ... |
| externalIPPool | String | ... `IPPool` ... `externalSubnet` ... `externalIPs` ... |
| selectors | []VpcEgressGatewaySelector | .../Pod ... |
| trafficPolicy | String | ... `Cluster` ... `Local` ... VPC ... |
| bfd | VpcEgressGatewayBFDConfig | BFD ... |
| policies | []VpcEgressGatewayPolicy | ... |
| nodeSelector | []VpcEgressGatewayNodeSelector | ... |
| tolerations | []Toleration | ... Kubernetes ... |
| resources | ResourceRequirements | ... |
| bandwidth | BandwidthLimit | .../... Mbps ... bit-rate quantity |
| observability | VpcEgressGatewayObservability | ...conntrack ...JSON ... ServiceMonitor ... |

#### VpcEgressGatewaySelector

| ... | ... | ... |
|---|---|---|
| namespaceSelector | LabelSelector | ... |
| podSelector | LabelSelector | Pod ... |

#### VpcEgressGatewayBFDConfig

| ... | ... | ... |
|---|---|---|
| enabled | Boolean | ... BFD... VPC ... `.spec.bfd.enabled=true` ... false |
| minRX | Int32 | BFD ...ms... 1000 |
| minTX | Int32 | BFD ...ms... 1000 |
| multiplier | Int32 | BFD ... 3 |



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#### VpcEgressGatewayPolicy

| ■■■■ | ■■ | ■■ |
| --- | --- | --- |
| snat | Boolean | ■■■■■■■■■■ SNAT/MASQUERADE■■■ false |
| ipBlocks | []String | ■■■ CIDR ■■ |
| subnets | []String | ■■■■■■■■■■ |

#### VpcEgressGatewayNodeSelector

| ■■■■ | ■■ | ■■ |
| --- | --- | --- |
| matchLabels | map[String]String | ■■■■■■■ |
| matchExpressions | []NodeSelectorRequirement | ■■■■■■■■ |
| matchFields | []NodeSelectorRequirement | ■■■■■■■■ |

#### BandwidthLimit

| ■■■■ | ■■ | ■■ |
| --- | --- | --- |
| ingress | Int64 ■ String | ■■■■■■■■ VPC ■■■■■■■■■■■■■■■■■■■■■■ Mbps■quantity ■ `M` ■ `Mi` ■ `G` ■ `Gi` bit/s |
| egress | Int64 ■ String | ■ VPC ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ Mbps■quantity ■ `M` ■ `Mi` ■ `G` ■ `Gi` bit/s |

#### VpcEgressGatewayObservability

■■ Kubernetes 1.29 ■■■■■■■■■■■■■■■■■■■■■■■■ schema ■■■■■■■■■■ VPC Egress Gateway ■■■■■

| ■■■■ | ■■ | ■■ |
| --- | --- | --- |
| resources | ResourceRequirements | ■■■observer sidecar ■■■■■■ request ■ `20m` CPU ■ `64Mi` ■■■limit ■ `200m` CPU ■ `256Mi` ■■ |
| interfaceMetrics | VpcEgressGatewayObservabilityFeature | ■■■■■■■■■■ |
| conntrack | VpcEgressGatewayConntrackObservability | conntrack ■■■■■■■■■ |
| serviceMonitor | VpcEgressGatewayServiceMonitor | ■■■■■■■■ ServiceMonitor ■■■■■ |

#### VpcEgressGatewayObservabilityFeature

| ■■■■ | ■■ | ■■ |
| --- | --- | --- |
| enabled | Boolean | ■■■■■■■■■■■ false |

#### VpcEgressGatewayConntrackObservability

| ■■■■ | ■■ | ■■ |
| --- | --- | --- |
| metrics | VpcEgressGatewayObservabilityFeature | ■■■■ NAT conntrack ■■■■■ |
| log | VpcEgressGatewayConntrackLog | JSON Lines NAT ■■■■■■ |



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**VpcEgressGatewayConntrackLog**

| ... | ... | ... |
| --- | --- | --- |
| enabled | Boolean | ... JSON ... observer ... false |
| events | []String | ... `start` ... `end` ... |
| rateLimit | VpcEgressGatewayConntrackLogRateLimit | ... Pod ... |
| filters | VpcEgressGatewayConntrackLogFilters | ...include ... exclude ... |

**VpcEgressGatewayConntrackLogRateLimit**

| ... | ... | ... |
| --- | --- | --- |
| recordsPerSecond | Int32 | ... 1 ... 100000... 100 |
| burst | Int32 | ... 1 ... 1000000... 1000 |

**VpcEgressGatewayConntrackLogFilters**

| ... | ... | ... |
| --- | --- | --- |
| include | []VpcEgressGatewayConntrackLogFilter | ... 64 ... include ... NAT ... |
| exclude | []VpcEgressGatewayConntrackLogFilter | ... 64 ... exclude ...exclude ... |

**VpcEgressGatewayConntrackLogFilter**

... AND ... OR ...

| ... | ... | ... |
| --- | --- | --- |
| addressFamilies | []String | ... `ipv4` ... `ipv6` |
| protocols | []String | ... `tcp` ... `udp` ... `sctp` ... `icmp` ... `icmpv6` ... `other` |
| natTypes | []String | ... `snat` ... `dnat` ... `snat_dnat` |
| original | VpcEgressGatewayConntrackTupleFilter | ... tuple ... |
| translated | VpcEgressGatewayConntrackTupleFilter | ... tuple ... |

**VpcEgressGatewayConntrackTupleFilter**

| ... | ... | ... |
| --- | --- | --- |
| sourceCIDRs | []String | ... CIDR... 64 ... |
| destinationCIDRs | []String | ... CIDR... 64 ... |
| sourcePorts | []VpcEgressGatewayPortRange | ... 64 ... |
| destinationPorts | []VpcEgressGatewayPortRange | ... 64 ... |

**VpcEgressGatewayPortRange**

| ... | ... | ... |
| --- | --- | --- |
| start | Int32 | ... 0 ... 65535 |
| end | Int32 | ... `start` ... 65535 |



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VpcEgressGatewayServiceMonitor

| ... | ... | ... |
| --- | --- | --- |
| labels | map[String]String | ...ServiceMonitor ... selector ... |
| annotations | map[String]String | ...ServiceMonitor annotations |

VPCEGRESSGATEWAYSTATUS

| ... | ... | ... |
| --- | --- | --- |
| replicas | Int32 | ... scale ... |
| labelSelector | String | ... scale ... |
| ready | Boolean | ... |
| phase | String | ... `Pending` ... `Processing` ... `Completed` |
| internalIPs | []String | ... IP ... |
| externalIPs | []String | ... IP ... |
| conditions | []Condition | ... |
| workload | VpcEgressWorkload | ... |

VpcEgressWorkload

| ... | ... | ... |
| --- | --- | --- |
| apiVersion | String | ... API ... |
| kind | String | ... `Deployment` |
| name | String | ... |
| nodes | []String | ... |

### BgpConf

...cluster-scoped... `VpcEgressGateway.spec.bgpConf` ...

| ... | ... | ... |
| --- | --- | --- |
| apiVersion | String | ... Kubernetes ... kubeovn.io/v1 |
| kind | String | ... Kubernetes ...`BgpConf` |
| metadata | ObjectMeta | ... Kubernetes ... |
| spec | BgpConfSpec | BgpConf ... |



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BGPCONFSPEC

| ■■■■ | ■■ | ■■ |
|---|---|---|
| localASN | UInt32 | ■■ BGP AS Number |
| peerASN | UInt32 | ■■ BGP AS Number |
| routerId | String | ■■■BGP Router ID |
| neighbours | []String | BGP ■■ IP ■■ |
| password | String | ■■■TCP MD5 ■■■■ |
| holdTime | Duration | ■■■BGP Hold time |
| keepaliveTime | Duration | ■■■BGP Keepalive ■■ |
| connectTime | Duration | ■■■BGP Connect ■■ |
| ebgpMultiHop | Boolean | ■■■■■■■ eBGP MultiHop |
| gracefulRestart | Boolean | ■■■■■■■ BGP Graceful Restart |

**EvpnConf**

■■■■cluster-scoped■■■■■ `VpcEgressGateway.spec.evpnConf` ■■■■■■ EVPN ■■■■■ VNI ■ Route Target■

| ■■■■ | ■■ | ■■ |
|---|---|---|
| apiVersion | String | ■■ Kubernetes ■■■■■■■■■■■■■■■■■■■■ kubeovn.io/v1 |
| kind | String | ■■ Kubernetes ■■■■■■■■■■■■■■■■■■■■■ `EvpnConf` |
| metadata | ObjectMeta | ■■ Kubernetes ■■■■■■■■ |
| spec | EvpnConfSpec | EvpnConf ■■■■■■■■■ |

EVPNCONFSPEC

| ■■■■ | ■■ | ■■ |
|---|---|---|
| vni | UInt32 | EVPN VNI |
| routeTargets | []String | Route Target ■■ |

**DNSNameResolver**

■■■■cluster-scoped■■■■■ `enable-dns-name-resolver` ■■■ ANP/CNP ■■■■■■■■■■■■ DNS ■■■■■

| ■■■■ | ■■ | ■■ |
|---|---|---|
| apiVersion | String | ■■ Kubernetes ■■■■■■■■■■■■■■■■■■■■ kubeovn.io/v1 |
| kind | String | ■■ Kubernetes ■■■■■■■■■■■■■■■■■■■■■ `DNSNameResolver` |
| metadata | ObjectMeta | ■■ Kubernetes ■■■■■■■■ |
| spec | DNSNameResolverSpec | DNSNameResolver ■■■■■■■■■ |
| status | DNSNameResolverStatus | DNSNameResolver ■■■■■■■ |



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#### DNSNAMERESOLVERSPEC

| ... | ... | ... |
| --- | --- | --- |
| name | String | ... DNS ... `*.` ... |

#### DNSNAMERESOLVERSTATUS

| ... | ... | ... |
| --- | --- | --- |
| resolvedNames | []DNSNameResolverResolvedName | ... DNS ... IP ... |

#### DNSNameResolverResolvedName

| ... | ... | ... |
| --- | --- | --- |
| dnsName | String | ... DNS ... |
| resolvedAddresses | []DNSNameResolverResolvedAddress | ... IP、TTL ... |
| conditions | []Condition | ... `Degraded` ... |
| resolutionFailures | Int32 | ... 5 ... TTL ... |

#### DNSNameResolverResolvedAddress

| ... | ... | ... |
| --- | --- | --- |
| ip | String | ... IP ... |
| ttlSeconds | Int32 | TTL ... `lastLookupTime + ttlSeconds` ... |
| lastLookupTime | Time | ... |

### VpcDns

| ... | ... | ... |
| --- | --- | --- |
| apiVersion | String | ... Kubernetes ... kubeovn.io/v1 |
| kind | String | ... Kubernetes ... `VpcDns` |
| metadata | ObjectMeta | ... Kubernetes ... |
| spec | VpcDNSSpec | VpcDns ... |
| status | VpcDNSStatus | VpcDns ... |

#### VPCDNSSPEC

| ... | ... | ... |
| --- | --- | --- |
| replicas | Int32 | ... |
| vpc | String | ... VPC |
| subnet | String | ... |

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## 8.11.9 ⬚⬚

# 8.12 Annotation □□□□

Kube-OVN □□□□□□ Pod □ Node Annotation □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ Annotation □□□□□□□□□□□□□□□□□□

> □□□□□ Annotation □□□□□□□□□□□□□□□□



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## 8.12.1 Pod Annotation

| Key | Value | Description |
| --- | --- | --- |
| ovn.kubernetes.io/allocated | `true` or `false` | Pod [illegible] |
| ovn.kubernetes.io/routed | `true` or `false` | Pod [illegible] OVN [illegible] |
| ovn.kubernetes.io/routes | String | Pod [illegible] |
| ovn.kubernetes.io/mac_address | String | Pod [illegible] Mac [illegible] Pod [illegible] Annotation [illegible] Mac [illegible] |
| ovn.kubernetes.io/ip_address | String | Pod [illegible] IP [illegible] Pod [illegible] Annotation [illegible] IP [illegible] |
| ovn.kubernetes.io/ip_family | `ipv4` or `ipv6` | [illegible] Pod [illegible] IP [illegible] |
| \<nadName>.\<nadNamespace>.kubernetes.io/ip_address.\<interfaceName> | String | [illegible] IP[illegible]interfaceName [illegible] Multus [illegible] interface [illegible] ovn.kubernetes.io/ip_address [illegible] |
| \<nadName>.\<nadNamespace>.kubernetes.io/mac_address.\<interfaceName> | String | [illegible] MAC[illegible]interfaceName [illegible] Multus [illegible] interface [illegible] |
| ovn.kubernetes.io/cidr | String | Pod [illegible] CIDR |
| ovn.kubernetes.io/gateway | String | Pod [illegible] Gateway [illegible] |
| ovn.kubernetes.io/ip_pool | IP [illegible] | Pod [illegible] Workload [illegible] IP [illegible] |
| ovn.kubernetes.io/bgp | `true`, `cluster`, `local` | [illegible] BGP [illegible] Pod [illegible] |
| ovn.kubernetes.io/snat | String | Pod [illegible] SNAT [illegible] |
| ovn.kubernetes.io/eip | String | Pod [illegible] EIP [illegible] |
| ovn.kubernetes.io/vip | String | Pod [illegible] VIP[illegible] Annotation [illegible] VIP [illegible] |
| ovn.kubernetes.io/aaps | String | Pod [illegible] AAPs (Additional Allowed Addresses Pairs) [illegible] |
| ovn.kubernetes.io/virtualmachine | String | Pod [illegible] VirtualMachineInstance |
| ovn.kubernetes.io/activation_strategy | String | Pod [illegible] |
| ovn.kubernetes.io/logical_router | String | Pod [illegible] VPC |
| ovn.kubernetes.io/layer2_forward | `true` or `false` | Pod [illegible] OVN LSP [illegible] `unknown` [illegible] |
| ovn.kubernetes.io/port_security | `true` or `false` | Pod [illegible] Port Security |
| ovn.kubernetes.io/logical_switch | String | Pod [illegible] |
| ovn.kubernetes.io/vlan_id | Int | Pod [illegible] Vlan ID |
| ovn.kubernetes.io/ingress_rate | Int | Pod [illegible] Mbits/s |
| ovn.kubernetes.io/egress_rate | Int | Pod [illegible] Mbits/s |
| ovn.kubernetes.io/security_groups | String [illegible] | Pod [illegible] Security Group |
| ovn.kubernetes.io/default_route | `true` or `false` | [illegible] |
| ovn.kubernetes.io/provider_network | String | Pod [illegible] ProviderNetwork |
| ovn.kubernetes.io/mirror | `true` or `false` | Pod [illegible] |
| ovn.kubernetes.io/north_gateway | String | Pod [illegible] |
| ovn.kubernetes.io/latency | Int | Pod [illegible] ms |
| ovn.kubernetes.io/limit | Int | Pod [illegible] qdisc [illegible] |
| ovn.kubernetes.io/loss | Float | Pod [illegible] |

| Key | Value | Description |
| --- | --- | --- |
| ovn.kubernetes.io/jitter | Int | Pod [illegible] ms |
| ovn.kubernetes.io/generate-hash | `true` or `false` | [illegible] Pod [illegible] |
| ovn.kubernetes.io/attachmentprovider | String | Pod [illegible] |
| ovn.kubernetes.io/ingress_burst | Int | Pod [illegible] burst [illegible] |
| ovn.kubernetes.io/vpc_nat_gw | String | VPC NAT Gateway Pod [illegible] |
| ovn.kubernetes.io/vpc_nat_gw_init | `true` or `false` | VPC NAT Gateway Pod [illegible] |
| ovn.kubernetes.io/vpc_nat_gw_container_restarted | `true` or `false` | VPC NAT Gateway [illegible] |
| ovn.kubernetes.io/vpc_nat_gw_activated | `true` or `false` | VPC NAT Gateway [illegible] |
| \<provider>.kubernetes.io/ip_pool | IP [illegible] | [illegible] provider [illegible] IP [illegible] |
| \<provider>.kubernetes.io/ip_family | `ipv4` or `ipv6` | [illegible] provider [illegible] IP [illegible] |
| \<provider>.kubernetes.io/dhcp-v4-options | String | [illegible] provider [illegible] DHCPv4 [illegible] |
| \<provider>.kubernetes.io/port_security | `true` or `false` | [illegible] provider [illegible] |

## 8.12.2 Node Annotation

| Key | Value | Description |
| --- | --- | --- |
| ovn.kubernetes.io/allocated | `true` or `false` | [illegible] `ovn0` [illegible] `join` [illegible] |
| ovn.kubernetes.io/mac_address | String | Node `ovn0` [illegible] Mac [illegible] |
| ovn.kubernetes.io/ip_address | String | Node `ovn0` [illegible] IP [illegible] |
| ovn.kubernetes.io/cidr | String | Node `ovn0` [illegible] `join` [illegible] CIDR |
| ovn.kubernetes.io/gateway | String | Node `ovn0` [illegible] `join` [illegible] Gateway [illegible] |
| ovn.kubernetes.io/chassis | String | Node [illegible] OVN-SouthBoundDB [illegible] Chassis ID |
| ovn.kubernetes.io/port_name | String | Node `ovn0` [illegible] OVN-NorthboundDB [illegible] LSP [illegible] |
| ovn.kubernetes.io/logical_switch | String | Node `ovn0` [illegible] |
| ovn.kubernetes.io/tunnel_interface | String | [illegible] |
| ovn.kubernetes.io/node_networks | String | [illegible] Pod [illegible] attachment networks [illegible] NetworkAttachmentDefinition [illegible] |

## 8.12.3 Namespace Annotation

| Key | Value | Description |
| --- | --- | --- |
| ovn.kubernetes.io/cidr | CIDR [illegible] | [illegible] Namespace [illegible] CIDR |
| ovn.kubernetes.io/exclude_ips | excludeIPs [illegible] | [illegible] Namespace [illegible] excludeIPs |

## 8.12.4 Subnet Annotation

| Key | Value | Description |
| --- | --- | --- |
| ovn.kubernetes.io/bgp | `true`, `cluster`, `local` | [illegible] BGP [illegible] |
| ovn.kubernetes.io/vpc | String | [illegible] VPC[illegible] VPC [illegible] |



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## 8.12.5 Service Annotation

| Key | Value | Description |
| --- | --- | --- |
| ovn.kubernetes.io/bgp | `true`, `cluster`, `local` | □□□□□□ BGP □□ Service □□□□ ClusterIP Service □□□□□□ Pod/Subnet □□□ |
| ovn.kubernetes.io/switch_lb_vip | String | Service □ Kube-OVN □□□□□□ VIP □□ |
| ovn.kubernetes.io/vpc | String | Service □□□ VPC |
| ovn.kubernetes.io/service_external_ip_from_subnet | `true` or `false` | Service □□ IP □□□□□□□ |
| ovn.kubernetes.io/service_health_check | `true` or `false` | Service □□□□□□□□ |
| ovn.kubernetes.io/lb_svc_img | String | □□□□□□□□□□□ |

## 8.12.6 Networkpolicy Annotation

| Key | Value | Description |
| --- | --- | --- |
| ovn.kubernetes.io/enable_log | `true` or `false` | □□□□ NetworkPolicy □□ |
| ovn.kubernetes.io/log_acl_actions | "allow,drop,pass" □□□□□□□□ □□ | □□□□ Action ACL □□□ |
| ovn.kubernetes.io/acl_log_meter_rate | Int | NetworkPolicy □□□□□□□□□□□□□□□/□ |
| ovn.kubernetes.io/network_policy_enforcement | `standard` or `lax` | □□ NetworkPolicy □□□□□□□□□□□□□□ |
| ovn.kubernetes.io/network_policy_for | String | □□□ NetworkPolicy □□□□□□□ provider□□□ `<nadName>.<nadNamespace>.ovn` □ |

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## 8.13.1 □□

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| --- | --- |
| Kube-OVN □□□□□□□□□□□□□□ 1.10 □□ Kube-OVN□ | Kube-OVN □□□□□□□□□□□□□□ 1.10 □□ Kube-OVN□ |

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| `wget 127.0.0.1` | `wget 127.0.0.1` |

## 8.13.2 □□□

yaml □□□□□□□□ yaml□

| Bad | Good |
| --- | --- |
| ````<br>apiVersion: kubeovn.io/v1<br>kind: Subnet<br>metadata:<br>    name: attach-subnet<br>```` | ````yaml<br>apiVersion: kubeovn.io/v1<br>kind: Subnet<br>metadata:<br>    name: attach-subnet<br>```` |

□□□□□□□□□□□□□□□□□ bash□

| Bad | Good |
| --- | --- |
| ````<br>wget 127.0.0.1<br>```` | ````bash<br>wget 127.0.0.1<br>```` |



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▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯ `#` ▯▯▯▯▯▯▯▯▯▯▯▯▯

| Bad | Good |
| --- | --- |
| oilbeater@macdeMac-3 ~ ping 114.114.114.114 -c 3<br>PING 114.114.114.114 (114.114.114.114): 56 data bytes<br>64 bytes from 114.114.114.114: icmp_seq=0 ttl=83 time=10.429 ms<br>64 bytes from 114.114.114.114: icmp_seq=1 ttl=79 time=11.360 ms<br>64 bytes from 114.114.114.114: icmp_seq=2 ttl=76 time=10.794 ms<br><br>--- 114.114.114.114 ping statistics ---<br>3 packets transmitted, 3 packets received, 0.0% packet loss<br>round-trip min/avg/max/stddev = 10.429/10.861/11.360/0.383 ms | # ping 114.114.114.114 -c 3<br>PING 114.114.114.114 (114.114.114.114): 56 data bytes<br>64 bytes from 114.114.114.114: icmp_seq=0 ttl=83 time=10.429 ms<br>64 bytes from 114.114.114.114: icmp_seq=1 ttl=79 time=11.360 ms<br>64 bytes from 114.114.114.114: icmp_seq=2 ttl=76 time=10.794 ms<br><br>--- 114.114.114.114 ping statistics ---<br>3 packets transmitted, 3 packets received, 0.0% packet loss<br>round-trip min/avg/max/stddev = 10.429/10.861/11.360/0.383 ms |

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| Bad | Good |
| --- | --- |
| # mv /etc/origin/ovn/ovnnb_db.db /tmp<br># mv /etc/origin/ovn/ovnsb_db.db /tmp | mv /etc/origin/ovn/ovnnb_db.db /tmp<br>mv /etc/origin/ovn/ovnsb_db.db /tmp |

## 8.13.3 ▯▯

▯▯▯▯▯▯▯▯▯ `md` ▯▯▯▯▯▯

| Bad | Good |
| --- | --- |
| [ ](http://kubeovn.github.io/prepare) | [ ](./prepare.md) |

| Bad | Good |
| --- | --- |
| [ Kubernetes ](http://kubernetes.io) | [ Kubernetes ](http://kubernetes.io){: target="_blank" } |

## 8.13.4 ▯▯

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| Bad | Good |
| --- | --- |
| ```bash<br>wget 127.0.0.1<br>``` | ```bash<br>wget 127.0.0.1<br>``` |

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| --- | --- |
| ```bash<br>wget 127.0.0.1<br>``` | ```bash<br>wget 127.0.0.1<br>``` |

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## 8.13.5 □□



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## 9.1 □□□□

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- Issue / □□□□ GitHub Issues □□□□□GitHub Discussions □□□□□□

## 9.2 □□□□

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## 9.3 □□

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## 9.4 □□



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